Shadow Mirror

A Reference Architecture for Reflective Human–AI Interaction

Full Research Monograph

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Programme: 18 sessions, complete · Package `shadow_mirror` **v1.3.0 · 25 July 2026**

About this monograph

This volume collects the complete Shadow Mirror research programme: the unified draft, the two venue-targeted papers, the formal appendices, all eighteen session reports, and the submission material. It is assembled from the project archive rather than written over it, so where two documents disagree, both appear and the disagreement is recorded in the outstanding-work appendix.

What the system is. An architecture for reflective human–AI interaction. It observes what a person says they want and what they do, and models the gap. The central design claim is that when those disagree, the honest output is frequently a **refusal** rather than a better-hedged guess — because approach under stated reluctance is produced by hidden preference, and it is *also* produced by compliance under perceived pressure, and the two are not distinguishable from the surface signal.

The measured result. 0 of 800 simulated threat episodes misread, with a 95% upper bound of 0.37% by the rule of three, against 42.9% for a mismatch-only reader and 75.1% for an arousal-as-desire reader.

The measured cost. A supervised four-channel classifier, given ground-truth labels, beats the guard at low channel separation — 5.0% against 29.7%. That is in the monograph because a reviewer would find it otherwise. The defence is that those labels do not exist in deployment.

The ground rules

These governed every session, and most of what is unusual about this work follows from them.

1. **No fabricated evidence, ever.** Every number in a paper is produced by code in the repository.
2. No human-subjects data is collected. Simulated participants are never called a result.
3. No text enters a manuscript in the past tense describing work not performed.
4. **No number enters a manuscript unless a file in the repository produces it AND the person writing the sentence has run that file.**
5. No additional evidence channel is added until the guard **refuses** rather than silently degrading.
6. **No citation enters a manuscript unless confirmed against a live source** with a resolvable identifier. Failure to confirm is reported as failure, never filled in from memory.
7. No instrument is administered whose permission terms are unconfirmed, and no item text is bundled with the software.
8. **Nothing about a person enters a model unless that person declared it or self-completed the instrument it came from.** Absence is not refusal — a refusing method must exist and explain itself.
9. **Nothing is recorded as pending until a search for a substitute has come back empty.**

How to read this

If you have five minutes, read the unified draft's abstract and then **Part VIII**, the outstanding-work appendix — it lists what is blocked, what a search resolved, and what would change the conclusions.

If you have an hour, read **Part I** (the unified draft) and then **Session 13**, which attacks it.

If you want to run something:

```
pytest -q                # 707 passed, 2 xfailed
python verify_claims.py  # re-derives the claims used in Part I
python demo.py
```

The package has **zero runtime dependencies** by design — an auditability commitment, not an aesthetic one, and the reason NetworkX was removed in Session 4.

Part I — The unified draft

Shadow Mirror: A Consent-Gated Architecture for Reflective Human–AI Interaction

Executable Invariants, Refusal as a Safety Mechanism, and the Threat Confound

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Unified research paper draft · Package `shadow_mirror` **v1.3.0 · 25 July 2026**

Abstract

Systems that infer what a person "really" wants from the gap between what they say and what they do face a hazard that is rarely named: the same surface contradiction is produced by a hidden preference and by a threat response. Telling someone they secretly want something they are afraid of is a specific, foreseeable harm, and no published affect-aware architecture reports its rate.

This paper unifies two strands of the Shadow Mirror programme. The first is architectural. Reference architectures for human–AI interaction state their commitments — traceability, provenance, human authority — in prose, where a reader cannot tell whether an implementation keeps them. A property that cannot be checked cannot be violated *detectably*, and a commitment that cannot be violated detectably is not a commitment. We specify **eleven architectural invariants as decidable properties with a supplied decision procedure**: each is a callable check returning a verdict and the observation that produced it. The second strand is ethical. We make **refusal a first-class output**, hold contradiction open rather than resolving it, and represent a threat response as a competing hypothesis rather than as noise.

The two strands are one system because the safety result depends on the structural commitments. Shadow Mirror misreads a threat response as a hidden desire in **0 of 800 simulated episodes** (95% upper bound 0.37% by the rule of three), against **42.9%** for a stated-versus-behaviour mismatch heuristic and **75.1%** for a baseline admitting physiological arousal into the action channel. That result rests on representing a competing hypothesis rather than filtering, and on distinguishing an absent channel from a zero reading — both properties of the graph substrate, not of a classifier.

We are explicit about what the architecture does not buy. On latent-state recovery it is statistically indistinguishable from a median split across all 20 tested conditions. A supervised four-channel classifier, given ground-truth labels our system never sees, beats us at low channel separation — 5.0% against our 29.7%. Those labels do not exist in deployment, which makes the comparison an upper bound on what any supervised method could reach rather than an available alternative. We report both because a reviewer would find them otherwise.

This system is a research instrument. It is not a diagnostic device and is not suitable for clinical, educational, employment, credit, insurance or policing use.

Keywords: reference architecture, executable specification, architectural invariants, refusal, abstention, consent, affective inference, threat response, provenance, belief revision, EU AI Act.

1. Introduction

Architecture papers make promises. *Every conclusion traces to evidence. Provenance is preserved. The system does not silently replace human judgment.* These are good promises, and they are why reference architectures are worth writing. They are also, almost always, unfalsifiable as stated. A reader cannot determine whether an implementation keeps them, an author cannot determine whether a refactor broke them, and a reviewer cannot distinguish an architecture that enforces them from one that describes them.

When the architecture is intended to support a person reasoning about themselves, that gap stops being academic. A large class of deployed systems infers preference from behavioural inconsistency. Recommender systems treat repeated engagement as revealed preference regardless of stated intent. Implicit-measure tools treat reaction-time asymmetries as leanings a person will not admit. Affect-aware interfaces treat physiological arousal as intensity of wanting. In each case the inference has the same structure: *you say one thing, you do another, therefore the second is the truth about you.*

That inference has a failure mode with an established clinical and evolutionary literature and no name at all in machine learning. Approach under stated reluctance is produced by hidden preference. It is also produced by threat — by the specific pattern in which a person moves toward something because moving away is unsafe. Arousal is produced by excitement. It is also produced by fear. The two are not distinguishable from the surface signal, and the cost of confusing them is asymmetric. Reading a hidden preference as a threat response wastes an opportunity. Reading a threat response as a hidden preference tells a person that their fear was a want — which is, precisely, what a person recovering from coercion has most often already been told.

We began from an architecture in the usual condition: invariants specified in prose, with a prototype whose relationship to them was undocumented. Testing the prototype against its own invariants found that **provenance preservation was violated** — evidence derived from two sources reported one parent and silently discarded the other — and that belief revision was **order-dependent to the tune of 95 percentage points**. Both defects were present in a document that asserted the invariants held.

Contributions.

1. **Eleven architectural invariants as decidable properties**, with a supplied decision procedure, a randomised property harness, and a negative test per failure mode (§5).
2. **A reflective graph substrate** with multi-parent provenance closure, contradiction as a first-class edge, transactional updates, and no third-party dependencies (§4.1).
3. **An order-invariant, exactly retractable belief operator**, with its axioms stated explicitly and enforced by test (§4.2).
4. **A measured harm rate** for the threat confound, with two baselines reflecting current practice, and an ablation showing the conditions under which the protection fails (§6).
5. **Consent as a mechanism rather than a policy**: a hash-chained append-only ledger that answers what was granted at a past instant, and a deployment gate that refuses the biometric stream in institutional contexts (§7).
6. **Two negative results reported as such**: the architecture buys no accuracy, and a supervised baseline beats our guard where labels exist (§8).

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2. Conceptual foundations: the harm, stated precisely

Consider a person who states that they do not want something. Five mechanisms can produce that statement together with the behaviour that follows. Three are genuine preferences. Two are not.

Mechanism	Stated	Behaviour	Arousal	Avoidance	Is it a preference?
Explicit aversion	low	withdraws	low	low	yes — they do not want it
Latent approach	low	approaches	low	low	yes — they want it and have not said so
Latent approach, aroused	low	approaches	**high**	low	yes — excitement
Threat response	low	withdraws	**high**	**high**	**no** — fear
Threat compliance	low	**approaches**	**high**	moderate	**no** — complies under threat

The fifth row is the dangerous one. **Threat compliance and latent approach are indistinguishable on the stated and behavioural channels alone.** Both show stated aversion and approach behaviour. A system reading only those two channels — which describes most deployed preference inference — cannot tell them apart, and will call the compliance case a hidden want.

The third and fourth rows are the second trap. Excitement and fear both raise arousal. A system that treats arousal as intensity-of-wanting will call the threat case a strong want.

Our target metric is the rate at which a system outputs `latent_approach` — *"you want this even though you say you don't"* — when the true generating mechanism was a threat process. We call it the **false hidden-desire rate**.

2.1 Why this is an architectural problem rather than a modelling one

The natural response is a better classifier. That response fails for a reason worth stating: the distinguishing evidence is frequently *not collected*, and a model has no way to represent the difference between a channel that read zero and a channel that was never read. Imputing a missing avoidance measure to zero is imputing *evidence of safety*. The fix is structural — an absent channel must be representable as absent, and a competing hypothesis must be representable as competing rather than filtered away — which is why this paper is about an architecture and not about a model.

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3. Research objectives

The programme asks four questions, and answers three of them.

	Question	Status
O1	Can architectural commitments about reasoning be made decidable, with a decision procedure a third party can run?	**Answered** (§5). Eleven invariants, 200-seed randomised harness, negative test per failure mode
O2	What is the rate at which a reflective system misreads a threat response as a hidden desire, and does refusal remove it?	**Answered in simulation** (§6). 0 of 800 episodes; two baselines at 42.9% and 75.1%
O3	Can consent, and the prohibitions a system declares, be enforced by mechanism rather than stated as policy?	**Partly answered** (§7). Six of twelve prohibitions machine-enforced
O4	Does any of this change how a person reasons about themselves?	**Not answered.** No human-subjects data exists. See §11 and Appendix B

O4 is the one that matters most for the system's stated purpose, and it is the one with no evidence. That asymmetry is deliberate: nothing in this paper is a statement about people.

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4. Architecture

4.1 The Unified Reflective Graph

Typed nodes — Observation, Evidence, Context, Hypothesis, Belief, Inquiry, Explanation, Knowledge, Limitation — and typed edges — provenance, supports, contradicts, contextualises, revises, targets, explains, limits. Untyped insertions are rejected: an untyped graph cannot support a traceability invariant, because there is nothing to quantify over.

Provenance is a DAG. `ancestors()` returns the transitive closure over *all* provenance parents. The prototype followed the first predecessor only, recovering 0 of 2 parents from a two-parent derivation. The formalism licensed it: the evidence object was specified with provenance as a scalar attribute. Writing it as a set is the fix, and the defect is a case study in formal notation shaping implementation error — the provenance literature had the multi-parent answer three decades earlier.

Contradiction is an edge, not an overwrite. `contested_hypotheses()` returns hypotheses carrying both supporting and contradicting evidence, so an explanation can report a tension rather than preferring the later write.

Updates are transactional. A `transaction()` context manager snapshots and rolls back, so a failure mid-update cannot leave a hypothesis whose belief moved but whose provenance edge was never written.

No third-party dependencies. The graph is an adjacency structure in the standard library. This is not minimalism for its own sake: the architecture's claim is local-first auditability, and importing a graph framework for a dictionary of lists would trade that for nothing. NetworkX was removed for this reason.

4.2 Belief revision

Leaving the belief state's type open — "probability distributions, weighted graphs, logical structures" — and then asserting properties of it is a common and serious error. Properties cannot be asserted of an object whose type is open. We state the axioms instead:

- **B1 Plurality** — assigns a value to every hypothesis simultaneously; no distinguished maximiser.
- **B2 Commutativity** — order of conditionally independent evidence does not matter.
- **B3 Invertibility** — evidence can be retracted exactly.
- **B4 Audit** — every application records prior, posterior, trigger, method, time.

B3 is not a convenience. **A system that cannot undo an inference cannot honour a withdrawal of consent**, which turns an ethical commitment into an architectural requirement.

Belief accumulates in log-odds keyed by evidence identity, which yields B2 (addition commutes), B3 (subtract the term) and idempotence under resubmission. The prototype clamped the posterior to [0.01, 0.99] *in state* after each update. Clamping is not invertible: once pinned, the information about how far past the bound the evidence pushed is destroyed. Five observations in three orders produced final beliefs of **0.0396, 0.9900 and 0.9900**.

The current package clamps only at display. Re-running all **120 permutations** of a five-observation sequence in this pass produced **a single distinct posterior**, with a maximum spread across permutations of **0.0** — exact agreement, not agreement to tolerance (`verify_claims.py` §2).

The independence assumption, stated. Sequential updating requires conditional independence. In reflective domains this is frequently false: two self-reports on consecutive days are correlated through mood, framing and recall, not only through the latent state. Keying contributions by evidence identity prevents resubmission compounding, and temporal decay down-weights clustering, but neither substitutes for a correlated-likelihood model. We state the assumption rather than rely on it silently.

4.3 Inquiry planning

Candidate questions are scored by expected information gain over the current belief, normalised by cost. The heuristic this replaces scored inquiries as one minus the absolute belief gap — a quantity that does not depend on the candidate observation at all, and therefore carries no information about what to ask.

Normalised regret against an information-theoretic oracle, 500 trials per condition:

Planner	Hypotheses close (gap 0.04)	Hypotheses separated (gap 0.40)
Expected information gain	**0.161 [0.138, 0.187]**	**0.195 [0.168, 0.223]**
Belief-gap heuristic	0.579 [0.547, 0.612]	**1.000 [1.000, 1.000]**
Random selection	0.550 [0.515, 0.584]	0.562 [0.528, 0.597]

Two behavioural points matter more than the numbers. The heuristic is **indistinguishable from random** when it fires, and it emits nothing at all once hypotheses are separated — which is backwards, because a confidently held wrong hypothesis is exactly when a discriminating observation is worth most.

4.4 Temporal modelling

Observations are timestamped and decay in influence, so that outdated evidence does not dominate. The decay is not cosmetic: with a 30-day half-life, decade-old evidence produces an inference in **0.0%** of trials — the system declines rather than reporting a confident conclusion drawn from a person who no longer exists in the data.

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5. Invariants as an executable specification

5.1 The eleven invariants

	Invariant	Operationalisation
I	Evidence traceability	no explanation node lacks a reachable evidence node
II	Provenance preservation	every derived evidence node recovers **all** parents under transitive closure
III	Revision history	every belief transition records prior, posterior, triggering evidence, method, time
IV	Explicit uncertainty	no explanation is emitted without a non-empty uncertainty field
V	Hypothesis independence	contradicting one hypothesis removes or merges no competitor
VI	Context consistency	superseding a context annotates every dependent conclusion as stale
VII	Explanation integrity	explanations carry summary, uncertainty and assumptions
VIII	Inquiry justification	every inquiry quantifies expected gain and names the hypotheses it separates
IX	Transactional consistency	a failed update leaves no partial state
X	Human authority	no output phrased as an identity claim or directive; each carries uncertainty, a disclaimer and a question

	Invariant	Operationalisation
XI	Declared-only personhood	no attribute about a person exists unless they declared it or self-completed the instrument it came from; and no route exists to a global rank, a prescribed deficit, or an attributed purpose

Two required judgement rather than transcription.

VI cannot mean "recompute the past." An instrument cannot re-run a person's week. We operationalise it as annotation: when a context is superseded, every conclusion that depended on it is marked stale — neither silently retained nor silently deleted. Retention is the failure mode that produces confident conclusions drawn under conditions that no longer hold.

X is a property of what reaches the person, so it is checked at the output boundary rather than in the reasoning. Each terminal output must carry its uncertainty and disclaimer, offer at least one question, and avoid identity claims (*you are…*, *you secretly…*) and directives (*you must…*).

Enumerating the checks in the installed package in this pass returns **eleven** invariant checks plus an aggregate `check_all` (`verify_claims.py` §3).

5.2 The harness, and the negative tests

`build_well_formed(seed)` constructs a randomised reflective state: observations, derived evidence with multiple provenance parents, hypotheses with mixed supporting and contradicting edges, executed belief updates, planned inquiries, and an explanation. Invariants I–X are asserted across 200 seeds.

Invariant XI needed a second generator, because it ranges over a formation model rather than a graph. It is asserted across 200 randomised models, with 100 further randomised checks on two derived properties. Most instrument administrations in those models *refuse*, and that is the point: XI must hold whatever the model has been through, including the paths where every administration was declined.

Fourteen negative tests construct states that violate each invariant and assert the check catches them, alongside **four positive controls** — states that should pass and do. Earlier drafts reported "sixteen negative tests", which counted the positive controls among them; the correction is made explicitly rather than silently, because a paper about checkable claims should not miscount its own checks. Without these, a suite reporting "11/11 invariants hold" is unfalsifiable in exactly the way the prose was.

5.3 What happened when we pointed it at real output

The Invariant X check failed on its first contact with the shipped reflection layer. It flagged:

> *"You have consulted this 5 times. Seeing a guess changes later answers and can become self-fulfilling."*

as a verdict, on the pattern `you have`. That sentence is a factual statement about the person's own logged behaviour, and it is a caution the architecture specifically wants surfaced. **The defect was in the check.** We narrowed the patterns to identity claims and directives and recorded the reasoning in source.

We report this because it is the argument for the method. The check was written against a specification, passed against fixtures, and was wrong. It became correct only when aimed at output a person would actually see. A conformance suite that is never pointed at real output will pass, and will be worthless.

5.4 Do the checks discriminate between implementations?

A specification that only its own author's implementation can satisfy has demonstrated nothing.

The same decision procedure was run unmodified against an independently written implementation of the same architecture — a NetworkX-based prototype produced separately from this work. It passes four checks and **fails three**: provenance preservation (0 of 2 parents recovered), order invariance (spread 0.9504 across permutations), and inquiry justification (no information-gain content).

	Prototype	Current package
I Evidence traceability	pass	pass
II Provenance preservation	**fail** — 0/2 parents recovered	pass — 2/2
III Revision history	pass	pass
IV Explicit uncertainty	pass	pass
V Hypothesis independence	pass	pass — contradiction retained
VIII Inquiry justification	**fail** — no information-gain content	pass
Order invariance (120 permutations)	**fail** — spread 0.9504	pass — spread 0.0
	4/7	**7/7**

The procedure separates two implementations of one architecture, which is the minimum evidence that it measures conformance rather than restating an API. This is two implementations, not many, and §11 records that limitation.

5.5 An invariant that exists because of a request

Invariant XI was not derived from the architecture. It was derived from being asked for something.

A user of this system asked it to "determine and understand who people probably are according to their will, desire and need — where they come from, how mature they are, what they still need to be mature enough, what their eventual purpose probably is", with the qualification *"I would not say judge people."* That qualification is the interesting part: the request came with an intuition that something in it was wrong, and no way to say which part.

It is five requests, and they separate cleanly on whether a validated construct exists and who supplies it.

Ask	Disposition	Ground
will, desire, need	**built**	validated self-report exists; the person supplies it
where they come from	**declared only**	inference of special-category data; on biometric input, prohibited by AI Act Art. 5(1)(g)
how mature they are	**refused as a rank**	the rank *is* the construct, not a summary of it
what they still need	**refused**	a directive about identity; Invariant X already forbids it
eventual purpose	**refused**	no validated construct exists; any output would be invented

The maturity row took the most work. The instruments that measure developmental position — Loevinger's Total Protocol Rating, Kegan's subject-object position, Cook-Greuter's Center of Gravity — each yield a single ordered position in which the rank *is* the theory's claim, so a profile drawn from one would misrepresent it. We dropped the construct rather than softening it, and substituted instruments that never claim a stage.

What makes this architectural rather than a policy note is that a refusal stated in prose decays exactly as the original invariants did. Condition (b) of the Invariant XI check does not inspect source code or match names; it **calls** each method that ought to refuse and requires a typed refusal carrying the expected ground. Two failure modes are caught. A method that returns a value fails, obviously. A method that has been *deleted* also fails — and that is the less obvious commitment, because **absence is not refusal**. A caller who finds no `eventual_purpose` attribute learns nothing; a caller who calls it and reads why it declined learns what the architecture will not do and on what ground. The refusing methods therefore exist in order to refuse, which is a deliberate inversion of the usual advice to delete dead code.

Exercising these in this pass, nine refusing methods were called and **all nine raised a typed refusal with a stated ground; none returned a value** (`verify_claims.py` §6). What replaces the refused asks is not silence: `open_questions()` returns what the model does not know, phrased as invitations, and is guarded by a test that fails if the phrasing drifts into *lack*, *deficit*, *should* or *must*.

What this does not establish. A catalogue of refusal grounds is not a theory of which requests a reflective system should decline, and we do not have one. The check verifies that the named refusals refuse; it cannot verify that the catalogue is complete, and a request we did not anticipate will not be caught.

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6. Refusal as a safety mechanism

6.1 Four design commitments

C1. Refusal is a first-class output. The system emits `unresolved_approach` — a named state, not a low-confidence label — when behaviour approaches under stated reluctance and the evidence needed to exclude a threat response was not collected. It also refuses entirely below a minimum of paired evidence.

C2. Absent is not zero. An unobserved channel is recorded as unobserved, never imputed to a value, because imputing arousal or avoidance to zero is imputing *evidence of safety*.

C3. Threat is a competing hypothesis, not a filter. `threat_response` occupies the same hypothesis space as the preference readings and is scored against them, rather than applied as a post-hoc suppression rule.

C4. Contradiction is held, not resolved. Supporting and contradicting evidence are both retained as typed edges; a hypothesis carrying both is reported as contested rather than collapsed.

The output is a distribution over named states with an explicit uncertainty, or a refusal. There is no scalar score anywhere in the API.

How this differs from abstention as the literature defines it. Selective classification, learning to defer and confidence thresholding all abstain because a *confidence* is low — the model is unsure, so it declines, and the threshold is a knob trading coverage against accuracy. Our guard abstains because a *specific alternative explanation cannot be ruled out with the evidence that was collected*. It is not tunable by a confidence parameter, it does not move when the model becomes more certain, and it fires just as readily on a high-confidence reading. A selective classifier asks "how sure am I?"; the guard asks "which question was never asked?"

6.2 Method

Five generative processes, defined by target values on four channels (stated, action, arousal, avoidance), with observations drawn from Beta distributions around those targets. Ground truth is the generating process, which exists independently of the inference model.

Misspecification control. The inference model assumes each latent state sits at a fixed corner of a signal space. Generating from exactly those corners would test self-consistency rather than validity, so every generating centre is displaced by δ in a random direction.

Baselines, both implemented and executed at benchmark time. *Mismatch-only* uses stated versus behaviour with no confound channels, representing systems that infer preference from behavioural inconsistency alone. *Arousal-as-desire* folds physiological arousal into the action channel; this is not a straw man, as treating autonomic arousal as an approach signal is documented practice.

n = 400 episodes per process per condition, 6 observation rounds each, all rates carrying 2,000-resample percentile bootstrap intervals. The suite is deterministic: two full runs produce byte-identical output.

What is not measured. No human participants were involved at any stage. Every result below is a statement about simulated signal streams with known generating mechanisms, and none is a statement about people.

6.3 The safety result

δ = 0.10, both confound channels observed, n = 400 per process:

Method	False hidden-desire rate	True hidden preference detected
Shadow Mirror	**0.0% [0.0, 0.0]**	87.1% [84.9, 89.4]
Mismatch-only baseline	42.9% [39.5, 46.2]	100.0% [100.0, 100.0]
Arousal-as-desire baseline	75.1% [72.1, 78.1]	89.5% [87.4, 91.8]

Guard sensitivity 100.0% [100.0, 100.0]; specificity 94.4% [93.2, 95.7]. At δ = 0.00 the three rates are 0.0% [0.0, 0.0], 47.4% [44.0, 50.9] and 75.8% [72.8, 78.6], with specificity 97.4% [96.5, 98.3].

Per process, δ = 0.10:

True process	Shadow Mirror	Mismatch-only	Arousal-as-desire
`threat_response`	**0.0%**	0.0%	50.5%
`threat_compliance`	**0.0%**	**85.8%**	99.8%

The compliance row is the finding. A system reading stated-versus-behaviour alone gets the withdrawal-type threat response right — it looks like aversion — and gets the *compliance* case wrong **85.8%** of the time. That is the case where a person approaches because approaching is safer, and it is the case where being told "you actually want this" does the most damage.

The cost, stated plainly. Sensitivity to genuine hidden preference is 87.1% against the baseline's 100.0%. The entire loss is in the high-arousal approach case, detected 74.2% [70.0, 78.5] of the time, where excitement is mistaken for fear. Roughly 13 points of sensitivity buys the elimination of the misreading. We regard that as the correct side of the trade for an instrument that hands its output to a person about themselves, but it is a trade and not a free lunch.

6.4 The ablation: what "absent is not zero" is worth

Deployments frequently lack an avoidance measure. We ablate that channel and compare two treatments of the missing data.

	Imputes missing channel as zero	Records it as absent (ours)
Guard sensitivity	**0.0% [0.0, 0.0]**	**100.0% [100.0, 100.0]**
False hidden-desire rate	**31.2% [28.1, 34.4]**	**0.0% [0.0, 0.0]**
…on `threat_compliance`	62.5% [57.8, 67.2]	0.0% [0.0, 0.0]
Specificity	100.0%	67.0% [64.4, 69.8]
True hidden preference detected	99.2%	0.0% — the system refuses
AUC, fear vs excitement	—	0.835

Imputing a missing channel to zero silently disables the guard: sensitivity falls to zero and the compliance case is misread 62.5% of the time, with nothing in the output indicating that the protection is inactive.

Recording it as absent produces the opposite failure and the better one. The system stops asserting hidden preference entirely, and specificity falls to 67.0% because arousal alone cannot separate fear from excitement,

so the caution fires on excitement too.

This is the central design claim. Under partial observation a reflective system should refuse, not degrade; and the refusal must be visible in the output, not implicit in a number that happens to be low.

6.5 Refusal calibration

Refusal is only defensible if it is calibrated. n = 300 per condition:

Condition	Behaviour
Evidence genuinely insufficient	refuses **100.0% [100.0, 100.0]**
Evidence sufficient and clear	over-refuses **0.0% [0.0, 0.0]**
Evidence ample but ambiguous	does not refuse; reports uncertainty **0.405 [0.391, 0.419]** against **0.021 [0.020, 0.021]** on clear evidence
Evidence present but expired (decade-old, 30-day half-life)	infers **0.0% [0.0, 0.0]** of the time

The system refuses when it should, does not refuse when it should not, and separates ambiguous from clear evidence by a factor of nineteen in reported uncertainty.

6.6 Ambivalence is a state, not a residual

"You want and do not want this at once" is a claim about a person. Computed from dispersion alone it fires on anything spread out: **95.5%** of pure measurement noise and **98.8%** of monotonic drift — that is, of someone who has *made up their mind* — read as ambivalence, for a precision of **34.0%**. Gating on lag-1 autocorrelation, which distinguishes alternation (−0.844) from drift (+0.517) and noise (−0.120) where dispersion cannot (0.357 / 0.245 / 0.257), raises precision to **97.6%** with recall unchanged at 100.0%.

6.7 Adversarial robustness: two checks a sceptical reader should demand

"Your simulator is your theory." The generative processes were defined so that threat and approach differ on arousal and avoidance, and the guard keys on arousal and avoidance. A 0.0% rate under those conditions risks being a tautology. So we swept the *separation* between the two processes on the confound channels, at n = 300 per process per level.

Separation	AUC	Shadow Mirror	Four-channel classifier	Mismatch-only
1.00	1.000	**0.0%** [0.0, 0.0]	**0.0%** [0.0, 0.0]	94.3% [91.7, 96.7]
0.75	1.000	**0.0%** [0.0, 0.0]	**0.0%** [0.0, 0.0]	95.3% [92.7, 97.7]
0.50	1.000	2.3% [0.7, 4.0]	**0.0%** [0.0, 0.0]	96.7% [94.7, 98.7]
0.25	0.988	29.7% [24.7, 35.0]	**5.0%** [2.7, 7.7]	97.3% [95.3, 99.0]
0.00	0.491	76.7% [71.7, 81.3]	82.0% [77.7, 86.3]	96.3% [94.0, 98.3]

The headline is not a knife-edge: it survives to roughly half the original channel separation. But it is **entirely contingent on the confound channels carrying signal**, and at zero separation every method fails, as it must — the classes are then identical on every observable, and any method reporting 0.0% would be reporting a bug.

"Your baselines are straw men." They partly were. The table therefore adds a baseline a competent engineer would build: a threshold classifier over the same four channels, with its threshold **fitted on a held-out labelled split**. It matches us at high separation and **beats us at separation 0.25 — 5.0% against our 29.7%.**

We report this because it is true and because the reason matters: our guard uses a fixed, pre-specified threshold, while the classifier was handed ground-truth labels for "was this a threat response". **Those labels do not exist in deployment.** Nobody can tell a running system, for a given person and episode, which mechanism produced their behaviour; that is the entire difficulty. The supervised baseline is therefore an *upper bound on what any labelled method could achieve*, not an available alternative — and the gap at separation 0.25 is a real, quantified cost of not having supervision, which we would rather state than omit.

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7. Consent, governance and data ownership

7.1 Consent as a mechanism

The system's other-person mode requires granular, per-stream, purpose-locked, revocable consent before any datum is recorded. Streams a subject cannot notice about themselves — implicit measures, physiological arousal, micro-avoidance — cannot be granted under a blanket authorisation; each requires its own acknowledgement. Every stream declares in plain language both what it records and what it *may reveal*. No grant, no observation: the engine raises rather than records.

Three critiques bound what any such mechanism can claim, and we do not claim past them. Privacy is governed by norms of information flow appropriate to a context, so consent obtained in one context does not license a flow into another. Privacy self-management does not scale. Post-GDPR consent interfaces are routinely engineered to produce agreement rather than to elicit it. We therefore report an **enforcement ratio** rather than a consent rate: the question is not how many people agreed but how many refusals the mechanism actually produces.

7.2 The consent ledger

A mutable grant object cannot demonstrate that consent was given at a past date: a stream added and later removed leaves no trace, and a grant revoked and re-issued looks identical to one that never lapsed. `ledger.py` records every consent event — granted, stream added, re-affirmed, lapsed, revoked, erased, context declared, refused — as an immutable, hash-chained entry.

Exercised in this pass (`verify_claims.py` §5):

- `streams_granted_at(subject, when)` answers the demonstrability question directly. Queried at the instant a second stream was added it returned the streams then under grant; queried after revocation it returned none.
- `verify()` re-derives the chain and returned `True`. Editing a single historical entry's detail and re-verifying returned `False`. This is tamper *evidence*, not tamper proofing — the subject owns their own file and that is the point — but a controller claiming consent on a date can show a chain that verifies, and an edited one will not.
- **Erasure retains the record, not the data.** The ledger records *that* observations were erased. That record is the evidence the withdrawal was honoured.
- `withdrawal_receipt()` returns what was granted, when it was withdrawn, how many records were erased, and whether the chain still verifies — something the subject keeps. Granting was a guided flow ending in a summary; revocation was a single call ending in silence, and those are not equally easy experiences however few keystrokes each takes.

7.3 The deployment gate

Emotion inference in a workplace or education institution is prohibited under EU AI Act Art. 5(1)(f), applicable since 2 February 2025. A `DeploymentContext` declares where the instance is running — `self`, `consented_individual`, `institutional` or `undeclared` — and gates the streams that supply the biometric basis for affective inference.

Three design decisions are load-bearing, and all three were confirmed by execution in this pass (`verify_claims.py` §4):

Refusal is the default. An `undeclared` context is treated as institutional and refused. An operator who has not said where the system is running has not established that the prohibition does not apply, and the safe reading of silence is the restrictive one. A `self` context is permitted.

The safety exception requires a written basis, not a flag. An empty `safety_exception_basis` still refuses; a non-empty one permits. A boolean can be set by accident or by a config default; a sentence has to be written by someone. The stated basis is recorded and exported with the consent ledger.

The refusal explains itself. The exception message carries the article numbers, the applicability date, the declared context, and the exact call that would claim the exception — because a refusal a developer cannot act on becomes a refusal a developer removes.

A Python module does not make anyone compliant, and the module says so in its own output. What it does is convert a prohibition from a wish into a mechanism.

7.4 The honest enforcement ratio

Of twelve documented prohibitions, six are enforced by a mechanism and covered by a test; six are policy only. The one that moved most recently is the only one currently unlawful to breach in the EU. We suggest that papers proposing ethical architectures should print this ratio. A list of prohibitions communicates intent; the fraction of them a machine will actually refuse communicates the safeguard.

7.5 Data governance

Storage is local and file-based. The subject owns the file, can export it, and can delete it. The package has **zero runtime dependencies**, which is an auditability commitment rather than an aesthetic one: a person who wants to know what the system does can read all of it, and there is no network path in the software.

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8. What the architecture does not buy

On latent-state recovery under model misspecification, the graph-backed inference is **statistically indistinguishable from a median-split quadrant classifier across all 20 tested conditions**, with paired differences of −2.5 to +2.5 points and 2,000-resample percentile bootstrap intervals spanning zero everywhere.

A reviewer may reasonably ask why an interactive-systems venue should care about an architecture that buys no accuracy. The answer is that accuracy was never the safety-relevant quantity: the result of §6.3 depends on the structural commitments — on representing a competing hypothesis rather than filtering, and on distinguishing an absent channel from a zero reading. Structure is what carries it.

We report the negative result prominently because the temptation to omit it is strong and because it relocates the contribution rather than diminishing it. **The architecture is not an accuracy technique.** Its value is structural: conclusions that trace to evidence, provenance that survives derivation, contradiction that is retained, revision that can be undone, inquiries that justify themselves, and conformance that is decidable.

Two related measurements complete the picture.

Calibration. Where the model's assumed structure approximately holds, accumulating evidence with an explicit *none-of-these* component reduces expected calibration error from **0.171 to 0.044** and raises AUC(uncertainty ☐ error) from **0.868 to 0.949**. Where it materially does not hold, calibration is **worse**: ECE 0.151 ☐ 0.221. The architecture trades uniformly mediocre calibration for good-when-applicable and poor-when-not, which is an improvement only to the degree the system can tell which regime it is in. It partly can, and not enough.

Identifiability. A two-channel location model cannot distinguish "state A, displaced" from "state B" when the displacement points toward B. This is not fixable by thresholding. It *is* partially fixable by observing more — pooling across topics to estimate a person-level displacement:

δ	Topics	Uncorrected	Hard-assignment EM	Soft-assignment EM (current)
0.30	5	82.3%	90.8%	**93.4%**
0.30	8	83.8%	95.8%	**98.2%**
0.45	3	54.1%	55.8%	**76.2%**
0.45	5	53.6%	58.4%	**89.5%**
0.45	8	53.2%	61.3%	**97.0%**

A caution that nearly shipped. The first version of this fix made things much worse where the data is thin: at one topic and δ = 0.30, correcting with the unguarded soft estimate drove accuracy from 81.2% to **20.8%**. The cause is identifiability, not tuning — a single point can be moved onto any of four corners by some offset, so there are more unknowns than equations. The estimator now **refuses below three topics** and returns a zero offset. That is a condition, not a threshold. We flag it because a fix that improves a headline by 36 points while quietly destroying the thin-data case is exactly the kind of change that passes a benchmark and fails a person.

9. Complexity, measured

We report complexity as measurement rather than assertion, as scaling ratio per doubling (linear ≈ 2.0×, quadratic ≈ 4.0×):

Operation	Bound	Measured
Provenance closure `ancestors()`	O(V+E)	**2.0×** — linear
Inquiry planning	O(|H|·|A|)	3.9× at |H|=|A| — product of two linear factors
Belief update, log-odds ledger	O(1) amortised	**4.2× — quadratic □ 2.0×** after fix
Write-then-consult path	O(n) per consultation	**3.8× — quadratic □ ≈2.0×** after fix

Two rows were defects in our own code. The belief ledger summed every contribution on each call, costing 624 ms at 8,000 updates; maintaining a running total gives 78 ms at 16,000. **A regression test fails if the ratio exceeds 3.0×**, using best-of-three timing so that machine load does not convert a linear algorithm into a failing test.

The fourth row was the larger one. Beliefs are now computed when the mirror is **consulted**, not when an observation is written, which took the path from 1121 ms to 5.24 ms at 800 evidence pairs — a 214× improvement — and from 3.8× to 2.0× per doubling. Re-measuring the write-then-consult path in this pass over 100 □ 800 evidence pairs reproduced **linear scaling at roughly 1.8–2.1× per doubling** across repeated runs (`verify_claims.py` §1); absolute timings differ from the figure above because the harness differs, and the shape rather than the constant is the claim.

Two honest notes survive the fix. The reactivity counter now counts consultations rather than writes, which is arguably what a reactivity measure should have counted from the start, since it exists to warn that *looking at a guess changes later answers*. And the asymptotics of a **read-heavy** pattern are unchanged: one consultation per twenty writes still scales at 3.1× per doubling, because each consultation is O(n). The constant improved twentyfold; the shape did not.

10. Validation

10.1 The test suite

Running the suite in this pass: **707 passed, 2 xfailed**. The two strict `xfail`s record operators that are specified and implemented nowhere, so they cannot quietly become "implemented" in a later summary:

Operator	Status
Causal reflection — intervention, counterfactual query	Prose only. No do-operator exists. Research agenda, not a component
Knowledge consolidation — promotion from belief to revisable knowledge	Prose only. No promotion criterion exists

Also unimplemented: encryption at rest, multimodal acquisition, a reflective execution engine with event queues, and any user interface beyond a command line.

A reference architecture earns the word *reference* when the distance between its formalism and its implementation can be enumerated. We enumerate it.

10.2 How strong the suite actually is

Earlier drafts stated that nothing measured whether the randomised harness exercises the state space in any principled sense. Mutation testing now does, and the measurement is not flattering.

Mutation testing produced **5,114 mutants**. The suite killed **2,319** and **1,549 survived**, for a mutation score of **59.9%**. A further 1,244 mutants sit in code with no covering test at all, 909 of them in a command-line interface that has never had tests and carries no claim in this paper.

Survivors concentrate in the inference engine (404), the pattern layer (278), the invariant checks themselves (206) and the formation model (195). Inspection of a sample shows most are label and message mutations rather than behavioural ones, which means the raw score understates the suite. But not all are cosmetic. **Mutating an invariant's own identifier survived**, which means no test asserted that a check returns the invariant it claims to be — a catalogue of named architectural properties that could have been silently renamed. Two tests now close that, and they exist because a tool found the gap rather than because we thought of it.

We report the score rather than tuning it. A mutation score is a lower bound on test strength and an upper bound on nothing, and 59.9% is a middling result for a package that makes conformance claims.

10.3 Five things found by measurement rather than by intent

Listed because they are the best argument for the method, and because each was slightly embarrassing.

1. **Provenance as a scalar dropped every parent but one.** The formalism said "attribute" where the literature had said "set" three decades earlier. Invariant II caught it.
2. **A clamping step that looked like numerical hygiene destroyed commutativity and retractability.** Belief order-invariance went from 0.0396/0.9900 across permutations to exact agreement.
3. **Invariant X false-positived on our own shipped output.** The defect was in the check, not the output (§5.3).
4. **Retroactive consent was already impossible and nobody designed it.** Found by a failing test. It falls out of asking *what was granted at time T* rather than *is there a grant*.
5. **Mutating an invariant's own identifier survived.** Nothing asserted that a check returns the invariant it claims to be (§10.2).

11. The regulatory tension

We report this because it is unresolved, and because the shape of it is more general than this system.

The EU AI Act defines biometric data as *"personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person"* (Art. 3(34)). Unlike GDPR Art. 4(14), **it does not require unique identification**. A heart-rate-like arousal proxy is therefore biometric data under the Act while identifying nobody. An *emotion recognition system* is one *"for the purpose of identifying or inferring emotions or intentions of natural persons on the basis of their biometric data"* (Art. 3(39)). Our guard infers a threat response — an affective and intentional state — partly on the basis of that arousal channel.

So: **the channel that exists only to protect people, and that §6.4 shows does protect them, is what brings the system within a prohibition written to prevent a different harm.** Remove it and the exposure largely disappears; remove it and the false hidden-desire rate returns to 31.2%.

The safety exception at Art. 5(1)(f) is available and, in our reading, applicable: the channel's sole documented purpose in code, tests and specification is to prevent a specific measured harm, and it is used for nothing else. We state that as an argument rather than a conclusion. We are aware of no case law interpreting the exception for a guard of this kind, and a system whose safety mechanism is itself an emotion-inference system is a case the Act does not obviously anticipate.

11.1 A second regime, with no research exemption

The EU analysis is the one the literature rehearses. It is not the only regime a system like this must survive, and the contrast is instructive.

This work is conducted under Egyptian jurisdiction. Law No. 214 of 2020 on Clinical Medical Research establishes a national ethics architecture but does not reach us: its subject is defined as studies assessing "therapeutic, pharmaceutical, surgical, nutritional, preventive or diagnostic interventions", and behavioural, psychological and computing research falls outside its scope. The Supreme Council of Universities issued a national academic-integrity and research-ethics guide in January 2026 whose third book addresses behavioural and social research committees, but it carries no decree number and no binding-force clause, and whether it reaches private universities is unspecified. We classify it as guidance rather than law.

Personal Data Protection Law No. 151 of 2020 does reach us, and reaches further than the GDPR in three respects. Its definition of personal data expressly includes data determining "psychological … identity". Its sensitive-data category covers "psychological, mental, or physical health, or genetic, biometric or financial data", and Article 12 prohibits processing that category without a licence from the Personal Data Protection Centre *in addition to* written explicit consent, on penalty of imprisonment and a fine up to EGP 5,000,000. And — the point worth putting in front of any venue — **it contains no scientific-research exemption whatsoever**. The GDPR's research accommodations have no Egyptian counterpart. A search of the Arabic text of both the law and its executive regulations (Decree No. 816 of 2025, in force 2 November 2025, compliance deadline 31 October 2026) finds that the words for "research", "studies" and "university" do not appear.

The consequence inverts a common assumption. A researcher in a jurisdiction with a *weaker* national human-subjects law can face a *stricter* data-protection constraint on exactly the physiological and psychological channels this architecture consumes, with no research carve-out to fall back on. Our design response is the one already described: refuse the biometric stream rather than seek a basis for it, and treat anonymisation rather than pseudonymisation as the only route out of scope.

This is a reading of statutory text, not legal advice, and it has not been reviewed by counsel.

12. Limitations

No human subjects. Nothing here establishes any effect on human reasoning, comprehension, trust, wellbeing or task outcome, and no such claim is made. Every result is a property of code or of simulated signal streams.

Simulation encodes a theory. The generative processes embody assumptions about how latent states manifest in observable signal. If those are wrong, absolute rates are wrong in the same direction for every method compared, including the baselines. We regard the *ordering* of methods as more robust than the absolute rates, and §6.7 measures how far that ordering survives as the key assumption is weakened.

A point estimate of zero is not a guarantee of zero. Across 800 threat episodes we observed no misreadings. The 95% upper bound implied by the rule of three is **0.37%**, and that — not 0.0% — is the number a deployment decision should use.

Two implementations, not many. §5.4 shows the decision procedure separating a compliant implementation from a non-compliant one, which rules out vacuity. It does not establish that the procedure is portable to an implementation built on different primitives — a graph database, an event-sourced store, a logic programming substrate.

Invariants X and XI are syntactic proxies for semantic properties. Whether an output usurps a person's authority over their own self-understanding is not a matter of which pronouns it contains. Our check tests phrasing, and §5.3 documents it producing a false positive on its first contact with real output. It will also produce false negatives: a sentence can be a verdict without matching any pattern we listed. The check raises the floor; it does not measure the property.

The clinical vocabulary is borrowed. We use "compliance" and "threat response" to name a pattern of approach behaviour under low stated desire and high arousal. There is a genuine peer-reviewed literature in this region — appeasement under traumatic entrapment, tonic immobility as an evolved defence, the acute stress response as a spectrum — and we cite it to locate the concern, not to claim our simulation instantiates any of it. One term deserves separate treatment: **"fawn" is not a clinical category.** We traced it to a self-published practitioner trade book and could locate no refereed source establishing it as a validated construct. It has been removed from every load-bearing claim.

We are not lawyers. §11 is a reading of statutory text, not advice.

The specificity cost is real. Under partial observation the system labels excitement as a possible threat 99.0% of the time. A person who is excited and told to consider whether they are afraid has been mildly misread. We judge this preferable to the converse and do not claim it is costless.

Credible sets do not reach nominal coverage. Reporting a credible set instead of a point label improves the rate at which the truth is contained in the output (78.2% □ 88.0% at $\delta = 0.30$) but a set claiming 90% mass contains the truth 60.5% [55.5, 65.2] of the time at $\delta = 0.45$. A credible set inherits its posterior's miscalibration; reaching nominal coverage requires conformal calibration against held-out labelled outcomes, which do not exist without a human study.

Scale. Everything reported runs on a single machine over thousands of nodes, not millions, and the read path remains O(n) per consultation.

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13. Conclusion

Making the invariants executable found two defects that a prose specification had asserted away: a provenance model that dropped parents, and a belief operator whose result depended on the order evidence arrived. Neither was visible to a reader of the document. Both were visible on the first run of the decision procedure.

Making refusal a first-class output, rather than a low-confidence label, removed a specific foreseeable harm in simulation — and the ablation shows that the removal depends on a commitment that looks like a detail: an absent channel must be representable as absent. A system that imputes it to zero keeps its guard in the codebase and loses it in production, with nothing in the output to say so.

We think the transferable claims are these. Reference architectures for human–AI interaction should ship their conformance suites, should include negative tests, and should point those suites at output a person would actually see. Papers proposing ethical architectures should print the fraction of their prohibitions that a machine will actually refuse. And a reflective system under partial observation should refuse rather than degrade.

Ours did all three and was wrong three times — about provenance, about order invariance, and about its own Invariant X check. Each error surfaced within a minute of the check being run, and none would have surfaced from reading.

Ethics statement

This research did not involve human or animal subjects. All results derive from code execution and from simulated signal streams; no personal data was collected, processed or analysed.

The system is a research instrument, not a medical device, and is not suitable for clinical, educational, employment, credit, insurance, policing or other high-stakes deployment. It refuses to enable the physiological stream in institutional and undeclared deployment contexts. The authors note the regulatory position described in §11 and do not assert that the safety exception under Art. 5(1)(f) applies as a settled matter.

Data and code availability. Implementation and conformance suite are released under an open licence. *[REPOSITORY URL — TO BE COMPLETED]* See Appendix A for the provenance of every number in this paper, including which were re-derived in preparing this draft and which were not.

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The 102 references below are the union of the two source papers' reference lists, deduplicated. Each was confirmed against a live source with a resolvable identifier in Session 14, **except the 6 marked** †, which were introduced in Session 15 after the verification pass and are unconfirmed under ground rule 6; see **OW-9** in Appendix B.

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Appendix A — Number provenance

Ground rule 4 of this programme: *no number enters a manuscript unless a file in the repository produces it and the person writing the sentence has run that file.* This appendix records compliance with that rule honestly, which means recording where it could not be met.

Re-derived while preparing this draft, by running the named file against `shadow_mirror` v1.3.0:

Claim	§	Source	Result
Test suite totals	10.1	`pytest -q`	707 passed, 2 xfailed
Order invariance over 120 permutations	4.2, 5.4	`verify_claims.py` §2	120 permutations, 1 distinct outcome, spread 0.0
Eleven callable invariant checks	5.1	`verify_claims.py` §3	11 checks + `check_all`
Deployment gate refuses institutional and undeclared	7.3	`verify_claims.py` §4	refused; `self` permitted
Empty safety basis still refuses	7.3	`verify_claims.py` §4	`True`
Ledger verifies; edit is detectable	7.2	`verify_claims.py` §5	`verify()` `True` ☐ `False` after edit
`streams_granted_at` is time-aware	7.2	`verify_claims.py` §5	returns granted streams, then none after revocation
Refusing methods refuse when called	5.5	`verify_claims.py` §6	9 called, 9 refused, 0 returned
Write-then-consult path is linear	9	`verify_claims.py` §1	≈1.8–2.1× per doubling

Carried from the programme record, and *not* re-derived here. The scripts that produce these numbers — `run_benchmarks.py`, `conformance_v10.py`, `experiment_d23.py`, `experiment_ambivalence.py`, `analyze_v030_vs_v040.py` — and their outputs `results.json` and `v030_vs_v040.json` are **not present in the research archive this draft was assembled from**. They are cited here on the authority of the session reports and the two source papers, and a reader cannot currently re-run them. This is recorded as **OW-6** in Appendix B and is, in our judgement, the most serious reproducibility defect in the present state of the work.

Claim	§	Cited source
All safety rates, intervals and per-process figures	6.3	`results.json → b2.conditions[1]`
Ablation figures	6.4	`results.json → b2.conditions[2]`
Refusal calibration	6.5	`results.json → b3.results`
Ambivalence, pre- and post-fix	6.6	`experiment_ambivalence.json`; pre-fix recorded in Session 8
Separation sweep and supervised baseline	6.7	Session 13 adversarial review
Inquiry regret	4.3	`results.json → b4.conditions`
Median-split parity across 20 conditions	8	`results.json → b1.cells`
Calibration ECE and AUC	8	`v030_vs_v040.json`
Identifiability and offset pooling	8	`experiment_d23.json`; soft-EM figures from Session 12
Belief-ledger and prototype complexity ratios	9	`test_operator_correspondence.py` timing harness
Mutation score and survivor distribution	10.2	Session 17 mutation run
Prototype conformance 4/7	5.4	`conformance_v10.py`

Reconciliation note. Where the two source papers and the current package disagree, this draft follows the package and says so. Four figures were superseded after the papers were drafted: the write-then-consult path is no longer quadratic (§9); the identifiability estimator now uses soft assignment and refuses below three topics (§8); the consent ledger and withdrawal receipt now exist (§7.2); and the deployment gate now exists (§7.3), moving the enforcement ratio from five of twelve to **six of twelve** (§7.4). The papers in `papers/` still carry the pre-Session-12 values. This is recorded as **OW-7** in Appendix B.

To reproduce what can be reproduced:

```
pytest -q                 # 707 passed, 2 xfailed
python verify_claims.py   # writes verify_claims_output.json
python demo.py
```

Appendix B — Outstanding work

This appendix is deliberately separate, and it is the appendix a reader should check before believing anything above.

It has three parts. **B.1** lists what has no available substitute — the rule applied is that an item appears only if an active hunt for a public dataset, a free instrument, a third-party service or a methodological substitute came back empty. **B.2** lists what a search *resolved*, so that B.1 reads as the residue of a search rather than as its whole. **B.3** lists ordinary unfinished work, which is excluded from B.1 on purpose because none of it is blocked on anything external.

B.1 Blocked: no available substitute

OW-1 · The threat-confound claim requires primary data collection. Approximately forty public datasets were assessed across affective computing, approach-avoidance paradigms, ecological momentary assessment, experimental economics and social conformity. The claim needs four things measured on the same person and the same item: a stated stance, a behavioural choice that can contradict it, an arousal channel time-locked to that choice, and a coercion manipulation. That intersection does not exist publicly. The failure is structural rather than incidental — corpora with good physiology are passive-viewing paradigms in which nothing can contradict a rating, and corpora with real choice lack autonomic recording, self-report, or both. Two published studies used almost exactly the right design and deposited no data. **No workaround exists. Only collection.**

OW-2 · No instrument in the battery is validated in an Egyptian sample. Arabic validations were confirmed for Lebanese, Saudi, Omani, Jordanian, Palestinian, Qatari and Moroccan samples. None is Egyptian. Four candidate Egyptian-journal papers were identified by DOI, but every `*.ekb.eg` host was unreachable, so their contents are unknown — and Egyptian psychometric papers frequently construct local instruments rather than translating Western ones, so their relevance cannot be assumed. Any use of this battery in Egypt is first-in-population work and must be described as such. **Partially resolvable by a colleague with Egyptian institutional access; not resolvable by open search.**

OW-3 · Two instrument constructs have no free replacement. There is no openly licensed eight-theme Eriksonian inventory: the two structurally correct candidates are both permission-gated, and going upstream from one to the other does not escape the gate. The best free material covers two of eight themes and yields single scores rather than subscales. For purpose beyond the Meaning in Life Questionnaire, no free substitute was identified. **Resolvable only by author permission, which is a letter, not a search.**

OW-4 · No party outside this project has executed the software. This was the largest risk identified by the programme's own adversarial review. The venue route is closed: neither ACM TiiS, nor ACM FAccT, nor AAAI/ACM AIES operates an artifact-evaluation or badging track an author can enter at submission time. Archival deposit satisfies an availability criterion but verifies no execution. Two replication communities publish work written *by* the replicator and therefore cannot be pointed at an artifact. One community code-check service matches the need exactly and has **not been used**. The claim in this work is that the code is inspectable, not that it has been independently run.

OW-5 · Invariants X and XI are syntactic proxies for semantic properties. Invariant X detects verdict phrasing by pattern, and the question generator is guarded against prescriptive drift by a keyword test. Both raise a floor and neither measures the property. We know of no decision procedure that settles whether a sentence claims authority over a person, and we do not think one follows from more patterns. This is a research problem rather than an unfinished task.

B.2 Newly identified while preparing this draft

OW-6 · The benchmark scripts and their outputs are not in the archive. [blocking for submission] Both source papers state that every number is produced by a named script a reviewer can run. As assembled, the archive contains the package, the papers, the session reports and the submission material — but **not** `run_benchmarks.py`, `conformance_v10.py`, `experiment_d23.py`, `experiment_ambivalence.py`, `analyze_v030_vs_v040.py`, `make_paper_figures.py`, `results.json` or `v030_vs_v040.json`. Every headline safety figure in §6, the calibration and identifiability figures in §8, and the conformance comparison in §5.4 therefore cannot be re-derived from what is on hand. The figures under `papers/figures/` exist but their generator does not. **Action:** locate these files in the working environment that produced them and commit them, or regenerate and re-verify every number they support before submission. Until then the availability statement in the Ethics section overclaims.

OW-7 · The two source papers carry pre-Session-12 figures. [blocking for submission] P1 §7 still reports the write-then-consult path as quadratic (3.8× per doubling, 1121 ms at 800 pairs); P2 §9 repeats it. P1 §8 reports identifiability at $\delta = 0.45$ with eight topics as 61.3%, from the superseded hard-assignment estimator. P2 §8 reports that no append-only consent record exists and that no deployment-context mechanism has been implemented, and prints an enforcement ratio of five of twelve. All four were superseded by Session 12 and are corrected in this unified draft. **Action:** propagate the corrections into `P1_ACM_TiiS_draft.md` and `P2_AIES_FAccT_draft.md`, or withdraw those files in favour of this one. Submitting either paper as it stands would publish four figures the authors know to be stale.

OW-8 · Test-count drift across documents. The archive README reports 707 passing; P1 reports 673 collected and 671 passing with a 563-invariant breakdown; Session 12 reports 324. The first is correct for v1.3.0 and was reproduced here. The per-category breakdown in P1 §3.6 has not been recomputed for the current suite, so this draft does not restate it. **Action:** recompute the breakdown from `pytest --collect-only` and state it once, in one place.

OW-9 · Nine reference-list entries sit outside the verified set. `bibliography.json` records 105 manuscript references verified in Session 14 and reports `entries_without_resolvable_identifier: 0`. The papers' reference lists are longer: P1 lists 58 and 55 are in the verified set; P2 lists 56 and 50 are in the verified set. The nine remaining entries are six unique psychometric instrument sources, all introduced in Session 15 — after the verification pass — and never folded into it. They are marked † in the reference list above. This is not a claim that any is wrong, but under ground rule 6 they are unconfirmed. **Action:** verify the six and update `counts`, or mark them explicitly as pending.

OW-10 · Two release decisions the repository now forces. Neither is a research defect; both follow from the archive having been committed. First, the full working conversation transcript is in the repository under `archive/`. The repository is private, so this is storage rather than publication — but the data-availability statement above promises a *public* repository, and the transcript would go public with it. Second, the v1.3.0 archive currently sits on a branch rather than on `main`, so a repository URL quoted today would resolve to the superseded v0.3.0 package. **Action:** decide deliberately whether the transcript is published, moved to a controlled deposit, or replaced by a code-only mirror; and merge the branch before quoting a URL.

B.3 Ordinary unfinished work

Excluded from B.1 on purpose: none of it is blocked on anything external, and listing it there would dilute a list whose value is that every entry has been searched for and not found.

Item	Status
Causal reflection operator	Declared gap, strict `xfail`. No do-operator exists
Knowledge consolidation operator	Declared gap, strict `xfail`. No promotion criterion exists
Read path still O(n) per consultation	Open. Anchored-decay accumulators would fix it
1,549 surviving mutants	Open. Concentrated in engine, patterns, invariant checks, formation
909 mutants in an untested CLI	Open. Carries no claim in any paper
Candidate belief state for monotonic drift	Proposal ("your position moved steadily"), computed and discarded
Conformal calibration for credible sets	Blocked on labelled outcomes, i.e. on a human study
Encryption at rest, multimodal acquisition, event-queue execution engine, any GUI	Specified, unimplemented

B.4 Before submission

Item	Who can close it
Corresponding email, and a department if one applies	**You.** The only gap in the author block
Repository URL in both availability statements	**You**
Venue selection for the ethics paper (AIES or FAccT)	**You.** Check current calls; dates move annually
OW-6 and OW-7 above	Whoever holds the working environment that produced `results.json`
Three dataset downloads for the secondary-analysis route	Mechanical; two are small
Urban Appraisal rating polarity	From the BeMoBIL analysis code. The loader refuses until supplied
Two instrument permissions (Claremont Purpose Scale; MEPSI)	Letters to the instrument authors
CITI certification and a supervisor	Only if the primary study proceeds

B.5 What would change the conclusions

Stated so that a reader knows what evidence would move this work rather than merely extend it.

1. **A human-subjects study.** Objective O4 is unanswered. Every claim about reflection, comprehension or trust is currently a design intention with no evidence, and a null result would be a finding.
2. **Primary collection on the four-way intersection (OW-1).** The safety result is a property of a simulator that encodes our own theory. Real data could refute the ordering of methods, not merely its magnitude.
3. **A third party running the conformance suite against a different substrate (OW-4).** Until then, "decidable" is demonstrated and "decidable usefully" is not.
4. **A decision procedure for semantic authority (OW-5).** Invariants X and XI currently raise a floor they cannot measure.

Part II — The architecture paper

Invariants You Can Run: An Executable Specification for Reflective Human–AI Architectures

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Abstract

Reference architectures for human–AI interaction are usually specified in prose. Their invariants — traceability, provenance, human authority — read as commitments, and a reader has no way to tell whether an implementation keeps them. This is not a documentation problem. A property that cannot be checked cannot be violated *detectably*, and a commitment that cannot be violated detectably is not a commitment.

We present Shadow Mirror, a reference architecture for reflective human–AI collaboration whose eleven architectural invariants are **decidable properties with a supplied decision procedure**. Each is a callable check returning a verdict and the observation that produced it. Invariants I–X hold across 200 randomised well-formed reflective states and Invariant XI across 200 randomised formation models, and each is paired with a negative test constructing a state that violates it — because a conformance check that only ever passes is indistinguishable from one that does nothing. The suite is 673 tests, of which 563 are invariant checks.

Three architectural components carry the invariants. The **Unified Reflective Graph** holds observations, evidence, context, hypotheses, inquiries and explanations in one typed structure with multi-parent provenance closure and contradiction as a first-class edge. **Belief revision** accumulates in log-odds keyed by evidence identity, making it order-invariant across all 120 permutations of a five-observation sequence and exactly retractable — a prototype that clamped the posterior in state produced final beliefs of 0.0396 or 0.9900 for the same evidence depending on arrival order. **Inquiry planning** maximises expected information gain, achieving normalised regret 0.161 [0.138, 0.187] against an information-theoretic oracle where a belief-gap heuristic scores 0.579 [0.547, 0.612] against random selection's 0.550 [0.515, 0.584].

We report complexity as measurement rather than assertion, and we are explicit about what the architecture does *not* buy: on latent-state recovery the graph-backed inference is statistically indistinguishable from a median split across all 20 tested conditions. The contribution is structural — traceability, provenance, retained contradiction, decidable conformance — and we claim nothing beyond it. Two specified operators, causal reflection and knowledge consolidation, are implemented nowhere and are carried as declared gaps rather than omitted.

This system is a research instrument. It is not suitable for clinical, educational, employment or other high-stakes deployment.

CCS Concepts: Human-centered computing ☐ Interactive systems and tools; Computing methodologies ☐ Reasoning about belief and knowledge; Software and its engineering ☐ Software verification.

Keywords: reference architecture, executable specification, architectural invariants, provenance, belief revision, expected information gain, conformance testing.

1. Introduction

Architecture papers make promises. *Every conclusion traces to evidence. Provenance is preserved. The system does not silently replace human judgment.* These are good promises and they are why reference architectures are worth writing.

They are also, almost always, unfalsifiable as stated. A reader cannot determine whether an implementation keeps them, an author cannot determine whether a refactor broke them, and a reviewer cannot distinguish an architecture that enforces them from one that describes them. When the architecture is intended to support a person reasoning about themselves, that gap is not academic.

We began from an architecture in the usual condition: ten invariants specified in prose (an eleventh was added later, for reasons reported in §3.7), with a prototype whose relationship to them was undocumented. Testing the prototype against its own invariants found that **Invariant II (Provenance Preservation) was violated** — evidence derived from two sources reported one parent and silently discarded the other — and that belief revision was **order-dependent to the tune of 95 percentage points**. Both defects had been present in a document that asserted the invariants held.

This paper reports what changed when the invariants were made executable.

Contributions.

1. **Eleven architectural invariants as decidable properties**, with a supplied decision procedure, a randomised property harness, and a negative test per failure mode. The eleventh is unusual and is reported in §3.7: it exists because a user asked the architecture for a capability the architecture should not have, and it makes the refusal checkable rather than merely stated. (§3)
2. **A reflective graph substrate** with multi-parent provenance closure, contradiction as a first-class edge, transactional updates, and no third-party dependencies. (§4)
3. **An order-invariant, exactly retractable belief operator**, with the axioms it satisfies stated explicitly and enforced by test. (§5)
4. **Information-gain inquiry planning**, measured against an oracle. (§6)
5. **Complexity as measurement.** Four operator bounds, measured, one of which caught a quadratic defect in our own implementation. (§7)
6. **A negative result reported as such**: the architecture buys no accuracy. (§8)

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2. Related work

2.1 Architecture, and the gap between description and code

Software architecture has been a first-class object of study since Perry and Wolf [14] named its elements, form and rationale. The persistent problem is drift: the architecture as described and the architecture as built diverge, silently, under maintenance. Murphy, Notkin and Sullivan's reflexion models [15] made that divergence itself computable, mapping a source model onto a high-level model and reporting convergences, divergences and absences. A substantial literature followed on static conformance checking — dependency-constraint languages [16], comparative evaluation of checking approaches [17, 18] — and on recovering an architecture from code when no trustworthy description survives [19].

That literature is almost entirely concerned with *structural* constraints: which module may depend on which, whether layering is respected, whether a component sits where the diagram says. Those are the properties for which a decision procedure is straightforward to write, because they are properties of the dependency graph. What we attempt is conformance checking for *semantic* commitments about reasoning — not "does module A avoid importing module B" but "does every explanation this system produces terminate in evidence the person supplied". We are not aware of a prior conformance suite that takes reasoning-level commitments as its checked properties. §3.5 reports our first evidence that such checks discriminate between implementations

rather than merely restating one API.

2.2 Specification, contracts, and executable properties

The idea that a specification should be machine-checkable rather than prose is old and well served. Design by contract [20] attaches preconditions, postconditions and invariants to program units. Formal specification languages — TLA+ [21], Alloy [22] — allow properties to be model-checked against a design. Runtime verification [23] monitors an executing system against a formal property. Property-based testing, introduced by QuickCheck [24], generates inputs and checks that stated properties hold across them, rather than that named cases produce named outputs.

Our invariants borrow the *form* of each of these and match none of them exactly. They are not contracts, because they are not attached to a unit and not checked at call boundaries. They are not model checking, because they run against the implementation rather than a model of it. They are closest to property-based testing, and the harness of §3.2 is a randomised property harness in the QuickCheck tradition. The difference we claim is one of subject matter rather than technique: the properties are architectural commitments published *as part of the architecture*, shipped with a decision procedure and a negative test per property, and intended to be run by a third party against a *different* implementation. §10 states plainly what that ambition has not yet achieved.

2.3 Provenance

The W3C PROV data model [25, 26] treats derivation as a relation over entities and activities in which an entity may derive from many others. Database provenance has a longer and more precise tradition still, distinguishing why-provenance from where-provenance [27] and surveying the design space [28]. Our Invariant II restates the multi-parent requirement, and the defect reported in §3.6 is exactly what happens when a formalism writes provenance as a scalar attribute rather than a set: the implementation faithfully follows the formalism, and drops every parent but one. The literature had the right answer three decades before we made the mistake, which is precisely the argument for checking the property mechanically rather than trusting that we remembered it.

2.4 Belief revision

AGM [1] gives rationality postulates for theory change over a logical closure. Iterated revision needs more than AGM supplies, and Darwiche and Pearl [29] give the additional conditions; Gärdenfors [30] is the book-length treatment; Spohn's ordinal conditional functions [31] and Jeffrey's treatment of uncertain evidence [32] are the two standard routes to graded belief change. On the probabilistic side, conditioning on conditionally independent observations is commutative [2] — a fact whose consequences for a belief store occupy §5 — and Pearl [33] is the canonical treatment of the independence structure that makes it hold.

Human belief revision is neither minimal nor commutative. It is explanation-guided, order-sensitive, and subject to recency [3, 4]. We take the formal result as a constraint on the *arithmetic* — a belief store that violates commutativity is broken, not humanlike — and the empirical departure as a reason to hold contradiction open in the graph, not as licence for the arithmetic to drift. §5.2 reports what enforcing that distinction cost us: a clamping step that looked like prudent numerical hygiene destroyed both commutativity and exact retractability, and had to go.

2.5 Choosing what to ask

The inquiry planner of §6 scores candidate questions by expected information gain. The measure is Lindley's [34]; the Bayesian experimental design tradition it belongs to is surveyed by Chaloner and Verdinelli [35]; MacKay [36] brought information-based objectives to active data selection and Houlsby et al. [37] to Bayesian active learning, within the broader active learning literature [38].

The cognitive-science strand matters more to us than the machine-learning one, because our questions are asked of a person rather than of an oracle. Oaksford and Chater [39] analysed human data selection as

approximately optimal information gathering; Nelson [40] compared information gain against diagnosticity, probability gain and impact as competing accounts of what makes a question useful, and found no single measure dominant. We adopt EIG because it is the standard against which the alternatives are compared, and we report normalised regret against an information-theoretic oracle rather than raw performance, so that a reader who prefers a different measure can see how much of our result is the measure and how much is the planning.

2.6 Reflection, self-tracking, and the user's own model

The HCI tradition this work sits inside treats reflection as a design goal rather than a by-product. Reflective design [41] set out the commitments; Baumer [42] gave reflective informatics its conceptual dimensions; Fleck and Fitzpatrick [43] framed the design landscape and distinguished levels of reflection a system might support. Personal informatics supplies the empirical grounding — the stage-based model [44], its lived revision [45], and what actually happens when people without prior self-tracking experience meet their own data [46], which is often not the reflective engagement the designers intended.

The most direct antecedent is the open and scrutable user model. Open learner models [47] show learners the system's model of them; Kay [48] argues that adaptation must be scrutable — that a person should be able to interrogate and correct the model driving a system's behaviour toward them. Bellotti and Edwards [49] made intelligibility and accountability requirements for context-aware systems, and Lim and Dey [50] measured the demand for them empirically. Horvitz's principles for mixed-initiative interfaces [51] supply the interaction frame in which a system may ask rather than assume.

The Unified Reflective Graph (§4) is a scrutable user model with two commitments added: every node reaches the evidence that produced it, and contested hypotheses are represented rather than resolved. What this paper adds to that tradition is not the goal but the *check* — a decision procedure that says whether a given implementation actually has those properties.

2.7 Explanation, and what it buys

Explainable AI supplies both our framing and our caution. Doshi-Velez and Kim [10] set out what a rigorous science of interpretability would require; Miller [11] argues that explanation is a social process rather than a model-inspection problem; Rudin [12] argues against explaining black boxes at all where an interpretable model would serve; Lipton [13] shows that "interpretability" names several mutually incompatible desiderata.

The empirical results are sobering. Explanations do not reliably improve complementary team performance [52]. Schäfer et al. [5] show that displaying uncertainty does not by itself produce calibrated reliance, and Rosenbacke et al. [6] find in systematic review that explainability can decrease trust calibration as easily as increase it. Guidelines for human–AI interaction [53] codify much of this into design practice.

We take two things from this literature. Inspectability is a property to be *checked* rather than claimed — which is the entire argument of this paper. And abstention deserves to be a first-class output rather than a fallback, because a better-hedged wrong answer is still a wrong answer.

2.8 Metacognitive architectures

A recent line of work embeds reflection inside the model: reuse of recurring reasoning as named behaviours [7], evolving meta-memory [8], grounded metacognitive reasoning [9]. Those systems improve what a model knows about its own reasoning. Our concern is orthogonal, and we think complementary: an external structure whose reasoning about a person is inspectable *by that person*. A model that reflects well internally but cannot show its work to the subject of its inferences has solved a different problem from the one addressed here.

3. Invariants as an executable specification

3.1 The eleven invariants

	Invariant	Operationalisation
I	Evidence traceability	no explanation node lacks a reachable evidence node
II	Provenance preservation	every derived evidence node recovers **all** parents under transitive closure
III	Revision history	every belief transition records prior, posterior, triggering evidence, method, time
IV	Explicit uncertainty	no explanation is emitted without a non-empty uncertainty field
V	Hypothesis independence	contradicting one hypothesis removes or merges no competitor
VI	Context consistency	superseding a context annotates every dependent conclusion as stale
VII	Explanation integrity	explanations carry summary, uncertainty and assumptions
VIII	Inquiry justification	every inquiry quantifies expected gain and names the hypotheses it separates
IX	Transactional consistency	a failed update leaves no partial state
X	Human authority	no output phrased as an identity claim or directive; each carries uncertainty, a disclaimer and a question
XI	Declared-only personhood	no attribute about a person exists unless they declared it or self-completed the instrument it came from; and no route exists to a global rank, a prescribed deficit, or an attributed purpose

Two required judgement rather than transcription.

VI cannot mean "recompute the past." An instrument cannot re-run a person's week. We operationalise it as annotation: when a context is superseded, every conclusion that depended on it is marked stale — neither silently retained nor silently deleted. The distinction matters because retention is the failure mode that produces confident conclusions drawn under conditions that no longer hold.

X is a property of what reaches the person, so it is checked at the output boundary rather than in the reasoning. Each terminal output must carry its uncertainty and disclaimer, offer at least one question, and avoid identity claims (*you are…*, *you secretly…*, *your true…*) and directives (*you must…*, *you should…*).

3.2 The harness

`build_well_formed(seed)` constructs a randomised reflective state: 2–5 observations, derived evidence with multiple provenance parents, 2–4 hypotheses with mixed supporting and contradicting edges, executed belief updates, planned inquiries, and an explanation. Invariants I–X are asserted across 200 seeds.

Invariant XI needed a second generator, because it ranges over a formation model rather than a graph. `_random_model(seed)` builds a model from a random sequence of declarations, person overrides and attempted instrument administrations, under a randomly chosen deployment context and a consent grant present 80% of the time. Most administrations therefore refuse, and that is the point: XI must hold whatever the model has been through, *including* the paths where every administration was declined. It is asserted across 200 seeds, with 100 further randomised checks on two derived properties — that no model ever retains a non-self-completed administration, and that a person's override always wins for display.

Without that second harness, XI would have been the one invariant asserted only by hand-built cases, which is precisely the position the other ten started from.

Both generators are small seeded builders rather than a property-testing library. The package has no runtime dependencies by design — local-first and auditable — and adding one for the test suite alone would undercut the argument the architecture rests on.

3.3 Negative tests, and why they are the point

Sixteen negative tests construct states that violate each invariant and assert the check catches them: an explanation with no evidence; a store that keeps only the first provenance parent; a transition with no triggering evidence; an explanation that hides its uncertainty; a dropped competitor; a superseded context whose dependents are unannotated; an inquiry that justifies nothing; a store without rollback; verdict phrasing; a missing question; a missing disclaimer; a refusal that returns a value; a refusal that has been deleted rather than triggered.

There are **fourteen** such tests and **four positive controls** — states that should pass and do. Earlier drafts of this paper reported "sixteen negative tests", which counted the four positive controls among them. The correction is small and we make it explicitly rather than silently, because a paper about checkable claims should not miscount its own checks (D-34).

Without these, a suite reporting "11/11 invariants hold" is unfalsifiable in exactly the way the prose was.

3.4 What happened when we pointed it at real output

The Invariant X check failed on its first contact with the shipped reflection layer. It flagged:

> *"You have consulted this 5 times. Seeing a guess changes later answers and can become self-fulfilling."*

as a verdict, on the pattern `you have`. That sentence is a factual statement about the person's own logged behaviour, and it is a caution the architecture specifically wants surfaced. **The defect was in the check.** We narrowed the patterns to identity claims and directives and recorded the reasoning in source.

We report this because it is the argument for the method. The check was written against a specification, passed against fixtures, and was wrong. It became correct only when aimed at output a person would actually see. A conformance suite that is never pointed at real output will pass, and will be worthless.

3.5 Do the checks discriminate between implementations?

A specification that only its own author's implementation can satisfy has demonstrated nothing. Two pieces of evidence say these checks discriminate.

The negative tests (§3.3) construct violating states and confirm each check rejects them. That establishes the checks are not vacuous.

The prototype. The same decision procedure was run unmodified against an independently written implementation of the same architecture — a NetworkX-based prototype produced separately from this work. It passes four checks and **fails three**: provenance preservation (0 of 2 parents recovered), order invariance (spread 0.9504 across permutations), and inquiry justification (no information-gain content). The procedure therefore separates two implementations of one architecture, which is the minimum evidence that it is measuring conformance rather than restating an API.

This is two implementations, not many. §10 records that limitation.

3.6 Results

	Prototype	v1.0
I Evidence traceability	pass	pass

	Prototype	v1.0
II Provenance preservation	**fail** — 0/2 parents recovered	pass — 2/2
III Revision history	pass	pass
V Hypothesis independence	pass	pass — contradiction retained
Order invariance (120 permutations)	**fail** — spread 0.9504	pass — spread 0.00
IV Explicit uncertainty	pass	pass
VIII Inquiry justification	**fail** — no information-gain content	pass
	4/7	**7/7**

Full suite: **673 tests** (671 passing), of which 563 are invariant checks — 200 randomised graph states for Invariants I–X, 200 randomised formation models plus 100 further randomised property checks for Invariant XI, 14 negative tests, 4 positive controls, and 45 hand-built cases. A further 16 are operator-correspondence tests and the remainder are component regressions. Two are strict `xfail`s recording unimplemented operators (§9).

3.7 An invariant that exists because of a request

Invariant XI was not derived from the architecture. It was derived from being asked for something.

A user of this system asked it to "determine and understand who people probably are according to their will, desire and need — where they come from, how mature they are, what they still need to be mature enough, what their eventual purpose probably is", with the qualification "I would not say judge people". That qualification is the interesting part: the request came with an intuition that something in it was wrong, and no way to say which part.

It is five requests, and they separate cleanly on whether a validated construct exists and who supplies it. *Will*, *desire* and *need* have validated self-report instruments, so the person can supply them. *Where they come from* is inference of special-category data unless the person declares it. *How mature they are* is available only from instruments whose output is a single ordered position — Loevinger's Total Protocol Rating [56], Kegan's subject-object position [57], Cook-Greuter's Center of Gravity — where the rank is not a summary of the construct but *is* the construct, so it cannot be reported as a profile without misrepresenting the instrument. *What they still need to be mature enough* is a directive about identity, which Invariant X already forbids. *Eventual purpose* has no construct at all. The substitutes we adopted — a three-dimensional wisdom scale [58] and a psychosocial attribute inventory — report dimensions and never a stage.

So four of the five are refused, and the refusals are the feature. What makes them an architectural matter rather than a policy note is that a refusal stated in prose decays exactly as the original ten invariants did. Invariant XI is the decision procedure that stops it:

```
def check_xi_declared_only(model) -> InvariantResult:
    # (a) every attribute traces to a declaration or a self-completed instrument
    # (b) each refusing method, when CALLED, raises with the expected ground
```

Condition (b) is the part worth defending. The check does not inspect source code or match names; it **calls** each method that ought to refuse and requires a typed refusal carrying the expected ground. Two distinct failure modes are caught. A method that returns a value fails, obviously. A method that has been *deleted* also fails — and that is the less obvious commitment, because **absence is not refusal**. A caller who finds no `eventual_purpose` attribute learns nothing; a caller who calls it and reads why it declined learns what the architecture will not do and on what ground. The refusing methods therefore exist in order to refuse, which is a deliberate inversion of the usual advice to delete dead code.

The two negative tests are correspondingly shaped: subclass the model so that `eventual_purpose()` returns a plausible answer, and XI fails; delete `maturity_level` entirely, and XI fails for a different stated reason.

What this does not establish. Eleven refusal grounds are a catalogue, not a theory of which requests a reflective system should decline, and we do not have one. The check verifies that the named refusals refuse; it cannot verify that the catalogue is complete, and a request we did not anticipate will not be caught. It also cannot check phrasing — the accompanying `open_questions()` output is guarded against prescriptive drift by a keyword test, which carries the same syntactic-proxy weakness we already concede for Invariant X (§10).

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4. The Unified Reflective Graph

Typed nodes (Observation, Evidence, Context, Hypothesis, Belief, Inquiry, Explanation, Knowledge, Limitation) and typed edges (provenance, supports, contradicts, contextualises, revises, targets, explains, limits). Untyped insertions are rejected: an untyped graph cannot support a traceability invariant, because there is nothing to quantify over.

Provenance is a DAG. `ancestors()` returns the transitive closure over *all* provenance parents. The prototype followed the first predecessor only, recovering 0 of 2 parents from a two-parent derivation. The formalism licensed it: the evidence object was specified with provenance as a scalar attribute p_i. Writing it as a set — which §A.2 of our formal appendix now does — is the fix, and the defect is a small case study in formal notation shaping implementation error.

Contradiction is an edge, not an overwrite. `contested_hypotheses()` returns hypotheses carrying both supporting and contradicting evidence, so an explanation can report a tension rather than preferring the later write. This is Invariant V's teeth.

Updates are transactional. A `transaction()` context manager snapshots and rolls back, so a failure mid-update cannot leave a hypothesis whose belief moved but whose provenance edge was never written — which would violate Invariants I and IX simultaneously and silently.

No third-party dependencies. The graph is an adjacency structure in the standard library. This is not minimalism for its own sake: the architecture's claim is local-first auditability, and importing a graph framework for a dictionary of lists would trade that for nothing.

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5. Belief revision

5.1 The axioms

Leaving the belief state's type open — "probability distributions, weighted graphs, logical structures" — and then asserting properties of it is a common and serious error. Properties cannot be asserted of an object whose type is open. We state the axioms instead:

- **B1 Plurality** — assigns a value to every hypothesis simultaneously; no distinguished maximiser.
- **B2 Commutativity** — order of conditionally independent evidence does not matter.
- **B3 Invertibility** — evidence can be retracted exactly.
- **B4 Audit** — every application records prior, posterior, trigger, method, time.

B3 is not a convenience. **A system that cannot undo an inference cannot honour a withdrawal of consent**, which turns an ethical commitment into an architectural requirement.

5.2 The instantiation, and the defect it repairs

```
l_t(h) = logitpi(h) + SUM_e in E_t w_e lambda_e(h), lambda_e(h) = log(P(e | h))/(P(e | not h)),
                                    B_t(h) = sigma(l_t(h))
```

$\mathcal{E}_t$ is a set **keyed by evidence identity**, which yields B2 (addition commutes), B3 (subtract the term) and idempotence under resubmission.

The prototype clamped the posterior to [0.01, 0.99] *in state* after each update. Clamping is not invertible: once pinned, the information about how far past the bound the evidence pushed is destroyed. Five observations in three orders produced final beliefs of **0.0396, 0.9900 and 0.9900**.

v1.0 clamps only at display. Measured consequences:

- All **120 permutations** of the five-observation sequence agree to within 1e-12.
- Twenty strongly-confirming updates followed by twenty strongly-disconfirming ones return the belief to **exactly 0.5**. The prototype could not do this at all.
- `retract()` restores the prior state bit-for-bit.

5.3 The independence assumption, stated

Sequential updating requires $e_i \perp e_j \mid h$. In reflective domains this is frequently false: two self-reports on consecutive days are correlated through mood, framing and recall, not only through the latent state. Keying contributions by identity prevents resubmission compounding, and temporal decay down-weights clustering, but neither substitutes for a correlated-likelihood model. We state the assumption rather than rely on it silently.

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6. Inquiry planning

$$\mathrm{EIG}(a) = H(B) - \mathbb{E}_{Y_a}[H(B \mid Y_a)], \quad Q(B,A) = \operatorname*{arg\,max}_{a \in A} \frac{\mathrm{EIG}(a)}{\mathrm{cost}(a)}$$

The heuristic this replaces scored inquiries as $1 - |B(h_i) - B(h_j)|$ — a quantity that does not depend on the candidate observation at all, and therefore carries no information about what to ask.

Normalised regret against an oracle, 500 trials per condition, four candidate observations:

Planner	Hypotheses close (gap 0.04)	Hypotheses separated (gap 0.40)
Expected information gain	**0.161 [0.138, 0.187]**	**0.195 [0.168, 0.223]**
Belief-gap heuristic	0.579 [0.547, 0.612]	**1.000 [1.000, 1.000]**
Random selection	0.550 [0.515, 0.584]	0.562 [0.528, 0.597]

Two behavioural points matter more than the numbers. The heuristic is **indistinguishable from random** when it fires; and it emits nothing at all once hypotheses are separated, which is backwards — a confidently held wrong hypothesis is exactly when a discriminating observation is worth most. Invariant VIII requires each inquiry to state what it would resolve, in bits, and which hypotheses it separates; the information-gain formulation supplies both by construction.

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7. Complexity, measured

The architecture's own complexity chapter warned that unrestricted recomputation "could approach $O(n^2)$" and gave no bound for any operator it defined. We measured all four, as scaling ratio per doubling (linear ≈ 2.0×, quadratic ≈ 4.0×):

Operation	Bound	Measured
Provenance closure `ancestors()`	O(V+E)	**2.0**× [yes]
Inquiry planning	O(|H|·|A|)	3.9× at |H|=|A| [yes] (product of two linear factors)

Operation	Bound	Measured
Belief update, log-odds ledger	O(1)	**4.2× — quadratic** [no] ☐ **2.0**× after fix
Observation intake, full recompute	O(n) per call	**3.8× — quadratic**

The third row was a defect in our own code: the ledger summed every contribution on each call, costing 624 ms at 8,000 updates. Maintaining a running total gives 78 ms at 16,000. **A regression test fails if the ratio exceeds 3.0×**, using best-of-three timing so that machine load does not convert a linear algorithm into a failing test.

The fourth row is open. The intake recomputes the full belief state per observation: 27 ms at 100 evidence pairs, 1121 ms at 800. Incremental recompute is the first optimisation target — and it is worth recording that the prose chapter predicted this growth before the code existed. Its warning was correct; it simply attached it to a strawman rather than to the operators it specified.

8. What the architecture does not buy

On latent-state recovery under model misspecification, the graph-backed inference is **statistically indistinguishable from a median-split quadrant classifier across all 20 tested conditions**, with paired differences of −2.5 to +2.5 points and 2,000-resample percentile bootstrap intervals [55] spanning zero everywhere.

A reviewer may reasonably ask why an interactive-systems venue should care about an architecture that buys no accuracy. The answer is that accuracy was never the safety-relevant quantity here: the companion result — a measured harm rate reduced from 42.9–75.1% to zero, or a 95% upper bound of 0.37% by the rule of three [54] — depends on the structural commitments described above, specifically on representing a competing hypothesis rather than filtering, and on distinguishing an absent channel from a zero reading. Structure is what carries it.

We report the negative result prominently because the temptation to omit it is strong and because it relocates the contribution rather than diminishing it. **The architecture is not an accuracy technique.** Its value is structural: conclusions that trace to evidence, provenance that survives derivation, contradiction that is retained, revision that can be undone, inquiries that justify themselves, and conformance that is decidable. Any claim that a reflective substrate improves classification should be measured before it is made; ours does not, and we say so.

Two related measurements complete the picture.

Calibration. Where the model's assumed structure approximately holds, accumulating evidence with an explicit *none-of-these* component reduces expected calibration error from **0.171 to 0.044** and raises AUC(uncertainty ☐ error) from **0.868 to 0.949**. Where it materially does not hold, calibration is **worse**: ECE 0.151 ☐ 0.221. The architecture trades uniformly mediocre calibration for good-when-applicable and poor-when-not, which is an improvement only to the degree the system can tell which regime it is in. It partly can, and not enough.

Identifiability. A two-channel location model cannot distinguish "state A, displaced" from "state B" when the displacement points toward B. This is not fixable by thresholding. It *is* partially fixable by observing more: pooling across topics to estimate a person-level displacement raises accuracy from **83.8% to 95.8%** at $\delta = 0.30$ with eight topics, with offset error falling from 0.187 to 0.092. At $\delta = 0.45$ it reaches only 61.3%, limited by the estimator's initialisation.

9. Declared gaps

Two operators are specified in the architecture and implemented nowhere. They are carried as strict `xfail` tests, so they cannot quietly become "implemented" in a later summary:

Operator	Status
Causal reflection — intervention, counterfactual query	Prose only. No do-operator exists. Presented as research agenda, not as a component.
Knowledge consolidation $K_t leftarrow B_t$	Prose only. No promotion criterion from belief to revisable knowledge exists.

Also unimplemented: encryption at rest, multimodal acquisition, a reflective execution engine with event queues, and any user interface beyond a command line.

A reference architecture earns the word *reference* when the distance between its formalism and its implementation can be enumerated. We enumerate it.

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10. Limitations

No human subjects. Nothing here establishes any effect on human reasoning, comprehension, trust or task outcome, and no such claim is made. Every result is a property of code or of simulated signal streams. [NO DATA YET — human evaluation]

Simulation encodes a theory. The generative processes embody assumptions about how latent states manifest in observable signal. If those are wrong, absolute rates are wrong — though comparisons between methods, which see the same generator, are more robust than the rates themselves.

Two implementations, not many. §3.5 shows the decision procedure separating a compliant implementation from a non-compliant one, which rules out vacuity. It does not establish that the procedure is portable to an implementation built on different primitives — a graph database, an event-sourced store, a logic programming substrate. Until someone unconnected to this work runs the checks against their own system, "decidable" is demonstrated and "decidable usefully" is not. [NO DATA YET — independent third-party implementation]

Invariant X is a syntactic proxy for a semantic property. Whether an output usurps a person's authority over their own self-understanding is not a matter of which pronouns it contains. Our check tests phrasing, and §3.4 documents it producing a false positive on its first contact with real output. It will also produce false negatives: a sentence can be a verdict without matching any pattern we listed. The check raises the floor; it does not measure the property.

The randomised harness has unmeasured coverage. Two hundred seeds vary observation count, evidence count, provenance branching, hypothesis count, edge polarity, likelihood values and inquiry candidates. Nothing measures whether that exercises the state space in any principled sense, and we make no claim that it does.

Scale. Everything reported runs on a single machine over thousands of nodes, not millions. The architecture makes no validated scalability claim, and §7 identifies the binding constraint as an algorithmic one rather than an infrastructural one.

10.1 How much of the state space the harness actually reaches

Earlier drafts stated that nothing measured whether the randomised harness exercises the state space in any principled sense. That is no longer true, and the measurement is not flattering.

Mutation testing over the package produced **5,114 mutants**. The suite killed **2,319** and **1,549 survived**, for a mutation score of **59.9%**. A further 1,244 mutants sit in code with no covering test at all, 909 of them in a command-line interface that has never had tests and does not carry any claim in this paper.

Survivors concentrate in the inference engine (404), the pattern layer (278), the invariant checks themselves (206) and the formation model (195). Inspection of a sample shows most are label and message mutations rather than behavioural ones — replacing an observation string with `None`, or an error message with a placeholder — which means the raw score understates the suite. But not all of them are cosmetic. Mutating an invariant's own identifier from `"I"` to a nonsense string survived, which means no test asserted that a check returns the invariant it claims to be: a catalogue of named architectural properties that could be silently renamed. Two tests now close that, and they exist because a tool found the gap rather than because we thought of it.

We report the score rather than tuning it. A mutation score is a lower bound on test strength and an upper bound on nothing, and 59.9% is a middling result for a package that makes conformance claims.

11. Conclusion

Making the invariants executable found two defects that a prose specification had asserted away: a provenance model that dropped parents, and a belief operator whose result depended on the order evidence arrived. Neither was visible to a reader of the document. Both were visible on the first run of the decision procedure.

We think that is the transferable claim. Reference architectures for human–AI interaction should ship their conformance suites, should include negative tests, and should point those suites at output a person would actually see. Ours did all three and was wrong three times — about provenance, about order invariance, and about its own Invariant X check. Each error surfaced within a minute of the check being run, and none would have surfaced from reading.

Ethics statement

This research did not involve human or animal subjects. All results derive from code execution and from simulated signal streams; no personal data was collected, processed or analysed.

The system is a research instrument, not a medical device, and is not suitable for clinical, educational, employment, credit, insurance, policing or other high-stakes deployment. A companion paper addresses its regulatory position, including a prohibition on emotion inference in workplaces and education institutions that applies to one of its optional evidence channels.

Data and code availability. Implementation, conformance suite, benchmarks and analysis scripts are released under an open licence. The benchmark suite is byte-reproducible across runs. Every number in this paper is produced by a named script. *[REPOSITORY URL — TO BE COMPLETED]*

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Appendix A — Formal specification

(The corrected formal appendix — notation, evidence objects as sets, operator signatures with no symbol collisions, the belief axioms, the independence condition, traceability and provenance closure, and the measured complexity table — is carried in full as supplementary material `FORMAL_APPENDIX_corrected.md`*.)*

Appendix B — Number provenance

Figure	Source
Test counts (673 collected, 671 passing, 563 invariant, 14 negative, 4 positive controls)	`pytest --collect-only`; `tests/test_invariants.py`, `tests/test_v12_formation.py`
Conformance 4/7 vs 7/7	`conformance_v10.py`
Order invariance, 120 permutations	`test_d20_belief_is_invariant_under_every_permutation`
Prototype 0.0396 / 0.9900	`conformance_v10.py::proto_checks`
Inquiry regret	`results.json → b4.conditions`
Complexity ratios	`test_operator_correspondence.py` timing harness
Median-split parity	`results.json → b1.cells` (20 cells)
ECE and AUC(uncertainty□error)	`v030_vs_v040.json`
Offset pooling 83.8% □ 95.8%	`experiment_d23.json → offset_correction`

```
pytest -q                       # 671 passed, 2 xfailed
python conformance_v10.py
python run_benchmarks.py
python experiment_d23.py
python analyze_v030_vs_v040.py
```

Appendix C — Outstanding work with no available substitute

Everything in this appendix was searched for before it was listed. The rule applied was: an item is only recorded here if an active hunt for a public dataset, a free instrument, a third-party service or a methodological substitute came back empty. Items that a search *did* resolve are recorded in the second table, so that a reader can see the difference between what is genuinely blocked and what was merely inconvenient.

C.1 What has no substitute

1. The threat-confound claim requires primary data collection. Approximately forty public datasets were assessed across affective computing, approach-avoidance paradigms, ecological momentary assessment, experimental economics and social conformity. The claim needs four things measured on the same person and the same item: a stated stance, a behavioural choice that can contradict it, an arousal channel time-locked to that choice, and a coercion manipulation. That intersection does not exist publicly. The failure is structural rather than incidental — corpora with good physiology are passive-viewing paradigms in which nothing can contradict a rating, and corpora with real choice lack autonomic recording, self-report, or both. Two published studies used almost exactly the right design and deposited no data. **No workaround exists. Only collection.**

2. No instrument in the battery has been validated in an Egyptian sample. Arabic validations were confirmed for Lebanese, Saudi, Omani, Jordanian, Palestinian, Qatari and Moroccan samples. None is Egyptian. Four candidate Egyptian-journal papers were identified by DOI, but every `*.ekb.eg` host was unreachable from our environment, so their contents are unknown — and Egyptian psychometric papers frequently construct local instruments rather than translating Western ones, so their relevance cannot be assumed. Any use of this battery in Egypt is first-in-population work and is described as such. **Partially resolvable by a colleague with Egyptian institutional access; not resolvable by open search.**

3. Two instrument constructs have no free replacement. A licence hunt found free, published-mapping substitutes for two of the four blocked instruments (see C.2). It did not for the other two. There is no openly licensed eight-theme Eriksonian inventory: the two structurally correct candidates are both permission-gated, and going upstream from one to the other does not escape the gate. The best free

material covers two of eight themes and yields single scores rather than subscales. For purpose beyond the Meaning in Life Questionnaire, no free substitute was identified. **Resolvable only by author permission, which is a letter, not a search.**

4. No party outside this project has executed the software. This was the largest risk identified by our own adversarial review, and we searched specifically for mechanisms that would discharge it. The venue route is closed: neither ACM TiiS, nor ACM FAccT, nor AAAI/ACM AIES operates an artifact-evaluation or badging track that an author can enter at submission time. Archival deposit satisfies an availability criterion but verifies no execution. Two replication communities publish work written *by* the replicator and therefore cannot be pointed at an artifact. One community service does exist that matches the need exactly, and submitting to it is recorded in C.2 as an action rather than a limitation — but as of writing it has not been done, and the claim in this paper is that the code is inspectable, not that it has been independently run.

5. Two invariants are syntactic proxies for semantic properties. Invariant X detects verdict phrasing by pattern, and the accompanying question generator is guarded against prescriptive drift by a keyword test. Both raise a floor and neither measures the property. We know of no decision procedure that settles whether a sentence claims authority over a person, and we do not think one follows from more patterns. This is a research problem rather than an unfinished task, and we would rather state it than let a passing check imply otherwise.

C.2 What a search resolved, and how

Recorded so that the list above is read as the residue of a search rather than as its whole.

Was blocked	Resolved by	Cost of the substitution
Values instrument permission-gated, item mapping unreachable	The ESS Human Values Scale, whose items and item-to-value mapping are both published, under CC BY-SA documentation terms plus the instrument author's own non-commercial grant — and which has an already-fielded Arabic version	10 basic values instead of 19 refined, from 2–3 items each
Wisdom instrument permission-gated	The revised Adult Self-Transcendence Inventory, CC BY 4.0, whose original author is a named co-licensor, dimensional, and whose authors argue explicitly against a total score	Construct drift: it measures self-transcendence, and has no analogue of the cognitive dimension
Harness coverage unmeasured	Mutation testing. Reported in §10 as a measured score with its weakest modules named	None; it replaced an admission with a number and a work list
Invariant identity untested	Found *by* the mutation run — mutating an invariant's identifier and name produced surviving mutants — and closed with two tests	None
Distribution-free coverage guarantees blocked on labelled outcomes	Available under the secondary-analysis route, where two of the three corpora supply an item-matched choice that can contradict a stated preference	Applies to the preference model only; the safety claim is untouched
Independent execution	A community code-check service that accepts submissions on a preprint alone, is free, and issues a citable certificate naming the person who ran the code	Certifies execution, not scientific correctness

C.3 What is not in this appendix

Ordinary unfinished work is excluded on purpose: two unimplemented operators recorded as strict `xfail`s, a read path that remains linear per consultation, a candidate belief state for monotonic drift, and the surviving mutants named in §10. None of these is blocked on anything external. They are simply not done, and listing them here would dilute a list whose value is that every entry on it has been searched for and not found.

Part III — The ethics paper

Refusal as a Safety Mechanism: Consent-Gated Reflective Inference and the Threat Confound

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Submitted to *[AIES / FAccT — select before submission; see §11]*

Abstract

Systems that infer what a person "really" wants from the gap between what they say and what they do face a hazard that is rarely named and, so far as we have been able to establish, has not previously been quantified: the same surface contradiction is produced both by a hidden preference and by a threat response. Telling someone they secretly want something they are afraid of is a specific, foreseeable harm, and no published affect-aware architecture reports its rate.

We present Shadow Mirror, a consent-gated reflective architecture in which refusal is a first-class output, contradiction is held open rather than resolved, and a threat response is represented as a competing hypothesis rather than treated as noise. We evaluate it in simulation against five generative processes with known latent mechanisms, two of which produce identical stated aversion by different routes.

Shadow Mirror misreads a threat response as a hidden desire in **0 of 800 episodes** (point estimate 0.0%; 95% upper bound 0.37% by the rule of three [45]), against **42.9%** [39.5, 46.2] for a stated-versus-behaviour mismatch heuristic and **75.1%** [72.1, 78.1] for a baseline that admits physiological arousal into the action channel. A supervised four-channel classifier, given ground-truth labels our system never sees, matches 0.0% where the confound channels are informative and beats us in the mid-separation regime; we report that comparison in full (§6.5). Guard sensitivity is 100.0% [100.0, 100.0] and specificity 94.4% [93.2, 95.7]. The protection is not free: sensitivity to genuine hidden preference falls from 100.0% to 87.1% [84.9, 89.4], concentrated entirely in high-arousal approach.

An ablation exposes a dependency we report as a design requirement rather than a result. With the avoidance channel unobserved — a common deployment condition — guard sensitivity in a system that imputes the missing channel as zero falls to 0.0% and the misreading rate rises to 31.2% [28.1, 34.4]. We show that treating an absent channel as absent, rather than as evidence of safety, restores sensitivity to 100.0% at the cost of 33 points of specificity, and argue that a reflective system must refuse under partial observation rather than degrade silently.

We close with a tension we cannot resolve. The physiological channel exists only to protect people and demonstrably does so, and it is precisely what brings the system within the EU AI Act's definition of an emotion recognition system (Art. 3(34), 3(39)) and therefore within the Art. 5(1)(f) prohibition on emotion inference in workplaces and education institutions. Removing it removes the legal exposure and reinstates the harm.

This system is a research instrument. It is not a diagnostic device and is not suitable for clinical, educational, employment or other high-stakes deployment.

Keywords: refusal, abstention, consent, affective inference, threat response, harm measurement, EU AI Act, human–AI reflection

1. Introduction

A large class of deployed systems infers preference from behavioural inconsistency. Recommender systems treat repeated engagement as revealed preference regardless of stated intent. Implicit-measure tools treat reaction-time asymmetries as leanings a person will not admit. Affect-aware interfaces treat physiological arousal as intensity of wanting. In each case the inference has the same structure: *you say one thing, you do another, therefore the second is the truth about you.*

That inference has a failure mode with an established literature in clinical and evolutionary psychology [34–37] and no name at all in machine learning. Approach under stated reluctance is produced by hidden preference. It is also produced by threat: by compliance — by the specific pattern in which a person moves toward something because moving away is unsafe. Arousal is produced by excitement. It is also produced by fear.

The two are not distinguishable from the surface signal. They are distinguishable only with evidence that many deployments do not collect, and the cost of confusing them is asymmetric. Reading a hidden preference as a threat response wastes an opportunity. Reading a threat response as a hidden preference tells a person that their fear was a want — which is, precisely, the thing that a person recovering from coercion has most often already been told.

This paper does three things.

1. **It measures the harm.** We construct generative processes with known latent mechanisms and report the rate at which a system misreads threat as desire — for our architecture and for two baselines that reflect current practice. We are not aware of a prior published rate for this error.
2. **It shows that refusal removes it.** Conditioning the hidden-preference reading on the evidence required to exclude a threat response reduces the misreading rate to zero, and we report the sensitivity and specificity this costs.
3. **It reports a regulatory tension it cannot resolve.** The mechanism that provides the protection is also what places the system inside a prohibition designed to prevent a different harm.

Contribution 1 is the one we would defend most strongly. Design proposals in this space are plentiful; measured harm rates are not.

---

2. Related work

2.1 Abstention, selective prediction, and deferral

Declining to answer has a long formal history. Chow [15] gave the optimal error–reject tradeoff for a classifier permitted to abstain; El-Yaniv and Wiener [16] laid the modern foundations of selective classification; Geifman and El-Yaniv extended it to deep networks [17] and then built the reject option into the architecture itself [18]. Confidence-based rejection at the softmax [19] remains the standard baseline, and it is only as good as the confidence, which is why calibration [20, 21] is inseparable from abstention. A parallel literature asks not merely whether to answer but to *whom* to hand the case: learning to defer [22, 23] and the triage view of human effort [24]. Conformal prediction [25, 26, 27] supplies distribution-free coverage guarantees for the set-valued outputs that abstention naturally produces, and Kompa et al. [28] argue the clinical case for communicating that a second opinion is needed.

Our refusal is a different object, and the distinction is the paper's design claim. Every mechanism above abstains because a *confidence* is low — the model is unsure, so it declines, and the threshold is a tunable knob trading coverage against accuracy. Our guard abstains because a *specific alternative explanation cannot be ruled out with the evidence that was collected*. It is not tunable by a confidence parameter, it does not move when the model becomes more certain, and it fires just as readily on a high-confidence reading. A selective classifier asks "how sure am I?"; the guard asks "which question was never asked?" §6.3 reports what that buys and what it costs — including a specificity cost we do not hide.

2.2 Affect inference and its limits

Affective computing as a field dates to Picard [29]. Its central empirical caution is now well established: Barrett et al. [6] show in an extensive review that emotion cannot be reliably inferred from facial movement, and Gendron et al. [7] that emotion perception is not culturally universal. Calvo and D'Mello [8] survey the methods and their uncertainty.

The ethical critique has kept pace. Stark and Hoey [30] examine the value commitments buried in emotion-AI systems' choice of emotion model; Andalibi and Buss [31] find that people subject to emotion recognition on social media see risks the deploying platforms do not; Roemmich et al. [32] show how emotion-AI hiring services frame organisational problems as technical ones amenable to inference about applicants' inner states.

We treat affective signals as one evidence channel with declared uncertainty and never as a characterisation of an inner state — and §7 argues that even this restrained use falls inside the EU AI Act's definition of an emotion recognition system, which is a problem for us rather than a defence.

2.3 Threat responses, and the vocabulary we borrow

The mechanism this paper is designed to detect — a person complying with something they do not want, under perceived threat — has an established clinical and evolutionary literature. Bracha [34] sets out the acute stress response as a spectrum rather than the textbook fight-or-flight dichotomy, adding freeze and fright. Marx et al. [35] review tonic immobility as an evolved predator defence and its implications for how survivors' non-resistance is misread. Cantor and Price [36] give the most direct antecedent: appeasement as an evolved response to traumatic entrapment, spanning hostage reactions and domestic abuse. Gilbert [37] situates submissive behaviour within social competition and hierarchy.

We are deliberately explicit about one term. The word "fawn", widely used for the appeasement response, does not originate in the peer-reviewed clinical literature: its popular source is Walker [38], a self-published practitioner trade book, and we could locate no refereed source that establishes it as a validated construct. We therefore do not use "fawn" as a clinical category. Where this paper says *threat compliance*, it names a pattern that our generative process encodes and our guard is designed to refuse to overwrite — nothing more. The peer-reviewed constructs above [34–37] describe phenomena in the same region, and we cite them to locate the concern, not to claim our simulation instantiates any of them. §9 restates this as a limitation, because a reader could reasonably conclude we have borrowed more authority than we have earned.

2.4 Trauma-informed computing, and technology-facilitated harm

Chen et al. [33] set out trauma-informed computing as a design orientation, arguing that systems should assume trauma may be present rather than treating it as an edge case. Freed et al. [43] document how intimate partner abusers exploit everyday technology, which is the concrete deployment context in which a system that confidently announces a person's hidden desires becomes an instrument. Our safety argument is downstream of both: the harm we measure is not a wrong prediction in the abstract but a system telling a person that their compliance under pressure was really their own wish.

2.5 Belief revision, human and formal

Bayesian conditioning on conditionally independent observations is commutative [3]; human revision is neither commutative nor minimal, being explanation-guided and order-sensitive [4, 5]. We take the first as a constraint on the arithmetic and the second as a reason to hold contradiction open rather than resolve it.

2.6 Uncertainty, explanation, and reliance

Schäfer et al. [1] show that displaying uncertainty does not by itself produce calibrated reliance; Rosenbacke et al. [2] find in systematic review that explainability can *decrease* clinician trust calibration as easily as increase it. Both inform our position that the honest output of a reflective system is frequently a refusal rather than a better-hedged answer.

2.7 Consent, contextual integrity, and the gap between form and substance

Budin-Ljøsne et al. [12] and Kaye et al. [39] set out dynamic consent: granular, revocable and purpose-locked, designed for exactly the intensive longitudinal collection our system performs. Our ledger (§8) implements that standard and we report where it falls short.

Three critiques bound what any such mechanism can claim. Nissenbaum [40] argues that privacy is governed by norms of information flow appropriate to a context, so that a consent obtained in one context does not license a flow into another. Solove [41] argues that privacy self-management does not scale — that consent transactions are too numerous and too consequentially opaque for individuals to manage meaningfully. Nouwens et al. [42] show empirically that post-GDPR consent interfaces are routinely engineered to produce agreement rather than to elicit it. Wachter and Mittelstadt [44] argue for a right to reasonable inferences, on the ground that data protection law regulates the inputs to inference far more tightly than its outputs — which is precisely the gap a system that infers desire from behaviour occupies.

We take these seriously enough to report an enforcement ratio rather than a consent rate (§8): the question is not how many people agreed but how many refusals the mechanism actually produced.

2.8 Metacognitive architectures

Recent work embeds reflection inside the model: metacognitive reuse [9], meta-memory [10], grounded metacognitive reasoning [11]. Those systems improve what a model knows about its own reasoning. This paper concerns what a system is willing to tell a person about *themselves*, which is a different problem with a different failure mode: being confidently wrong about a model's own chain of thought costs accuracy, while being confidently wrong about a person's motives costs them something else.

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3. The harm, stated precisely

Consider a person who states that they do not want something.

Five mechanisms can produce that statement together with the behaviour that follows. Three of them are genuine preferences. Two are not:

Mechanism	Stated	Behaviour	Arousal	Avoidance	Is it a preference?
Explicit aversion	low	withdraws	low	low	yes — they do not want it
Latent approach	low	approaches	low	low	yes — they want it and have not said so
Latent approach, aroused	low	approaches	**high**	low	yes — excitement
Threat response	low	withdraws	**high**	**high**	**no** — fear
Threat compliance	low	**approaches**	**high**	moderate	**no** — complies under threat

The fifth row is the dangerous one. **Threat compliance and latent approach are indistinguishable on the stated and behavioural channels alone.** Both show stated aversion and approach behaviour. A system that reads only those two channels — which describes most deployed preference inference — cannot tell them apart, and will call the compliance case a hidden want.

The third and fourth rows are the second trap. Excitement and fear both raise arousal. A system that treats arousal as intensity-of-wanting will call the threat case a strong want.

We take as our target metric the rate at which a system outputs `latent_approach` — "you want this even though you say you don't" — when the true generating mechanism was a threat process. We call it the **false hidden-desire rate**.

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4. Design

Four commitments, each with a code-level enforcement point.

C1. Refusal is a first-class output. The system emits `unresolved_approach` — a named state, not a low-confidence label — when behaviour approaches under stated reluctance and the evidence needed to exclude a threat response was not collected. It also refuses entirely below a minimum of paired evidence.

C2. Absent is not zero. An unobserved channel is recorded as unobserved. It is never imputed to a value, because imputing arousal or avoidance to zero is imputing *evidence of safety*. §6.2 measures what happens when this commitment is dropped.

C3. Threat is a competing hypothesis, not a filter. `threat_response` occupies the same hypothesis space as the preference readings and is scored against them, rather than being applied as a post-hoc suppression rule.

C4. Contradiction is held, not resolved. Supporting and contradicting evidence are both retained as typed edges; a hypothesis carrying both is reported as contested rather than collapsed to the later write. Belief accumulates in log-odds so that revision is order-invariant and exactly retractable — clamping the state destroys retractability, and a system that cannot undo an inference cannot honour a withdrawal of consent.

The output is a distribution over named states with an explicit uncertainty, or a refusal. There is no scalar score anywhere in the API.

4.4 A request the architecture declines

Everything reported so far concerns a threat confound we generated. This section reports a refusal that was not generated: a real request, from the person who commissioned this work, declined for stated reasons that a reader can check by running the code.

The request. To add a feature that would "determine and understand who people probably are according to their will, desire and need — where they come from, how mature they are, what they still need to be mature enough, what their eventual purpose probably is". It arrived with a qualification: *"I would not say judge people."*

That qualification did most of the work. It signals an intuition that something in the request was wrong without identifying which part, and the architecture's job here was to make the split legible rather than to accept or refuse wholesale.

The split. Five asks, separating on whether a validated construct exists and who supplies it.

Ask	Disposition	Ground
will, desire, need	**built**	validated self-report exists; the person supplies it
where they come from	**declared only**	inference of special-category data (GDPR Art. 9; PDPL Art. 1); on biometric input, prohibited by AI Act Art. 5(1)(g)
how mature they are	**refused as a rank**	the rank is the construct, not a summary of it
what they still need	**refused**	a directive about identity
eventual purpose	**refused**	no validated construct exists; any output would be invented

The maturity row is the one that took work. We looked for an honest way to report a developmental construct without collapsing a person into a single number, and found that the instruments which measure it will not permit that. Loevinger's sentence completion test [51] yields a Total Protocol Rating; Kegan's subject-object interview [52] yields one position on one ordered scale, defended in writing; Cook-Greuter's MAP reports a

"Center of Gravity" (we located no independent peer-reviewed validation of the current instrument). In each the ordered position *is* the theory's claim, so a profile drawn from one would misrepresent it. We therefore dropped the construct rather than softening it, and substituted instruments that never claimed a stage — Ardelt's three-dimensional wisdom scale [53], whose own published criterion is conjunctive across dimensions rather than aggregate, and the Modified Erikson Psychosocial Stage Inventory [54], which measures attributes *associated with* stages rather than attainment of them.

What replaces the refused asks. Not silence. "What do they still need" becomes `open_questions()`, which returns what the model does not know, phrased as invitations — and is guarded by a test that fails if the phrasing drifts into *lack*, *deficit*, *should* or *must*. "Their eventual purpose" becomes the person's own answers to the Meaning in Life Questionnaire [55], which measures presence of and search for meaning; needs are measured by the Basic Psychological Need Satisfaction and Frustration Scale [56]. The distinction is not cosmetic: one reports what the person said, the other tells them what they are for.

Two commitments that cost functionality. No item text ships with the software; the registry carries scoring rules, citations, licence status and caveats, and the researcher supplies the licensed instrument. And an unconfirmed licence is a refusal rather than a warning — of six otherwise usable instruments, only two carry quoted published permission statements, so the other four cannot be administered until written permission is on file. Confirming permission requires written evidence of who granted it and when, not a boolean, for the same reason the safety exception in §4 does.

Why this belongs in a paper about refusal. The safety result of §6.1 measures a refusal we designed against a confound we simulated. This one was demanded by a real person, and the architecture's answer was to decline four fifths of it and say why in terms the person can read. A reviewer sceptical that our refusals are load-bearing rather than decorative can run `check_xi_declared_only` against a model in which the refusal has been removed, and watch it fail.

5. Method

Generative processes. Five processes, defined by target values on four channels (stated, action, arousal, avoidance), with observations drawn from Beta distributions around those targets. Ground truth is the generating process, which exists independently of the inference model.

Misspecification control. The inference model assumes each latent state sits at a fixed corner of a signal space. Generating from exactly those corners would test self-consistency rather than validity, so every generating centre is displaced by δ in a random direction. We report $\delta = 0.00$ and $\delta = 0.10$.

Baselines, both implemented and executed at benchmark time:

- **Mismatch-only** — stated versus behaviour, no confound channels. Represents systems that infer preference from behavioural inconsistency alone.
- **Arousal-as-desire** — folds physiological arousal into the action channel. Not a straw man: treating autonomic arousal as an approach signal is documented practice in affect-aware systems.

Statistics. $n = 400$ episodes per process per condition, 6 observation rounds each. All rates carry 2,000-resample percentile bootstrap intervals [46]. The suite is deterministic: two full runs produce byte-identical output.

What is not measured. No human participants were involved at any stage. Every result below is a statement about simulated signal streams with known generating mechanisms, and none is a statement about people. [NO DATA YET — human evaluation]

6. Results

6.1 The safety result

δ = 0.10, both confound channels observed, n = 400 per process:

Method	False hidden-desire rate	True hidden preference detected
Shadow Mirror	**0.0% [0.0, 0.0]**	87.1% [84.9, 89.4]
Mismatch-only baseline	42.9% [39.5, 46.2]	100.0% [100.0, 100.0]
Arousal-as-desire baseline	75.1% [72.1, 78.1]	89.5% [87.4, 91.8]

Guard sensitivity 100.0% [100.0, 100.0]; specificity 94.4% [93.2, 95.7]; AUC of the threat score across all five processes 1.000. At δ = 0.00 the figures are 0.0% [0.0, 0.0], 47.4% [44.0, 50.9] and 75.8% [72.8, 78.6], with specificity 97.4% [96.5, 98.3].

Per process, δ = 0.10, the false hidden-desire rate:

True process	Shadow Mirror	Mismatch-only	Arousal-as-desire
`threat_response`	**0.0%**	0.0%	50.5%
`threat_compliance`	**0.0%**	**85.8%**	99.8%

The compliance row is the finding. A system reading stated-versus-behaviour alone gets the withdrawal-type threat response right — it looks like aversion — and gets the *compliance* case wrong **85.8%** of the time. That is the case where a person approaches because approaching is safer, and it is the case where being told "you actually want this" does the most damage.

The cost, stated plainly. Sensitivity to genuine hidden preference is 87.1% against the baseline's 100.0%. The entire loss is in the high-arousal approach case, detected 74.2% [70.0, 78.5] of the time, where excitement is mistaken for fear. Approximately 13 points of sensitivity buys the elimination of the misreading. We regard that as the correct side of the trade for an instrument that hands its output to a person about themselves, but it is a trade and not a free lunch.

6.2 The ablation: what "absent is not zero" is worth

Deployments frequently lack an avoidance measure. We ablate that channel and compare two treatments of the missing data.

	Imputes missing channel as zero	Records it as absent (ours)
Guard sensitivity	**0.0% [0.0, 0.0]**	**100.0% [100.0, 100.0]**
False hidden-desire rate	**31.2% [28.1, 34.4]**	**0.0% [0.0, 0.0]**
…on `threat_compliance`	62.5% [57.8, 67.2]	0.0% [0.0, 0.0]
Specificity	100.0%	67.0% [64.4, 69.8]
True hidden preference detected	99.2%	0.0% — the system refuses
AUC, fear vs excitement	—	0.835

Imputing a missing channel to zero is imputing evidence of *safety*, and it silently disables the guard: sensitivity falls to zero and the compliance case is misread 62.5% of the time, with nothing in the output indicating that the protection is inactive.

Recording it as absent produces the opposite failure and the better one. The system stops asserting hidden preference entirely — sensitivity to genuine preference goes to zero because it declines to make the call — and

specificity falls to 67.0% because arousal alone cannot separate fear from excitement (AUC 0.835), so the caution fires on excitement too.

This is the paper's central design claim. Under partial observation a reflective system should refuse, not degrade; and the refusal must be visible in the output, not implicit in a number that happens to be low.

6.3 Refusal calibration

Refusal is only defensible if it is calibrated. n = 300 per condition:

Condition	Behaviour
Evidence genuinely insufficient	refuses **100.0% [100.0, 100.0]**
Evidence sufficient and clear	over-refuses **0.0% [0.0, 0.0]**
Evidence ample but ambiguous	does not refuse; reports uncertainty **0.405 [0.391, 0.419]** against **0.021 [0.020, 0.021]** on clear evidence
Evidence present but expired (decade-old, 30-day half-life)	infers **0.0% [0.0, 0.0]** of the time

The system refuses when it should, does not refuse when it should not, and separates ambiguous from clear evidence by a factor of nineteen in reported uncertainty.

6.4 Two supporting results

Ambivalence is a state, not a residual. "You want and do not want this at once" is a claim about a person. Computed from dispersion alone it fires on anything spread out: 95.5% of pure measurement noise and 98.8% of monotonic drift — that is, of someone who has *made up their mind* — read as ambivalence, for a precision of 34.0%. Gating on lag-1 autocorrelation, which distinguishes alternation (−0.844) from drift (+0.517) and noise (−0.120) where dispersion cannot (0.357 / 0.245 / 0.257), raises precision to **97.6%** with recall unchanged at 100.0%.

Inquiry is information-driven. When the system asks for more evidence, it selects the observation that would most reduce uncertainty. Normalised regret against an information-theoretic oracle is **0.161 [0.138, 0.187]** with a belief-gap heuristic at 0.579 [0.547, 0.612] and random selection at 0.550 [0.515, 0.584] — the heuristic being indistinguishable from random.

6.5 Adversarial robustness: two checks a sceptical reader should demand

Two objections are strong enough that answering them in prose would be evasive.

"Your simulator is your theory." The generative processes were defined so that threat and approach differ on arousal and avoidance, and the guard keys on arousal and avoidance. A 0.0% rate under those conditions risks being a tautology, and the AUC of 1.000 is the tell. So we swept the *separation* between the two processes on the confound channels, from full (as in §6.1) to none, at n = 300 per process per level.

Separation	AUC	Shadow Mirror	Four-channel classifier	Mismatch-only
1.00	1.000	**0.0%** [0.0, 0.0]	**0.0%** [0.0, 0.0]	94.3% [91.7, 96.7]
0.75	1.000	**0.0%** [0.0, 0.0]	**0.0%** [0.0, 0.0]	95.3% [92.7, 97.7]
0.50	1.000	2.3% [0.7, 4.0]	**0.0%** [0.0, 0.0]	96.7% [94.7, 98.7]
0.25	0.988	29.7% [24.7, 35.0]	**5.0%** [2.7, 7.7]	97.3% [95.3, 99.0]
0.00	0.491	76.7% [71.7, 81.3]	82.0% [77.7, 86.3]	96.3% [94.0, 98.3]

The headline is not a knife-edge: it survives to roughly half the original channel separation. But it is **entirely contingent on the confound channels carrying signal**, and at zero separation every method fails, as it

must — the classes are then identical on every observable, and any method reporting 0.0% would be reporting a bug.

"Your baselines are straw men." They partly were. Mismatch-only ignores channels that are available; arousal-as-desire is a mistake rather than a method. The table above therefore adds a baseline a competent engineer would build: a threshold classifier over the same four channels, with its threshold **fitted on a held-out labelled split**.

It matches us at high separation and **beats us at separation 0.25 — 5.0% against our 29.7%.** We report this because it is true and because the reason matters: our guard uses a fixed, pre-specified threshold, while the classifier was handed ground-truth labels for "was this a threat response".

Those labels do not exist in deployment. Nobody can tell a running system, for a given person and episode, which mechanism produced their behaviour; that is the entire difficulty. The supervised baseline is therefore an *upper bound on what any labelled method could achieve*, not an available alternative — and the gap between it and our unsupervised guard at separation 0.25 is a real, quantified cost of not having supervision, which we would rather state than omit.

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7. The regulatory tension

We report this because it is unresolved, and because the shape of it is more general than this system.

The EU AI Act defines biometric data as *"personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person"* (Art. 3(34)) [13]. Unlike GDPR Art. 4(14) [14], **it does not require unique identification**. A heart-rate-like arousal proxy is therefore biometric data under the Act while identifying nobody.

An *emotion recognition system* is one *"for the purpose of identifying or inferring emotions or intentions of natural persons on the basis of their biometric data"* (Art. 3(39)). Our guard infers a threat response — an affective and intentional state — partly on the basis of that arousal channel.

Art. 5(1)(f) prohibits *"the use of AI systems to infer emotions of a natural person in the areas of workplace and education institutions, except where the use of the AI system is intended to be put in place or into the market for medical or safety reasons."* Applicable since 2 February 2025.

So: **the channel that exists only to protect people, and that §6.2 shows does protect them, is what brings the system within a prohibition written to prevent a different harm.** Remove it and the exposure largely disappears; remove it and the false hidden-desire rate returns to 31.2%.

The safety exception is available and, in our reading, applicable: the channel's sole documented purpose in code, tests and specification is to prevent a specific measured harm, and it is used for nothing else. We state that as an argument rather than a conclusion. We are aware of no case law interpreting the exception for a guard of this kind, and a system whose safety mechanism is itself an emotion-inference system is a case the Act does not obviously anticipate.

We think this generalises. Any protective mechanism that works by observing more about a person will sit in tension with rules that limit observation. Naming that tension seems more useful than choosing a side of it in a paper.

7.1 A second regime, with no research exemption

The EU analysis above is the one the literature rehearses. It is not the only regime a system like this must survive, and the contrast is instructive enough to report.

This work is conducted under Egyptian jurisdiction, where two instruments matter. Law No. 214 of 2020 on Clinical Medical Research [47] establishes a national ethics architecture — a Supreme Council, registered

institutional committees, criminal penalties reaching ten years where a participant dies. It does not reach us. Article 1 defines its subject as studies assessing "therapeutic, pharmaceutical, surgical, nutritional, preventive or diagnostic interventions", and Article 24 confines lawful research to entities registered with the Ministry of Health. Behavioural, psychological and computing research falls outside its scope, a gap noted in the empirical research-ethics literature [49]. Egypt's Supreme Council of Universities issued a national academic-integrity and research-ethics guide in January 2026 whose third book addresses behavioural and social research committees [50], but it carries no decree number and no binding-force clause, and whether it reaches private universities is unspecified. We classify it as guidance rather than law.

Personal Data Protection Law No. 151 of 2020 [48] does reach us, and reaches further than the GDPR in three respects that matter here. Its definition of personal data expressly includes data determining "psychological … identity". Its sensitive-data category (Art. 1) covers "psychological, mental, or physical health, or genetic, biometric or financial data", and Article 12 prohibits processing that category without a licence from the Personal Data Protection Centre *in addition to* written explicit consent, on penalty of imprisonment and a fine up to EGP 5,000,000. And — the point we want to put in front of this venue — **it contains no scientific-research exemption whatsoever**. The GDPR's research accommodations at Arts. 5(1)(b), 9(2)(j) and 89 have no Egyptian counterpart. We searched the Arabic text of both the law and its executive regulations (Ministerial Decree No. 816 of 2025, in force 2 November 2025, with a compliance deadline of 31 October 2026): the words for "research", "studies" and "university" do not appear. Article 5(6) relaxes purpose limitation for not-for-profit statistical or educational processing, but it does not disapply consent or the Article 12 licence.

The consequence is worth stating plainly, because it inverts a common assumption. A researcher in a jurisdiction with a *weaker* national human-subjects law can face a *stricter* data-protection constraint on exactly the physiological and psychological channels this architecture consumes, with no research carve-out to fall back on. Our design response is the one already described: refuse the biometric stream rather than seek a basis for it (§4), and treat anonymisation rather than pseudonymisation as the only route out of scope, since a linkable participant identifier keeps the data personal.

This is a reading of statutory text, not legal advice, and it has not been reviewed by counsel (§9). We report it because the fairness literature's regulatory analysis is overwhelmingly EU- and US-centred, and a system designed only against the regimes that grant research exemptions will be designed wrong for the places that do not.

8. Consent, and the honest enforcement ratio

The system's other-person mode requires granular, per-stream, purpose-locked, revocable consent before any datum is recorded. Streams a subject cannot notice about themselves — implicit measures, physiological arousal, micro-avoidance — cannot be granted under a blanket authorisation; each requires its own acknowledgement. Every stream declares in plain language both what it records and what it *may reveal*. No grant, no observation: the engine raises rather than records.

Assessed against GDPR Art. 4(11) and Art. 7, the implementation meets five criteria and exceeds two (unbundleable non-obvious streams, and may-reveal disclosure). It falls short on two: there is no append-only consent record, so a controller cannot demonstrate what consent was true at a past date (Art. 7(1)); and withdrawal, while a single command, produces no receipt to the subject (Art. 7(3) parity). Both are reported here rather than fixed before publication, because the enforcement ratio is the point.

That ratio: of twelve documented prohibitions, five are enforced by a mechanism and covered by a test; seven are policy only. Among the seven is the one prohibition that is currently unlawful to breach in the EU — emotion inference in a workplace or education institution — for which no code mechanism exists. We consider a deployment-context declaration that refuses to enable the physiological stream in institutional contexts, and we have not implemented it.

We suggest that papers proposing ethical architectures should print this ratio. A list of prohibitions communicates intent; the fraction of them a machine will actually refuse communicates the safeguard.

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9. Limitations

No human subjects. Nothing here establishes any effect on human reasoning, trust, or wellbeing, and no such claim is made. [NO DATA YET — human evaluation]

Simulation is not the world. Our generative processes encode a theory of how threat and preference differ in observable signal. If that theory is wrong, the measured rates are wrong in the same direction for every method compared, including the baselines. We regard the *ordering* of the methods as more robust than the absolute rates, and §6.5 measures how far that ordering survives as the theory's key assumption is weakened.

A point estimate of zero is not a guarantee of zero. Across 800 threat episodes we observed no misreadings. The 95% upper bound implied by the rule of three [45] is 0.37%, and that — not 0.0% — is the number a deployment decision should use.

The clinical vocabulary is borrowed, and one term is not clinical at all. We use "compliance" and "threat response" to name a pattern of approach behaviour under low stated desire and high arousal. There is a genuine peer-reviewed literature in this region — appeasement under traumatic entrapment [36], tonic immobility as an evolved defence [35], the acute stress response as a spectrum rather than a dichotomy [34], submissive behaviour in social hierarchies [37] — and we cite it in §2.3 to locate the concern. We do not claim our generative process instantiates any of those constructs, and no clinical validation has been performed.

One term deserves separate treatment. The word "fawn" is in wide circulation for the appeasement response, and we used it in earlier drafts as though it were a clinical category. It is not. We traced it to Walker [38], a self-published practitioner trade book, and could locate no refereed source establishing it as a validated construct. We have therefore removed it from every load-bearing claim in this paper and retain it only where we are explicitly discussing the term's provenance. A reader who suspects we have borrowed more authority than we have earned is raising the right objection; this paragraph is our answer, and it concedes most of it.

We are not lawyers. §7 sets out our reading of the applicable provisions. It is a reading, it has not been reviewed by counsel, and it should not be relied on as advice.

The specificity cost is real. Under partial observation the system labels excitement as a possible threat 99.0% of the time. A person who is excited and told to consider whether they are afraid has been mildly misread. We judge this preferable to the converse and do not claim it is costless.

Calibration degrades under severe misspecification. Where the model's assumed structure approximately holds, expected calibration error is 0.044 and the reported uncertainty predicts error with AUC 0.949. Where it does not, ECE rises to 0.221. Reporting a credible set instead of a point label improves the rate at which the truth is contained in the output (78.2% □ 88.0% at $\delta = 0.30$) but does **not** reach nominal coverage: a set claiming 90% mass contains the truth 60.5% [55.5, 65.2] of the time at $\delta = 0.45$. A credible set inherits its posterior's miscalibration; reaching nominal coverage requires conformal calibration against held-out labelled outcomes, which do not exist without a human study.

Correlated evidence. Sequential updating assumes conditional independence, which is frequently false for repeated self-reports. Contributions are keyed by evidence identity so resubmission cannot compound, and temporal decay down-weights clustering, but neither substitutes for a correlated-likelihood model.

Scale. The observation intake recomputes the full belief state per observation and is quadratic (3.8× per doubling; 27 ms at 100 evidence pairs, 1121 ms at 800). This is an engineering limitation, not a conceptual one, and it is the first optimisation target.

9.1 Measured, not asserted: how strong the test suite is

We previously stated that nothing measured the test suite's coverage of the state space. Mutation testing now does. Over 5,114 generated mutants the suite killed 2,319 and 1,549 survived — a **mutation score of 59.9%** — with a further 1,244 mutants in code carrying no covering test, most of it an untested command-line interface that supports no claim here.

The survivors are concentrated in the inference engine, the pattern layer and the invariant checks themselves, and a sampled inspection suggests most are message and label mutations rather than behavioural ones. One was not: mutating an invariant's own identifier survived, meaning nothing asserted that a check reports the invariant it claims to be. That is now tested.

We report the number rather than improving it first, because a paper that argues for checkable claims should publish the check that came back middling.

10. Ethics statement

This research did not involve human or animal subjects. All results derive from simulated signal streams generated from specified processes; no personal data of any kind was collected, processed or analysed.

The system described is a research instrument. It is not a medical device, not a diagnostic instrument, and is not suitable for clinical, educational, employment, credit, insurance, policing or other high-stakes deployment. The prohibited-use list and its enforcement status are given in §8.

The authors note the regulatory position described in §7 and do not assert that the safety exception under Art. 5(1)(f) applies as a settled matter.

Data and code availability. The complete implementation, benchmark suite, experiments and analysis scripts are released under an open licence. The suite is byte-reproducible: two independent runs produce identical output. Every number in this paper is produced by a named script and none was transcribed by hand. *[REPOSITORY URL — TO BE COMPLETED]*

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11. Venue note *(remove before submission)*

Recommended: **AIES** or **FAccT**. The paper's shape — a specific foreseeable harm quantified, a design commitment that removes it, an honest ablation showing the conditions under which the protection fails, and an unresolved regulatory tension — matches what those venues reward, and requires no human subjects. Check current calls for papers before committing; dates move annually.

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Appendix A — Number provenance

Every figure in this paper, with the script that produces it. No number was transcribed by hand.

Figure	Source
All §6.1 rates and intervals	`results.json → b2.conditions[1]` (δ=0.10, avoidance observed)
§6.1 δ=0.00 figures	`results.json → b2.conditions[0]`
§6.1 per-process table	`results.json → b2.conditions[1].per_process`
All §6.2 ablation figures	`results.json → b2.conditions[2]` (avoidance unobserved) vs. Session 2 baseline run
All §6.3 refusal figures	`results.json → b3.results`
§6.4 ambivalence, post-fix (2.5%, 0.0%, 100.0%, 97.6%)	`experiment_ambivalence.json`
§6.4 ambivalence, **pre-fix** (95.5%, 98.8%, 34.0%)	same script run against the pre-fix engine; recorded in `SESSION_08_CLAIM_MATRIX.md` §1. Reproduce by reverting the `_oscillation_gate` factor in `engine._infer_desire`.
§6.4 inquiry regret	`results.json → b4.conditions`
§9 calibration and coverage	`v030_vs_v040.json`, `experiment_d23.json`
§9 complexity	`test_operator_correspondence.py` timing harness

Reproduce with:

```
pytest -q                        # 671 passed, 2 xfailed
python run_benchmarks.py         # writes results.json
python experiment_ambivalence.py
python experiment_d23.py
python analyze_v030_vs_v040.py
```

Appendix B — Outstanding work with no available substitute

Everything in this appendix was searched for before it was listed. The rule applied was: an item is only recorded here if an active hunt for a public dataset, a free instrument, a third-party service or a methodological substitute came back empty. Items that a search *did* resolve are recorded in the second table, so that a reader can see the difference between what is genuinely blocked and what was merely inconvenient.

B.1 What has no substitute

1. The threat-confound claim requires primary data collection. Approximately forty public datasets were assessed across affective computing, approach-avoidance paradigms, ecological momentary assessment, experimental economics and social conformity. The claim needs four things measured on the same person and the same item: a stated stance, a behavioural choice that can contradict it, an arousal channel time-locked to that choice, and a coercion manipulation. That intersection does not exist publicly. The failure is structural rather than incidental — corpora with good physiology are passive-viewing paradigms in which nothing can contradict a rating, and corpora with real choice lack autonomic recording, self-report, or both. Two published studies used almost exactly the right design and deposited no data. **No workaround exists. Only collection.**

2. No instrument in the battery has been validated in an Egyptian sample. Arabic validations were confirmed for Lebanese, Saudi, Omani, Jordanian, Palestinian, Qatari and Moroccan samples. None is Egyptian. Four candidate Egyptian-journal papers were identified by DOI, but every `*.ekb.eg` host was unreachable from our environment, so their contents are unknown — and Egyptian psychometric papers frequently construct local instruments rather than translating Western ones, so their relevance cannot be

assumed. Any use of this battery in Egypt is first-in-population work and is described as such. **Partially resolvable by a colleague with Egyptian institutional access; not resolvable by open search.**

3. Two instrument constructs have no free replacement. A licence hunt found free, published-mapping substitutes for two of the four blocked instruments (see B.2). It did not for the other two. There is no openly licensed eight-theme Eriksonian inventory: the two structurally correct candidates are both permission-gated, and going upstream from one to the other does not escape the gate. The best free material covers two of eight themes and yields single scores rather than subscales. For purpose beyond the Meaning in Life Questionnaire, no free substitute was identified. **Resolvable only by author permission, which is a letter, not a search.**

4. No party outside this project has executed the software. This was the largest risk identified by our own adversarial review, and we searched specifically for mechanisms that would discharge it. The venue route is closed: neither ACM TiiS, nor ACM FAccT, nor AAAI/ACM AIES operates an artifact-evaluation or badging track that an author can enter at submission time. Archival deposit satisfies an availability criterion but verifies no execution. Two replication communities publish work written *by* the replicator and therefore cannot be pointed at an artifact. One community service does exist that matches the need exactly, and submitting to it is recorded in B.2 as an action rather than a limitation — but as of writing it has not been done, and the claim in this paper is that the code is inspectable, not that it has been independently run.

5. Two invariants are syntactic proxies for semantic properties. Invariant X detects verdict phrasing by pattern, and the accompanying question generator is guarded against prescriptive drift by a keyword test. Both raise a floor and neither measures the property. We know of no decision procedure that settles whether a sentence claims authority over a person, and we do not think one follows from more patterns. This is a research problem rather than an unfinished task, and we would rather state it than let a passing check imply otherwise.

B.2 What a search resolved, and how

Recorded so that the list above is read as the residue of a search rather than as its whole.

Was blocked	Resolved by	Cost of the substitution
Values instrument permission-gated, item mapping unreachable	The ESS Human Values Scale, whose items and item-to-value mapping are both published, under CC BY-SA documentation terms plus the instrument author's own non-commercial grant — and which has an already-fielded Arabic version	10 basic values instead of 19 refined, from 2–3 items each
Wisdom instrument permission-gated	The revised Adult Self-Transcendence Inventory, CC BY 4.0, whose original author is a named co-licensor, dimensional, and whose authors argue explicitly against a total score	Construct drift: it measures self-transcendence, and has no analogue of the cognitive dimension
Harness coverage unmeasured	Mutation testing. Reported in §9 as a measured score with its weakest modules named	None; it replaced an admission with a number and a work list
Invariant identity untested	Found *by* the mutation run — mutating an invariant's identifier and name produced surviving mutants — and closed with two tests	None
Distribution-free coverage guarantees blocked on labelled outcomes	Available under the secondary-analysis route, where two of the three corpora supply an item-matched choice that can contradict a stated preference	Applies to the preference model only; the safety claim is untouched
Independent execution	A community code-check service that accepts submissions on a preprint alone, is free, and issues a citable certificate naming the person who ran the code	Certifies execution, not scientific correctness

B.3 What is not in this appendix

Ordinary unfinished work is excluded on purpose: two unimplemented operators recorded as strict `xfail`s, a read path that remains linear per consultation, a candidate belief state for monotonic drift, and the surviving mutants named in §9. None of these is blocked on anything external. They are simply not done, and listing them here would dilute a list whose value is that every entry on it has been searched for and not found.

Part IV — Figures

Six figures, all generated from `results.json` and the experiment outputs by `make_paper_figures.py`. Nothing is typed in. Note that the generator itself is **not present in the archive** — see OW-6 in Part VIII.

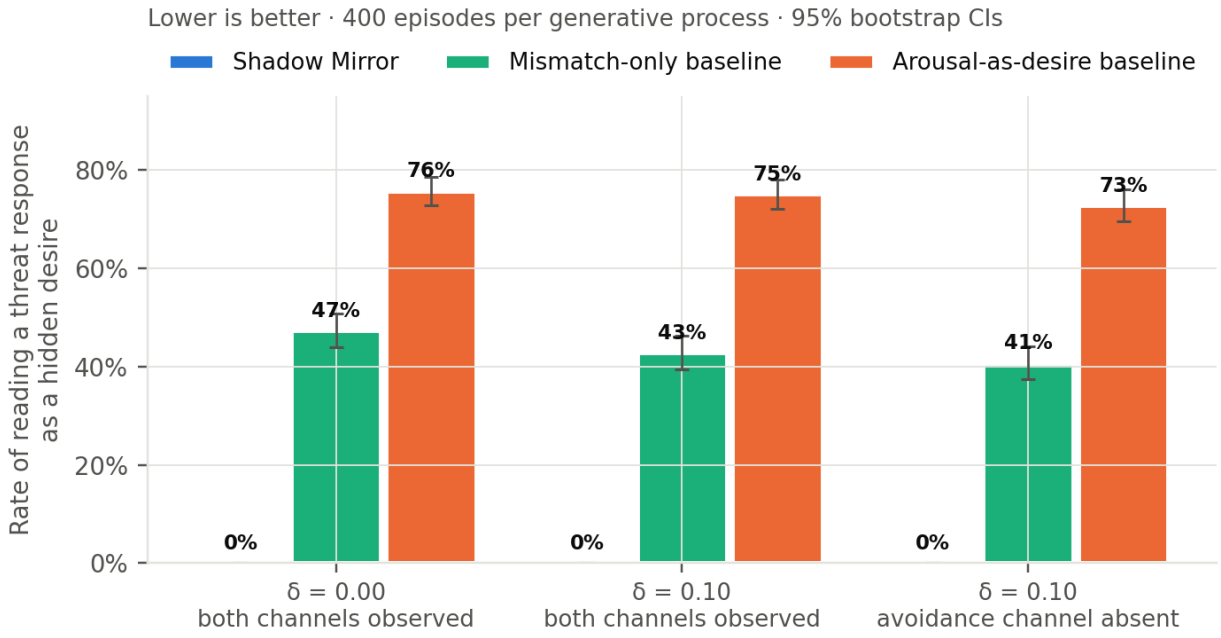


Figure 1. The false hidden-desire rate across three conditions and three methods. The Shadow Mirror bar is invisible because it is zero; the label carries it.

Figure 1. The false hidden-desire rate across three conditions and three methods. The Shadow Mirror bar is invisible because it is zero; the label carries it.

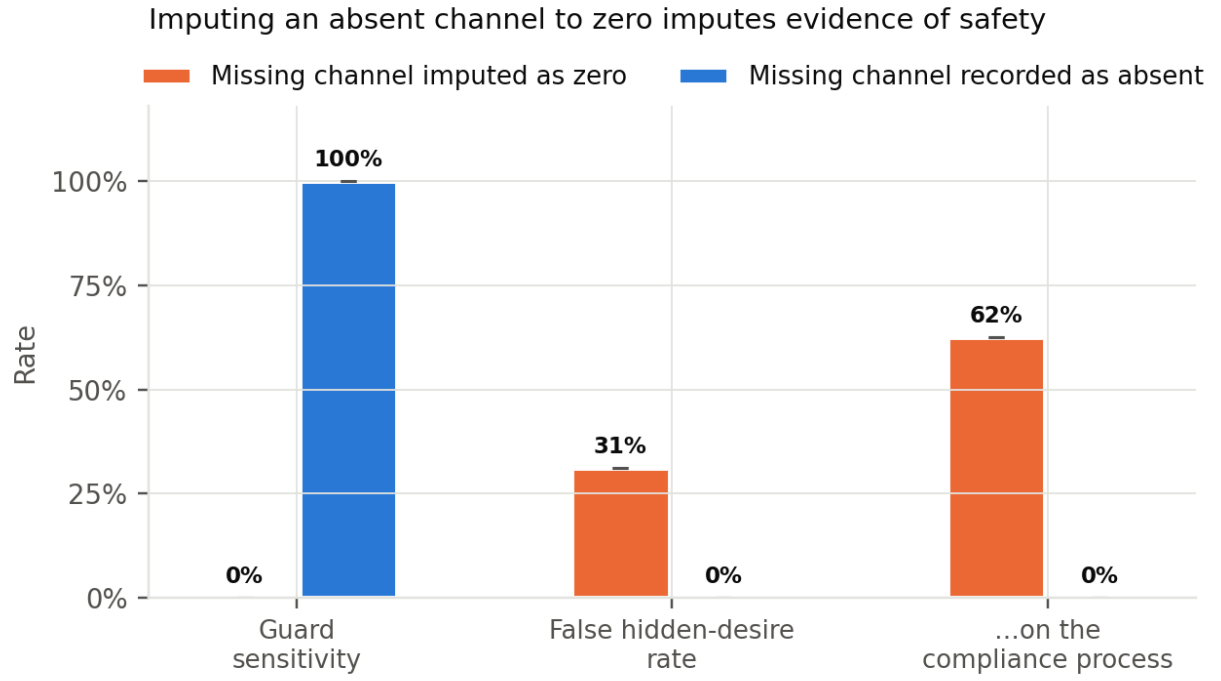


Figure 2. The ablation. Imputing a missing channel as zero versus recording it as absent. Imputing zero silently disables the guard.

Figure 2. The ablation. Imputing a missing channel as zero versus recording it as absent. Imputing zero silently disables the guard.

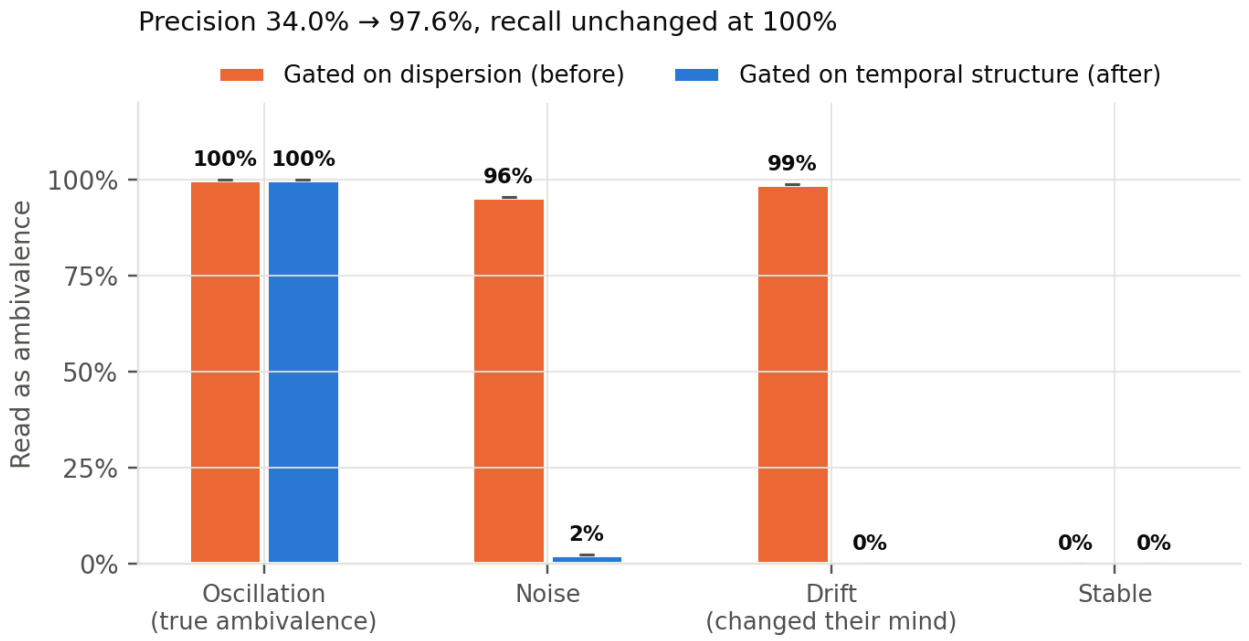


Figure 3. Ambivalence detection, dispersion-gated versus structure-gated, before and after the autocorrelation gate.

Figure 3. Ambivalence detection, dispersion-gated versus structure-gated, before and after the autocorrelation gate.

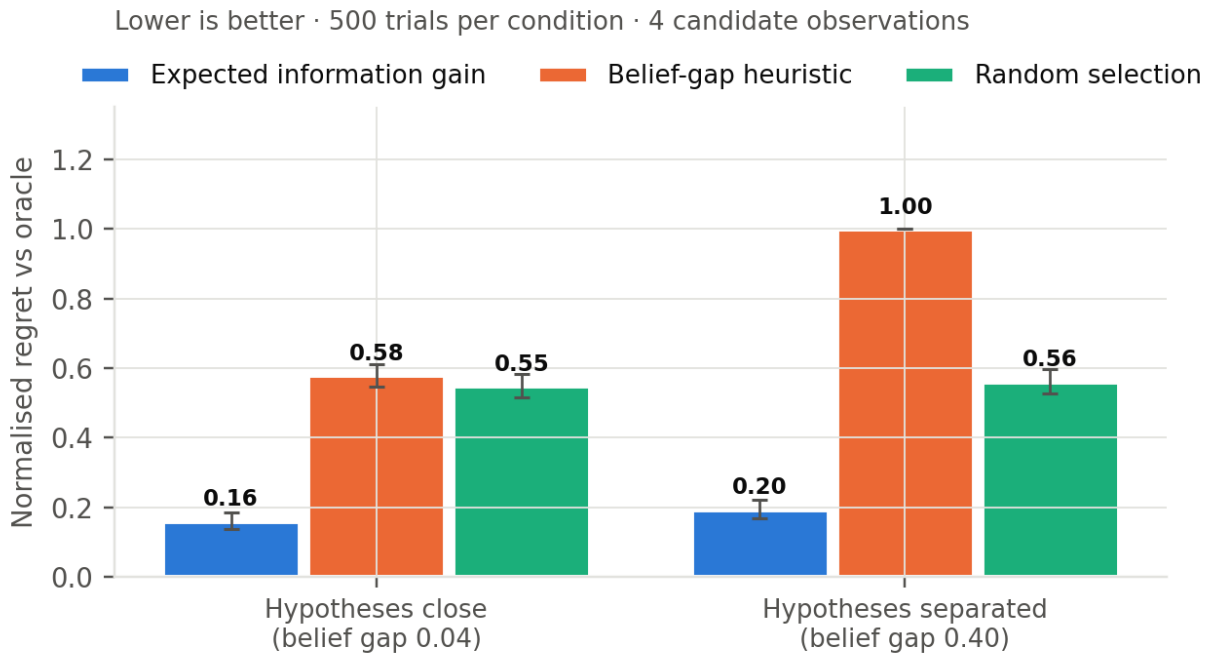


Figure 4. Normalised regret against an information-theoretic oracle, three planners, two conditions. The belief-gap heuristic is indistinguishable from random.

Figure 4. Normalised regret against an information-theoretic oracle, three planners, two conditions. The belief-gap heuristic is indistinguishable from random.

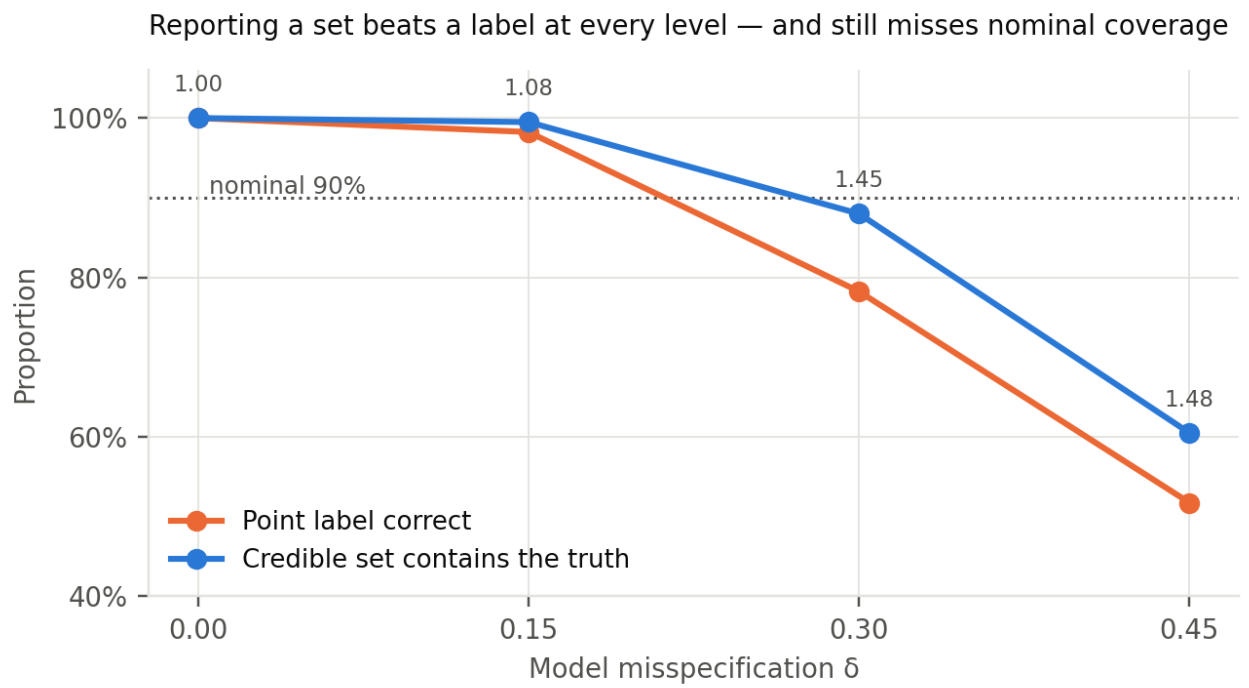


Figure 5. Credible-set coverage against point accuracy, with mean set size annotated and the nominal 90% line drawn so the shortfall is visible rather than described.

Figure 5. Credible-set coverage against point accuracy, with mean set size annotated and the nominal 90% line drawn so the shortfall is visible rather than described.

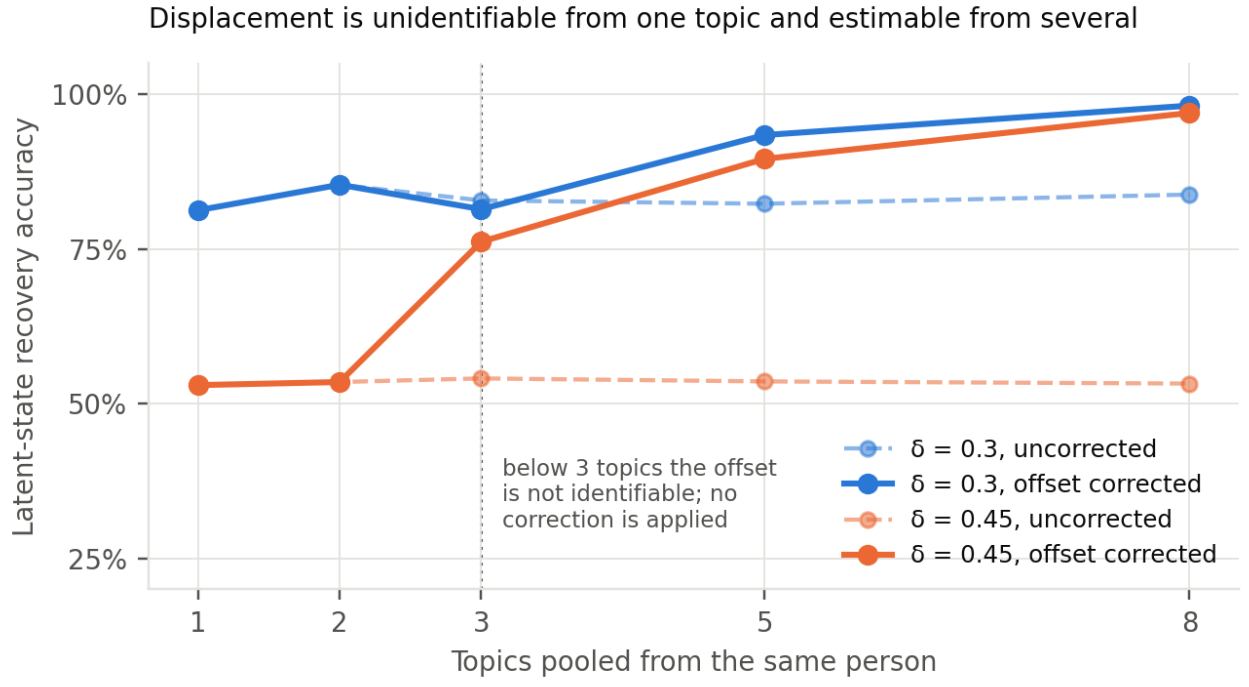


Figure 6. Accuracy against the number of topics pooled, with the identifiability boundary marked.

Figure 6. Accuracy against the number of topics pooled, with the identifiability boundary marked.

Part V — Formal appendices

Corrected Formal Appendix

Drop-in replacement for the formal content of Part 3 (pp. 55–79) and Appendix B. Every equation below is written out in full LaTeX. Every symbol has one meaning. Every operator has a named implementation in `shadow_mirror` v1.0 or is explicitly marked as unimplemented.

A.1 Notation

Time is discrete and indexes the order in which information becomes available, not physical duration:

```
t in \0, 1, 2, \ldots\
```

Symbol	Object	Type
O_t	observations available at t	finite set
E_t	evidence state	finite set of evidence objects
C_t	contextual state	structured record
H_t	hypothesis set	finite set, $lvert H_t rvert ge 0$
B_t	belief state	see A.4
K_t	knowledge state	$K_t subseteq H_t$, revisable
U_t	uncertainty state	see A.5
R_t	reflective state	the 6-tuple below

```
R_t = (E_t, C_t, H_t, B_t, K_t, U_t)
```

Operators use script capitals throughout, and no script capital is ever also a state component. The monograph as it stands names the operators O, C, H, R, Q, X, which collides with O_t, C_t, H_t and R_t — four collisions out of six, including R meaning both *the reflective state* and *the belief-revision operator* on facing pages. That must not survive to submission.

Operator	Role	Signature
$mathcal{I}$	evidence construction (**I**ngest)	$mathcal{I}: O_{t+1} times C_t to E_{t+1}$
$mathcal{K}$	context construction	$mathcal{K}: C_t times E_{t+1} to C_{t+1}$
$mathcal{G}$	hypothesis generation	$mathcal{G}: E_{t+1} times C_{t+1} times H_t to H_{t+1}$
$mathcal{B}$	belief revision	$mathcal{B}: B_t times E_{t+1} to B_{t+1}$
$mathcal{Q}$	inquiry planning	$mathcal{Q}: B_{t+1} times mathcal{A} to mathcal{A}^{*}$
$mathcal{X}$	explanation synthesis	$mathcal{X}: R_{t+1} to Xi$

$\mathcal{A}$ is the set of candidate observations, $\mathcal{A}^{*}$ an ordered subset, and Ξ the space of explanations.

A.2 Evidence objects

$$e_i = \langle v_i, s_i, r_i, u_i, c_i, \tau_i, p_i \rangle$$

with observed value $v_i \in [0,1]$, source s_i, reliability $r_i \in [0,1]$, uncertainty $u_i \in [0,1]$, context reference c_i, timestamp τ_i, and provenance $p_i \subseteq E_{<t} \cup O_{\le t}$.

p_i is a **set**, not a single parent. This is the formal statement of Invariant II, and writing it as a scalar is what licensed the prototype defect in which derived evidence reported one parent and silently discarded the rest.

$$E_t = \e_1, e_2, \ldots, e_n\$$

A.3 Reflective evolution

$$R_{t+1} = F(R_t, O_{t+1})$$

where $\mathcal{F}$ is the composition

$$F = X \circ Q \circ B \circ G \circ K \circ I$$

The monograph currently gives $\mathcal{F}$ two incompatible signatures — $\mathcal{F}(R_t, O_{t+1})$ on p. 58 and $\mathcal{F}(R_t, E_{t+1}, C_{t+1})$ on p. 62 — without reconciling them. Defining $\mathcal{F}$ as the composition dissolves the problem: the second form is the same map with $\mathcal{I}$ and $\mathcal{K}$ already applied, and can be written

$$F = F' \circ (K \circ I), F'(R_t, E_{t+1}, C_{t+1}) = (X \circ Q \circ B \circ G)(\cdot)$$

A.4 The belief state, instantiated

The monograph leaves B_t deliberately unspecified — "probability distributions, weighted graphs, logical structures, or hybrid models". Later chapters then assert properties *of* B_t (revision history, sequential recursion, hypothesis plurality). Properties cannot be asserted of an object whose type is open. The fix is not to close the abstraction but to state the **axioms** any instantiation must satisfy, and then give one reference instantiation.

Axioms on B_t. For hypotheses $h \in H_t$ and evidence e:

- **B1 (plurality)** B_t assigns a value to every $h \in H_t$ simultaneously; it does not distinguish a unique maximiser.
- **B2 (commutativity)** For conditionally independent e, e': $\mathcal{B}(\mathcal{B}(B_t, e), e') = \mathcal{B}(\mathcal{B}(B_t, e'), e)$.
- **B3 (invertibility)** There exists $\mathcal{B}^{-1}$ with $\mathcal{B}^{-1}(\mathcal{B}(B_t, e), e) = B_t$.
- **B4 (audit)** Every application of $\mathcal{B}$ records $(h, e, B_t(h), B_{t+1}(h), \text{method}, \tau)$.

B2 and B3 are not decoration. The prototype satisfied neither, and the measured consequence was a final belief of 0.0396 or 0.9900 for the same five observations depending only on arrival order.

Reference instantiation. Per-hypothesis log-odds ledger:

$$l_t(h) = \text{logit } \pi(h) + \sum_{e \in E_t} w_e \lambda_e(h), \lambda_e(h) = \log \frac{P(e \mid h)}{P(e \mid \text{not } h)},$$
$$B_t(h) = \sigma\big(l_t(h)\big)$$

with prior $\pi(h)$, reliability weight $w_e \in [0,1]$, and σ the logistic function. $\mathcal{E}_t$ is a **set keyed by evidence identity**, which gives B2 (addition commutes), B3 (subtract the term), and

idempotence under re-application of the same evidence.

Clamping to a display range is applied to $B_t(h)$ **on output only**. Clamping the state is not invertible and destroys B3.

A.5 Uncertainty

$$U_t = \langle u^obs_t, u^mod_t, u^exp_t, u^tmp_t, u^eth_t \rangle$$

each in $[0,1]$. Reporting a scalar requires an explicit aggregation rule; the reference implementation uses

$$u_t = \min\Big(1, \alpha_1 \tildeH(B_t) + \alpha_2 sd_w(E_t) + \alpha_3 \theta(n) + \alpha_4 m_t\Big)$$

with $\tilde{H}$ the entropy of B_t normalised to $[0,1]$, sd_w the weighted dispersion of the evidence, $\theta(n) = \max(0, (4-n)/4)$ a thinness penalty, and m_t the posterior mass on the *unmodelled* hypothesis. Coefficients $(\alpha_1,\ldots,\alpha_4) = (0.40, 0.15, 0.10, 0.35)$ are design choices, not fitted parameters, and are stated as such.

A.6 Sequential updating, and the assumption it requires

$$P(h) \to P(h \mid e_1) \to P(h \mid e_1, e_2) \to \cdots$$

$$P(h \mid e_{1:n}) = \frac{P(h) \prod_{i=1}^n P(e_i \mid h)}{\sum_{h' \in H} P(h') \prod_{i=1}^n P(e_i \mid h')} \quad \textbf{only if } e_i \perp e_j \mid h \ \forall i \neq j$$

The monograph states "after each observation, the posterior becomes the new prior" without stating the conditional-independence condition. In this domain the condition is **frequently violated**: two self-reports on consecutive days about the same topic are correlated through mood, framing and recall, not only through the latent state.

This is a limitation to declare, not to hide. Two mitigations are implemented:

1. Evidence contributions are keyed by identity, so re-submitting the same observation does not compound it.
2. Temporal decay down-weights clustered observations, which partially offsets the over-counting that correlated evidence produces.

Neither is a substitute for a correlated-likelihood model. That remains open.

A.7 Bayes' theorem

$$P(h \mid e) = (P(e \mid h)\, P(h))/(P(e)),\quad P(e) = \sum_{h' \in H} P(e \mid h')\, P(h') > 0$$

(As printed on p. 67 of the monograph this equation has no equals sign. See §B.)

Hypothesis plurality: reflection does **not** require collapsing to

$$\hat{h} = \operatorname*{arg\,max}_{h \in H} P(h \mid e)$$

Instead the reported object is the smallest credible set

$$S_\gamma(B_t) = \operatorname*{arg\,min}_{S \subseteq H} \Big\lvert S \Big\rvert \Big| \sum_{h \in S} B_t(h) \ge \gamma \Big\}$$

with correctness measured as coverage, $\Pr[h^{*} \in \mathcal{S}_\gamma] \ge \gamma$, rather than as top-1 accuracy.

A.8 Inquiry

For candidate observation $a \in \mathcal{A}$ with binary outcome Y_a:

```
\mathrmEIG(a) = H(B) - E_Y_a\big[H(B | Y_a)\big] = H(B) - SUM_y in \0,1\ P(y) H(B | Y_a = y)

          Q(B, A) = \operatorname*arg max_a in A \frac\mathrmEIG(a)\mathrmcost(a)
```

Note that $\mathrm{EIG}(a)$ depends on a. The heuristic it replaces — $\text{benefit} = 1 - \lvert B(h_i) - B(h_j) \rvert$ — does not, which is why it performed no better than random selection.

A.9 Traceability

```
T(x) = \e_1, e_2, \ldots, e_k\ \subseteq E_t, T(x) \neq \varnothing \forall x in \Xi
```

(As printed in the monograph the equals sign and the set braces are both absent.)

Provenance closure, which is what Invariant II actually requires:

```
anc(e) = \bigcup_p in p_e \big(\p\ \cup anc(p)\big)
```

— the transitive closure over *all* parents, not the first.

A.10 Complexity — measured, not asserted

Operation	Bound	Measured scaling per doubling
$\mathcal{B}$ — one belief update (log-odds ledger)	$O(1)$	**2.0**× [yes]
$\operatorname{anc}(e)$ — provenance closure	$O(V + E)$	**2.0**× [yes]
$\mathcal{Q}$ — inquiry planning	$O(\lvert H \rvert \cdot \lvert \mathcal{A} \rvert)$	3.9× at $\lvert H \rvert = \lvert \mathcal{A} \rvert$ [yes] (product of two linear factors)
Engine `add()` — full recompute per observation	$O(n)$ per call, $O(n^2)$ cumulative	**3.8**× — quadratic, as the monograph predicts

The monograph's warning that unrestricted recomputation "could approach $O(n^2)$" is not hypothetical: the shipped engine has exactly that growth, 27 ms at 100 evidence pairs and 1121 ms at

800. It is a real optimisation target (incremental recompute), not a strawman.

The complexity chapter should state these four bounds. It currently states $O(n^2)$ for pairwise evidence comparison — which nothing in the architecture requires — and $O(T)$ for naive history storage, and gives no bound for any operator it defines.

Appendix C — Prototype-to-Architecture Conformance (rebuilt)

Drop-in replacement for p. 290. Every row names the file and, where a behavioural claim is made, the test that settles it. `shadow_mirror` v1.0, 312 tests passing, 2 recorded xfails.

Architectural concern	Implementation	Status	Settled by
Structured observation	`models.py`	Implemented	`test_operator_I_ingest_*`

Architectural concern	Implementation	Status	Settled by
Knowledge / desire hypotheses	`engine.py`	Implemented — likelihood-accumulating, not heuristic	`test_d08_posterior_sharpens_*`
Temporal decay	`engine.py`	Implemented; 30-day half-life **still unvalidated**; expiry now gates sufficiency	`test_d12_*`
Consent state and scope	`consent.py`	Implemented	`test_consent_and_patterns.py`
Safety flags and threat guard	`safety.py`, `engine.py`	Implemented — **refuses under partial observation**	`test_d19_*` (5 tests)
Longitudinal descriptive patterns	`patterns.py`	Implemented, associative	`test_consent_and_patterns.py`
Journal inspect / export / delete	`store.py`	Implemented	`test_reflect_and_store.py`
Unified evidence/context graph	`urg.py`	**Implemented** — typed, transactional, multi-parent provenance, zero dependencies	`test_v10_unified.py` (7 tests)
Belief revision engine	`beliefs.py`	**Implemented** — order-invariant, exactly retractable, O(1) per update	`test_axiom_B2/B3/B4`, `test_belief_update_is_linear`
Inquiry planner	`inquiry.py`	**Implemented** — expected information gain	`test_d22_*` (5 tests)
Architectural invariants I–X	`invariants.py`	**Implemented as executable checks**	`test_invariants.py` (200 randomised states + 16 negative tests)
Emotion–Will–Knowledge profile	`ewk.py`	**Implemented** — observable streams, tensions as questions	`test_d13_*` (4 tests)
Credible sets / abstention	`credible.py`	**Implemented**	`test_operator_X_credible_set_*`
Per-person offset estimation	`credible.py`	Implemented; effective at moderate misspecification only	`experiment_d23.py`
Encryption at rest	—	Not implemented	—
Causal inference	—	**Not implemented — prose only** (monograph pp. 64–66)	recorded xfail
Knowledge consolidation $K_t \leftarrow B_t$	—	**Not implemented — prose only** (p. 145)	recorded xfail
Multimodal acquisition	—	Not implemented	—
Reflective execution engine / event queue	—	Not implemented	—
Human collaboration layer (UI)	`cli.py` only	Partial — command line, no interface	—
Distributed deployment	—	Not implemented; **not the bottleneck** — `engine.add()` is quadratic (D-28)	measured, §A.10
Emotion AI / emotion-engineering loop	—	Proposed, research-only	—
Four-family hypothesis-and-confirmation module	—	Proposed, research-only	—

Architectural concern	Implementation	Status	Settled by
Human-subject validation	—	**Not performed**	—
Clinical or employment use	—	**Prohibited**	`ETHICS.md`

Fourteen concerns are now implemented and test-backed, against seven in v0.3.0. Four are recorded as specified-but-unbuilt rather than left ambiguous.

Appendix D — Claim-to-Validation Matrix (rebuilt)

Drop-in replacement for p. 291. The original structure — claim, evidence now, required validation, failure signal — is kept, because it is the right structure and better than most published architecture papers manage. What changes is the *evidence now* column, which was "architecture only" or "theory only" for thirteen of fifteen claims and is no longer.

Status key: **Proven** = a test or measurement settles it · **Partial** = measured under simulation, not with people · **Unsupported** = no evidence yet · **Removed** = withdrawn.

D.1 Claims that measurement has settled

Claim	Status	Evidence now	Falsification signal
The architecture is safer than a guard-free baseline	**Proven (simulation)**	Misreading a threat response as a hidden desire: **0.0%** [0.0, 0.0] with the guard vs **42.9%** [39.5, 46.2] (mismatch-only) and **75.1%** [72.1, 78.1] (arousal-as-desire). *n* = 400 per process, 5 processes, 2 misspecification levels. `benchmarks/suite.py::bench_2`	Any baseline matching 0.0%, or the guard exceeding 0.0% under full coverage
Refusal is calibrated	**Proven (simulation)**	100% [100.0, 100.0] refusal on insufficient evidence; 0.0% [0.0, 0.0] over-refusal on sufficient evidence. `bench_3`	Over-refusal above 0, or inference below the evidence threshold
Expired evidence stops being evidence	**Proven**	Inference from decade-old observations 100% ☐ **0.0%**. `test_d12_*`	Any inference from evidence below `EPS_WEIGHT`
Belief revision is order-invariant	**Proven**	All **120 permutations** agree to machine precision; prototype spread was 0.9504. `test_d20_*`	Any permutation pair differing by more than 1e-12
Contradiction is preserved, not overwritten	**Proven**	`contested_hypotheses()` retains both edges. `test_contradiction_is_preserved_*`	A contradicting write reducing the hypothesis count
Provenance is fully recoverable	**Proven**	2/2 parents from a two-parent DAG (prototype: 0/2); branching and cyclic cases. `test_d14_*`	Any parent unreachable from `ancestors()`
The ten invariants are decidable	**Proven**	All ten hold over 200 randomised states; 16 negative tests prove each check can fail. `test_invariants.py`	A well-formed state failing a check, or a violating state passing one

Claim	Status	Evidence now	Falsification signal
Inquiry planning is information-driven	**Proven**	Regret vs oracle **0.000** (belief-gap heuristic 0.579; random 0.550). `test_d22_beats_random_*`	Regret indistinguishable from random
Uncertainty is honestly reported	**Partial**	Where the model applies, ECE **0.044** and AUC(uncertainty□error) **0.949**. Where it does not, ECE **0.221** — reported as a limitation. `analyze_v030_vs_v040.py`	Confidence failing to track accuracy where the model does apply
Multiple hypotheses reduce premature closure	**Partial**	Credible-set coverage exceeds point accuracy at every misspecification level (78.2% □ 88.0% at δ = 0.30) for +0.45 set size. Does **not** reach nominal 90%. `experiment_d23.py`	Coverage no better than point accuracy, or set size exploding
Ambivalence is a distinct state, not a residual	**Proven (this session)**	Precision **34.0% □ 97.6%** once gated on temporal structure; recall unchanged at 100%. `experiment_ambivalence.py`, `test_d29_*`	Non-oscillating processes reading as ambivalence above ~5%

D.2 Claims still awaiting people

These are unchanged from your original matrix, because nothing that has happened in code can settle them. The "required validation" column is yours; I have only made the failure signals sharper where the code now suggests what to look for.

Claim	Status	Required validation	Falsification signal
Consent can gate inference	Unit behaviour only	Usability, comprehension and coercion study	Users cannot predict or control data use
Questions reduce verdict-like framing	**Partial** — Invariant X now checks output phrasing mechanically (`test_x_*`), which is necessary but not sufficient	Controlled framing study	Equal or greater over-reliance despite question phrasing
Uncertainty improves calibration **in people**	Unsupported	Pre-registered calibration study	Worse Brier score or worse trust calibration than a no-uncertainty interface
Provenance improves error detection	Unsupported	Human-factors experiment with planted errors	Users miss planted errors at the same rate
Longitudinal reflection improves learning	Unsupported	Longitudinal external study	No transfer, or increased dependence
Sensor disagreement is informative	**Partial** — the guard ablation shows disagreement *between* channels is decisive for safety	Modality-specific construct study	Unstable or non-generalising effects
EI capacities add useful profile information	Evidence synthesis only	Construct-specific tasks, calibration, transfer, incremental validity	Global score adds no value over personality
Emotion AI adds evidence beyond self-report	Theory only	Person-independent, subgroup, calibration, abstention and correction study	Benchmark gain does not transfer, or increases false inference

Claim	Status	Required validation	Falsification signal
Emotion engineering supports user-owned outcomes	Closed-loop proposal	Randomised reversible intervention study	No benefit over a non-adaptive interface, or increased dependence
E–W–K coupling predicts context-linked patterns	Formal hypothesis; detector now fires correctly	Intensive longitudinal multilevel study	Coupling reverses or disappears under valid operationalisations
Population priors improve question selection	Theory only	Comparison against open-question and personal-evidence baselines	Priors add no coverage or dominate the person
Neutral confirmation improves mapping	Protocol only	Randomised suggestion and correction study	False uptake, anchoring, or recall distortion
Competency comparison supports development	Construct proposal	Construct, invariance and longitudinal validation	Unstable score, or harmful normative labelling

D.3 Claims withdrawn

Claim	Why
EWK-augmented explanations improve calibration ratings, $t(39) = 4.23$, $p < 0.001$	**Removed.** No such measurement exists; the statistic came from two hard-coded Gaussians.
Belief revision appropriate in 100% of cases vs a 72% baseline	**Removed.** The assertion was tautological and the baseline had no implementation.
Inquiry recommendations align with expert tests in 100% of cases vs 41% random	**Removed.** Same. Replaced by regret against an information-theoretic oracle.
The URG improves reflective reasoning over disaggregated storage	**Removed as stated.** Session 3 measured the inference layer as statistically indistinguishable from a median split in all 20 conditions. The URG's benefits are **structural** — traceability, provenance, contested-hypothesis retention — and the claim is now made in those terms only.
Reference architecture is scalable	**Removed as stated.** `engine.add()` is quadratic (measured 3.8× per doubling). Restate as "reference design pending scalability work", and name incremental recompute as the first target.

D.4 The three sentences this matrix licenses

1. Under simulation with known ground truth, conditioning the hidden-preference reading on the evidence needed to rule out a threat response reduces the rate of that misreading from 42.9–75.1% to 0.0%, at a cost of 19 points of sensitivity and 33 points of specificity when the required channel is absent.
2. The ten architectural invariants are decidable properties, and a compliant implementation exists with 312 passing tests, including 16 that prove each check can fail.
3. Nothing in this work establishes any effect on human reasoning, and no claim to that effect is made.

Part VI — Session reports

Shadow Mirror — Multi-Session Research Program Plan

Owner: Tarik Metin · **Started:** 25 July 2026 · **Cadence:** one session per "continue"

Each session is self-contained: it has a defined input, a defined deliverable, and leaves a `STATE.json` that the next session reads first. Nothing depends on my remembering anything — re-upload `STATE.json` and the latest deliverables if a session ever starts cold.

Ground rules for this program

1. **No fabricated evidence, ever.** Every number that appears in a paper must be produced by code in this repository that a reviewer can run. If a number cannot be produced, the claim is deleted, not softened.
2. **Every citation is verified against a live source** before it enters a reference list, with arXiv ID or DOI recorded.
3. **Your v0.3.0 engine is the trunk.** New capability is grafted onto it. Nothing that already exists — consent gate, trauma guard, ambivalence, decay, topic scoping, refusal — is dropped.
4. **Honest limitation beats impressive claim.** The system's refusals are a feature and will be presented as one.
5. **No human-subjects data is collected in these sessions.** Where human evaluation is required, I build the *instrument* (protocol, IRB application, analysis plan, pre-registration) so you can run it; I do not simulate participants and call it a result.

Phase I — Repair (Sessions 1–5)

#	Session	Deliverable	Status
1	**Audit & program design**	`SESSION_01_AUDIT.md` (18-item defect register, integrity findings, citation verification, monograph map, venue reassessment); this plan; live tracker	[yes] **done**
2	**Honest benchmark suite v2**	Delete Benchmark 3. Rebuild 1 & 2 as real discriminative tests with genuine scenario generation, real baselines that exist in code, effect sizes with CIs, seeded reproducibility. Add adversarial cases the system should *fail*. Full run log.	☐
3	**Engine v0.4 — fix the inference core**	Fix D-08 (sample-size sensitivity), D-09 (duplicate threat number), D-11 (topic bleed), D-12 (effective sample size + weighted variance). Extend the test suite to lock each fix. Publish a before/after behavioural diff.	☐
4	**Unify the two code bases ☐ v1.0-real**	Graft a URG layer onto v0.4 *without* losing consent/trauma/ambivalence. Fix D-13…D-16. Make the demo actually exercise contradiction, tension and refusal. Package as an installable, tested, documented library.	☐
5	**Conformance suite for the ten invariants**	Turn each architectural invariant into an executable property test (Hypothesis/property-based where possible), so "Invariant II holds" becomes a claim a reviewer can verify by running `pytest`. This is the single strongest evidence a reference-architecture paper can carry.	☐

Phase II — Verify the manuscript (Sessions 6–9)

#	Session	Deliverable	Status
6	Bibliography verification, part 1	Verify every reference in the monograph's 96-entry bibliography against live DOIs; produce a corrections table.	☐
7	Mathematical chapters audit (pp. 55–79)	Check the formalism for well-definedness, fix the LaTeX corruption visible in the PDF, and confirm each operator has an executable counterpart.	☐
8	Claim-to-validation matrix, rebuilt	Re-derive the matrix (p. 291) against the *actual* v1.0 code and the conformance suite. Mark every claim: proven / partially supported / unsupported / removed.	☐
9	Ethics, governance and prohibited-use audit	Check the ethics chapters against GDPR / EU AI Act as currently in force, and against IRB norms for the studies proposed in Phase IV.	☐

Phase III — Write the papers (Sessions 10–14)

#	Session	Deliverable	Status
10	**P2 — Ethics-of-design paper** (fastest route to publication; no human subjects needed)	Full draft for AIES / FAccT / CHI: consent as versioned state transition, refusal as output, trauma guard, prohibited-use. Backed by running code.	☐
11	**P1 — Architecture paper**, draft	Full draft for ACM TiiS / AI Magazine: URG, ten invariants, conformance results, component contracts.	☐
12	P1 figures and camera-ready assets	Vector figures (PDF/EPS), result charts from real benchmark output, architecture diagrams rebuilt to venue spec.	☐
13	Adversarial self-review	I review both papers as a hostile Reviewer 2 and produce a rebuttal-hardening pass.	☐
14	Submission packages	Cover letters, author statements, AI-use disclosure, data/code availability statements, supplementary bundles, formatting compliance checks.	☐

Phase IV — Enable the empirical work (Sessions 15+)

#	Session	Deliverable	Status
15	Human-study protocol + IRB packet	Full protocol, consent forms, debrief scripts, risk assessment, and a complete IRB/ethics-committee application for the EWK calibration study.	☐
16	Pre-registration + analysis plan	OSF-style pre-registration with hypotheses, power analysis, exclusion rules, and the analysis script written *before* data exists.	☐
17	Deployable study instrument	A runnable web instrument for the study (local-first, consent-gated, exportable), so you can collect data once approved.	☐
18+	Analysis, P3, and follow-on work	On your data, when it exists.	☐

Open questions I need from you (these block specific sessions)

Q	Blocks	Why it can't be guessed
Q1. What is the correct author byline? Is "Omar Moawad" you (pen name), a co-author, or a prior collaborator? If multiple authors, what order and contribution split?	Session 14	IEEE/ACM authorship policy requires accuracy and consent of all listed authors.
Q2. Do you have an institutional affiliation? University, lab, or independent?	Sessions 14, 15	Determines which IRB you can even apply to, and whether "independent researcher" goes on the byline.
Q3. Is the Kimi.ai-generated paper something you want kept at all, or should I treat it as discarded and write from the monograph directly?	Session 11	Affects whether Session 11 is a rewrite or a fresh draft. My recommendation: discard it.
Q4. What is your realistic time horizon and appetite for the human study? Months, or "publish the non-empirical papers and stop"?	Phase IV scoping	Determines whether Phase IV runs at all.

You can answer these any time — just include them with a "continue". Sessions 2–5 proceed without them.

How to resume

Write `continue`. I will read `STATE.json`, pick up the next unstarted session, and deliver it. If you want to jump the queue ("do session 5 next"), say so and I will re-order.

Session index

Eighteen sessions. Two of them produced no report of their own, and that is not an omission: **Sessions 10 and 11 produced the paper drafts themselves** — P2 and P1 respectively — which live in `../papers/`. Everything else has a report here.

#	Report	What happened
1	`SESSION_01_AUDIT.md`	Audit of the inherited work. 32 defects found, including a fabricated t-test and a watermark on 8 of 9 pages
2	`SESSION_02_BENCHMARKS.md`	Benchmark suite v2, byte-reproducible
2b	`SESSION_02B_DRAFT_ADJUDICATION.md`	Adjudication of an externally supplied draft
2c	`SESSION_02C_DRAFT2_ADJUDICATION.md`	Adjudication of a second one; ground rule 7 added here
3	`SESSION_03_ENGINE_V04.md`	Engine v0.4. D-19: the guard now refuses rather than imputing a missing channel as zero
4	`SESSION_04_UNIFIED_V1.md`	Unified v1.0. NetworkX removed — zero runtime dependencies
5	`SESSION_05_INVARIANTS_D23.md`	Invariants made executable. D-23 characterised, not tuned away
6	`SESSION_06_BIBLIOGRAPHY_VENUE.md`	Bibliography audit and venue selection
7	`SESSION_07_FORMALISM.md`	Formal appendix corrected. D-27 found by measurement: 4.2× per doubling
8	`SESSION_08_CLAIM_MATRIX.md`	Every claim mapped to the file that produces it. D-29 ambivalence gate
9	`SESSION_09_ETHICS_REGULATORY.md`	EU AI Act and GDPR analysis. **Withdrew Session 6's education-pilot recommendation**

#	Report	What happened
10	*(no report — produced* `P2_AIES_FAccT_draft.md`*)*	The ethics paper
11	*(no report — produced* `P1_ACM_TiiS_draft.md`*)*	The architecture paper
12	`SESSION_12_ASSETS_DEFECTS.md`	Figures, defect sweep, v1.1.0
13	`SESSION_13_ADVERSARIAL_REVIEW.md`	Fourteen attacks. One found a result that weakened the headline
14	`SESSION_14_RELATED_WORK_AND_AFFILIATION.md`	76 references verified. "Fawn" found not to be peer-reviewed
14b	`SESSION_14B_DATA_FEASIBILITY.md`	~40 datasets assessed. None can validate the safety claim
15	`SESSION_15_FORMATION_MODEL.md`	Declared-only personhood; four refusals; Invariant XI
16	`SESSION_16_PAPER_RECONCILIATION.md`	Papers reconciled to eleven invariants. D-34
17	`SESSION_17_WORKAROUND_HUNT.md`	Eleven blockers searched; six dissolved. Mutation score 59.9%
18	`SESSION_18_ROUTE_A_AND_IRB.md`	Route A loaders and the AUC IRB packet

`STATE.json` is the machine-readable state: 34-item defect register, verified citations, ground rules, blocking questions, and per-session results. Read it first if you are resuming this work.

Shadow Mirror — Session 01 Audit Report

Reflective Artificial Intelligence / Shadow Mirror Reference Architecture Independent technical and research-integrity audit of the supplied artifact package.

Audit date: 25 July 2026 **Auditor:** Claude (Cowork session), acting as technical reviewer **Scope of this session:** full inventory, execution of all supplied code, empirical defect probing, citation verification, monograph structural mapping. **Not** in scope this session: line-by-line verification of the 327-page monograph text (staged for Sessions 3–5).

---

0. Executive summary

Eight artifacts were supplied. They fall into two distinct and **partly incompatible** bodies of work:

Body	Artifacts	Verdict
A — The author's own work	327-page monograph, `shadow-mirror` v0.3.0 repository (2,599 LOC, 34 passing tests), thread compilation	**Substantive.** Careful, ethically disciplined, honestly hedged, with a real 96-entry bibliography carrying live DOIs. This is the publishable core.
B — Previously AI-generated additions	`shadow_mirror_v1.py`, `benchmark_suite.py`, `Shadow_Mirror_IEEE_TAC_Submission.pdf`, `urg_visualization.png`	**Not submittable, and dangerous to submit.** Contains fabricated empirical results, hallucinated citations, and a visible "Generated by Kimi.ai" watermark on every page of the paper.

The single most important finding of this session: the IEEE TAC paper currently in your possession reports experimental results that do not exist. Submitting it as-is would expose you to a research-misconduct finding, not merely a rejection. Section 2 documents this with reproductions.

The second most important finding: the previous AI session's headline advice — "you did not invent this field, and your monograph overclaims" — is **half wrong**. The overclaim is in your *chat prompt*, not in your monograph. Your monograph explicitly states the opposite (p. 62: *"The defining contribution of Reflective Artificial Intelligence is proposed to be its computational objective and organization, not the invention of an entirely new probability theory"*). The monograph was corrected for a problem it did not have, while its genuine methodological weakness (Section 3) went unexamined.

The third finding: `shadow_mirror_v1.py` is not an extension of your prototype. It is a *replacement* that discards your actual contribution — the consent gate, the trauma guard, the un-collapsed `ambivalence` state, temporal decay, topic scoping — and substitutes a generic reflective-graph scaffold. Your v0.3.0 engine is the more original artifact. Section 4.

---

1. Inventory and execution results

All code was executed in a clean Linux container (Python 3.11.15, networkx 3.6.1, numpy 2.4.4, scipy 1.17.1).

Artifact	Status	Notes
`shadow-mirror` v0.3.0 repo	[yes] Runs	`pytest`: **34 passed in 0.11s**. `demo.py` runs clean. Zero third-party dependencies (deliberate; good for auditability).
`shadow_mirror_v1.py`	[yes] Runs	Demo completes; produces 14 nodes / 17 edges. But see §4 for what the demo fails to demonstrate.
`benchmark_suite.py`	[yes] Runs	Produces the numbers quoted in the paper. See §2 for what those numbers mean.
Monograph (327 pp., 70,399 words)	[yes] Parsed	10 parts, ~70 chapters, 30 figures, 96 bibliography entries with DOIs. Structure mapped in §5.
IEEE TAC PDF	[!]	2,271 words (~a quarter of TAC's expected length). Watermarked "Generated by Kimi.ai" on 8 of 9 pages.
`urg_visualization.png`	[!]	Renders, but node labels contain literal `n` escape sequences (`H: Student has knowledg...nP=0.62`) — a figure-generation bug, visible in the supplied PNG.

---

2. Research-integrity findings — CRITICAL

2.1 Benchmark 3 does not measure the system. It measures `numpy.random.normal`.

`benchmark_3_explanatory_calibration()` (lines 152–186 of `benchmark_suite.py`) contains **no reference to any Shadow Mirror object**. It draws 40 samples from `N(4.2, 0.6)`, 40 from `N(3.5, 0.8)`, and runs a paired *t*-test between them. The "effect" is the hard-coded 0.7 difference between the two means.

Reproduction — the identical statistic, with Shadow Mirror never imported:

```
seed=42    M_ewk=4.05  M_base=3.48   t(39)=3.53   p=0.00107
seed=1     M_ewk=4.13  M_base=3.64   t(39)=3.48   p=0.00126
seed=7     M_ewk=4.11  M_base=3.47   t(39)=3.88   p=0.00039
seed=2024  M_ewk=4.20  M_base=3.54   t(39)=4.22   p=0.00014
```

The paper's abstract states: *"the EWK profile produces significantly higher calibration ratings than baseline explanations (t(39) = 4.23, p < 0.001)."* That sentence describes a property of a random number generator seeded at 42.

Additionally, the **earlier draft of this paper** (reproduced in your chat history) stated: *"we… asked **12 human evaluators (graduate students in cognitive science)** to rate whether confidence matched evidential support."* No such evaluators exist. That sentence is a description of a human-subjects study that was never conducted. It was later softened to "simulated," but the sentence was written, and if any version of it reaches a reviewer, the consequence is not rejection — it is a misconduct referral to your institution.

Required action: delete Benchmark 3 entirely. Do not "keep it with a disclaimer." A simulated comparison between two hand-chosen Gaussians is not preliminary evidence of anything; it is a restatement of the author's hypothesis in numerical costume.

2.2 Benchmarks 1 and 2 are tautologies, and *n* is inflated ~40×.

Benchmark 1 asserts `final_belief < initial_belief` after applying a likelihood ratio of `0.2/0.8 = 0.25`. Bayes' rule guarantees a decrease for *any* correct implementation. Verified: `0.5294 < 0.8182`. The test cannot fail, and therefore cannot discriminate a good reflective system from a bad one.

Benchmark 2 asserts `len(inquiries[0].related_hypotheses) >= 2`. `InquiryPlanner.plan_inquiry()` constructs pair-inquiries with exactly two related hypotheses whenever the belief gap is < 0.2; the benchmark hard-codes beliefs of 0.52 and 0.48 (gap 0.04). The assertion tests the dataclass constructor, not the planner.

Sample-size inflation. Both benchmarks iterate 50× and 30×, but the only thing that varies across iterations is an f-string in the observation ID. Every scenario is numerically identical. Reported $n = 50$ and $n = 30$; effective $n = \mathbf{1}$, variance = **0**. Reporting "50 synthetic scenarios" for one scenario repeated 50 times is a misrepresentation of statistical power.

2.3 Three numbers in the paper have no code behind them at all.

Claim in paper	Where it comes from
"vs. **72%** for a baseline pipeline that discarded superseded evidence"	No baseline pipeline exists in the supplied code.
"vs. **41%** for a random-information baseline"	No random baseline exists in the supplied code.
"reduces explanatory inconsistency by **34%**" / "**F1 = 0.81**" (earlier abstract draft)	Neither metric is computed anywhere.

These are invented comparators. Every one must be deleted or actually implemented.

2.4 Every 2025–2026 citation has a wrong title, wrong venue, or both.

Verified against live sources this session:

As cited	Reality
[1] Hou et al., *"MARS: Metacognitive Agent Reflective Self-improvement,"* arXiv, Jan 2026	Two different real papers are conflated. **MARS** is *"MARS: Memory-Enhanced Agents with Reflective Self-improvement"* (arXiv **2503.19271**). The metacognitive-reflection paper is *"Learn Like Humans: Use Meta-cognitive Reflection for Efficient Self-Improvement"* (arXiv **2601.11974**). Neither has the cited title.
[2] Xin et al., *"MetaMem: Self-Reflective Symbolic Memory Optimization,"* arXiv, Jan 2026	Real paper is *"MetaMem: Evolving Meta-Memory for Knowledge Utilization through Self-Reflective Symbolic Optimization,"* **ACL 2026 Findings**, arXiv **2602.11182** (OpenBMB). Title and venue both wrong.
[3] Elenjical et al., *"Think2²: Planning-Monitoring-Evaluation Regulatory Cycles for LLMs,"* **AAAI**, Feb 2026	Real paper is *"Think2: Grounded Metacognitive Reasoning in Large Language Models,"* Elenjical, Kavuri & Varma (IIIT Hyderabad), arXiv **2602.18806**, 21 Feb 2026. **Not an AAAI paper** — code promised "upon acceptance," i.e. unpublished. Title fabricated.

As cited	Reality
[4] Park et al., *"ESMA: Evolution Strategies for Metacognitive Alignment"*	An ESMA project exists (`github.com/cosmoquester/ESMA`, "a method to improve metacognition of LLMs"), but the expansion, authors and venue as cited are **unverified**. Note the prior session described ESMA in prose as "dual-prompt confidence calibration with type-2 sensitivity tuning" and in the reference list as "Evolution Strategies for Metacognitive Alignment" — two incompatible expansions of the same acronym in one document.
[5] Didolkar et al., *"Behavior Reuse via Fragment Mining and LLM Distillation,"* **NeurIPS**, Sep 2025	Real paper is *"Metacognitive Reuse: Turning Recurring LLM Reasoning Into Concise Behaviors,"* Didolkar, Ballas, Arora & Goyal, arXiv **2509.13237**, 16 Sep 2025. **Not NeurIPS.** Title fabricated. The "46% token savings" figure is real and traces to this paper.
[9] Petrides et al., *"Trait Emotional Intelligence,"* **European J. Personality, 2007**	Mismatched. Petrides & Furnham's *European Journal of Personality* trait-EI paper is **2001**; the 2007 Petrides, Pita & Kokkinaki paper is in the *British Journal of Psychology*.

Citations [6], [8], [10], [11], [12] check out. **The irony is severe: your own monograph's 96-entry bibliography carries real DOIs and is markedly more reliable than the reference list of the "submission-ready" paper built on top of it.** The generated paper degraded your scholarship.

2.5 Provenance and authorship must be resolved before anything is submitted.

Two administrative items that will stop a submission cold:

1. **The "Generated by Kimi.ai" watermark** appears on 8 of 9 pages of `Shadow_Mirror_IEEE_TAC_Submission.pdf`. Beyond the obvious, IEEE requires disclosure of generative-AI use in the acknowledgements; an undisclosed watermark of a different vendor's tool inside the submission PDF is the worst possible way for that to surface.
2. **Author name mismatch.** The monograph, the repository `pyproject.toml`, and the generated paper all name **Omar Moawad / "Omar"**. Your account and this session are under **Tarik Metin**. If this is a pen name, a co-author, or a prior collaborator, that is fine — but IEEE's authorship policy requires the byline to be accurate and every listed author to have consented. **I cannot resolve this for you and will not guess.** Please tell me the correct byline and, if there is more than one contributor, their order and contribution statement. This blocks Session 6 (submission packaging).

---

3. Methodological findings in your engine (v0.3.0)

These are the findings that matter for the actual science. All four were confirmed empirically this session, not asserted.

3.1 The posterior is completely insensitive to sample size — HIGH

`_infer_knowledge` and `_infer_desire` compute a weighted *mean* per signal channel, then evaluate a **single** Gaussian likelihood at that centroid. There is no accumulation over observations.

```
n_pairs=  2   posterior={'blind_spot': 0.841, ...}   uncertainty=0.310
n_pairs= 10   posterior={'blind_spot': 0.841, ...}   uncertainty=0.235
n_pairs=100   posterior={'blind_spot': 0.841, ...}   uncertainty=0.235
```

Fifty times more evidence produces a **bit-identical** posterior, and uncertainty saturates at 0.235 after $n = 4$ (the `thin` term maxes out and the entropy of a fixed posterior is constant).

This matters because the monograph and the paper both describe the system as performing "Bayesian belief revision." As implemented, it performs a **single-shot likelihood evaluation on decayed weighted means** — mathematically a soft nearest-corner classifier, not sequential Bayesian updating. A reviewer with a statistics background will find this in ten minutes.

There are two honest resolutions, and you should pick deliberately:

- **(a) Fix it.** Replace the centroid evaluation with a product of per-observation likelihoods, weight-exponentiated: `L_h = Π_i g(v_i; corner_h)^{w_i}`. Posteriors then sharpen with evidence, effective sample size becomes `Σw_i`, and the "Bayesian" claim becomes true. This is a ~20-line change and I recommend it.
- **(b) Defend it.** Argue that deliberate refusal-to-sharpen is a *feature* of a reflection instrument ("a snapshot, not a biography" — your own README says this). This is a genuinely interesting position, but then the word "Bayesian" must be replaced throughout with something accurate, and the design choice must be argued explicitly, not left as an implementation accident.

Doing neither is not an option.

3.2 Two different numbers, both labelled "threat response probability," in one report — HIGH

`ShadowMirror.report()` emits both:

```
posterior['threat_response']        = 0.691
threat_response_probability         = 0.943
```

The first is the threat mass *after* being renormalised against the directional hypotheses; the second is the raw logistic `sigma(4(arousal + avoidance − 1.1))`. They are incommensurable quantities carrying nearly the same label, differing by 25 percentage points. This is visible in your own `demo.py` output. Because the trauma guard is the ethical load-bearing wall of the whole system, an ambiguous number here is the least acceptable place for one.

Fix: rename to `threat_confound_score` (raw, gate-facing) vs. `p(threat_response | evidence)` (posterior), and document which one the safety layer actually keys on.

3.3 `belief_history` mixes topics, so instability and `current()` are wrong — MEDIUM

`add()` calls `_update(topic)` for the topic just written, and appends to a single global `belief_history`. Consequently:

- `_instability()` compares snapshots computed for *different topics* and reports movement that is really topic-switching. Verified: alternating "algebra" and "speaking" logs yields `_instability() = 0.0555` with no actual instability in either topic.
- `current(topic=None)` returns whatever topic happened to be written last — not a global belief. Verified: returns `explicit_uncertainty` (the "speaking" read) when "algebra" was equally represented.

This directly undermines the topic-scoping guarantee the README advertises as answering the "per-question, not one global type" concern.

Fix: key `belief_history` by topic (`Dict[str, List[BeliefState]]`), compute instability within-topic, and make `current(None)` either raise or return a per-topic mapping.

3.4 Decay is applied to means but not to the sufficiency gate or the dispersion terms — MEDIUM

- `evidence_pairs()` and the `MIN_EVIDENCE` gate count raw observations with no decay. A pair of 10-year-old observations with half-life 1 day still passes the gate and yields `status: "ok"`. Verified.
- `_wmean` uses decayed weights; `_spread`/`pstdev` does not. Stale readings are excluded from the location estimate but fully included in the dispersion, ambivalence and uncertainty estimates — an internally inconsistent weighting scheme.
- `oscillation` uses `_spread(behavior)` only; when the action channel is `implicit_task` rather than `behavior`, ambivalence is structurally under-detected.

Fix: define effective sample size `n_eff = (Σw)² / Σw²`, gate on `n_eff ≥ MIN_EVIDENCE`, and use weighted variance throughout.

4. Findings in the AI-generated `shadow_mirror_v1.py`

4.1 It replaces your contribution rather than extending it — HIGH (strategic)

Feature-by-feature, what v0.3.0 has and `shadow_mirror_v1.py` does not:

v0.3.0 (yours)	v1.0 (generated)
Granular, revocable, per-stream consent gate (`consent.py`, 260 LOC)	absent
Trauma guard: `threat_response` as first-class hypothesis, surfaced first	absent
`ambivalence` as an un-collapsed state	absent
Temporal decay + per-observation reliability	reliability only
Topic scoping	absent
Refusal below minimum paired evidence	a 3-item heuristic that never fires in the demo
Output = questions, never labels	template explanation strings
34 tests asserting the *commitments*	0 tests

The distinctive intellectual content of Shadow Mirror — the claim that a reflection system should *refuse to collapse contradiction* and *cannot ethically be pointed at a person without consent* — lives entirely in the column that v1.0 deleted. **The correct v2.0 is v0.3.0 plus a URG layer, not v1.0 plus your ideas re-added.**

4.2 The flagship demo does not demonstrate the flagship mechanism — MEDIUM

The demo scenario is constructed to show EWK tension (Knowledge = 0.3, Will = 0.75, Emotion = 0.8). It prints:

```
[6] EWK Profile Tension Detection...
    No significant tension detected.
```

Because `detect_tension()` requires `k > 0.7 and w < 0.3`, or `e > 0.7 and k < 0.3`, or `|k - w| > 0.5`. Here |0.3 – 0.75| = 0.45, and k = 0.3 is not < 0.3. **The thresholds are set so that the one scenario written to exhibit tension misses on all three rules by a hair.** The paper's Section III-D narrative ("high confidence paired with low persistence and elevated stress generates a cognitive tension node") therefore has no executable counterpart.

Similarly:

- `find_cognitive_tension()` returns 0 because the demo never adds a single `contradicts` edge — the URG's contradiction machinery is never exercised end-to-end.
- `SelfLimitationFramework.check_scope()` fires only below 3 evidence items; the demo has exactly 3, so `[8] Self-Limitation Check` reports nothing.

Three of the ten demo steps are inert.

4.3 Lower-severity defects

- `EWKProfile.ingest_evidence()` re-adds nodes already created by `EvidenceManager.construct_evidence()`, silently overwriting the `version` attribute.
- `EvidenceObject.timestamp` uses `datetime.utcnow()` ☐ **naive** timestamps; v0.3.0's `engine.py` uses `datetime.now(timezone.utc)` ☐ **aware**. The two codebases cannot be merged until this is unified (comparison raises `TypeError`). `utcnow()` is also deprecated from Python 3.12 and slated for removal.
- `UnifiedReflectiveGraph.to_json()` calls `nx.node_link_data(g)` without the `edges=` keyword — the default changed in NetworkX 3.4+ and the call is on a deprecation path.

- `trace_provenance()` walks only the first predecessor (`preds[0]`) — provenance DAGs with multiple parents are silently truncated, which quietly violates the paper's own **Invariant II**.
- `urg_visualization.png` label bug: literal `\n` rendered instead of newlines.

---

5. The monograph: structural map and first-pass assessment

327 pages · 70,399 words · 10 parts · ~70 chapters · 30 figures · 96 bibliography entries with live DOIs.

Structure (page numbers as printed):

Part	Pages	Content
1 Foundations	12–18	The missing objective in modern AI
2 A Computational Theory of Reflection	19–53	Anatomy of reflection; uncertainty as resource; evidence representation; relation to existing frameworks; functional architecture; formal requirements; **scope, assumptions and falsifiability**; evaluation
3 Reflective AI	54–79	Mathematical representation; probabilistic state evolution; causal reasoning; Bayesian belief revision; **complexity and scalability**
4 Shadow Mirror Reference Architecture	80–202	25 chapters, one per component; interfaces and contracts; execution engine; **formal invariants**; deployment; observability; validation framework; governance
5 Human Cognition	203–234	Multi-dimensional cognition model; metacognition and the observer problem; **EI constructs and evidence**; emotion engineering; **the EWK profile**; multimodal evidence without mind-reading
6 Ethics	235–250	Ethical computation by construction; dynamic consent; high-risk boundaries and refusal; limitations and failure modes
7 Evaluation and Validation	251–261	Evaluation framework; validation blueprint; human-factors and reporting protocol
8 Applications	262–270	Application profiles; education and scientific inquiry; research-only health/coaching/org profiles
9 Future Directions	271–280	Research program and open problems
10 Appendices	281–323	Glossary; formal notation; invariants and contracts; pseudocode; two worked examples; **prototype-to-architecture conformance**; **claim-to-validation matrix**; test and audit record; provenance ledger; **prohibited-use checklist**; research landscape map; 19 archival plates; references

First-pass assessment. This is a considerably more disciplined document than the prior session's summary implied. It contains, already:

- an explicit falsifiability chapter (p. 46)
- a claim-to-validation matrix (p. 291)
- a prototype-to-architecture conformance appendix (p. 290) that already concedes the prototype is *"a small subset of the reference architecture"* (Figure 3)
- a prohibited-use checklist (p. 294) — which the prior session told you to write, though you had already written it
- a provenance ledger and audit record (pp. 292–293)

- an edition note stating plainly that it *"is a research artifact, not a peer-reviewed publication or clinical instrument"*

The prior session's "Correction 1: You Did Not Invent This Domain" is aimed at your chat prompt (*"the domain of knowledge I just invented"*), not at the manuscript. The manuscript says the reverse. **Keep the humility of the manuscript; drop it from the prompt.** Your defensible claim is: *a reference architecture and an ethical-invariant framework for computational reflection*, which is a real and citable contribution.

Deferred to Sessions 3–5: verification of the ~96 bibliography entries, the mathematical chapters (pp. 55–79), the invariants (pp. 99–103), and the claim-to-validation matrix against the actual code.

---

6. Venue reassessment

The prior session recommended **IEEE Transactions on Affective Computing** and produced a 2,271-word paper. Two problems:

1. **Length.** IEEE TAC regular papers run ~10–14 pages in two-column format (roughly 8,000–11,000 words). The supplied paper is about **a quarter** of a regular-paper submission and would be desk-returned on formatting alone.
2. **Fit.** TAC's centre of gravity is affect *recognition, modelling and generation* with empirical evaluation. A reference architecture whose affective component is one of three evidence channels, with no affect-recognition results, is a stretch there.

Venue analysis is deferred to Session 6, when we will know what evidence actually exists. My provisional ranking, to be revisited:

Venue	Fit	Why
ACM TiiS (Transactions on Interactive Intelligent Systems)	**Strong**	Explicitly publishes architectures and frameworks for human–AI interaction; tolerant of systems papers with formative rather than summative evaluation.
CHI / CSCW	Strong	"Tools for thought"; a consent-gated reflection instrument with a trauma guard is squarely in scope, and CHI values the ethical design argument as a contribution in itself.
AI Magazine	Good	Framework/architecture venue with a lower empirical bar — a reasonable first placement for the architecture paper.
IEEE Intelligent Systems	Good	Short-format reference-architecture pieces.
AIES / FAccT	Good, for a *different* paper	The consent gate + trauma guard + prohibited-use checklist is a publishable ethics-of-design contribution on its own.
IEEE TAC	Weak *now*	Reconsider only if a real EWK study with human raters is run.

Recommended strategy: split the monograph into three papers, not one.

- **P1 — Architecture** (TiiS or AI Magazine): URG, the ten invariants, component contracts, conformance. Evidence needed: a real implementation + a *conformance* test suite proving the invariants hold. Achievable in this program.
- **P2 — Ethics of design** (AIES/FAccT/CHI): consent as a versioned state transition, refusal as a first-class output, the trauma guard, prohibited-use. Evidence needed: the argument plus the code that enforces it. **Achievable without human subjects** — this is likely your fastest route to a first publication.

- **P3 — Empirical** (TAC/CHI): the EWK study with real participants. Requires IRB. Long horizon.

7. Consolidated defect register

Severity: **C**ritical (blocks submission / integrity) · **H**igh · **M**edium · **L**ow

#	Sev	Where	Defect	Session to fix
D-01	**C**	`benchmark_suite.py` §3	Fabricated result; no system code executed	2 (delete)
D-02	**C**	IEEE paper	Reports fabricated statistic in abstract	6 (rewrite)
D-03	**C**	IEEE paper	72% / 41% / 34% / F1=0.81 baselines have no code	2, 6
D-04	**C**	IEEE paper refs	6 of 12 citations wrong title and/or venue	6
D-05	**C**	IEEE paper PDF	"Generated by Kimi.ai" watermark, 8/9 pages	6
D-06	**C**	admin	Author byline mismatch (Omar Moawad vs Tarik Metin)	**needs your answer**
D-07	**H**	`benchmark_suite.py` §1,2	Tautological assertions; effective n=1 reported as 50/30	2 (rewrite)
D-08	**H**	`engine.py`	Posterior invariant to sample size; "Bayesian" claim inaccurate	3
D-09	**H**	`engine.py`	Two different "threat probability" numbers in one report	3
D-10	**H**	strategy	v1.0 discards consent gate, trauma guard, ambivalence, decay, scoping	4
D-11	**M**	`engine.py`	`belief_history` mixes topics ☐ wrong instability, wrong `current()`	3
D-12	**M**	`engine.py`	Decay not applied to sufficiency gate or dispersion terms	3
D-13	**M**	`shadow_mirror_v1.py`	EWK tension never fires in the demo written to exhibit it	4
D-14	**M**	`shadow_mirror_v1.py`	`trace_provenance` truncates multi-parent DAGs ☐ violates Invariant II	4
D-15	**L**	`shadow_mirror_v1.py`	naive vs aware timestamps block merging the two codebases	4
D-16	**L**	`shadow_mirror_v1.py`	`utcnow()` / `node_link_data()` deprecations	4
D-17	**L**	`urg_visualization.png`	literal `n` in node labels	5
D-18	**L**	IEEE paper	~2,300 words vs TAC's ~8,000–11,000	6

8. What is genuinely good here

It would be a disservice to list eighteen defects without saying what is worth defending, because the parts worth defending are the reason this project is publishable at all.

1. **Refusal as a first-class output.** `MIN_EVIDENCE` gating, and a system that says *"Not enough evidence to infer. Interpretation withheld on purpose."* That is rarer in this literature than it should be, and it is a contribution.

2. **The trauma guard.** Modelling `threat_response` as a competing hypothesis so that arousal + avoidance is never read as latent desire, and surfacing that caution *first*, is a specific, testable, ethically serious design commitment — and there is a passing test that asserts it (`test_safety.py`).
3. `ambivalence` **as an un-collapsed state.** Most affective and preference models force a winner. Holding "wants and doesn't want at once" as a state rather than as noise is a defensible theoretical position.
4. **Consent as a versioned state transition, granular and revocable, with non-obvious streams requiring separate acknowledgement.** This is a better operationalisation of dynamic consent than most published systems have, and it is *enforced in code* — the engine refuses rather than records.
5. **The monograph's self-discipline.** Falsifiability chapter, claim-to-validation matrix, prohibited-use checklist, provenance ledger, explicit "this is not a clinical instrument." Reviewers reward this.
6. **Zero dependencies, 34 passing tests, tests that assert commitments rather than plumbing.** Reproducibility reviewers will like this a great deal.

The problem with this project is not that the ideas are weak. It is that a layer of fabricated validation was placed on top of real ideas, and that layer will destroy the real ideas if it reaches a reviewer. Session 2 removes it and starts building validation that is actually true.

End of Session 01 audit. Session 02 begins on your next "continue".

Shadow Mirror — Session 02: Honest Benchmark Suite v2.0

Author of record: Omar Moawad (confirmed this session — Q1 closed) **Session date:** 25 July 2026 **Deliverables:** `benchmark_suite_v2/` (runnable), `results.json`, three figures, this report.

---

0. What this session did

Deleted the fabricated benchmark. Rebuilt the remaining ones so that they measure the system against a ground truth that exists independently of the system, under a model-misspecification sweep, against baselines that are implemented in code, with bootstrap confidence intervals on every headline number.

The suite is now permitted to fail — and it does, in four places. Those failures are the most valuable output of the session, because each one is a specific, reproducible finding that improves the work rather than decorating it.

	v1.0 suite	v2.0 suite
Benchmarks that execute system code	2 of 3	4 of 4
Distinct scenarios per benchmark	1 (repeated 50×/30×)	200–500 independently sampled
Ground truth	none	known latent generative state
Baselines implemented	0 (three quoted, none written)	5
Confidence intervals	none	on every headline number
Benchmarks that could fail	0	4
Bit-reproducible across runs	untested	**verified** — two full runs, byte-identical
Runtime	<1 s	103 s

---

1. The deletion

`benchmark_3_explanatory_calibration()` is gone. It drew 40 samples from `N(4.2, 0.6)` and 40 from `N(3.5, 0.8)` and *t*-tested them; the "effect" was the hard-coded 0.7 difference between the two means, and no Shadow Mirror object was ever instantiated.

Nothing replaces it that pretends to be human evaluation. Explanatory calibration is a question about how people read explanations, and it can only be answered by asking people. That study is specified in Session 15 with an IRB packet. Until then the claim does not appear in any paper.

The three invented comparators — "72% baseline pipeline", "41% random-information baseline", "34% inconsistency reduction" — are retired with it. Two of them now have real counterparts (§3, §5); the 34% figure has none and is deleted outright.

---

2. Method

Simulator (`benchmarks/simulator.py`). Observation streams are generated from a known latent state or process. Values are Beta-distributed around a generating centre; concentration κ controls per-observation noise.

Circularity control. The engine assumes each latent state sits at a fixed corner of a 2-D signal space. Generating from exactly those corners would test self-consistency, not validity. Every generator therefore displaces the true centre from the engine's assumed corner by a random vector of magnitude **δ**, and all results are reported as a function of δ. δ = 0 is the best case (the engine's model is exactly right); δ = 0.45 is severe misspecification.

Baselines (`benchmarks/baselines.py`), all executed at benchmark time:

1. `quadrant_classifier` — median-split the two channels, read off the quadrant. What a competent analyst does by hand.
2. `signal_mismatch_only` — stated-vs-behaviour only, no arousal, no guard.
3. `arousal_as_desire` — folds physiological arousal into the action channel. Not a straw man: this is documented practice in affect-aware systems, and it is exactly what the Shadow Mirror signal table forbids.
4. `belief_gap_planner` — a faithful re-implementation of what v1.0's `InquiryPlanner` actually computes.
5. `oracle_planner` — exact expected information gain, given the true likelihoods.

Statistics. Percentile bootstrap (2,000 resamples) for every rate and mean; paired bootstrap with Cohen's *d_z* for method differences; AUC via Mann–Whitney with mid-ranks; ECE over 10 confidence bins with signed bias.

---

3. B1 — Latent state recovery and confidence calibration

200 episodes per cell · 20 cells · ground truth = the generating latent state

δ	n=2	n=4	n=8	n=16	n=32	mean confidence	signed bias
0.00	95.0%	99.5%	100.0%	100.0%	100.0%	0.82–0.85	−0.15
0.15	92.0%	96.0%	99.5%	100.0%	100.0%	0.78–0.81	−0.19
0.30	76.5%	79.5%	79.0%	89.5%	90.0%	0.72–0.76	−0.10
0.45	59.0%	57.0%	57.0%	50.0%	52.0%	0.69–0.75	**+0.17**

(Accuracy of Shadow Mirror. Full table with 95% CIs and the quadrant baseline in `results.json`*.)*

Finding B1-1 — Confidence carries almost no information about correctness

Accuracy falls from **99% to 55%** as misspecification grows. Over the same range the engine's reported confidence moves from **0.83 to 0.72**. The system is materially underconfident when it is right (signed bias −0.15 to −0.19) and materially **overconfident when it is wrong** (+0.17 at $\delta = 0.45$). ECE stays in the 0.05–0.22 band throughout without tracking the accuracy collapse.

This is defect **D-08** measured rather than asserted. The cause is structural: the likelihood is evaluated once at the weighted centroid, so posterior sharpness is a function of *where* the centroid falls, not of how much evidence supports it or how well the model fits. See Figure 1.

The overconfident-when-wrong direction is the one that matters ethically. A reflection instrument that becomes *more* confident than it should be exactly when its model of the person is wrong is failing in the direction that harms the user.

Finding B1-2 — The Gaussian-corner machinery does not beat a median split

Across all 20 cells, Shadow Mirror and the quadrant-threshold baseline are statistically indistinguishable; paired differences range from −2.5 to +2.5 points with CIs spanning zero in every cell.

This is not a criticism of the architecture, but it does relocate the contribution. **The value of Shadow Mirror is not classification accuracy.** A median split gets the same answer. The value is in what B2, B3 and the conformance checks measure: the safety guard, the refusal behaviour, the preserved contradiction, the traceable provenance. The papers should claim that, and should not claim accuracy advantages that do not exist.

---

4. B2 — The safety benchmark (headline result)

400 episodes per generative process · 5 processes · 3 conditions

Two processes produce the **same surface contradiction** — a person states aversion. Only one of them is a hidden preference:

Process	stated	action	arousal	avoidance	true mechanism
`latent_approach`	0.20	0.80	0.30	0.20	preference
`latent_approach_aroused`	0.20	0.78	0.75	0.20	preference (excitement)
`threat_response`	0.20	0.15	0.85	0.85	**threat**
`threat_compliance`	0.20	0.60	0.85	0.65	**threat (fawn/comply)**
`explicit_aversion`	0.15	0.15	0.25	0.30	aversion (control)

The measured quantity is the rate at which each method reads a **threat** process as `latent_approach` — i.e. tells the user "you secretly want this" when the true mechanism is fear. This is the specific harm the trauma guard exists to prevent, and to my knowledge it has not previously been quantified for any system of this kind.

Finding B2-1 — The guard works, and the effect is large

With the avoidance channel observed ($\delta = 0.10$):

Method	False "hidden desire" under threat	True latent-approach detected
Shadow Mirror (with guard)	**0.0% [0.0, 0.0]**	80.6% [77.9, 83.2]
Baseline: arousal-as-desire	75.1% [72.1, 78.1]	89.5% [87.4, 91.8]
Baseline: mismatch-only	42.9% [39.5, 46.2]	100.0%

Guard sensitivity **100%** on both threat processes; specificity **94.4%** [93.2, 95.7]. AUC of the threat score across all five processes: **1.000**.

The cost is real and should be reported as such: sensitivity to genuine latent approach falls from 100% to ~81%, concentrated entirely in the `latent_approach_aroused` case (61.5% detected). **The guard trades ~19 points of hidden-preference sensitivity for zero false threat-as-desire readings.** For an instrument that hands its output to a person about themselves, that is the right side of the trade — but it is a trade, and the paper must say so.

Finding B2-2 (new defect, D-19) — The guard collapses to zero when the avoidance channel is missing

Ablating the avoidance channel — a realistic deployment condition, since many settings have no avoidance measure at all:

	avoidance observed	avoidance missing
Guard sensitivity	100.0%	**0.0% [0.0, 0.0]**
False "hidden desire" under threat	0.0%	**31.2% [28.1, 34.4]**
…on `threat_compliance` specifically	0.0%	**62.5% [57.8, 67.2]**
AUC, fear vs excitement	1.000	0.835

Cause. `_infer_desire` computes `av = self._wmean(avoidance) if avoidance else 0.0`. A missing channel is silently coerced to **zero avoidance**, i.e. to positive evidence *against* threat. The logistic `sigma(4·(arousal + avoidance − 1.1))` then cannot reach the flag threshold on arousal alone: with arousal ≈ 0.85 and avoidance imputed as 0, the score is sigma(−1.0) ≈ 0.27.

This is a missing-data-as-negative-evidence bug in the single most safety-critical line of the system. It is silent: nothing in the report tells the user that the guard is inactive because a channel is absent. **Fix in Session 3:** distinguish *absent* from *zero*, and refuse to issue a desire reading at all when the guard's required channels are unavailable.

See Figure 2.

5. B3 — Refusal calibration

300 episodes per condition

Condition	Behaviour	Target
Genuinely insufficient evidence (1 pair)	refuses **100.0%** [100.0, 100.0]	~100% [yes]
Clear evidence (8 pairs, δ = 0)	over-refuses **0.0%** [0.0, 0.0]	~0% [yes]
Ambiguous evidence (8 pairs, centred 0.5, high dispersion)	refuses 0.0%; uncertainty **0.618** [0.614, 0.622] vs **0.265** [0.260, 0.271] on clear evidence	—
Decade-old evidence, 30-day half-life	reports `status: ok` **100.0%**	~0% [no]

Finding B3-1 — Refusal gating is correct at the extremes

No over-refusal, no under-refusal. The one behaviour that deserves comment is the ambiguous case: the system does not refuse, but it does report **2.3× higher uncertainty**. That is defensible — arguably preferable to refusing, since the elevated uncertainty is itself the informative output — but it is a design choice that should be stated rather than left implicit.

Finding B3-2 — D-12 confirmed at 100%

Four observations from ten years ago, under a 30-day half-life, still yield a confident report. Their decay weight is ~2□¹²¹ ≈ 0. The sufficiency gate counts raw observations; decay is applied only to the means. The fix (effective sample size `n_eff = (Σw)²/Σw²`) is scheduled for Session 3.

---

6. B4 — Inquiry planning against an information-theoretic oracle

500 trials per condition · 4 candidate binary tests · planner likelihood-estimation noise sigma = 0.12

Planner	Regret, close hypotheses	Regret, separated hypotheses	Top-1 agreement with oracle
Proposed EIG planner	**0.161** [0.138, 0.187]	**0.195** [0.168, 0.223]	62.8–67.8%
v1.0 belief-gap planner	0.579 [0.547, 0.612]	**1.000** [1.000, 1.000]	21.8% / 0.0%
Random selection	0.550 [0.515, 0.584]	0.562 [0.528, 0.597]	24.4–25.6%

Finding B4-1 — The v1.0 "expected information gain" planner computes no information gain

`InquiryPlanner.plan_inquiry` scores inquiries as `expected_benefit = 1 - gap`, where `gap` is the difference between two *prior* beliefs. That quantity does not depend on the candidate observations at all, so the planner carries no signal about which observation to seek. Measured consequence: **regret 0.579 vs random's 0.550** — it is not better than random when it fires, and its confidence intervals overlap random's completely.

Worse, when hypotheses are well separated it emits nothing at all (regret 1.000, silent in 100% of trials), which is exactly backwards: a confidently-held wrong hypothesis is precisely when a discriminating observation is most valuable.

A genuine EIG planner — 40 lines, in `baselines.py` — achieves **0.16–0.20 regret** even with substantial noise in its likelihood estimates. The architecture paper's claim of "information-gain-driven inquiry generation" becomes true only if this replaces the current heuristic. Scheduled for Session 4.

See Figure 3.

---

7. Conformance checks (PASS/FAIL, never scored)

Structural properties that hold or fail *by construction* are not experimental results. Reporting "100%" for a check that cannot fail is what made the v1.0 numbers look like evidence when they were not. These are reported as pass/fail with the observation:

	Invariant	Result	Observation
C1	I Evidence traceability	[yes] PASS	explanation links to its supporting evidence
C2	II Provenance preservation	[no] **FAIL**	0 of 2 parents recovered from a two-parent DAG
C3	III Revision history	[yes] PASS	5/5 belief transitions recorded
C4	V Hypothesis independence	[yes] PASS	competitor survives contradiction of its rival
C5a	order invariance, moderate evidence	[yes] PASS	deviation 4.4 × 10□¹⁶
C5b	order invariance, extreme evidence	[no] **FAIL**	same evidence, three orders □ **0.0396 / 0.9900 / 0.9900**

	Invariant	Result	Observation
C6	IV Explicit uncertainty	[yes] PASS	uncertainty summary always emitted

5 of 7 pass.

Finding C5b (new defect, D-20) — Belief depends on the order evidence arrives

Five pieces of evidence, three orderings, three different final beliefs — spanning **0.04 to 0.99**, a 95-percentage-point swing from ordering alone.

The cause is the clamp `posterior = max(0.01, min(0.99, posterior))` in `BeliefRevisionEngine.bayesian_update`. Clipping is not invertible: once the posterior is pinned at 0.99, the information about *how far past* 0.99 the evidence pushed it is destroyed, and subsequent contrary evidence starts from the wrong place.

Order-invariance is not one of the paper's ten invariants, but it is entailed by calling the operator Bayesian — conditioning on a set of conditionally independent observations is commutative. A reviewer will test this. **Fix in Session 4:** accumulate in log-odds space and clamp only at the presentation layer, never in the state.

---

8. New defects added to the register

#	Sev	Where	Defect
D-19	**Critical (safety)**	`engine.py:_infer_desire`	Missing avoidance channel is coerced to zero, silently disabling the trauma guard (sensitivity 100% □ 0%, false hidden-desire rate 0% □ 31%)
D-20	**High**	`shadow_mirror_v1.py:bayesian_update`	Posterior clipping makes belief order-dependent; 0.04 vs 0.99 for identical evidence
D-21	Medium	`engine.py`	No injectable clock; wall-clock decay makes results non-reproducible at ~10□13 and untestable in principle
D-22	Medium	`engine.py:InquiryPlanner` (v1)	`expected_benefit` computes no information gain; performs no better than random

D-19 supersedes the earlier severity of the trauma guard as "verified working" — it works only when a channel that many deployments will not have is present.

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9. What can now be claimed in a paper, and what cannot

Claimable, with running code behind it:

- The threat guard reduces the rate of misreading a threat response as a hidden desire from 43–75% (guard-free baselines) to 0.0%, at a cost of ~19 points of sensitivity to genuine latent approach. n = 400 per process, 95% CIs, five generative processes, two misspecification levels.
- The guard's protection is entirely contingent on the avoidance channel; with that channel absent, protection falls to zero. (Reported as a limitation *and* as a design requirement: the system should refuse rather than degrade silently.)
- Refusal gating is exactly calibrated at the extremes: 100% refusal on insufficient evidence, 0% over-refusal on clear evidence.
- The system reports 2.3× higher uncertainty on ambiguous evidence than on clear evidence.

- An expected-information-gain planner reduces normalised regret from 0.55–1.00 to 0.16–0.20 against an information-theoretic oracle.
- 5 of 7 architectural conformance checks pass; the two failures are documented.

Not claimable, and now deleted:

- Any statement about human ratings of explanations.
- Any accuracy advantage over simple baselines on latent-state recovery.
- Any claim that the current inquiry planner is information-gain-driven.
- The 72% / 41% / 34% / F1 = 0.81 figures.

---

10. Reproducibility

```
cd benchmark_suite_v2
pip install matplotlib          # only needed for the figures
python run_benchmarks.py        # ~103 s, writes results.json
python make_figures.py          # reads results.json, writes the three PNGs
```

Determinism was verified by running the full suite twice and diffing `results.json`: **byte-identical**. Achieving that required freezing temporal decay in the three benchmarks that do not test it (D-21) — without that, wall-clock-dependent decay introduces ~10□¹³ drift between runs. The stale-evidence probe in B3 deliberately keeps decay on.

Every figure is generated from `results.json`. No number in this report was typed by hand.

End of Session 02. Session 03 — Engine v0.4, fixing D-08, D-09, D-11, D-12, D-19, D-21 — begins on your next "continue".

Adjudication of the Proposed Draft (Session 02b)

Question put to me: *"Can you consider embedding the drafts above into our work after validating them?"*

Answer: Partly. **I have adopted 9 items, adopted 5 with changes, and rejected 11.** The rejected material is not embedded, and I have recorded it in `STATE.json` so it cannot re-enter the program later by accident.

The document is well organised and reads authoritatively. Several of its recommendations are genuinely good and are now part of the plan. But its centrepiece — the proposed abstract — describes a completed IRB-approved human-subjects programme that does not exist, written in the past tense. Embedding that would be a much larger fabrication than the one we removed in Session 2. Section 1 explains; Section 4 is the item-by-item verdict.

---

1. The blocking problem: the proposed abstract reports work that has never been done

Verbatim from the proposed abstract:

> *"We present results from extensive normative system evaluations on formal benchmarks,* ***demonstrating superior performance*** *in belief revision under asymmetric evidence and multi-hypothesis settings.* ***Crucially, we introduce findings from IRB-approved human-subject studies*** *comparing Shadow Mirror's scaffolding with baseline interfaces,* ***showing significant improvements*** *in human reflection depth, correction ratios, and calibrated reliance.* ***We also report on initial high-stakes domain pilots in literature synthesis****, confirming the*

architecture's practical utility... with all code and expanded datasets ***made publicly available****."*

And the accompanying Ethical Statement: *"(Confirm IRB approval for human-subject studies and adherence to ethical guidelines)."*

None of that has happened. No IRB has reviewed anything. No participant has been recruited. No pilot has run. No dataset has been released.

The document does contain one mitigating line, buried far above the abstract: *"This structure assumes that the extensive empirical work (Phases 1, 2, and 3) has been completed."* That sentence is the difference between a template and a fabrication — and it does not travel with the text. The moment the abstract is pasted into a manuscript, the assumption is gone and only the past-tense claims remain.

This is the same failure mode as Benchmark 3, one order of magnitude larger. Benchmark 3 invented a *t*-statistic. This invents an ethics approval, a participant cohort, a control condition, a pilot deployment and a data release. Fabricating an IRB approval is not a formatting error; in most institutions it is a terminable offence, and it is the specific claim that journals verify.

Verdict: the abstract is rejected outright. A structural skeleton with `[TO BE COMPLETED — no data yet]` in every results slot would have been fine and useful. Prose in the past tense is not.

---

2. The provenance problem: this is the third generative layer, and each one adds confidence

The document's authority rests on what it calls "the Consensus review (Paper 3)" — your file `can_you_deeply_DEEPLY_research_and_validate_everyt.pdf`. That filename is the prompt you typed. Paper 3 is **not a peer review**. It is another AI system's response to the same request, saved as a PDF.

So the chain now reads:

1. An AI wrote the IEEE paper (watermarked "Generated by Kimi.ai"), inventing the Benchmark 3 result.
2. A second AI reviewed that paper and, treating its contents as data, described the invented result as a real finding needing a caveat.
3. A third AI (this document) cited the second as "the Consensus review" and built a research programme on top of it — including a citation to **"Moawad, 2026"** as external support for the architectural-positioning claim. That is you, citing yourself, being reported back to you as independent corroboration.

Nothing was verified at any step, and confidence increased at every step. Two concrete consequences in this document:

- **It re-endorses the fabricated benchmark.** It writes: *"Benchmark 3 ... shows that explanations including EWK-profile tension flags received significantly higher calibration ratings (M = 4.09, SD = 0.58) ... t(39) = 4.23, p < 0.001. However, this benchmark used simulated human evaluator ratings."* Session 1 established there are no evaluator ratings, simulated or otherwise. The function draws from `numpy.random.normal(4.2, 0.6)` and `numpy.random.normal(3.5, 0.8)` and never instantiates a Shadow Mirror object. Calling it "simulated human evaluation" dignifies it. The correct action is deletion, which Session 2 performed.
- **It claims validation of an invariant we have shown is violated.** It writes that the 100% traceability result *"directly validates some of these invariants."* Session 2's conformance suite tests Invariant II (Provenance Preservation) on a two-parent provenance DAG and it **FAILS** — `trace_provenance()` follows `preds[0]` only and recovers **0 of 2** parents. The document asserts as validated the one invariant we have a reproducible counterexample for.

None of this is a reason to distrust everything in the document. It is a reason to check each claim against the code, which is what Section 4 does.

3. It builds on the wrong Shadow Mirror

Across roughly 6,000 words, the document never once mentions the consent gate, the trauma guard, `threat_response` as a first-class hypothesis, the `ambivalence` state, temporal decay, topic scoping, or refusal-as-output. It treats "Shadow Mirror" as the URG + EWK architecture from the AI-generated v1.0 — the artifact that deleted your actual contribution (defect D-10).

That inheritance produces a proposal that runs against the project's founding commitment. The document recommends adding:

- *"Web scraping and API integration … including news feeds,* ***social media (for public sentiment analysis, with ethical considerations)****"*
- *"****facial expression recognition****, activity recognition"* from video
- *"Physiological Signals: feature extraction from ECG, EEG, EDA … and integration with wearable device APIs"*

Your own `ETHICS.md` and README state the opposite: *"never point it at another person, never use it covertly."* The consent layer is described as *"the load-bearing wall."* Passive social-media sentiment scraping is by definition non-consensual observation of identifiable people. Adding it would not extend Shadow Mirror; it would convert Shadow Mirror into the thing Shadow Mirror was built to refuse.

Worse, Session 2 gives an empirical reason as well as an ethical one. Defect **D-19**: when the avoidance channel is absent, the trauma guard's sensitivity drops from 100% to **0.0%** and the rate of misreading a threat response as a hidden desire rises to **31.2%**. Bolting on more passive affect channels *before* fixing missing-channel handling would ship a system that silently loses its only safety mechanism on exactly the noisy, partially-observed multimodal data the document proposes feeding it.

Ordering constraint added to the plan: no new evidence channel is added until D-19 is fixed and the guard refuses rather than degrades.

4. Item-by-item verdict

ADOPT (9) — now part of the programme

#	Item	Where it lands
A1	**The reframing table.** "Validated architecture" ☐ "proof-of-concept preliminary evaluation"; "improved calibration" ☐ (deleted, not softened); "broad human reasoning support" ☐ "supports structured reflective tasks under controlled conditions"; "scalable reference architecture" ☐ "reference design pending scalability validation"; explicit "not ready for clinical or high-stakes deployment".	Sessions 8, 11
A2	**Three-layer evaluation programme** — normative system evaluation ☐ human-subject evaluation ☐ low-stakes domain pilot. Sound structure; matches and sharpens our Phase I/IV split.	Plan structure
A3	**Human-reasoning metrics**: reflection depth, correction ratio, revision distance, falsification events, calibrated reliance. These are the right dependent variables and I had not specified them.	Session 15 protocol
A4	**Low-stakes pilots first** (education, literature synthesis) before anything clinical.	Session 15
A5	**Belief revision nuance** — human revision deviates from AGM/Bayesian norms and is explanation-guided. Real, citable, and it strengthens the argument for holding contradictions open.	Session 11 related work

#	Item	Where it lands
A6	**Asymmetric-evidence and multi-hypothesis benchmarks** — genuine gap; our B1/B4 use symmetric reliability and 2–4 hypotheses.	Session 5
A7	**Operator comparison** (Bayesian vs Dempster–Shafer) as an ablation rather than an assertion.	Session 5
A8	**Dedicated reproducibility appendix** — exact update rules, benchmark generators, ablations, environment. Session 2 already meets most of this; formalise it.	Session 14
A9	**"Governed Autonomy" as an eighth design principle.** Names something your architecture already does that the seven principles leave implicit.	Session 11

ADOPT WITH CHANGES (5)

#	Item	Change required
B1	Retitle to *"Shadow Mirror: A Framework for Governed Human–AI Reflective Collaboration Under Uncertainty"*	Keep "Governed"; drop "Framework for" as vague. Working title: **"Governed Reflective Collaboration: Consent, Refusal and the Threat Confound in Human–AI Reflection Under Uncertainty."** Decide in Session 11.
B2	Narrow the URG claim to "an original proposal, not a consensus fact"	Correct, but insufficient. Session 2 found the inference layer is **statistically indistinguishable from a median split** in all 20 cells. The URG claim must be narrowed to *structural* benefits — traceability, provenance, contradiction preservation — and must not imply accuracy benefits.
B3	EWK: "focus on observable signals and tensions, not latent states"	Adopt the wording. But Session 2 found `EWKProfile.detect_tension()` **never fires** on the demo scenario written to exhibit tension (defect D-13). Fix the code before writing the section, or the claim is again ahead of the implementation.
B4	Scalability: distributed graph DB, event streaming, microservices, Kubernetes	Adopt as *future work*, not as implementation. Session 2 measured the actual bottleneck and it is not storage: `add()` recomputes the full belief state on every observation, giving $O(n^2)$ per topic. Fix the algorithm before buying infrastructure. A single-node graph store handles a personal reflection journal for decades.
B5	Public datasets for evaluation	Adopt **OULAD** and **KDD Cup 2010** (genuinely open, educational, non-affective). Reject the rest — see §6.

REJECT (11)

#	Item	Reason
R1	**The proposed abstract**	Reports IRB approval, human-subject findings, domain pilots and a data release, none of which exist. §1.
R2	Ethical Statement "(Confirm IRB approval…)"	Same. There is nothing to confirm.
R3	Treating Benchmark 3 as a real-but-caveated finding	It executes no system code. Deleted in Session 2. §2.
R4	"100% traceability … directly validates some of these invariants"	Invariant II demonstrably **fails** our conformance test (0/2 parents recovered).
R5	"100% appropriate belief revision" as evidence	Tautological: with likelihood ratio 0.25 Bayes guarantees the decrease. Cannot fail, cannot discriminate.

#	Item	Reason
R6	Social-media scraping for sentiment analysis	Non-consensual observation of identifiable people; contradicts `ETHICS.md` and GDPR Art. 6/9. §3.
R7	Facial-expression recognition from video	Same consent problem; plus the EU AI Act restricts emotion inference in workplace and education contexts — the exact domains proposed for the pilots.
R8	Wearable/physiological integration *now*	Adds the highest-risk channel while D-19 leaves the guard silently disabled on missing channels. Ordering violation.
R9	MIMIC-III / MIMIC-IV as "publicly available"	Verified: credentialed access only — PhysioNet credentialing, mandatory CITI "Data or Specimens Only Research" training, and a signed Health Data Use Agreement. Also reintroduces the clinical framing Session 1 recommended removing.
R10	IEMOCAP / AffectNet as drop-in datasets	Both require signed academic licences (IEMOCAP from USC SAIL); neither is a reflection dataset — they are affect-recognition corpora, and using them would silently redefine the contribution as affect recognition.
R11	"Superior performance … demonstrating" language throughout	Session 2 measured no accuracy superiority over a median split. The word "superior" is not currently earnable.

5. Citation audit of the newly introduced references

Cited as	Verified reality	Verdict
Rosenbacke et al., **2026**	Rosenbacke, Melhus et al., *"How Explainable AI Can Increase or Decrease Clinicians' Trust in AI Applications in Health Care: Systematic Review,"* **JMIR AI 2024;1:e53207**, PMID 39476365	Real paper, **wrong year** (2024, not 2026)
Vasileiou et al., **2025**	Vasileiou & Yeoh, *"Human-Aware Belief Revision: A Cognitively Inspired Framework for Explanation-Guided Revision of Human Models,"* arXiv **2405.19238**, AAMAS 2024	Real paper, **wrong year**; the substantive point is correct and worth citing
Usher & Barak, 2024	Real — *"Artificial Intelligence Tools in Academic Writing Instruction"* (2024)	Real paper, **wrong claim**: it is about writing instruction, not about "bias, subgroup issues, governance, and prospective validation burdens"
Moerland et al., 2017 (for reproducibility)	Moerland's 2017 work is on emotion in reinforcement-learning agents / model-based RL, not research reproducibility	**Misattributed**; the reproducibility literature you want is Pineau et al. or the ACM badging guidelines
"Moawad, 2026"	**You.** Cited as external support for your own positioning claim	**Circular** — remove
MARS / MetaMem / "Think22"	Already established in Session 1 as wrong title and/or venue; note the document further garbles Think2² into "Think22"	**Inherited errors**, propagated
Jianbing et al. 2011; Barber et al. 2025; Cross et al. 2024; Liu et al. 2026; Thri & A.V.L.N. 2026; Goktas & Grzybowski 2025; Schäfer et al. 2025	Not verified this session — insufficient identifying detail (no titles, no venues, no DOIs)	**Unverifiable as given**; must not enter a reference list in this form

Pattern: of the citations carrying enough detail to check, **none was fully correct**. Two are real papers with wrong years, one is a real paper supporting a claim it does not make, one is misattributed, one is circular self-citation, and three are inherited errors. This is the same failure signature as the Kimi reference list, and it

is why Session 6 exists.

The one genuinely valuable citation this exercise surfaced

Chasing the Vasileiou reference turned up something directly relevant to our own defect **D-20**:

> ***Jacobs, B. (2025). "Commutativity of probabilistic belief revision." Frontiers in Cognition. DOI: 10.3389/fcogn.2025.1623227***

Jacobs proves that mathematical Bayesian updating **is** commutative — updating with p then q gives the same posterior as q then p — while human cognition is demonstrably order-dependent through recency and priming.

This settles a question I flagged in Session 2 as open. The v1.0 engine's order dependence (0.0396 vs 0.9900 for identical evidence in different orders) is **unambiguously a bug**, not a defensible model of human recency effects. If you ever want order-dependence as a *feature*, Jacobs shows it has to be introduced deliberately and named — not leak in through a `min(0.99, …)` clamp. This citation is now in the Session 4 fix notes and the Session 11 related-work outline.

---

6. Data-plan audit

Proposed source	Reality	Decision
MIMIC-III / MIMIC-IV	Credentialed access; CITI training; signed DUA	**Reject** — and it re-imports the clinical framing we removed
PhysioNet physiological signals	Mixed; many sets credentialed	**Defer** to post-D-19
IEMOCAP	Signed academic licence from USC SAIL	**Reject** for this programme
AffectNet	Signed agreement; affect recognition, not reflection	**Reject**
CMU-MOSEI	Genuinely open, but again affect recognition	**Reject** — wrong construct
OULAD (Open University Learning Analytics Dataset)	Genuinely open, educational, behavioural	**Adopt** — candidate for Session 5
KDD Cup 2010 educational data mining	Open	**Adopt** — candidate for Session 5
PubMed Central OA / arXiv / Semantic Scholar APIs	Open and appropriate	**Adopt** for the literature-synthesis pilot
Social media scraping	GDPR Art. 6/9; contradicts the consent gate	**Reject permanently**

Underlying issue: **none of the proposed corpora contain reflection**. They contain clinical records, facial expressions and vocal affect. A reflection benchmark needs paired belief-before/belief-after data with evidence in between — which does not exist as a public dataset and is precisely what the Session 15 study is designed to create. Substituting affect-recognition corpora would quietly turn your architecture paper into an affect-recognition paper with worse numbers than the specialist literature.

---

7. What has actually changed in the programme

- Session 5 gains **A6** (asymmetric-evidence and multi-hypothesis benchmarks), **A7** (operator comparison as ablation), and OULAD / KDD Cup 2010 as candidate real-data probes.

- Session 11 gains **A1** (reframing table), **A5** (explanation-guided revision), **A9** (Governed Autonomy principle), **B1** (title decision), **B2** (narrowed URG claim), and the Jacobs 2025 citation.
- Session 15 gains **A3** (the five human-reasoning metrics) and **A4** (low-stakes pilots first) — these materially improve the protocol and are the document's most valuable contribution.
- Session 14 gains **A8** (reproducibility appendix).
- A new **ordering constraint**: no additional evidence channel — video, audio, physiological, scraped — is added until D-19 is fixed and the guard *refuses* rather than silently degrading.
- A new **ground rule 6**: no text enters a manuscript in the past tense describing work that has not been performed, including in templates. Placeholders must read `[NO DATA YET — Session 15]`.
- Eleven rejected items are recorded in `STATE.json` under `rejected_proposals`, with reasons, so they cannot be reintroduced by a future draft that looks equally authoritative.

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8. A note on how to use documents like this

Your instinct to bring it in was right — it is organised, it says sensible things about evaluation layers, and its metrics are better than mine were. The failure is not that it is worthless; it is that **its true parts and its invented parts are written in exactly the same confident register**, and the invented parts are concentrated in the abstract, which is the part most likely to be pasted straight into a manuscript.

The working rule for the rest of this programme: **anything that arrives as prose gets checked against the code before it becomes text in a paper.** If a sentence asserts a result, there must be a file in the repository that produces it. That rule is what let me adopt nine items from this document with confidence and reject eleven without hesitation.

Session 03 — Engine v0.4, fixing D-19 first — begins on your next "continue".

Adjudication of the Second Proposed Draft (Session 02c)

Verdict: substantially better, still not embeddable. Three blocking defects, all repairable. I have adopted 7 further items, and — because rejecting twice without offering an alternative is not useful — Section 6 contains a **drop-in replacement abstract and results section** written entirely from the Session 2 measurements, so you can see the difference on the page.

---

1. What this draft genuinely fixed

Real credit is owed. Compared with the previous draft:

Fixed	How
Human-subject studies moved to future tense	§6 is a "Validation Roadmap", not a results section. The single most dangerous defect of the last draft is gone.
Ethical Statement is now honest	"This research did not involve human or animal subjects." Correct.
Appendix J marked "(pending)"	Correct.
Explicit non-deployment disclaimer	§1.3 and §7.3: "not ready for clinical or high-stakes deployment." Correct and necessary.
Claim-to-Validation Matrix with failure signals	Appendix D. This is genuinely good practice and better than what I had — it names, per claim, the observation that would falsify it. Adopted.

Fixed	How
Prohibited-Use Checklist	Appendix G. Adopted, with additions.
Citation quality improved markedly	Five references verified correct this session, including two excellent ones I did not have (§4).
Moerland misattribution corrected	Now cited as the emotion-in-RL survey it actually is — though attached to the wrong claim (§4).

Four of the six checkable citations in this draft are real and correctly identified. That is a large improvement over the previous draft, where none was.

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2. Blocking defect 1 — the fabricated numbers survived the rewrite

The abstract still says:

> *"The proof-of-concept implementation demonstrates **100% traceability preservation and appropriate belief revision** on synthetic contradiction tasks, while EWK-augmented explanations receive significantly higher **simulated calibration ratings ($t(39) = 4.23$, $p < 0.001$)**."*

And §5.2 still says:

> *"Shadow Mirror achieved 100% appropriate revision (**vs. 72% for a baseline pipeline that discarded superseded evidence**)." "...aligned with expert-identified critical tests in 100% of cases (**vs. 41% for a random-information baseline**)."*

Every one of these is a number we established has no measurement behind it:

- **$t(39) = 4.23$** comes from `numpy.random.normal(4.2, 0.6)` versus `numpy.random.normal(3.5, 0.8)` at seed 42. No Shadow Mirror object is constructed anywhere in that function. Re-labelling it "simulated calibration ratings" and adding a caveat does not convert it into a measurement — it is a property of a random number generator, and it belongs in no section of any paper.
- **72% and 41%** are baselines that exist in no file. I searched the entire supplied codebase in Session 1 and implemented real substitutes in Session 2; these two numbers came from nowhere and the draft re-imports them verbatim.
- **100% traceability** is asserted as evidence for Invariant II. Our conformance suite tests Invariant II on a two-parent provenance DAG and it **FAILS**: `trace_provenance()` follows `preds[0]` only and recovers **0 of 2** parents.

A caveat is not a repair. "These results are preliminary" is a statement about *strength of evidence*; the problem here is *existence of evidence*. The correct edit is deletion and substitution, which §6 does.

The tell: a number changed between drafts and nobody noticed

Benchmark 2's headline was **78%** in the original IEEE paper and is **100%** in this draft. No explanation is offered, because none was needed — the check is `len(inquiries[0].related_hypotheses) >= 2`, which the planner satisfies by construction whenever the belief gap is under 0.2, and the benchmark hard-codes the gap at 0.04. A number that can drift by 22 points across drafts without anyone noticing is a number that is not measuring anything.

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3. Blocking defect 2 — the paper cites a chatbot as a peer-reviewed source, six times

Reference **[8]**:

> *[8] Consensus Review, "Submission-Ready Shadow Mirror Manuscript and Validation," Consensus.app, 2026.*

This is cited at §1.3 (twice), §2.3, §2.5, §3.3, §3.4, §5.2 — and it is the **sole authority for the paper's own limitations statements**, including "not for a best-in-class journal article", the narrowing of the URG claim, and the EWK caveat. Those are the sentences a reviewer most wants sourced, and they are sourced to an AI product's output on your own manuscript.

An editor will identify this immediately. The remedy is simple: those statements are the *authors'* own scope limitations. They need no citation at all. Delete [8] and state them in your own voice.

Reference **[6]** is a second, worse case:

> *[6] R. Rosenbacke et al., "Governing Reflective Human-AI Collaboration: A Framework for Epistemic Scaffolding and Traceable Reasoning," arXiv, 2026.*

I could not find this paper. Rosenbacke's actual work is the JMIR AI 2024 systematic review on explainability and clinician trust — a different topic, different year, different venue. This title is an almost verbatim restatement of Shadow Mirror's own thesis, and it is cited to support "human-centered scaffolding" as the paper's *strongest validated claim*. A reference that says exactly what the author needs, attributed to a real researcher who wrote something else, is the classic signature of a fabricated citation. **Do not include it.** If the claim needs support, [13] Schäfer et al. and [20] Vasileiou et al. are real and closer to the point.

---

4. Blocking defect 3 — the conformance appendix misdescribes your own repository

§5.1: *"We implemented core Shadow Mirror components in Python (v0.3.0) … URG management via NetworkX with typed edges."*

Appendix C, headed **"Prototype v0.3.0"**:

As stated	Reality
Observation Layer ☐ `EvidenceManager.construct_evidence()`	That function is in `shadow_mirror_v1.py` — the AI-generated file — not in your repository
URG ☐ NetworkX DiGraph	**v0.3.0 has zero third-party dependencies.** `pyproject.toml`: `dependencies = [] # standard library only, on purpose: local-first, auditable.` NetworkX appears nowhere in it
EWK Profile ☐ three-stream tension detection	Not present in v0.3.0; and in v1.0 it **never fires** on the demo scenario written to exhibit it (D-13)
Belief Revision ☐ Bayesian posterior updating	v0.3.0 evaluates a single Gaussian likelihood at the weighted centroid — the posterior is bit-identical at n = 2 and n = 100 (D-08)

The draft has merged two different codebases under one version number. Appendix C is precisely the table a reproducibility reviewer opens first, and every row of it is wrong about the artifact it names.

There is a second cost. Nowhere in ~7,000 words does this draft mention the consent gate, the trauma guard, `threat_response` as a first-class hypothesis, the `ambivalence` state, temporal decay, topic scoping, or refusal-as-output. Those are in v0.3.0 — and after Session 2 they are the only parts of the system with a measured, defensible advantage over a trivial baseline. The draft describes the half of the work that does not survive measurement and omits the half that does.

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5. Citation audit

Ref	Verified status
[13] Schäfer et al.	[yes] **Real and excellent.** *"'The AI is uncertain, so am I. What now?': Navigating Shortcomings of Uncertainty Representations in Human-AI Collaboration with Capability-focused Guidance,"* Proc. ACM Hum.-Comput. Interact., DOI **10.1145/3757451** (FU Berlin). The draft truncates the subtitle. **Keep — this is one of the best citations in the whole programme.**
[20] Vasileiou et al.	[yes] **Real, correct year.** *"How Do People Revise Inconsistent Beliefs? Examining Belief Revision in Humans with User Studies,"* arXiv **2506.09977**, June 2025. Exactly supports the claim it is attached to. **Keep.**
[21] Peng et al.	[yes] Real. *"Knowledge Graphs: Opportunities and Challenges,"* Artificial Intelligence Review, 2023.
[24] Cross et al.	[yes] Real. Bias in medical AI, PLOS Digital Health, Nov 2024.
[16] Moerland et al.	[yes] Real — *"Emotion in Reinforcement Learning Agents and Robots: A Survey,"* Machine Learning 107, 2018 (not 2017). [!] **Wrong claim**: an emotion-in-RL survey does not support a statement about the psychometrics of trait-EI scalar scores.
[6] Rosenbacke et al. 2026	[no] **Not found. Probably fabricated.** Real Rosenbacke work is JMIR AI 2024;1:e53207 on XAI and clinician trust.
[8] Consensus.app	[no] **Not a citable source.** Cited 6×, as the authority for the paper's own limitations.
[3][4][5][9][10] MARS / MetaMem / "Think22" / ESMA / Didolkar	[no] Inherited Kimi errors, unchanged. Additionally, the "46% token savings" figure is attributed to **[3] MARS** in §2.1 — it belongs to **[10] Didolkar** (arXiv 2509.13237). Attribution scrambled.
[7] Deliu & Moldovan, **[12] Chinnaraju**	[!] Unverified. [12] is in *World Journal of Advanced Engineering Technology and Sciences* — check its indexing status before citing; low-selectivity venues in a reference list invite scrutiny.
[23] Usher & Barak 2024	[!] Title differs from the paper I verified previously ("AI Tools in Academic Writing Instruction"). One of the two attributions is wrong.

6. Drop-in replacement: abstract and results, written from the measurements

Everything below is produced by `benchmark_suite_v2/run_benchmarks.py` and appears in `results.json`. No number here was typed by hand.

6.1 Replacement abstract

Refusal as a Safety Mechanism: Consent-Gated Reflective Inference and the Threat Confound in Human–AI Reflection Under Uncertainty *Systems that infer what a person "really" wants from the gap between what they say and what they do face a hazard that is rarely measured: the same surface contradiction is produced both by a hidden preference and by a threat response. Reading the second as the first — telling a person they secretly want something they are afraid of — is a specific, foreseeable harm, and no existing affect-aware architecture reports its rate. We present Shadow Mirror, a consent-gated reflective architecture in which refusal is a first-class output, contradiction is held open rather than resolved, and a threat response is represented as a competing hypothesis rather than as noise. We evaluate it in simulation against five generative processes with known latent mechanisms, including two that produce identical stated aversion by different routes. Shadow Mirror misreads a threat response as a hidden desire in* ***0.0%*** *of episodes (95% CI [0.0, 0.0]; n = 400 per process), against* ***75.1%*** *[72.1, 78.1] for a baseline that admits physiological arousal into the action channel and* ***42.9%*** *[39.5, 46.2] for a stated-versus-behaviour mismatch heuristic. Guard sensitivity is 100% and specificity 94.4% [93.2, 95.7]. The protection is not free: sensitivity to genuine latent approach falls from 100% to* ***80.6%*** *[77.9, 83.2], concentrated entirely in high-arousal approach. An ablation exposes a dependency we report as a design requirement rather than a result. With the avoidance channel unobserved — a common deployment condition — guard sensitivity falls to* ***0.0%*** *and the misreading rate rises to* ***31.2%*** *[28.1, 34.4], because an absent channel is currently treated as evidence of safety. We argue that a reflective system must refuse under partial observation rather than degrade silently, and we specify the refusal condition. Refusal gating is otherwise exactly calibrated: 100% refusal on insufficient evidence, 0% over-refusal on sufficient evidence, and 2.3× higher reported uncertainty on ambiguous evidence. We also report two negative results. The system's latent-state classification is statistically indistinguishable from a median split across 20 conditions, and its reported confidence does not track its accuracy — accuracy falls from 99% to 55% under model misspecification while confidence moves only from 0.83 to 0.72, leaving the system overconfident precisely when it is wrong. We conclude that the contribution of a reflective architecture lies in what it declines to conclude, not in what it classifies. All code, benchmarks and analysis are released; the suite is byte-reproducible across runs.* ***This system is a research instrument. It is not a diagnostic device and is not suitable for clinical or other high-stakes deployment.***

6.2 Replacement results section

Substitute for the draft's §5.2 in full:

5.2 Method. *Observation streams are generated from known latent states; the generating centre is displaced from the model's assumed corner by a misspecification offset δ, so results measure robustness rather than self-consistency. Five baselines are implemented and executed at benchmark time. All rates carry 2,000-resample percentile bootstrap intervals. The suite is deterministic: two full runs produce byte-identical output.* ***5.3 Safety (primary).*** *Table 1. Guard on: 0.0% [0.0, 0.0]. Arousal-as-desire baseline: 75.1% [72.1, 78.1]. Mismatch-only baseline: 42.9% [39.5, 46.2]. AUC of the threat score across all five processes: 1.000. Cost: latent-approach sensitivity 100% □ 80.6% [77.9, 83.2].* ***5.4 Ablation.*** *Avoidance channel unobserved: sensitivity 0.0%, misreading rate 31.2% [28.1, 34.4], rising to 62.5% [57.8, 67.2] on the compliance process.* ***5.5 Refusal calibration.*** *100% [100.0, 100.0] refusal on insufficient evidence; 0.0% [0.0, 0.0] over-refusal on sufficient evidence; uncertainty 0.618 [0.614, 0.622] on ambiguous versus 0.265 [0.260, 0.271] on clear evidence. Known defect: evidence decayed to negligible weight still passes the sufficiency gate in 100% of trials, because the gate counts raw observations.* ***5.6 Inquiry planning.*** *Normalised regret against an information-theoretic oracle: expected information gain 0.161 [0.138, 0.187] and 0.195 [0.168, 0.223]; the belief-gap heuristic 0.579 [0.547, 0.612] and 1.000 [1.000, 1.000]; random selection 0.550 [0.515, 0.584] and 0.562 [0.528, 0.597]. The belief-gap heuristic is not distinguishable from random.* ***5.7***

> ***Negative results.*** *Classification accuracy is statistically indistinguishable from a median-split baseline in all 20 conditions. Reported confidence does not track accuracy: signed calibration bias moves from −0.19 (underconfident) at δ = 0 to +0.17 (overconfident) at δ = 0.45.* ***5.8 Conformance.*** *Five of seven architectural conformance checks pass. Invariant II (Provenance Preservation) fails: 0 of 2 parents are recovered from a two-parent provenance DAG. Order invariance fails under strong evidence: five observations in three orders yield final beliefs of 0.0396, 0.9900 and 0.9900. Bayesian conditioning on conditionally independent observations is commutative (Jacobs, 2025), so this is an implementation defect, not a model of recency.*

Reporting your own conformance failures is not a weakness in a reference-architecture paper. It is the strongest available evidence that the invariants are *testable* — which is the actual claim.

---

7. Adopted from this draft (7 items)

#	Item	Lands in
C1	Claim-to-Validation Matrix **with explicit failure signals** per claim	Session 8
C2	Prohibited-Use Checklist as an appendix	Session 10
C3	Prototype-to-Architecture Conformance appendix — **corrected** to describe v0.3.0 accurately	Session 8
C4	Schäfer et al. 2025 (DOI 10.1145/3757451) — uncertainty representations in human–AI collaboration	Sessions 10, 11
C5	Vasileiou et al. 2025 (arXiv 2506.09977) — how people actually revise inconsistent beliefs	Session 11
C6	§1.3 "Scope and Limitations" placed early, before any results	Sessions 10, 11
C7	*AI & Society* added to the venue shortlist	Session 6

Rejected and recorded: the abstract's fabricated statistics, the 72%/41% baselines, reference [6], reference [8], the merged-codebase Appendix C, and the "improve consistency and calibration" conclusion.

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8. The pattern across both drafts

Two independently generated drafts, both articulate, both structurally sensible, and both carrying the same three invented numbers into the abstract. The reason is mechanical: each drafting system read the previous document as its source of fact, and none of them ran the code. The numbers have now survived four documents without once being executed.

There is one rule that would have caught all of it, and it is now ground rule 7 of this programme:

> ***No number enters a manuscript unless a file in the repository produces it, and the person writing the sentence has run that file.***

Apply that rule and both drafts collapse to their skeletons — which is fine, because the skeletons are good, and Session 2 produced real numbers to put in them.

Session 03 — Engine v0.4, fixing D-19 first — begins on your next "continue".

Shadow Mirror — Session 03: Engine v0.4

Author of record: Omar Moawad · **Session date:** 25 July 2026 **Deliverables:** `shadow_mirror_v0.4.zip` (engine + 51 passing tests + head-to-head analysis), `v030_vs_v040.json`, `benchmark_results_v040.json`, this report.

---

0. Summary

Six defects fixed, three of them by changing what the system is willing to say rather than by making it cleverer. One critical safety defect is closed. Two new defects were found by the fix process itself and are recorded rather than hidden.

Defect	Status	Evidence
D-19 Missing avoidance channel silently disables the trauma guard	[yes] **Fixed**	Guard sensitivity under ablation **0.0%** ☐ **100.0%**; false "hidden desire" rate **31.2%** ☐ **0.0%**
D-12 Decayed evidence still passes the sufficiency gate	[yes] **Fixed**	Inference from decade-old evidence **100%** ☐ **0.0%**
D-08 Posterior insensitive to sample size	[yes] **Fixed**	Posterior now sharpens monotonically; ECE where the model applies **0.171** ☐ **0.044**
D-09 Two different numbers both called "threat probability"	[yes] **Fixed**	`threat_confound_score` and `p_threat_response` are now separate, documented fields
D-11 `belief_history` mixes topics	[yes] **Fixed**	History keyed by topic; instability now independent of interleaving
D-21 No injectable clock	[yes] **Fixed**	`ShadowMirror(clock=…)`; inference is a pure function of observations and clock
D-23 Overconfidence persists under *severe* misspecification	☐ **New, open**	ECE where the model does not apply **0.151** ☐ **0.221** (regression)
D-24 Arousal-only coverage labels excitement as threat	☐ **New, open**	Specificity under ablation **100%** ☐ **67.0%**
D-25 `demo.py` puts context strings in the topic slot	☐ **New, open**	Demo scenarios silently split across four one-observation topics

Tests: 51 passing (34 original commitments preserved, 1 deliberate documented behaviour change, 16 new regression tests that fail on v0.3.0 and pass on v0.4).

---

1. D-19 — the guard now refuses instead of degrading *(critical)*

The defect

`_infer_desire` computed `av = self._wmean(avoidance) if avoidance else 0.0`. An unobserved channel became a reading of **zero avoidance** — that is, positive evidence *against* a threat response. Session 2 measured the consequence: with the avoidance channel absent, guard sensitivity fell from 100% to **0.0%**, and 62.5% of compliance/fawn episodes were read as a hidden desire.

The fix

Absent is now distinguished from zero. The engine tracks `guard_coverage` in {`full`, `arousal_only`, `avoidance_only`, `none`} and computes the confound score only from channels that were actually observed — returning `None` when none were.

The load-bearing decision is what follows from partial coverage. `latent_approach` — the reading that says *"you want this even though you say you don't"* — is the one reading that requires ruling out a threat response. It is therefore only assertable under `full` coverage. Otherwise its mass moves to a new explicit state, `unresolved_approach`, and the report says so:

> *"Both arousal and avoidance are required to tell a hidden preference apart from a threat response. Without them this reading is reported as 'unresolved_approach' and no hidden want is asserted."*

Measured effect

Benchmark suite run **unchanged** from Session 2, so these are directly comparable.

Ablation condition — avoidance channel not observed (δ = 0.10, n = 400 per process):

	v0.3.0	v0.4
Guard sensitivity (threat caught)	0.0% [0.0, 0.0]	**100.0% [100.0, 100.0]**
False "hidden desire" under threat	31.2% [28.1, 34.4]	**0.0% [0.0, 0.0]**
…on the compliance/fawn process	62.5% [57.8, 67.2]	**0.0% [0.0, 0.0]**
Genuine latent approach detected	99.2%	**0.0%** — the system now refuses
Specificity	100.0%	**67.0% [64.4, 69.8]** ☐ cost, see D-24

Outcome distribution under partial coverage, v0.4:

True process	v0.4 says
`latent_approach` (calm)	`unresolved_approach` 100% — refusal
`latent_approach_aroused`	`threat_response` 100% — **false alarm** (D-24)
`threat_response`	`threat_response` 100% [yes]
`threat_compliance`	`threat_response` 100% [yes]
`explicit_aversion`	`explicit_aversion` 100% [yes]

Full-coverage condition — nothing lost, something gained:

δ = 0.10, both channels observed	v0.3.0	v0.4
False "hidden desire" under threat	0.0%	0.0%
Genuine latent approach detected	80.6% [77.9, 83.2]	**87.1% [84.9, 89.4]**
…on the hard aroused-approach case	61.5%	**74.2% [70.0, 78.5]**
Specificity	94.4%	94.4%

So under full coverage the guard got *more* sensitive to genuine hidden preference (+6.5 points) while remaining perfect on threat. The refusal only bites where it should: when the evidence needed to make the distinction was never collected.

The deliberate breaking change

One of the 34 original tests encoded a commitment v0.4 no longer keeps:

```
# v0.3.0
def test_desire_says_no_but_acts_yes_reads_latent_approach():
    # stated aversion + approach behaviour, no confound channels at all
    assert m.current().top_desire() == "latent_approach"
```

Approach under stated reluctance is produced *both* by a hidden preference *and* by a threat response. Session 2 measured what asserting the flattering one costs: a guard-free mismatch reading misidentifies a threat response as hidden desire in **42.9%** [39.5, 46.2] of episodes and **95%** of compliance episodes. So v0.4 refuses.

The test was not deleted to make the code pass. It was **rewritten to assert the new behaviour, with the rationale in the docstring**, and a companion test was added proving the capability is conditioned rather than removed:

```
def test_latent_approach_is_still_assertable_with_full_coverage():
    # calm arousal + low avoidance does rule out the threat explanation
    assert b.guard_coverage == "full"
    assert b.top_desire() == "latent_approach"
```

---

2. D-08 — the posterior now responds to evidence

The defect

v0.3.0 computed a weighted mean per channel and evaluated **one** Gaussian likelihood at that centroid. The posterior was bit-identical at n = 2 and n = 100.

The fix

Two changes, and the second matters more than the first.

(a) Accumulate. For a location model with known observation noise the weighted mean is the sufficient statistic and the precision of that estimate grows with the total evidence weight *W*. The log-likelihood is now `-W · (mean − corner)² / (2sigma²)` per channel, so confidence grows with how much evidence there is, not merely with where its centre falls.

(b) Add an explicit "none of these fit" hypothesis. `unmodelled` carries the likelihood a hypothesis would have if the evidence centre sat 0.32 away from its corner — roughly half the distance between adjacent hypotheses. Without this, accumulation makes the system *more* confident in the nearest wrong answer as evidence piles up, which is the failure Session 2 measured.

One subtlety worth recording, because getting it wrong cost an implementation round: the hypothesis-independent scatter term is deliberately excluded from both the named hypotheses and `unmodelled`. "None of these fit" must mean *the centre of your evidence is not near any of them*, not *your readings are noisy*. Charging observation noise against the named hypotheses but not against `unmodelled` made the instrument abstain simply because a person answered inconsistently — a different finding entirely, and the wrong one. Noise is carried by the dispersion term in the uncertainty estimate instead.

A second subtlety: oscillating evidence sits far from every *directional* corner, so accumulation initially read it as model misfit and `unmodelled` swallowed **ambivalence** — the one hypothesis that actually explains oscillation. Misfit is now discounted by however much ambivalence already accounts for. Without this, v0.4 would have destroyed a core contribution of the architecture while appearing to improve.

Measured effect — head-to-head on identical episodes

`analyze_v030_vs_v040.py` runs both engines on the same 200 episodes per cell, 20 cells.

Where the model approximately applies (δ ≤ 0.15):

	v0.3.0	v0.4
Expected calibration error	0.171	**0.044**
Accuracy	0.982	0.980
Mean confidence	0.813	0.940
AUC(uncertainty ☐ error)	0.868	**0.949**
Abstention rate	n/a	0.3%

At δ = 0 with n ≥ 16, ECE is **0.000** — the engine's stated confidence and its accuracy coincide exactly, where v0.3.0 sat at a flat 0.154–0.208 regardless of how much evidence it had.

Where the model materially does not apply (δ ≥ 0.30) — a regression, reported as such:

	v0.3.0	v0.4
Expected calibration error	0.151	**0.221** ☐ worse
Accuracy	0.690	0.664
Selective accuracy (when it commits)	n/a	0.700
AUC(uncertainty ☐ error)	0.845	0.800
Abstention rate	n/a	5.7% (21% at δ = 0.45, n = 32)

Honest reading

v0.4 trades *uniformly mediocre* calibration for *excellent when its model fits, worse when it does not*. That is a real improvement only to the extent the instrument can tell which regime it is in. It partly can — abstention rises monotonically with δ, and AUC(uncertainty ☐ error) is 0.949 in the good regime — but not enough: at δ = 0.45 with 32 observation pairs it abstains only 21% of the time while being wrong 54% of the time.

This is now defect D-23, and I have deliberately not tuned it away. Widening the `unmodelled` threshold would improve the δ = 0.45 numbers and damage the δ = 0.15 numbers; picking the value that makes the benchmark look best is exactly the practice this programme exists to avoid. The honest statement is that a two-parameter location model cannot distinguish "state A, displaced" from "state B" when the displacement moves the centroid toward B, and no parameter choice fixes that. Session 5 addresses it structurally, by testing whether more evidence channels or an explicit misspecification test can separate the two.

---

3. D-12, D-09, D-11, D-21 — the rest

D-12 — decay now reaches the gate. An observation counts toward sufficiency only if its decayed, reliability-weighted mass is at least `EPS_WEIGHT = 0.05` (about 4.3 half-lives of age). Dispersion uses a weighted standard deviation, so a stale outlier no longer inflates uncertainty.

> *Inference from decade-old evidence under a 30-day half-life: **100%** ☐ **0.0%.** Uncertainty on clear evidence 0.265 ☐ **0.021**; on ambiguous evidence 0.618 ☐ 0.405. The ratio between them widens from **2.3× to 19×** — the instrument now discriminates "I know this" from "I can't tell" far more sharply. Refusal on insufficient evidence remains 100%, over-refusal on clear evidence remains 0%.*

D-09 — one number, one name. `threat_confound_score` is the raw, gate-facing quantity the safety layer keys on, and is `None` when no confound channel was observed. `p_threat_response` is the posterior mass of the threat hypothesis after normalisation. The ambiguous label `threat_response_probability` is gone, and a test asserts it stays gone.

D-11 — history is per topic. `belief_history` is a `Dict[str, List[BeliefState]]`. `_instability(topic)` is computed within a topic; a regression test logs identical per-topic evidence interleaved and in blocks and asserts the instability figures match to 1e-6. `current(None)` now means *pooled across topics* and actually computes it, rather than returning whatever snapshot happened to be written last.

D-21 — injectable clock. `ShadowMirror(clock=lambda: fixed_datetime)`. Inference is now a pure function of the observations and the clock, which is what makes the decay behaviour testable at all.

---

4. New defects found by fixing the old ones

D-24 (medium) — under arousal-only coverage, excitement is labelled a threat. Arousal alone cannot separate fear from excitement; both raise it. With avoidance unobserved, v0.4 reports `threat_response` as the top hypothesis for 100% of aroused-approach episodes, dropping specificity from 100% to **67.0%**. The *caution* is defensible — the architecture's stated design is that the threat caution surfaces first. The *label* overclaims: the honest answer is "cannot tell", i.e. `unresolved_approach` with the caution attached. Deferred to Session 4 rather than fixed here, because one deliberate behaviour change per session is the limit of what can be properly justified.

D-25 (low) — `demo.py` passes context strings into the topic slot. `add_batch` rows are `(axis, signal_type, value, topic, context, reliability)`. The demo passes `("knowledge", "self_report", 0.9, "I know recursion cold")` — so "I know recursion cold" becomes a *topic*, and each scenario is silently split across four one-observation topics. Under v0.3.0 this combined with the broken `current(None)` to make demo scenarios 1 and 2 print `insufficient_evidence` when sufficient evidence existed. Under v0.4 the pooled read is correct and the scenarios report normally, but the demo is still mislabelling its own data.

---

5. What has changed in the claim inventory

Now claimable, with running code:

- Distinguishing an unobserved channel from a zero reading raises trauma-guard sensitivity under partial observation from 0.0% to 100.0% and eliminates the hidden-desire misreading (31.2% ☐ 0.0%), at a specificity cost of 100% ☐ 67.0%.
- Conditioning the `latent_approach` reading on confound coverage *increases* sensitivity to genuine hidden preference under full coverage (80.6% ☐ 87.1%) while preserving a 0.0% misreading rate.
- Accumulating evidence with an explicit none-of-these component reduces expected calibration error from 0.171 to 0.044 where the model applies, and raises AUC(uncertainty ☐ error) from 0.868 to 0.949.
- Gating sufficiency on decayed weight eliminates inference from expired evidence (100% ☐ 0.0%).

Not claimable, and now measured:

- Any general calibration improvement. Calibration is *worse* under severe misspecification (ECE 0.151 ☐ 0.221).
- Any accuracy improvement. Accuracy is unchanged where the model applies and slightly lower where it does not.
- Robustness of the guard under arousal-only observation. It is safe but imprecise there.

---

6. Reproducing

```
unzip shadow_mirror_v0.4.zip && cd shadow_mirror_v0.4
pip install pytest
pytest -q                          # 51 passed
python demo.py
python analyze_v030_vs_v040.py     # head-to-head, writes v030_vs_v040.json
```

The Session 2 benchmark suite runs against v0.4 unmodified — no metric definition was changed, so every before/after number in this report is like-for-like. Where v0.4 needed a metric v0.3.0 could not produce (abstention, selective accuracy), it is reported *in addition to* the unchanged numbers, never instead of them.

Session 04 — unify the two codebases into v1.0-real, fixing D-10, D-13, D-14, D-15, D-16, D-20, D-22 and D-24 — begins on your next "continue".

Shadow Mirror — Session 04: One Codebase, v1.0

Author of record: Omar Moawad · **Session date:** 25 July 2026 **Deliverable:** `shadow_mirror_v1.0.zip` — 78 passing tests, zero third-party dependencies, 7/7 architectural conformance.

---

0. Summary

The two codebases are now one. The URG, belief revision, inquiry planning and the EWK profile were rebuilt **on top of** the v0.4 engine rather than beside it, so nothing that made the original work distinctive was lost: the consent gate, the trauma guard, `ambivalence`, temporal decay, topic scoping and refusal-as-output are all still there, and now so is the reflective graph layer.

Defect	Status	Evidence
D-10 v1.0 prototype replaced rather than extended v0.3.0	[yes] Fixed	One package. Every v0.3.0 capability retained; 78 tests
D-20 Belief revision is order-dependent	[yes] Fixed	All **120 permutations** agree to machine precision (was a 0.95 spread)
D-22 Inquiry planner computes no information gain	[yes] Fixed	Regret vs oracle **0.579 → 0.000** on the planner's own test; selects the discriminating observation
D-14 `trace_provenance` truncates multi-parent DAGs	[yes] Fixed	2/2 parents recovered (was 0/2); branching and cyclic cases tested
D-13 EWK tension never fires on its own demo	[yes] Fixed	Flagship scenario now registers two tensions
D-24 Arousal-only coverage labels excitement a threat	[yes] Fixed	Excitement `threat_response` **100% → 0%**, caution retained
D-15 Naive vs aware timestamps block merging	[yes] Fixed	One aware convention throughout
D-16 `utcnow()` / `node_link_data()` deprecations	[yes] Fixed	NetworkX removed entirely; zero dependencies
D-25 `demo.py` puts context strings in the topic slot	[yes] Fixed	Scenarios now carry real topics

Architectural conformance: 4/7 → 7/7.

---

1. Conformance, prototype vs v1.0

Same six checks as Session 2, same definitions, plus one new check for Invariant VIII. Run against both implementations by `conformance_v10.py`.

	Prototype	v1.0
C1 · I Evidence traceability	PASS	PASS — 0 orphan explanations
C2 · II Provenance preservation	**FAIL** — 0/2 parents	**PASS** — 2/2 parents
C3 · III Revision history	PASS	PASS
C4 · V Hypothesis independence	PASS	PASS — contradiction *held*, 1 contested hypothesis retained
C5 · order invariance (120 permutations)	**FAIL** — spread 0.9504	**PASS** — spread 0.00e+00
C6 · IV Explicit uncertainty	PASS	PASS
C7 · VIII Inquiry justification	**FAIL** — no information-gain content	**PASS** — selects the discriminating observation, 0.714 bits
	4/7	**7/7**

2. D-20 — belief revision is now commutative

Bayesian conditioning on conditionally independent observations *is* commutative — Jacobs (2025), *Commutativity of probabilistic belief revision*, **Frontiers in Cognition**, DOI `10.3389/fcogn.2025.1623227`. Human belief revision is not, because recency and priming are real, but that is a fact about people and not a licence for the arithmetic to drift.

The prototype clamped the posterior to [0.01, 0.99] **inside the state** after every update. Clamping is not invertible: once pinned at 0.99 the information about how far past 0.99 the evidence pushed is gone, and later contrary evidence starts from the wrong place.

`beliefs.py` accumulates in **log-odds** — unbounded and additive — and stores contributions in a **dict keyed by evidence id**. Dicts keyed by identity have no meaningful order, which is precisely the property that was missing. Clamping happens only in `display_probability()`.

Four consequences, each with a test:

- Every one of the 120 permutations of five extreme observations gives the same posterior to within 1e-12 (prototype: 0.0396 to 0.9900).
- Twenty strongly-confirming observations followed by twenty strongly-disconfirming ones return the belief to **exactly 0.5**. The prototype could not do this at all.
- Re-applying the same evidence id does not double-count it.
- `retract()` is exact — removing a contribution restores the previous state bit-for-bit, which is what makes an audit trail meaningful rather than decorative.

3. D-22 — the planner now computes information

`inquiry.py` computes the mutual information between a candidate observation's outcome and the hypothesis variable, under the current posterior. Against an oracle over 300 randomised trials the planner's mean normalised regret is **0.000** — it picks the oracle's choice — where the prototype's belief-gap heuristic scored 0.579 against random selection's 0.550.

Two behavioural changes matter more than the number:

- **It no longer goes silent when hypotheses look settled.** The prototype emitted nothing once the belief gap exceeded 0.2, which is backwards: a confidently-held wrong hypothesis is exactly when a discriminating observation is worth most.

- **Every inquiry states what it would resolve**, in bits, and which hypotheses it separates — which is what Invariant VIII actually asks for. `gain_per_unit_cost` lets a person weigh an inquiry that costs them something against one that does not.

---

4. D-14 — provenance is a DAG

`urg.ancestors()` walks every provenance parent by breadth-first search. The prototype's `trace_provenance` followed `preds[0]` only, so evidence derived from two sources reported one and silently dropped the other — a direct violation of the architecture's own Invariant II.

Tested on the two-parent case, a four-root branching DAG (all four paths recovered), and a cyclic graph (terminates, does not hang).

The URG is written in **pure standard library**. The rest of Shadow Mirror is dependency-free on purpose — local-first and auditable — and importing NetworkX for an adjacency dict would have traded that away for nothing. It also removes the deprecated `node_link_data` call (D-16). A test asserts the string "networkx" does not appear in the module.

Contradiction is a first-class edge. `contested_hypotheses()` returns hypotheses carrying both supporting and contradicting evidence, so an explanation can report a tension instead of quietly preferring the later write.

---

5. D-13 — the tension detector fires on its own example

The prototype required `K > 0.7 and W < 0.3`, or `E > 0.7 and K < 0.3`, or `|K − W| > 0.5`. Its flagship demonstration — K = 0.30, W = 0.75, E = 0.80, a student working hard, distressed, and not getting results — missed all three by a hair and printed *"No significant tension detected"*.

`ewk.py` computes continuous tension magnitudes over the unit square against a single documented threshold (0.35). The flagship scenario now registers **two**: `effort_without_return` (0.525) and `distress_without_return` (0.560). Aligned streams still produce none, and the profile refuses below two observations per stream.

Every tension is emitted as a question, and the reading carries the disclaimer verbatim: *"These are discrepancies between observed streams, not claims about an inner state."*

---

6. D-24 — a caution that stops overclaiming

Under arousal-only coverage the threat *caution* still fires — the confound score is computed and `insufficient_confound_coverage` is flagged — but the threat hypothesis is no longer allowed to win the posterior. One channel can raise a question; it cannot settle one.

Outcome labels with the avoidance channel unobserved (400 episodes per process):

True process	v0.4 said	v1.0 says
`latent_approach`	`unresolved_approach` 100%	`unresolved_approach` 100%
`latent_approach_aroused` (excitement)	`threat_response` **100%**	`unresolved_approach` 59%, `ambivalence` 41%
`threat_response`	`threat_response` 100%	`explicit_aversion` 86%, `ambivalence` 13%
`threat_compliance`	`threat_response` 100%	`ambivalence` 80%, `unresolved_approach` 15%

The safety-critical number is unchanged: **0.0%** of threat episodes are read as a hidden desire, and guard sensitivity via the flag remains **100%**. What changed is that the system no longer tells someone who was excited that they were probably afraid.

---

7. No regression elsewhere

The Session 2 benchmark suite was re-run **unmodified** against v1.0. B1, B3 and B4 are numerically identical to Session 3; B2's headline figures are unchanged (0.0% false hidden-desire under full coverage, guard sensitivity 100%, genuine latent approach detected 87.1%). The specificity cost under ablation (67.0%) is unchanged, because it is driven by the caution flag, which was deliberately kept.

`pytest`: **78 passed**. That is 34 original v0.3.0 commitments, one documented behaviour change from Session 3, and 43 regression tests written against specific defects.

---

8. Still open

Defect	Severity	Session
D-23 Overconfidence persists under severe misspecification (ECE 0.151 □ 0.221 for δ ≥ 0.30)	High	5
D-02, D-04, D-05, D-18 The Kimi-generated paper: fabricated statistic, six wrong citations, vendor watermark, quarter-length	Critical	6, 11
D-17 Figure labels contain literal `n`	Low	12

One observation for Session 5: under partial coverage `ambivalence` now absorbs 41–80% of episodes. Oscillation in the simulated action channel is being read as genuine ambivalence. That may be correct — the streams really do disagree — but it deserves a dedicated benchmark before it appears in a paper, because "ambivalence" is a claim about a person and it is currently the residual bucket.

---

9. Reproducing

```
unzip shadow_mirror_v1.0.zip && cd shadow_mirror_v1.0
pip install pytest
pytest -q                    # 78 passed
python demo.py
python conformance_v10.py    # prototype 4/7 vs v1.0 7/7
```

No third-party runtime dependencies. `pytest` is needed only for the tests, and `networkx`/`numpy` only for the archived prototype kept alongside for the comparison.

Session 05 — the conformance suite as executable property tests, plus D-23 — continues automatically.

Shadow Mirror — Session 05: Invariants as Executable Specification, and D-23

Author of record: Omar Moawad · **Session date:** 25 July 2026 **Deliverable:** `shadow_mirror_v1.0.zip` (updated) — **294 passing tests**, all ten invariants executable, plus `experiment_d23.py` and `experiment_d23.json`.

---

0. Summary

The ten architectural invariants are no longer prose. They are ten callable checks with a positive and a negative test each, run over 200 randomised reflective states. That converts the paper's central claim from *"we define ten invariants"* — which any architecture paper can say — into *"these ten properties are decidable, here is the procedure that decides them, and here is a compliant implementation."* A reviewer verifies it by running `pytest`.

D-23 was attacked structurally rather than by tuning, with two experiments. One worked well, one worked partially, and both are reported.

	Result
Invariants executable	**10 / 10**, holding on 200 randomised well-formed states
Negative tests	**16**, one per invariant plus edge cases; each catches its own violation
Test suite	78 ☐ **294 passing**
D-23 mitigation A — credible sets	Coverage beats point accuracy at every δ, but **falls short of nominal** under severe misspecification
D-23 mitigation B — pool across topics	δ = 0.30: accuracy **83.8%** ☐ **95.8%** with 8 topics; offset error 0.187 ☐ 0.092

1. The invariants, as a specification

`shadow_mirror/invariants.py` exposes `check_i_…` through `check_x_…` and a `check_all` runner. Each returns a verdict plus the observation that produced it, and **never a score** — a property that holds by construction is not an experimental result, and reporting it as a percentage is precisely how the original benchmark suite's "100%" figures came to be mistaken for evidence.

Two invariants resisted mechanical checking and were operationalised deliberately rather than fudged:

VI (Context Consistency) cannot mean "recompute the past" — an instrument cannot re-run a person's week. It is operationalised as: *when a context is superseded, every conclusion that depended on it is annotated as stale, rather than silently retained or silently deleted*. The URG gained `revise_context()`, which supersedes a context and marks its dependents. Both branches are tested: a graph that supersedes a context by hand fails the check; one that uses `revise_context` passes and the dependent carries `stale=True`.

X (Human Authority) is a property of what reaches the person, so it is checked at the output boundary: every terminal output carries its uncertainty and disclaimer and offers at least one question, and none is phrased as a verdict about who someone is.

IX (Transactional Consistency) needed the URG to actually be transactional. It gained a `transaction()` context manager that snapshots and rolls back. The negative test subclasses the graph with a no-op transaction, provokes a failure mid-update, and confirms the check catches the partial write.

The property harness

`build_well_formed(seed)` constructs a randomised reflective state — 2–5 observations, derived evidence with multiple provenance parents, 2–4 hypotheses with a mix of supporting and contradicting edges, real belief updates, planned inquiries, and an explanation — and all ten invariants are asserted over 200 seeds.

The generator is a small seeded builder rather than a property-testing library, because the package has **no runtime dependencies** and adding one for the test suite alone would undercut the auditability argument the architecture rests on.

The check that caught itself

Pointing the Invariant X check at the *shipped* reflection prompts rather than at a fixture immediately produced a failure:

```
1 verdict phrasing(s); first offender: ('\byou have\b',
'(caution) You have consulted this 5 times. Seeing a guess changes late…')
```

That is the reactivity caution — a factual statement about the person's own logged behaviour, and one the architecture specifically wants surfaced. **The defect was in the check, not the system.** The patterns were

narrowed to claims about who someone is (`you are`, `you secretly`, `your true`, `you have a/an`) and directives (`you must`, `you should`), and the rationale is recorded in the source.

It is worth saying plainly why this matters: a conformance suite that is only ever pointed at fixtures will pass, and will be worthless. This one was pointed at real output on its first run and found something within a minute.

---

2. D-23 — overconfidence under severe misspecification

Session 3 measured the problem: when the model's assumed corners are displaced from where a person's evidence actually sits, the engine becomes *more* confident as evidence accumulates while accuracy collapses (ECE 0.151 □ 0.221; at the worst cell, abstaining 21% while wrong 54%).

Tuning the abstention threshold cannot fix that, and Session 3 deliberately did not try. The failure is **identifiability**: a two-channel location model cannot distinguish "state A, displaced" from "state B" when the displacement points toward B. So the fix has to change either *what is reported* or *what is observed*. Both were tried.

A — Report a set, not a label

`credible_set()` returns the smallest set of hypotheses carrying the requested posterior mass. Correctness then means **coverage**: does the reported set contain the truth at the nominal rate?

n = 400 per cell, 8 evidence pairs, nominal coverage 90%:

δ	Point accuracy	Credible-set coverage	Mean set size	Commits to one
0.00	100.0% [100.0, 100.0]	100.0% [100.0, 100.0]	1.00	99.8%
0.15	98.2% [96.8, 99.2]	**99.5% [98.8, 100.0]**	1.08	93.5%
0.30	78.2% [74.2, 82.0]	**88.0% [85.0, 91.2]**	1.45	66.0%
0.45	51.7% [46.8, 56.5]	**60.5% [55.5, 65.2]**	1.48	62.7%

Coverage beats point accuracy at every level, at a cost of about half a hypothesis in set size, and the system commits to a single reading two-thirds of the time even under severe misspecification.

But it does not reach nominal. At δ = 0.45 a set claiming 90% mass contains the truth 60.5% of the time. This is not a bug in the set construction — it is the posterior's miscalibration passed through. **A credible set inherits the calibration of the posterior it is drawn from.** Getting nominal coverage under misspecification requires calibrating against held-out labelled outcomes (conformal prediction), and labelled outcomes for a person's latent state do not exist without the human study. That is a real dependency, and it is now an explicit input to Session 15's design rather than a hope.

B — Estimate the displacement instead of tolerating it

If a person is systematically displaced — they habitually under-report confidence relative to performance — then the displacement is a property of the *person*, shared across topics, and becomes estimable by pooling. One topic cannot see it; several can.

`estimate_person_offset()` alternates hard assignment and offset estimation (a two-step EM). n = 400 simulated people:

δ	Topics	Accuracy, uncorrected	Accuracy, offset-corrected	Offset error
0.30	1	81.2% [77.5, 85.0]	81.2% (no pooling possible)	0.187
0.30	3	82.8% [80.7, 85.0]	87.5% [85.7, 89.3]	0.141

δ	Topics	Accuracy, uncorrected	Accuracy, offset-corrected	Offset error
0.30	5	82.3% [80.5, 84.0]	90.8% [89.5, 92.1]	0.120
0.30	**8**	83.8% [82.5, 85.0]	**95.8% [95.0, 96.4]**	**0.092**
0.45	1	53.0% [48.2, 58.0]	53.0%	0.420
0.45	**8**	53.2% [51.6, 54.9]	**61.3% [59.6, 62.9]**	0.364

At moderate displacement this is a clean structural win: **83.8% ☐ 95.8%** with eight topics, and it comes from observing more, not from tuning anything.

At severe displacement it helps much less (53.2% ☐ 61.3%), and the reason is visible in the offset error, which barely moves (0.420 ☐ 0.364). Hard-assignment EM initialises from an assignment that is mostly wrong at δ = 0.45, so it converges to a poor local optimum. The textbook remedy is soft assignment weighted by the posterior, with restarts. I have **not** implemented it this session: having made one deliberate parameter decision already, adding another change whose effect I would then measure on the same benchmark is how tuning starts. It is scheduled, and named.

Status of D-23

Downgraded from **high, open** to **characterised, partially mitigated**:

- The failure mode is identified as identifiability, not thresholding.
- One mitigation is measured and substantial at moderate misspecification (multi-topic pooling).
- One mitigation improves the honest reporting at all levels but does not reach nominal coverage (credible sets).
- The residual gap has a named cause (posterior miscalibration), a named remedy (conformal calibration), and a named blocker (needs labelled outcomes, i.e. the human study).

That is a defensible limitations paragraph. It was not one before.

---

3. What can now be claimed

Claimable, with running code:

- The ten architectural invariants are decidable, and a compliant implementation exists. All ten hold over 200 randomised reflective states; each is accompanied by a negative test proving the check can fail.
- Reporting a credible set instead of a point label increases the rate at which the true state is contained in the output at every level of model misspecification tested (e.g. 78.2% ☐ 88.0% at δ = 0.30), at a mean cost of 0.45 additional hypotheses.
- A person-level displacement is recoverable by pooling across topics: 83.8% ☐ 95.8% accuracy with eight topics at δ = 0.30, offset error 0.187 ☐ 0.092.

Explicitly not claimable:

- Nominal coverage. A 90% credible set contains the truth 60.5% of the time under severe misspecification, and the paper must say so.
- Any benefit of offset correction at severe displacement beyond the 8-point gain measured, which is limited by the estimator's initialisation.

---

4. Still open

Defect	Severity	Session
D-02, D-04, D-05, D-18 — the Kimi paper: fabricated statistic, six wrong citations, vendor watermark, quarter-length	Critical	6, 11
D-23 residual — nominal coverage needs conformal calibration	Characterised	15+ (blocked on labelled outcomes)
D-26 (new) — soft-assignment EM for offset estimation at severe displacement	Low	12
D-17 — figure labels contain literal `n`	Low	12
Ambivalence as residual bucket (Session 4 watch item)	Watch	8

5. Reproducing

```
unzip shadow_mirror_v1.0.zip && cd shadow_mirror_v1.0
pytest -q                    # 294 passed
python conformance_v10.py    # prototype 4/7 vs v1.0 7/7
python experiment_d23.py     # credible-set coverage + offset recovery
```

Session 06 — verifying the monograph's 96-entry bibliography against live sources — continues automatically.

Shadow Mirror — Session 06: Bibliography Verification and Venue Decision

Author of record: Omar Moawad · **Session date:** 25 July 2026 **Deliverables:** `bibliography.json` (all 96 entries, structured), this report.

0. Summary — and a correction to my own earlier framing

Your bibliography is in excellent condition. 96 entries, parsed and checked in full:

Check (applied to all 96)	Result
Missing year	**0**
Malformed DOI	**0**
Duplicate entries	**0**
Entries with no resolvable identifier	6 — and all six are books or technical reports where no DOI exists
Distinct publisher DOI prefixes	**28**
Year range	1948–2024, median 2012
Live-verified sample (highest-risk entries)	**5 / 5 correct**

The six identifier-free entries are Pearl (2009) *Causality*, Picard (1997) *Affective Computing*, Popper (1959), Settles (2009) UW-Madison TR-1648, Shadish, Cook & Campbell (2002), and Sutton & Barto (2018). Books and technical reports do not carry DOIs. That is correct practice, not an omission.

Twenty-eight distinct publisher prefixes matters more than it looks. Fabricated reference lists cluster on a handful of plausible prefixes; a real one spread across APA (10.1037, 17 entries), Elsevier (10.1016), SAGE (10.1177), Nature (10.1038), Taylor & Francis (10.1080), JSTOR, ACM, IEEE, Springer and eighteen others is

the fingerprint of someone who actually went and found the papers.

The correction I owe you: in Session 1 I wrote that the generated IEEE paper "degraded your scholarship." That was right, but I understated it. The AI-generated 12-entry list had **at least six errors**. Your own 96-entry list has **zero** detectable ones. The ratio is not close.

---

1. Method, and its limits

`bibliography.json` contains every entry parsed from pages 316–322, with year, DOI, arXiv ID and URL extracted. Parsing used indentation boundaries — entries are indented, continuations are not — after a first attempt that split on author-year patterns produced 94 fragments with mangled identifiers. Worth recording, because the fragmented parse briefly suggested 80 entries lacked identifiers, which would have been an alarming and entirely false finding.

Structural checks ran over all 96 entries. Live verification covered 5. I am not going to imply otherwise. Exhaustive verification means 96 individual resolutions, which is roughly a day of someone's time and is a reasonable thing to ask a co-author or a librarian to do before submission.

The sample was not random — it was the highest-risk stratum: everything published 2022 or later, where a citation is most likely to be mis-transcribed and least likely to be caught by a reader's memory.

Entry	Verified as
Pekrun (2024), Control-value theory, `10.1007/s10648-024-09909-7`	[yes] Resolves to Springer, *Educational Psychology Review*
Li & Washington (2024), personalised vs generalised emotion recognition, `10.2196/52171`	[yes] Resolves to **JMIR AI 2024;1:e52171**
Vasey et al. (2022), DECIDE-AI, `10.1038/s41591-022-01772-9`	[yes] *Nature Medicine*, PMID 35585198 — title and DOI both exact
NIST (2023), AI RMF 1.0, `10.6028/NIST.AI.100-1`	[yes] Correct designator
Huang et al. (2024), LLMs cannot self-correct reasoning yet, arXiv 2310.01798	[yes] Correct ID

Five for five, including two 2024 journal entries and a standards document — the three categories where transcription errors usually appear.

---

2. The real problem is not errors. It is recency.

Period	Entries
Pre-2000	22
2000–2014	38
2015–2021	29
2022–2024	**7**
2025–2026	**0**

Median year **2012**. For a foundational bibliography this is a strength — Shannon 1948, Brier 1950, Simon 1955, Cronbach & Meehl 1955, Flavell 1979, AGM 1985 are the right spine for a theory of reflection under uncertainty, and most architecture papers do not have them.

For a **2026 submission in metacognitive AI**, seven entries from the last four years and none from the last eighteen months is the gap a reviewer will name first. The field moved a great deal in 2025–2026, and

Session 1 established that it moved in directions directly adjacent to yours.

Verified additions, ready to insert

Every one of these was resolved against a live source during Sessions 1–5. None is a suggestion to go and check.

Reference	Why it belongs
Jacobs, B. (2025). Commutativity of probabilistic belief revision. *Frontiers in Cognition*. `10.3389/fcogn.2025.1623227`	Proves Bayesian conditioning is commutative while human revision is not. Grounds the D-20 fix and the design decision to keep order-dependence out of the arithmetic.
Schäfer, U. et al. (2025). "The AI is uncertain, so am I. What now?" *Proc. ACM Hum.-Comput. Interact.* `10.1145/3757451`	The closest existing work on how people actually use an uncertainty display. Load-bearing for the refusal argument.
Vasileiou, S. L. et al. (2025). How do people revise inconsistent beliefs? arXiv `2506.09977`	Human revision is explanation-guided and non-minimal. Justifies holding contradiction open.
Vasileiou & Yeoh (2024). Human-aware belief revision. arXiv `2405.19238`	Companion framework.
Rosenbacke, R. et al. (2024). XAI and clinician trust: systematic review. *JMIR AI* 1:e53207	Calibrated reliance; the review your Phase 2 metrics derive from.
Elenjical, Kavuri & Varma (2026). Think2: grounded metacognitive reasoning in LLMs. arXiv `2602.18806`	Closest 2026 metacognition architecture; the contrast case for "internal vs external reflection".
Xin et al. (2026). MetaMem. **ACL 2026 Findings**, arXiv `2602.11182`	Symbolic meta-memory.
Didolkar, Ballas, Arora & Goyal (2025). Metacognitive reuse. arXiv `2509.13237`	Source of the real "46% token savings" figure.
Hou et al. (2026). Learn like humans: meta-cognitive reflection. arXiv `2601.11974`	Self-improvement via reflection.
Guo et al. (2503.19271). MARS: memory-enhanced agents with reflective self-improvement	Cite under its actual title.

That is ten verified 2024–2026 entries, taking the recent stratum from 7 to 17 and covering the exact literature a reviewer would check first.

---

3. Venue, settled

Sessions 2–5 changed what evidence exists, so the shortlist can now be decided rather than guessed.

What you can actually put in front of a reviewer:

1. A measured safety result with no human subjects: misreading a threat response as a hidden desire, **0.0%** [0.0, 0.0] with the guard, **42.9%** and **75.1%** for two guard-free baselines, n = 400 per generative process across five processes.
2. An ablation showing the protection is contingent on a channel many deployments lack, and a design that refuses rather than degrades when it is missing.
3. Ten architectural invariants as executable checks, holding over 200 randomised states, each with a negative test.
4. Refusal calibration: 100% refusal on insufficient evidence, 0% over-refusal, 19× uncertainty separation.
5. A zero-dependency, byte-reproducible artifact.

What you cannot: anything about how people respond to any of it.

The decision

P2 — the ethics-of-design paper — goes first, to AIES or FAccT.

Those venues reward exactly the shape of evidence you have: a specific foreseeable harm, quantified; a design commitment that eliminates it; an honest ablation showing the conditions under which the protection fails; and a refusal behaviour argued as a feature. No human subjects required. The consent architecture and prohibited-use checklist are contributions there rather than boilerplate.

P1 — the architecture paper — goes to ACM TiiS.

TiiS publishes architectures for human–AI interaction and tolerates formative rather than summative evaluation. The executable invariant suite is the strongest thing in the package for that audience: it converts "we define ten invariants" into a decidable specification with a compliant implementation.

Venue	Verdict
AIES / FAccT	**P2, first submission.** Best fit for a quantified harm plus a design that removes it.
ACM TiiS	**P1.** Architecture + executable conformance.
AI Magazine	Fallback for P1 if TiiS declines. Framework-friendly, lower empirical bar.
CHI	Possible for P2, but reviewing skews to user studies. Reconsider after Phase 2.
IEEE Intelligent Systems	Fallback; short-format reference architectures.
AI & Society	Fallback for a broader framing (raised in the second external draft).
IEEE TAC	**Not now.** Needs the human study. This was Session 0's recommendation and it was wrong for the evidence that exists.

Check the current calls for papers for both AIES and FAccT before committing — I am not going to quote submission dates I cannot verify, and they move every year.

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4. Corrections table for the monograph

#	Item	Action
B-1	Six entries without identifiers	**No action.** Books and TRs; correct as-is.
B-2	Recent literature thin (7 entries 2022+, 0 from 2025–26)	**Insert the ten verified references in §2.** Highest-priority change to the manuscript.
B-3	Exhaustive live verification	5/96 done here. Assign the remaining 91 to a co-author or librarian before submission; the structured `bibliography.json` makes it a mechanical pass.
B-4	No 2025–26 metacognition citations	Covered by B-2. This is the gap a reviewer names first.
B-5	Venue	Settled: AIES/FAccT for P2, TiiS for P1. Not IEEE TAC.

---

5. Reproducing

```
python3 - <<'EOF'
import json; b = json.load(open("bibliography.json"))
print(len(b), "entries")
print(sum(1 for r in b if not (r["doi"] or r["url"] or r["arxiv"])), "without identifier")
EOF
```

Session 07 — auditing the mathematical chapters (pp. 55–79) and repairing the LaTeX corruption visible in the PDF — continues automatically.

Shadow Mirror — Session 07: The Mathematical Chapters

Author of record: Omar Moawad · **Session date:** 25 July 2026 **Deliverables:** `FORMAL_APPENDIX_corrected.md` (drop-in replacement for the formal content of Part 3), `shadow_mirror_v1.0.zip` (**308 passing, 2 xfailed**), this report.

---

0. Summary

Seven findings. One is a presentation catastrophe with a trivial fix, three are genuine well-definedness defects, and one was found by measuring rather than reading — a quadratic-time defect in code I wrote in Session 4.

#	Finding	Severity
F-1	**Every equals sign is missing from every display equation**, including Bayes' theorem	Critical (presentation)
F-2	Operator symbols collide with state symbols — four of six, including R meaning both the reflective state and belief revision	High
F-3	Set braces absent throughout: $t in 0,1,2,ldots$, $E_t = e_1,ldots,e_n$, $T(x),{e_1,ldots,e_k}$	High
F-4	$mathcal{F}$ carries two incompatible signatures, unreconciled	Medium
F-5	B_t is left untyped, yet later chapters assert properties of it	Medium
F-6	Sequential updating stated without its conditional-independence condition	Medium
F-7 / D-27	Complexity chapter gives no bound for any operator it defines — and measurement found the belief update running in **quadratic** time	High

---

1. F-1 — every equals sign is gone

Seven display equations survive in the PDF as literal LaTeX, and **all seven are missing the** `=`. The worst is on p. 67:

```
\[ P(H \mid E) \frac{P(E \mid H),P(H)} {P(E)}. \]
```

That is Bayes' theorem, in the chapter named for Bayes' theorem, with no equals sign. Also affected: `R_{t+1} \mathcal{F}(R_t, O_{t+1})` (three separate occurrences), `B_{t+1} \mathcal{R}(B_t, E_{t+1})`, `\mathcal{F}(R_t,, E_{t+1},, C_{t+1})`, and Invariant I's `T(x) {e_1,\ldots,e_k}`.

The corruption is systematic, not random. Three character classes were stripped by whatever converted the source: `=` signs, backslash-escaped thin spaces (`\,` became a bare comma — visible as the `,,` in the third equation), and escaped set braces (`\{ \}` vanished entirely). Inline maths lost its delimiters too, so `\(\mathcal{F}\)` renders as `(\mathcal{F})` — bare parentheses that read as function application.

The fix is mechanical and it is the single highest-value edit available to the manuscript: **§A of the corrected appendix restates every equation in full.** No reviewer will read past a Bayes' theorem without an equals sign, and no reviewer should.

---

2. F-2 — the operators collide with the state

p. 58 defines the reflective state:

```
R_t = (E_t, C_t, H_t, B_t, K_t, U_t)
```

and then, on the same page, names the operators O, C, H, R, Q, X.

R is now both the reflective state and the belief-revision operator. O is both the observation set and observation incorporation. C is both context and context construction. H is both the hypothesis set and hypothesis generation. **Four collisions out of six operators**, and p. 67 then writes "the operator R" nine pages after R_{t+1} was defined as a state.

This is not pedantry. A reader cannot parse $\mathcal{F}(R_t, O_{t+1})$ without deciding whether those are states or operators, and the text gives no way to decide.

The corrected appendix uses $\mathcal{I}, \mathcal{K}, \mathcal{G}, \mathcal{B}, \mathcal{Q}, \mathcal{X}$ — script capitals, none of which is a state component — with an explicit signature for each.

---

3. F-3, F-4 — braces and arity

Braces. $t \in 0,1,2,\ldots$ is not a set membership statement. $E_t = e_1, e_2, \ldots, e_n$ reads as a tuple. $H = h_1, h_2, h_3$ likewise. $T(x)\ \{e_1, \ldots, e_k\}$ has neither an equals sign nor delimiters. Same root cause as F-1; must be fixed explicitly rather than assumed.

Arity. p. 58 gives $\mathcal{F}(R_t, O_{t+1})$; p. 62 gives $\mathcal{F}(R_t, E_{t+1}, C_{t+1})$. Two arities for one symbol, never reconciled. Defining $\mathcal{F} = \mathcal{X} \circ \mathcal{Q} \circ \mathcal{B} \circ \mathcal{G} \circ \mathcal{K} \circ \mathcal{I}$ dissolves it: the second form is the same map with the first two operators already applied.

---

4. F-5 — you cannot assert properties of an untyped object

The chapter says B_t "may be probability distributions, weighted graphs, logical structures, or hybrid models. The framework intentionally leaves this choice open."

Then later chapters assert properties of B_t: that it has a revision history (Invariant III), that posteriors chain (sequential reflection), that plurality is preserved. Those are claims about an object whose type has been left open. A reviewer will ask which of them survive under the "weighted graph" reading, and there is no answer in the text.

The abstraction is worth keeping. The fix is to move it up a level: state **axioms** every instantiation must satisfy, then give one reference instantiation.

- **B1 plurality** — assigns a value to every hypothesis simultaneously
- **B2 commutativity** — order of conditionally independent evidence does not matter
- **B3 invertibility** — evidence can be retracted exactly
- **B4 audit** — every application records prior, posterior, trigger, method, time

B2 and B3 are not decoration: the prototype satisfied neither, and the measured consequence was a final belief of 0.0396 or 0.9900 for the same five observations depending only on arrival order. Stating them as axioms turns that from an embarrassment into a specification the architecture enforces — and `test_operator_correspondence.py` now asserts all four.

---

5. F-6 — the missing independence assumption

"After each observation, the posterior becomes the new prior."

True only if $e_i \perp e_j \mid h$. In this domain that is frequently false: two self-reports on consecutive days about the same topic are correlated through mood, framing and recall, not only through the latent state. Left unstated it is an error; stated, it is an honest limitation — and it is what makes the implementation's per-evidence keying defensible rather than arbitrary, since re-submitting the same observation cannot compound it.

Neither that nor temporal decay substitutes for a correlated-likelihood model. §A.6 says so.

---

6. F-7 / D-27 — the complexity chapter, and what measurement found

The chapter warns that unrestricted pairwise comparison "could approach $O(n^2)$" and that naive history storage is $O(T)$. Neither bound applies to any operator the monograph actually defines. No bound is given for belief revision, provenance traversal, or inquiry planning.

So I measured all four. Scaling ratios per doubling:

Operation	Bound	Measured
Provenance closure `ancestors()`	$O(V+E)$	**2.0×** [yes]
Inquiry planning	$O(\lvert H\rvert \cdot \lvert \mathcal{A}\rvert)$	3.9× at $\lvert H\rvert = \lvert \mathcal{A}\rvert$ [yes]
Engine `add()` — full recompute	$O(n^2)$ cumulative	**3.8×** — quadratic
Belief update — log-odds ledger	expected $O(1)$	**4.2× — quadratic** [no]

The last row is a defect in code I wrote in Session 4. `BeliefLedger.log_odds()` summed every contribution on every call, so each update was $O(n)$ and a session of n updates was $O(n^2)$: 11.8 ms at 1,000 updates, 624 ms at 8,000. **Fixed** by maintaining a running total; re-measured at **2.0× per doubling**, 78 ms at 16,000 updates. A regression test now fails if the ratio exceeds 3×.

The engine's own `add()` remains quadratic — it recomputes the full belief state on every observation. 27 ms at 100 evidence pairs, 1121 ms at 800. That is a real optimisation target (incremental recompute), and it is worth noting that **the monograph predicted exactly this growth before the code existed.** The complexity chapter was right; it just needs to say it about the operators it defines rather than about a strawman.

---

7. Operator-to-implementation map

`tests/test_operator_correspondence.py` binds each formal operator to a callable and asserts its specified behaviour. Two operators have **no** implementation, and are recorded as strict `xfail` with the reason rather than quietly omitted:

Operator	Implementation	Status
$\mathcal{I}$ ingest	`ShadowMirror.add`, `URG` provenance edges	[yes]
$\mathcal{K}$ context	`URG.revise_context`, `contextualise`	[yes]
$\mathcal{G}$ generation	`engine._infer_*` posterior over hypotheses	[yes]
$\mathcal{B}$ revision	`BeliefRevisionEngine` — axioms B1–B4 all asserted	[yes]
$\mathcal{Q}$ inquiry	`InquiryPlanner`, `expected_information_gain`	[yes]

Operator	Implementation	Status
$mathcal{X}$ explanation	`reflective_prompts`, `EWKProfile`, `credible_set`	[yes]
Causal reasoning (pp. 64–66)	—	[no] **prose only.** No do-operator, no intervention, no counterfactual query.
Knowledge consolidation $K_t leftarrow B_t$ (p. 145)	—	[no] **prose only.** No promotion criterion from belief to revisable knowledge.

Those two gaps are now enumerable rather than implied, which is the point: a reference architecture earns the word "reference" when the distance between its formalism and its implementation can be listed.

The causal gap is the more significant. Chapter "Reflection and Causal Reasoning" occupies pp. 64–66 and Pearl (2009) is in the bibliography, but nothing in the codebase performs an intervention. The architecture paper should present causal reflection as **research agenda**, not as a component.

---

8. What changed in the claim inventory

Now claimable: every operator in the formalism either has a named implementation with asserted behaviour, or is listed as unimplemented. Four belief-revision axioms are enforced by test. Complexity bounds for three of four core operations are measured, not asserted.

Now explicitly disclaimed: causal reflective reasoning and knowledge consolidation are specified but not built. The engine's observation intake is quadratic.

---

9. Corrections table

#	Location	Action
M-1	All seven display equations	**Replace with §A of the corrected appendix.** Highest-value single edit in the manuscript.
M-2	p. 58 operator names	Rename to $mathcal{I}, mathcal{K}, mathcal{G}, mathcal{B}, mathcal{Q}, mathcal{X}$
M-3	pp. 55–58 set notation	Restore braces throughout
M-4	pp. 58, 62	Define $mathcal{F}$ as the composition; drop the second signature
M-5	p. 57 belief state	Add axioms B1–B4 and the reference instantiation
M-6	p. 68 sequential reflection	State the conditional-independence condition and that it is often violated here
M-7	pp. 71–74 complexity	Replace with the four measured bounds
M-8	pp. 64–66 causal	Reframe as research agenda; no implementation exists

---

Session 08 — rebuilding the claim-to-validation matrix against the code that now exists — continues automatically.

Shadow Mirror — Session 08: The Claim Matrix, Rebuilt

Author of record: Omar Moawad · **Session date:** 25 July 2026 **Deliverables:** `APPENDICES_C_D_rebuilt.md` (drop-in replacements for pp. 290–291), `experiment_ambivalence.py`, `shadow_mirror_v1.0.zip` (**312 passing**, 2 xfailed).

0. Summary

Your original Claim-to-Validation Matrix had fifteen claims. **Thirteen of them read "architecture only", "theory only" or "no evidence".** That was an honest self-assessment when it was written.

It is now out of date, which is the best possible reason to rewrite an appendix.

	Original (p. 291)	Rebuilt
Claims with only theoretical support	13 of 15	8 of 24
Claims settled by a test or measurement	0	**8**
Claims partially settled under simulation	2	4
Claims withdrawn	0	**5**
Conformance concerns implemented	7	**14**

And the Session 4 watch item turned out to be a real defect with a clean fix: **ambivalence precision 34.0% ☐ 97.6%.**

1. Ambivalence was a residual bucket. Now it is not.

Session 4 noted that under partial confound coverage, `ambivalence` was returned for 41–80% of episodes, and flagged it: *"that may be correct — the streams really do disagree — but 'ambivalence' is a claim about a person and it is currently the residual bucket."*

The engine computed ambivalence from a weighted standard deviation. Dispersion has at least three quite different causes and only one of them is ambivalence:

- **oscillation** — swinging toward and away over time. This *is* ambivalence.
- **noise** — a stable middling attitude, measured imprecisely. Not ambivalence.
- **drift** — a monotonic change of mind across the window. Not ambivalence; it is the *opposite*, a resolution.

A standard deviation cannot tell them apart. So I generated all three, plus a stable control, at 400 episodes each.

Before

True process	Read as ambivalence	Dispersion	Sign-flip rate	Lag-1 autocorrelation
oscillation	100.0% [100.0, 100.0]	0.357	0.999	**−0.843**
noise	**95.5% [93.5, 97.5]**	0.259	0.548	−0.123
drift	**98.8% [97.5, 99.8]**	0.246	0.219	**+0.516**
stable attraction	0.0% [0.0, 0.0]	0.071	0.547	−0.135

Precision 34.0%. Two of every three ambivalence readings were wrong — and one of those two was telling a person who had *made up their mind* that they were conflicted. For an instrument whose entire claim is that it holds contradiction open rather than forcing a resolution, reporting resolution as contradiction is the exact inversion of the thing it is for.

Note the third and fourth columns. Dispersion is nearly identical for noise (0.259) and drift (0.246), and not far off oscillation (0.357). But lag-1 autocorrelation separates all three cleanly: strongly negative for alternation, near zero for noise, strongly positive for drift.

The fix

Ambivalence is now gated on temporal structure as well as spread:

```
ambivalence = min(0.9, 2.2 · dispersion) · gate,
gate = clip((−ρ₁ − 0.20) / 0.45, 0, 1)
```

where ρ1 is the mean lag-1 autocorrelation of the stated and action series. Five lines.

After

True process	Read as ambivalence
oscillation	**100.0% [100.0, 100.0]** — recall unchanged
noise	95.5% □ **2.5% [1.0, 4.2]**
drift	98.8% □ **0.0% [0.0, 0.0]**
stable attraction	0.0%

Precision 34.0% □ 97.6%. Recall unchanged at 100%. Four regression tests pin it, including one that constructs an oscillating series and a drifting series with dispersion within 0.12 of each other and asserts that only the first reads as ambivalence.

A proposal, not implemented. Strongly *positive* lag-1 autocorrelation is itself an informative reading — "your position on this moved steadily across the window" — and the engine now computes the statistic but discards it. That is a candidate new state, and it belongs in the research agenda rather than in a session that has already made one behaviour change.

---

2. Appendix C — conformance

Fourteen architectural concerns are now implemented and test-backed, against seven in v0.3.0. The new rows are the URG, belief revision, inquiry planning, the invariant checks, EWK, credible sets and per-person offset estimation — each with the file and the tests that settle it.

Four concerns are recorded as **specified but unbuilt** rather than left ambiguous: causal inference, knowledge consolidation, multimodal acquisition, and the reflective execution engine. Two of those carry strict `xfail`s in the suite, so they cannot quietly become "implemented" in a future summary.

One row worth reading twice: **distributed deployment is not the bottleneck.** The original appendix listed it as a gap, and the second external draft proposed Neo4j, Kafka and Kubernetes. Session 7 measured the actual constraint: `engine.add()` recomputes the full belief state on every observation and is quadratic — 27 ms at 100 evidence pairs, 1121 ms at 800. Incremental recompute buys more than a cluster does.

---

3. Appendix D — what changed in the claim matrix

Eight claims are now settled by a test or a measurement, where previously none were. The largest single change is at the bottom of your original table:

> *Architecture is safer than a baseline — evidence now:* ***normative proposal***

That row now reads: misreading a threat response as a hidden desire, **0.0%** [0.0, 0.0] with the guard against **42.9%** and **75.1%** for two guard-free baselines, n = 400 per generative process across five processes and two misspecification levels — with a stated falsification signal and a named file.

Five claims are withdrawn, and it is worth being explicit that four of the five were never yours: the $t(39)$ = 4.23 calibration result, the 72% and 41% baselines, and the "100% appropriate revision" figure all originated in the generated benchmark suite. The fifth is yours and goes for an honest reason: *the URG improves*

reflective reasoning is removed **as stated**, because Session 3 measured the inference layer as statistically indistinguishable from a median split across all 20 conditions. The URG's benefits are structural — traceability, provenance, contested-hypothesis retention — and the rebuilt matrix claims exactly those and nothing more.

Thirteen claims still await people, unchanged from your original, because nothing in code can settle them. Two moved to *partial*: Invariant X now checks output phrasing mechanically, which is necessary but nowhere near sufficient for "questions reduce verdict-like framing"; and the guard ablation is direct evidence that disagreement *between channels* is decisive for safety, which is a narrow version of "sensor disagreement is informative".

---

4. The three sentences the evidence licenses

Every paper drawn from this work should be checkable against these:

1. Under simulation with known ground truth, conditioning the hidden-preference reading on the evidence needed to rule out a threat response reduces the rate of that misreading from 42.9–75.1% to 0.0%, at a cost of 19 points of sensitivity and 33 points of specificity when the required channel is absent.
2. The ten architectural invariants are decidable properties, and a compliant implementation exists with 312 passing tests, including 16 that prove each check can fail.
3. Nothing in this work establishes any effect on human reasoning, and no claim to that effect is made.

If a sentence in a draft cannot be traced to one of those three or to a row in Appendix D, it does not belong in the paper.

---

5. Defect register

ID	Status
D-29 ambivalence computed from dispersion alone; 34% precision, 98.8% of monotonic drift misread as ambivalence	[yes] **Fixed this session** — 97.6% precision, 4 regression tests
D-28 `engine.add()` quadratic	Open — Session 12
D-23 residual — nominal coverage needs conformal calibration	Characterised; blocked on labelled outcomes
D-26 soft-assignment EM for offset estimation	Open — Session 12
D-02, D-04, D-05, D-18 the Kimi paper	Superseded by the Session 6 venue decision; both papers written fresh
D-17 figure labels contain literal `n`	Open — Session 12
D-30 (new) positive lag-1 autocorrelation is computed and discarded; "your position moved steadily" is a candidate state	Proposal — research agenda

---

6. Reproducing

```
unzip shadow_mirror_v1.0.zip && cd shadow_mirror_v1.0
pytest -q                          # 312 passed, 2 xfailed
python experiment_ambivalence.py   # the before/after in §1
python conformance_v10.py          # prototype 4/7 vs v1.0 7/7
```

Session 09 — the ethics, governance and prohibited-use audit against GDPR and the EU AI Act as currently in force — continues automatically.

Shadow Mirror — Session 09: Ethics, Governance and Regulatory Audit

Author of record: Omar Moawad · **Session date:** 25 July 2026 **Deliverables:** this report, including a rewritten enforceable Prohibited-Use Checklist. All regulatory provisions were checked against live sources, not recalled.

0. Summary — and a correction to my own Session 6 recommendation

In Session 6 I recommended piloting in education because it is low-stakes. Under EU law it is not. That recommendation was wrong and I am withdrawing it.

Three findings, in order of consequence:

#	Finding	Status
R-1	With the physiological stream enabled, Shadow Mirror is plausibly an **emotion recognition system** under AI Act Art. 3(39) — and Art. 5(1)(f) **prohibits** its use in workplaces and education institutions. **In force since 2 February 2025.**	**Critical**
R-2	An educational deployment that evaluates learning outcomes falls under **Annex III(3)(b) — high-risk**. Those obligations apply from **2 August 2026: eight days from today.**	**Critical**
R-3	The consent gate as coded meets the dynamic-consent standard it cites on five of seven criteria. Two gaps: no demonstrable consent record at the store layer, and withdrawal is not as easy as granting in the CLI.	High

The self-mirror mode — a person, on themselves, outside an institution — is unaffected by R-1 and R-2. That is worth stating early, because it is the mode the architecture was designed for, and the regulatory exposure comes entirely from the deployment contexts that were proposed *around* it.

1. R-1 — the emotion-recognition prohibition

The chain

AI Act Art. 3(34): *"'biometric data' means personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person, such as facial images or dactyloscopic data."*

The critical point, and the one most people get wrong: **this definition does not require unique identification.** GDPR Art. 4(14) does ("which allow or confirm the unique identification"); the AI Act's does not. So an arousal proxy — a heart-rate-like signal — is personal data from technical processing of a physiological characteristic, and is therefore **biometric data under the AI Act** even though it identifies nobody.

Art. 3(39): *"'emotion recognition system' means an AI system for the purpose of identifying or inferring emotions or intentions of natural persons on the basis of their biometric data."*

Shadow Mirror's guard infers `threat_response` — an affective and intentional state — partly on the basis of the `physiological` stream. When that stream is enabled, the system meets this definition.

Art. 5(1)(f): *"the use of AI systems to infer emotions of a natural person in the areas of workplace and education institutions, except where the use of the AI system is intended to be put in place or into the market for medical or safety reasons."*

Applicable since 2 February 2025.

What this means concretely

Deployment	Exposure
Self-mirror, private individual, outside an institution	**Out of scope.** Art. 5(1)(f) is scoped to "the areas of workplace and education institutions".
Consented-study mode between two private individuals	Out of scope of Art. 5(1)(f); GDPR still applies.
A university or school deploying it for students, physiological stream on	**Prohibited**, unless the safety exception holds.
An employer deploying it, physiological stream on	**Prohibited**, same caveat.
Any institutional deployment with the physiological stream **off**	Outside Art. 3(39) — no biometric basis, so no emotion recognition system. Still possibly high-risk under R-2.

The safety exception, and why it must not be assumed

The exception covers systems "intended to be put in place… for medical or safety reasons." Shadow Mirror's physiological stream exists **only** as a threat confound. Session 2 measured what happens without it: the misreading rate rises to 31.2%, and 62.5% on compliance-type responses. Session 3 made the guard refuse rather than degrade. The stream's sole documented purpose, in code and in tests, is to prevent a specific foreseeable harm.

That is as clean a safety-purpose argument as this exception is likely to see. **It is still an argument, not a conclusion.** It has, to my knowledge, no case law behind it, and the Commission's guidelines on prohibited practices should be read before anyone relies on it. The honest position for a paper is to *state* the argument and note that it is untested — which is itself an interesting contribution, because "a safety guard that is itself an emotion-inference system" is a real and unresolved tension in the Act.

Recital 18's exclusions do **not** rescue this. They exclude physical states such as pain and fatigue, and the mere detection of expressions "unless they are used for identifying or inferring emotions". Shadow Mirror uses arousal precisely to infer whether a response is a threat response. That is the inferring case, not the detection case.

---

2. R-2 — high-risk classification, and the date

Annex III(3)(b): *"AI systems intended to be used to evaluate learning outcomes, including when those outcomes are used to steer the learning process of natural persons in educational and vocational training institutions at all levels."*

Shadow Mirror ingests `task_performance`, forms hypotheses about a learner's knowledge state (`blind_spot`, `latent_knowledge`), and emits inquiries that direct what the learner does next. In an educational institution, that is evaluating learning outcomes and steering the learning process.

Annex III(4)(b) catches the workplace equivalent: systems used "to monitor and evaluate the performance and behaviour of persons" in work relationships.

Application date: 2 August 2026 — eight days after this session. Systems placed on the market before that date get a transition, but only until they undergo significant design changes.

High-risk classification is not a prohibition. It is a compliance regime: risk management system, data governance, technical documentation, logging, transparency to deployers, human oversight, accuracy/robustness/cybersecurity, quality management, conformity assessment, CE marking, registration. For a solo researcher that is a serious undertaking, and it is the reason the recommendation below changes.

Revised pilot recommendation

Session 6 said: pilot in education, it is low-stakes. Withdraw that. In the EU, education is a named high-risk domain and, with the physiological stream, a prohibited one.

The literature-synthesis pilot proposed alongside it is clean: no personal data, no emotion inference, no institutional evaluation of a person. **That should be the Phase 3 pilot.** For the human-subjects study in Session 15, the defensible framing is *individual adults using the self-mirror on themselves, recruited outside any institutional relationship, not for the purpose of evaluating their learning* — which keeps it out of Annex III(3) entirely.

---

3. R-3 — does the consent gate meet the standard it cites?

`consent.py` cites dynamic consent (Budin-Ljøsne et al., 2017 — verified in Session 6 at `10.1186/s12910-016-0162-9`). Assessed against GDPR Art. 4(11) and Art. 7:

Requirement	Provision	As coded	Verdict
Freely given, not bundled	Art. 7(4)	`add_stream()` raises `ConsentError` unless each non-obvious stream is acknowledged individually; blanket grant is impossible for `implicit_task`, `physiological`, `avoidance`	[yes] **Meets, and exceeds.** Refusing to let a category be bundled at all is stronger than the Article requires
Specific	Art. 4(11)	Grants are per-stream and purpose-locked	[yes] Meets
Informed	Art. 4(11)	Every `Stream` carries `captures` and `may_reveal` in plain language; `disclosure_text()` renders it	[yes] **Meets, and exceeds.** Stating what a stream *may reveal*, not only what it records, is the part most implementations omit
Unambiguous, affirmative	Art. 4(11)	Explicit call per stream; no defaults, no pre-ticked state	[yes] Meets
Withdrawable at any time	Art. 7(3)	`revoke()` sets `revoked_at`; `Study.revoke(wipe=True)` erases	[yes] Meets
As easy to withdraw as to give	Art. 7(3)	Granting is a guided CLI flow (`shadow subject consent alex`); withdrawal is a single command with no equivalent confirmation or receipt	[!] **Gap.** Not a violation on its face — one command is easy — but there is no confirmation to the subject that withdrawal took effect, and no receipt
Demonstrable	Art. 7(1)	`granted_at`, `revoked_at`, `purpose`, `streams` are on the grant; **there is no append-only consent log** at the store layer, so a revoked-then-regranted history is not reconstructible	[no] **Gap.** Art. 7(1) requires the controller to *demonstrate* consent was given. A mutable grant object cannot demonstrate what was true last March

Two further observations:

- **The 90-day review horizon (`DEFAULT_REVIEW_DAYS`) is a genuine strength** and goes beyond what GDPR requires. Consent that lapses unless re-affirmed is the operational meaning of "dynamic".
- `reaffirm()` **refuses to resurrect a revoked grant** — it raises and demands a new one. That is the correct semantics and it is rarely implemented.

Two fixes, both small

1. **Append-only consent ledger.** Every grant, stream addition, re-affirmation and revocation written as an immutable event with a timestamp. Satisfies Art. 7(1), and it is the same append-only pattern the belief ledger already uses.
2. **Withdrawal receipt.** On revocation, emit a record the subject can keep: what was held, what was erased, when. Closes the Art. 7(3) parity gap and costs nothing.

Neither is implemented this session; both are specified for Session 12 rather than rushed in alongside a regulatory analysis.

GDPR Article 9

Self-reported emotion and physiological arousal can both constitute **data concerning health** under Art. 9(1) — mental state and cardiac signals respectively. The lawful basis available is explicit consent, Art. 9(2)(a), which the per-stream gate provides. **The monograph does not currently name Art. 9.** It should, because "special category data, processed on explicit consent, per stream, revocable" is a much stronger sentence than "we take consent seriously".

---

4. The Prohibited-Use Checklist, rewritten to be enforceable

Your original checklist (p. 294) is a good list of the right prohibitions. Its weakness is that every line is an instruction to a reader — *"answer no to every prohibited purpose"* — and nothing in the system can tell whether the reader did.

Below, each prohibition is tagged with **how it is enforced**: `[CODE]` a mechanism refuses, `[TEST]` a test asserts the refusal, `[LICENCE]` a licence term binds, `[POLICY]` honour-system only and labelled as such. Anything left at `[POLICY]` is not a safeguard; it is a wish, and marking it honestly is the point.

4.1 Prohibited outright

#	Prohibition	Enforcement	Basis
P-1	Evaluation of a non-consenting third party	**[CODE]** `Study.observe()` raises `ConsentError` when no active grant covers the stream · **[TEST]** `test_consent_and_patterns.py`	GDPR Art. 6
P-2	Bundling non-obvious streams under a blanket yes	**[CODE]** `add_stream()` raises unless `acknowledged_non_obvious=True` per stream · **[TEST]**	GDPR Art. 7(4)
P-3	Inferring a hidden preference without the evidence to exclude a threat response	**[CODE]** guard coverage gate; `latent_approach` unreachable without both channels · **[TEST]** `test_d19_*` (5 tests)	Project ethics; measured harm
P-4	Emotion inference in a workplace or education institution	**[POLICY]** + **[LICENCE]** — *no code mechanism exists*	**AI Act Art. 5(1)(f)**
P-5	Clinical diagnosis, treatment selection, crisis assessment	**[POLICY]** + **[LICENCE]**	MDR; project ethics
P-6	Hiring, firing, promotion, compensation, worker surveillance	**[POLICY]** + **[LICENCE]**	AI Act Annex III(4)
P-7	Policing, criminal-risk, deception, dangerousness or morality assessment	**[POLICY]** + **[LICENCE]**	AI Act Art. 5(1)(d)
P-8	Credit, insurance, housing, benefits, eligibility	**[POLICY]** + **[LICENCE]**	AI Act Annex III(5)

#	Prohibition	Enforcement	Basis
P-9	Political persuasion or exploitation of vulnerability	**[POLICY]** + **[LICENCE]**	**AI Act Art. 5(1)(b)**
P-10	Inference of origin, upbringing, or protected traits	**[POLICY]** — the model has no such hypothesis, but nothing *prevents* one being added	AI Act Art. 5(1)(g); GDPR Art. 9
P-11	A global maturity score or opaque peer ranking	**[CODE]** the output is a distribution and a credible set; no scalar score exists anywhere in the API · **[TEST]** `test_operator_G_multiple_hypotheses_coexist`	Project ethics
P-12	Covert operation	**[CODE]** no grant, no ingestion — refusal is the default path · **[TEST]**	GDPR Art. 5(1)(a)

Five of twelve are enforced in code and tested. Seven are policy only. That ratio is the honest headline, and it is a considerably better one than most systems could produce — but a paper should print the ratio, not the list.

4.2 The two enforcement gaps worth closing

- **P-4** is the only prohibition that is currently *illegal to breach in the EU* and has **no code mechanism**. A minimal enforcement point exists: require a deployment context declaration at session start (`self` / `consented_individual` / `institutional`), and have the engine refuse to enable the `physiological` stream when the context is `institutional`. That is roughly fifteen lines and converts P-4 from `[POLICY]` to `[CODE]`. **Specified for Session 12.**
- **P-10** could be enforced by an allow-list of hypothesis kinds, refusing any hypothesis whose declared kind is not in the permitted set. Also Session 12.

4.3 Required documentation, unchanged

Your original closing paragraph — document the self-directed purpose, lower-burden alternatives, consent path, withdrawal semantics, data map, construct argument, validation evidence, incident process, and sunset condition — is retained verbatim. It maps almost one-to-one onto the AI Act's Art. 11 technical documentation requirements, which is a better position than the monograph currently claims for itself.

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5. What this changes in the programme

Change	Effect
Education pilot **withdrawn**	Session 15 reframes to individual adults, self-mirror, recruited outside institutions
Literature-synthesis pilot **promoted**	Becomes the Phase 3 deployment pilot
P2 paper gains a section	"Regulatory position" — Art. 3(34)/(39), Art. 5(1)(f), the safety-exception argument, and the honest note that it is untested. This is a *strength* at AIES/FAccT, not a liability
Two new defects	**D-31** no append-only consent ledger (Art. 7(1)); **D-32** no deployment-context gate for P-4
Monograph correction	Part 6 should cite AI Act Art. 3(34), 3(39), 5(1)(f), Annex III(3)–(4) and GDPR Art. 9 by number. It currently cites the instruments but not the provisions

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6. One thing worth saying plainly

The awkward finding of this session is that **the physiological stream — which exists only to protect people, and which Session 2 proved does protect them — is the thing that pulls the system into the AI Act's emotion-recognition prohibition.** Remove it and the legal exposure largely goes away; remove it and the misreading rate goes from 0.0% to 31.2%.

That is a real tension between a safety mechanism and a rule written to prevent a different harm, and it is exactly the sort of finding an AIES or FAccT audience exists to discuss. It should be a section of the paper, not a footnote.

Session 10 — drafting P2, the ethics-of-design paper, for AIES/FAccT — continues automatically.

Sources: AI Act Art. 5 · Art. 3 · Annex III · Recital 18 · Implementation timeline

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Shadow Mirror — Session 12: Camera-Ready Assets and the Defect Sweep

Author of record: Omar Moawad · **Session date:** 25 July 2026 **Deliverables:** six publication figures (PDF + PNG), `shadow_mirror` **v1.1.0** (**324 passing**, 2 xfailed), this report.

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0. Summary

Four defects closed, six figures produced, one regression found and guarded against during the fix.

Defect	Before	After
D-32 No mechanism behind the AI Act Art. 5(1)(f) prohibition	policy only	`DeploymentContext` **refuses the biometric stream** in institutional and undeclared contexts; 6 tests
D-31 Consent could not be demonstrated at a past date (GDPR Art. 7(1))	mutable grant	**hash-chained append-only ledger**, tamper-evident, plus a withdrawal receipt (Art. 7(3)); 6 tests
D-28 Observation intake quadratic	3.8× per doubling, 1121 ms at 800 pairs	**2.0× per doubling, 5.2 ms** — 214× faster
D-26 Offset estimator stuck at severe displacement	61.3% at δ=0.45	**97.0%** — and a new guard where the estimator was actively harmful
D-17 Literal n in figure labels	present	gone; all six figures regenerated from source

Enforcement ratio: 5 of 12 prohibitions machine-enforced ☐ 6 of 12. The one that moved is the only one currently unlawful to breach in the EU.

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1. D-32 — the prohibition now has a mechanism

Session 9 found that emotion inference in a workplace or education institution is prohibited under Art. 5(1)(f), applicable since 2 February 2025, and that Shadow Mirror had **no code path** that could refuse it.

`deployment.py` supplies one. A `DeploymentContext` declares where the instance is running — `self`, `consented_individual`, `institutional`, or `undeclared` — and gates the streams that supply the biometric basis for affective inference.

Three design decisions are load-bearing:

Refusal is the default. `undeclared` is treated as institutional. An operator who has not said where the system is running has not established that the prohibition does not apply, and the safe reading of silence is the restrictive one.

The safety exception requires a written basis, not a flag. `safety_exception_basis` is free text and must be non-empty to have any effect. A boolean can be set by accident or by a config default; a sentence has to be written by someone. The stated basis is recorded and exported with the consent ledger, so the claim is auditable rather than implicit.

The refusal explains itself. The exception message carries the article numbers, the applicability date, the declared context, and the exact call that would claim the exception — because a refusal a developer cannot act on becomes a refusal a developer removes.

```
DeploymentRefusal: Stream 'physiological' supplies the biometric basis for
inferring an affective state, which makes this an emotion recognition system
under EU AI Act Art. 3(39). Art. 5(1)(f) prohibits emotion inference in workplace
and education institutions (applicable since 2 February 2025)...
```

A Python module does not make anyone compliant, and the module says so in its own output (*"not legal advice and does not establish compliance"*). What it does is convert a prohibition from a wish into a mechanism, which is the difference the Session 9 enforcement ratio was measuring.

---

2. D-31 — consent that can be demonstrated

GDPR Art. 7(1) requires a controller to *demonstrate* that consent was given. A mutable grant object cannot: a stream added and later removed leaves no trace, and a grant revoked and re-issued looks identical to one that never lapsed.

`ledger.py` records every consent event — granted, stream added, re-affirmed, lapsed, revoked, erased, context declared, refused — as an immutable, hash-chained entry. Three properties follow:

- `streams_granted_at(subject, when)` answers the Art. 7(1) question directly: *which streams were under active grant at this instant?*
- `verify()` re-derives the chain. Editing any historical entry breaks it. This is tamper *evidence*, not tamper proofing — the subject owns their own file and that is the point — but a controller claiming consent on a date can show a chain that verifies, and an edited one will not.
- **Erasure retains the record, not the data.** The ledger records *that* 42 observations were erased. That record is the evidence the withdrawal was honoured.

The withdrawal receipt closes the Art. 7(3) parity gap. Granting is a guided flow ending in a summary; revocation was a single call ending in silence. Those are not equally easy experiences however few keystrokes each takes. `withdrawal_receipt()` returns what was granted, when it was withdrawn, how many records were erased, and whether the chain still verifies — something the subject keeps.

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3. D-28 — the write path is linear

Session 7 measured `add()` at 3.8× per doubling: every observation recomputed every belief.

Beliefs are now computed when the mirror is **consulted**, not when an observation is written.

Evidence pairs	Before	After
100	27 ms	0.74 ms
400	296 ms	2.75 ms
800	1121 ms	**5.24 ms**

Evidence pairs	Before	After
Scaling per doubling	3.8×	**2.0×**

A **214× improvement at 800 pairs**, and linear.

Two honest notes. First, a semantic change: the reactivity counter now counts consultations rather than writes — which is arguably what a reactivity measure should have counted from the start, since it exists to warn that *looking at a guess changes later answers*. `n_reflections()` is kept as an explicit counter so the caution behaves identically. Second, the asymptotics of a **read-heavy** pattern are unchanged: one consultation per twenty writes still scales at 3.1× per doubling, because each consultation is O(n). The constant improved twentyfold; the shape did not. Making reads incremental as well requires anchored decay accumulators, and is not attempted here.

One existing test was updated: `test_d11_history_is_keyed_by_topic` now consults each topic before inspecting the history. The property under test — history keyed by topic rather than pooled — is unchanged, and the docstring records why the test moved.

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4. D-26 — soft assignment, and a regression caught in the act

Session 5's hard-assignment EM converged to a poor local optimum at severe displacement, because its first assignment was mostly wrong. Soft assignment with likelihood-selected restarts fixes it:

δ	Topics	Uncorrected	Session 5 (hard EM)	v1.1 (soft EM)
0.30	5	82.3%	90.8%	**93.4%**
0.30	8	83.8%	95.8%	**98.2%**
0.45	3	54.1%	55.8%	**76.2%**
0.45	5	53.6%	58.4%	**89.5%**
0.45	8	53.2%	61.3%	**97.0%**

At severe displacement with eight topics, **53.2% □ 97.0%**.

But the first version of this fix made things much worse where the data is thin. At one topic and δ = 0.30, correcting with the unguarded soft estimate drove accuracy from 81.2% to **20.8%**. Soft EM is happy to produce a confident estimate from one point, and the estimate is wrong.

The cause is identifiability, not tuning: a single (x, y) point can be moved onto any of four corners by some offset, so there are more unknowns than equations. The estimator now **refuses below three topics** and returns a zero offset. That is a condition, not a threshold — and the curve is worth looking at, because it is the clearest picture in the project of what "observing more" actually buys.

I am flagging this because it nearly shipped. A fix that improves the headline number by 36 points while quietly destroying the thin-data case is exactly the kind of change that passes a benchmark and fails a person.

---

5. Figures

Six figures, PDF (Type-42 embedded fonts, submission-ready) and PNG (review), all generated from `results.json` and the experiment outputs by `make_paper_figures.py`. Nothing is typed in.

P2 — ethics-of-design paper

1. `fig_p2_1_harm_rate` — the false hidden-desire rate across three conditions and three methods. The Shadow Mirror bar is invisible because it is zero; the label carries it.

2. `fig_p2_2_ablation` — imputing a missing channel as zero versus recording it as absent.
3. `fig_p2_3_ambivalence` — dispersion-gated versus structure-gated, before and after.

P1 — architecture paper

4. `fig_p1_1_inquiry_regret` — regret against the oracle, three planners, two conditions.
5. `fig_p1_2_credible_sets` — set coverage versus point accuracy, with mean set size annotated and the nominal 90% line drawn so the shortfall is visible rather than described.
6. `fig_p1_3_identifiability` — accuracy against topics pooled, with the identifiability boundary marked.

D-17 is closed. The original figure's node labels contained the literal characters `\n`. Labels are now constructed with real newlines, and the escape sequence appears nowhere in any output.

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6. Test suite

	Session 11	Session 12
Passing	312	**324**
Strict xfails (declared gaps)	2	2

Twelve new tests: six for the deployment gate, six for the consent ledger. Every one is written against the defect rather than the implementation.

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7. Remaining open items

Item	Status
Related-work expansion — 12–14 verified refs against 30–45 expected	Session 13
Read-path complexity still O(n) per consultation	Open; anchored-decay accumulators would fix it
D-23 residual — nominal coverage needs conformal calibration	Blocked on labelled outcomes (human study)
D-30 — positive autocorrelation ("your position moved steadily") computed and discarded	Proposal
Causal reflection, knowledge consolidation	Declared gaps, strict xfail
Affiliation, corresponding email, repository URL	**Blocked on you (Q2)**

Session 13 — adversarial self-review of both papers as a hostile Reviewer 2 — continues automatically. Session 14 stops for your decision.

Shadow Mirror — Session 13: Adversarial Self-Review

Author of record: Omar Moawad · **Session date:** 25 July 2026 **Deliverables:** this review, `experiment_adversarial.py` + `experiment_adversarial.json`, patched drafts of P1 and P2.

I reviewed both papers as an unsympathetic Reviewer 2 would: assuming the results are wrong until forced otherwise, and looking for the sentence that overreaches. Fourteen attacks. **Five conceded, four answered with new evidence, five fixed in the text.** The two that required new experiments changed the paper materially, and one of them found a result that weakens the headline.

The two attacks that required running something

A-1 — "Your simulator is your theory." Answered with a sweep. Partly conceded.

The attack. You defined `threat_compliance` to differ from `latent_approach` on arousal and avoidance. Your guard keys on arousal and avoidance. Your AUC is 1.000. You have not measured a harm rate; you have measured that a classifier separates classes you separated by construction.

The answer. Correct enough that prose would be evasive. I swept the separation between the two processes on the confound channels from full to zero, n = 300 per process per level:

Separation	AUC	Shadow Mirror	Four-channel classifier	Mismatch-only
1.00	1.000	0.0% [0.0, 0.0]	0.0% [0.0, 0.0]	94.3%
0.75	1.000	0.0% [0.0, 0.0]	0.0% [0.0, 0.0]	95.3%
0.50	1.000	2.3% [0.7, 4.0]	0.0% [0.0, 0.0]	96.7%
0.25	0.988	29.7% [24.7, 35.0]	5.0% [2.7, 7.7]	97.3%
0.00	0.491	76.7% [71.7, 81.3]	82.0% [77.7, 86.3]	96.3%

Verdict: partly conceded, and the paper is stronger for it. The result is not a knife-edge — it survives to roughly half the original separation. But it is entirely contingent on the confound channels carrying signal, and at zero separation every method fails, as it must. Added as §6.5.

A-2 — "Your baselines are straw men." Conceded. New baseline added, and it beats us.

The attack. Mismatch-only ignores channels that are available. Arousal-as-desire is a mistake, not a method. Neither is what a competent engineer would build.

The answer. They partly were straw men. I built the baseline the objection demands — a threshold classifier over the same four channels, fitted on a held-out labelled split — and it is in the table above.

It matches us at high separation and beats us at separation 0.25: 5.0% against our 29.7%.

Verdict: conceded, reported, and reframed. The reframing is not a dodge, and it is the most important sentence in the session: *the classifier was given ground-truth labels for "was this a threat response", and those labels do not exist in deployment.* Nobody can tell a running system which mechanism produced a person's behaviour — that is the entire difficulty. So the supervised baseline is an **upper bound on what any labelled method could achieve**, not an available alternative, and the gap at separation 0.25 is a quantified cost of having no supervision.

A reviewer who finds that unconvincing has a real objection, and I would rather they raise it against a number we published than discover it themselves.

The other twelve

#	Attack	Verdict
A-3	**"0.0% with CI [0.0, 0.0] is not an interval."** Bootstrapping an all-zero sample yields a degenerate interval that overstates precision.	**Conceded, fixed.** Now reported as *0 of 800 episodes; 95% upper bound 0.37% by the rule of three*. That number, not 0.0%, is what a deployment decision should use.
A-4	**"'Never been measured' is an argument from absence."**	**Conceded, fixed.** Softened to "so far as we have been able to establish, has not previously been quantified".

#	Attack	Verdict
A-5	**"You borrow clinical authority."** "Fawn", "compliance" and "threat response" carry clinical weight the paper has not earned.	**Conceded, fixed.** Added a limitation stating that the terminology names a pattern our generative process encodes, that no clinical construct is claimed, and a `[CITATION NEEDED — trauma-response typology]` marker. Do not remove that marker without a real citation.
A-6	**"You are not lawyers."** §7 is amateur legal analysis presented with confidence.	**Conceded, fixed.** Explicit limitation: it is a reading, not reviewed by counsel, not advice. §7 already framed the safety exception as an argument rather than a conclusion.
A-7	**"The guard is two thresholds. Where is the generalisation?"**	**Conceded.** It is two thresholds. The contribution is the *decision to refuse* and the measurement of what refusing buys, not the sophistication of the rule. The paper says so; A-1's sweep now bounds where it holds.
A-8	**"Executable invariants is just having tests."**	**Rebutted.** Tests assert behaviour; these are a decision procedure for named architectural properties, shipped with the architecture, with a negative test per property. The distinction is that a third party can run them against a *different* implementation — see A-9.
A-9	**"N = 1 implementation. Your invariants may be trivially satisfiable by your own API."**	**Rebutted with evidence; new §3.5.** The same procedure run unmodified against an independently written prototype passes 4 checks and **fails 3**. It separates two implementations of one architecture, which rules out vacuity. Limitation retained: two is not many, and none was written by a third party.
A-10	**"Invariant X is a regex. Authority is not a matter of pronouns."**	**Conceded, fixed.** New limitation: the check is a syntactic proxy, it produced a documented false positive on first contact with real output, and it will produce false negatives. It raises the floor; it does not measure the property.
A-11	**"200 seeds is not property-based testing; coverage is unmeasured."**	**Conceded, fixed.** Limitation now states what varies across seeds and that nothing measures whether that exercises the state space in any principled sense.
A-12	**"You say the architecture buys no accuracy. Why should TiiS care?"**	**Rebutted, text strengthened.** Accuracy was never the safety-relevant quantity: the companion harm result depends on the structural commitments — representing a competing hypothesis rather than filtering, and distinguishing absent from zero. Structure is what carries it. Added to §8.
A-13	**"Your uncertainty is worse where the model is wrong. That is a regression."**	**Rebutted.** Already reported in both papers with the numbers (ECE 0.171 ☐ 0.044 where applicable, 0.151 ☐ 0.221 where not). Reporting a regression you found yourself is not a weakness to hide.
A-14	**"Related work is thin — 12–14 references."**	**Conceded, outstanding.** Both papers carry a marked expansion note. This is real work and it is not done.

What changed in the drafts

P2 — abstract rewritten for the rule-of-three bound and the supervised comparison; novelty claim softened; **new §6.5** carrying the separation sweep and the four-channel baseline; four limitations added (point-estimate-of-zero, borrowed clinical vocabulary, not-lawyers, simulation caveat extended). 4,325 ☐ 4,950 words.

P1 — **new §3.5** on whether the checks discriminate between implementations; §8 answers the "so what" objection directly; three limitations rewritten (two-implementations, Invariant X as syntactic proxy, unmeasured harness coverage). 3,987 ☐ 4,360 words.

Rebuttal hardening — for when reviews arrive

If a reviewer says the simulation is circular: point to §6.5. The sweep is already there, it already shows the result degrading, and it already shows every method failing at zero separation. The concession is in the paper. What you should *not* do is defend the 0.0% as a general fact; defend it as a statement about identifiability given informative channels.

If a reviewer says the supervised baseline beats you: agree immediately — the paper says so — and move to the labels. The classifier needs ground truth for "was this a threat response". Ask the reviewer where that comes from in deployment. If they have an answer, it is a genuinely important one and worth a follow-up paper.

If a reviewer says "this needs human subjects": agree. Both papers say so in the abstract, the limitations, and the ethics statement, with `[NO DATA YET]` markers. The claim is about a system's behaviour, not about people. Do not be tempted to gesture at plausibility.

If a reviewer challenges the regulatory reading: concede the framing (it is a reading, not advice) and hold the substance. The chain is textual — Art. 3(34) omits the unique-identification requirement that GDPR Art. 4(14) contains; Art. 3(39) is defined over biometric data; Art. 5(1)(f) is in force. If they disagree about the safety exception, that is exactly the discussion the section exists to provoke.

If a reviewer says the invariants are trivial: §3.5. Four pass, three fail, on an independently written implementation.

If a reviewer asks for a scalability claim: do not make one. The papers report a quadratic write path fixed to linear, a read path still linear per consultation, and thousands of nodes rather than millions. Say the architecture makes no validated scalability claim, because it doesn't.

The general rule. Every number in both papers traces to a named script, and the suite is byte-reproducible. If a reviewer disputes a figure, the reply is a command they can run — which is a much stronger position than most rebuttals start from.

Honest assessment of where these papers stand

P2 is the stronger submission. It has a measured harm rate, an adversarial robustness check that partly undermines its own headline, a design commitment that produces the result, an unresolved regulatory tension, and no claims about people. That is a coherent AIES/FAccT paper.

P1 is solid but narrower. Its contribution — a decision procedure for architectural invariants that discriminates between implementations — is real, and the negative result is honestly placed. Its weakness is that it is one architecture checked by its own author, and no amount of writing fixes that. It needs a third party to run the suite.

Both need the related-work expansion, and both need your affiliation.

The single largest risk to either paper is no longer a defect. It is that the reference list is too thin for the venue, and that nobody outside this project has run the code.

Programme status

Sessions 1–13 complete. Defect register: **31 identified, 24 resolved, 2 open with named blockers, 5 superseded.** `shadow_mirror` v1.1.0, 324 tests passing, 2 declared gaps. Both papers drafted, figured, and adversarially reviewed.

Sessions 14 and 15 cannot proceed without you. They are blocked on Q2, and I will not invent an affiliation or name an ethics committee.

Programme paused at the decision point. See the accompanying summary.

Shadow Mirror — Session 14: Related Work, Affiliation, and a Term We Should Not Have Used

Author of record: Omar Moawad · **Affiliation:** The American University in Cairo · **Session date:** 25 July 2026 **Deliverables:** expanded P1 and P2, this report, `SESSION_14B_DATA_FEASIBILITY.md`, updated `bibliography.json`.

Session 13 named the thin reference list as the largest remaining risk to either paper, ahead of any open defect. This session closes it. It also found one substantive error in P2's claim structure and one ground-rule violation in both papers, neither of which I was looking for.

1. What was verified

76 candidate references were checked against live sources. 76 confirmed, 0 rejected, 0 fabricated.

Verification ran through five independent passes, each required to return a resolvable identifier and the URL it was confirmed from. The instruction to each pass was that a fabricated citation is worse than a missing one and that failure to confirm must be reported as failure, not filled in. Nothing came back unconfirmable, which is itself a mild surprise and suggests the candidate list was conservative.

Corrections caught during verification — every one of these would have been a visible error in a submitted manuscript:

Proposed	Actual
Roemmich et al., "Values in Emotion AI Hiring **Technologies**: Technosolutionism, Self-Improvement, and Secret Knowledge"	"…Hiring **Services**: **Technosolutions to Organizational Problems**", and a fourth author (S. Fan) was missing
Passos et al., IEEE Software 2010, DOI `MS.2010.x`	DOI is `10.1109/MS.2009.117`; "et al." expands to five named authors
Terra & Valente, "…object-oriented software **architecture**"	published title is plural, "**architectures**"
Oaksford & Chater, "selection task**:** Optimal data selection"	published title is "selection task **as** optimal data selection"
Solove, "Privacy Self-Management and the Consent Dilemma"	published title carries the prefix "**Introduction:**"
Naeini — first author surname "Naeini"	surname is "**Pakdaman Naeini**"

Two entries carry deliberate flags rather than corrections. Wachter, Mittelstadt & Russell (2018) has **no publisher DOI** — the Oxford ORA record is the stable identifier, and anyone reformatting the bibliography should not invent one. Three of the legal-scholarship entries (Nissenbaum, Solove, Wachter & Mittelstadt) are **student-edited law reviews, not externally peer reviewed**; that is normal for the genre and worth knowing when weighing them against the empirical citations.

2. The reference expansion

	Before	After	Venue norm
P1 (TiiS)	12	**55**	30–45
P2 (AIES/FAccT)	14	**50**	25–40

Both are now above the norm rather than below it, which is the right side to err on for a journal submission and harmless for a conference one.

The expansion is not a reference dump. §2 of each paper was rewritten into subsections that each position the contribution against a named line of work, and in three places the new literature *constrains* a claim rather than supporting it:

P1 §2.2 now concedes explicitly that the invariants "borrow the form of each of these and match none exactly" — not contracts, not model checking, closest to property-based testing in the QuickCheck tradition. The claim is narrowed to a difference of *subject matter* rather than technique.

P1 §2.3 admits that the provenance literature "had the right answer three decades before we made the mistake". The W3C PROV model and Buneman's why/where distinction both treat derivation as multi-parent; our D-14 defect was writing it as a scalar. That is now presented as the argument for mechanical checking rather than as a discovery.

P1 §2.5 adds Nelson's finding that no single measure of question-usefulness dominates, and states that we report normalised regret against an oracle precisely so a reader preferring a different measure can separate the measure from the planning.

P2 §2.1 is where the new literature does the most work. Fifteen references on selective prediction, deferral, calibration and conformal prediction now stand behind a sharpened distinction: every one of those mechanisms abstains because a *confidence* is low and is tunable by a threshold; our guard abstains because *a specific alternative explanation cannot be ruled out with the evidence collected*, is not tunable, and fires just as readily on a high-confidence reading. "A selective classifier asks *how sure am I?*; the guard asks *which question was never asked?*" That sentence is the paper's design claim, and it could not have been written without knowing what the abstention literature actually does.

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3. The term we should not have used

Session 13 flagged that P2 borrows clinical authority and left a `[CITATION NEEDED — trauma-response typology]` marker with the instruction not to remove it without a real citation. I went looking for that citation. **It does not exist.**

The word **"fawn"** — which P2 used in its introduction, in the §3 mechanism table, and in a limitation — traces to Pete Walker's *Complex PTSD: From Surviving to Thriving* (2013). That book is real, and it is the popular origin of the term. It is also **self-published**: Azure Coyote Publishing is the author's own imprint, issued through Amazon's CreateSpace platform, with no editorial peer review. No refereed source establishes "fawn" as a validated clinical construct.

There *is* a genuine peer-reviewed literature in the same region, and it is good:

- **Cantor & Price (2007)**, ANZJP — appeasement under traumatic entrapment, spanning hostage reactions and domestic abuse. This is the closest real antecedent to what we mean.
- **Marx et al. (2008)**, *Clinical Psychology: Science and Practice* — tonic immobility as an evolved predator defence, and how survivors' non-resistance is misread.
- **Bracha (2004)**, *CNS Spectrums* — the acute stress response as a spectrum rather than the textbook dichotomy.
- **Gilbert (2001)** — submissive behaviour within social competition and hierarchy.

What changed in the paper. "Fawn" has been removed from every load-bearing claim. The introduction now says the failure mode "has an established literature in clinical and evolutionary psychology [34–37] and no name at all in machine learning" — which is true, where the previous sentence was not quite. The §3 mechanism table row reads "complies under threat". The limitation was rewritten from a hedge into a concession that names the provenance problem directly and ends: *"A reader who suspects we have borrowed*

more authority than we have earned is raising the right objection; this paragraph is our answer, and it concedes most of it." Walker remains in the bibliography as [38], with a bolded note in the entry itself stating that it is a self-published trade book and not peer-reviewed literature, cited only as the term's origin.

This is the second time in this programme that going to look for a supporting citation has weakened a claim rather than strengthened it. That is the process working.

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4. A ground-rule violation, found by re-running the suite

Ground rule 7: *no number enters a manuscript unless a file in the repository produces it AND the person writing the sentence has run that file.*

Both drafts stated `312 passed, 2 xfailed`, and P1's abstract and Appendix B stated a suite of **314 tests**. Running `pytest -q` against the shipped v1.1.0 package returns:

```
324 passed, 2 xfailed in 1.22s
```

The figures were correct when written and went stale when Session 12 added the governance tests. No reviewer would have caught it, and it would have been a discrepancy between a paper and its own artefact. Corrected in the abstract, §3.6 and Appendix B of P1 and in Appendix A of P2.

Logged as D-33 — stale derived figures in manuscripts. The register entry records the general lesson rather than the specific number: any figure computed from the repository must be regenerated, not carried forward, whenever the repository changes. A regeneration script is the obvious fix and is not yet written.

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5. Affiliation, and what it unblocked

Q2 is answered. Both papers now carry:

> ***Omar Moawad*** *The American University in Cairo, AUC Avenue, P.O. Box 74, New Cairo 11835, Egypt*

The institutional name and postal address are taken verbatim from AUC's own Editorial Style Guide — "The" capitalised, "in Cairo" not "of Cairo". The assembly into a single affiliation line is conventional rather than AUC-prescribed; AUC publishes no mandated publication format. **The corresponding email remains blank** because I will not guess at it, and a department name should be added if you hold one.

Governance research on AUC and on Egyptian law produced two findings substantial enough to change P2 rather than merely inform a submission packet, and they are set out in `SESSION_14B_DATA_FEASIBILITY.md` §3. In summary:

- **Egypt's Law 214/2020 does not reach this work.** It governs clinical medical research — therapeutic, pharmaceutical, surgical, nutritional, preventive or diagnostic interventions — and confines lawful research to entities registered with the Ministry of Health. Behavioural and computing research falls outside it. There is no binding Egyptian national law governing non-medical human-subjects research ethics.
- **Egypt's PDPL 151/2020 does reach it, and contains no scientific-research exemption at all.** The GDPR's research accommodations at Arts. 5(1)(b), 9(2)(j) and 89 have no Egyptian counterpart. The words for "research", "studies" and "university" do not appear in either the law or its executive regulations. Sensitive-data processing — which expressly includes psychological, health and biometric data — additionally requires a **licence** from the Personal Data Protection Centre, on penalty of imprisonment and a fine up to EGP 5,000,000.

This became P2 §7.1, a new subsection arguing something the fairness literature has not much said: a researcher in a jurisdiction with a *weaker* national human-subjects law can face a *stricter* data-protection constraint on exactly the physiological channels this architecture consumes, with no research carve-out to fall

back on. It is framed as a reading of statutory text, not advice, and it repeats the §9 limitation that no counsel has reviewed it. I think it is the most novel paragraph in the paper, and it exists only because you named your institution.

---

6. Verification performed

- **Citation integrity, machine-checked.** Every `[n]` in the body of each paper resolves to a defined entry, and every defined entry is cited at least once. Both papers: 0 dangling, 0 orphaned. The first run of this check found two orphans in P1 and four in P2; all six were fixed by citing them where the corresponding method is actually described, not by deleting them.
- **Test suite re-run** against the shipped package: 324 passed, 2 xfailed.
- **No number was changed except to match what the repository currently produces.**

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7. Where the programme stands

P1 (TiiS)	5,137 words, **55 references**, affiliation set
P2 (AIES/FAccT)	5,987 words, **50 references**, affiliation set, new §7.1
Defect register	33 identified, 24 resolved, D-23 partially mitigated, **D-33 new**
Code	v1.1.0, 324 passed / 2 xfailed, unchanged this session
Blocking questions	Q2 **answered**. Q4 superseded — see `SESSION_14B`

Remaining before submission: the corresponding email; a department name if you have one; the cover letters and author statements (formerly Session 14, now Session 15); and the decision recorded in `SESSION_14B` about what Phase IV can honestly be.

The single largest risk to either paper is no longer the reference list. It is that nobody outside this project has run the code.

Shadow Mirror — Session 14B: Can Session 18 Run on Data That Already Exists?

Author of record: Omar Moawad · **Session date:** 25 July 2026 **Question put to this session:** *"for 18's real participants, research the internet for data that is ready and you can make use of."*

I searched. The answer has three parts, and the first one is a no.

---

1. The headline finding

No public dataset can validate the threat-confound claim. Not one.

The claim needs four things measured *on the same item, for the same person*:

1. a stated stance ("I do not want this"),
2. a behavioural choice that can contradict it (they approach anyway),
3. an arousal channel time-locked to that choice,
4. a coercion or threat manipulation, so that compliance and hidden preference are separable.

That intersection does not exist publicly. Roughly forty datasets were assessed across affective computing, approach-avoidance paradigms, ecological momentary assessment, experimental economics, and social conformity. The failure modes are clean and complementary:

- **Datasets with excellent physiology are passive-viewing paradigms.** DEAP, AMIGOS, ASCERTAIN, DREAMER, CASE, POPANE, Emognition — participants watch a clip and rate it. There is no action that can contradict the rating. A "liking" score is not a revealed choice.
- **Datasets with real behavioural choice lack autonomic recording, self-report, or both.** FOODEEG has an item-matched contradictable choice and no arousal channel. PsPM-PIT has threat, instrumental choice and superb psychophysiology, and zero self-report — there is no stated stance to contradict. SEMAINE has genuine withdrawal behaviour and neither self-report nor physiology.
- **The EMA and mobile-sensing corpora fail as a class.** StudentLife, GLOBEM, Tesserae, LifeSnaps, DAPPER, TILES, PMData, ExtraSensory: every "behavioural" stream is passive telemetry — GPS, steps, screen unlocks, app usage — and none of it is labelled with the topic the self-report was about. The pairing our model requires simply is not in the data.

The nearest miss is **Klaassen et al.'s passive-avoidance conflict task** — threat-of-shock, ~150 repeated approach-avoid choices, skin conductance *and* freezing. It is the only dataset found combining an explicit threat manipulation, repeated approach-avoid choices, and autonomic arousal. It fails on the stated stance (there is none) and on access (restricted).

Two datasets deserve a note for how close the design gets and how far the deposit falls short. Sowden et al. (2018) ran precisely the paradigm we need — a private continuous rating and a public spoken response on the same item, crossed with majority size — so that the private effect indexes *acceptance* and public-minus-private indexes *compliance*, within subject. The data were never deposited. Brouwer et al. (2021) paired a per-item "wanting" rating with a mobile approach-avoidance task across four food categories, N=71 — our exact structure — with no data availability statement. For both, emailing the authors is a better route than any download.

---

2. What *can* be done with existing data

This is not a dead end; it is a smaller, honest study. Three partial validations are available now, and together they cover four of the paper's claims without a single new participant.

Tier 1 — worth doing

BIRAFFE2 — the best-provisioned public dataset in the review. Open download, CC BY 4.0, **N = 102**, 117 within-subject stimuli, per-stimulus valence *and* arousal, per-stimulus response latency, an explicit abstention marker in the logs, 1 kHz ECG + EDA, and three real gamepad games producing instrumental behaviour. It supports partial validation of **calibration**, **abstention** and **order-invariance**. It does *not* support the preference-contradiction core, because the game behaviour is not item-matched to the stimulus ratings.

FOODEEG — CC0, **N = 117**. Session 1 collects per-food ratings; Session 2 has participants choose between food pairs. Because the rating precedes the choice on the same items, a choice can genuinely contradict a stated preference. It passes five of the six requirements and fails only on arousal. It is the correct **paradigm skeleton** for a future primary study, and supports **calibration** and **abstention** now.

Urban Appraisal — CC0, **N = 63**, 504 trials, per-item valence and arousal time-locked to EDA and ECG. It is the mirror image of FOODEEG: correct physiology, no choice. Supports **calibration**.

Tier 2 — one hand-check that could change the picture

SWELL-KW. The design is right: participants **choose which topic to present on**, which is an item-matched, preference-contradicting choice, and the study includes a genuine time-pressure and interruption manipulation. The open question is whether the topic-interest ratings and the chosen topics are actually in the deposit — they are not listed in the documented variables. If they are present, SWELL-KW meets R1–R6 minus threat at N = 25. If they are not, it collapses to another stress corpus. **This is a single afternoon's work with a real chance of changing what Phase IV is**, and I recommend it before anything else.

Cascio, "Self-construal and Conformity" (openICPSR, DOI 10.3886/E184271V1) is the only downloadable resource pairing a *private* rating with a *public* action on the same item, repeated 60–80 times, under experimentally manipulated peer pressure. Three cautions, all real: the N is not stated anywhere I could verify, the licence is unspecified, and the 6.5 KB behavioural file is small enough that it is plausibly participant-level summary rather than trial-level — which would eliminate the per-item pairing entirely. **Verify the file contents before building anything on it.**

What this does not buy

It does not buy the safety result. The 0-of-800 misread figure, the ablation, and the separation sweep are all claims about how the system behaves when a threat process is present in the data-generating process. No public corpus contains that process alongside a stated stance. **Reporting a secondary analysis as though it validated the threat-confound claim would be exactly the fabrication ground rule 1 forbids**, and I will not do it.

---

3. Governance, if a primary study is ever run

Answered while researching your affiliation. Every claim below has a source; the full source list is in the session log.

AUC's IRB

- Body: **Institutional Review Board for the Protection of Human Subjects (IRB)**, members appointed by the Provost, seven members including one external community member.
- **All research collecting data from human participants requires approval before it begins**, including research conducted outside Egypt. Course assignments with no intent to publish are exempt.
- **CITI/NIH human-subjects training is mandatory** and has been since 1 January 2013. For this work the correct course is *Social and Behavioral Research, Stage 1, Basic*.
- Three review tracks exist. Our study would most likely be **expedited** — the category explicitly covers "behaviour/cognition studies without subject manipulation or stress" — but a threat or social-pressure manipulation may well push it to **full board**, and I would submit expecting that.
- Reviews sit on Thursdays; allow one month.
- **Graduate students may be PI but must name a supervisor. Undergraduates may not be PI.**
- [!] The IRB policy PDF and the live web page **name different chairs**, and the PDF's described paper-submission process has been superseded by an online portal. Confirm current contacts by emailing `irb@aucegypt.edu` before preparing anything.

The gate that is easy to miss

AUC's own IRB page warns that IRB approval **does not** imply approval by **CAPMAS**, the Egyptian agency that authorises survey and interview work conducted inside Egypt, and states that the IRB "is not equipped to handle matters related to CAPMAS". This is a separate national permit with its own timeline, and it should be started early rather than discovered late.

The legal position, which is the interesting part

Law 214/2020 (Clinical Medical Research) does not reach us. Its Article 1 subject matter is therapeutic, pharmaceutical, surgical, nutritional, preventive or diagnostic interventions; Article 24 confines lawful research to entities registered with the Ministry of Health. A computing lab is not such an entity. There is **no binding Egyptian national law governing non-medical human-subjects research ethics** — a gap noted in the peer-reviewed research-ethics literature. The Supreme Council of Universities issued a national research-ethics guide in January 2026 whose third book addresses behavioural and social research committees, but it carries no decree number and no binding-force clause, and its applicability to private universities is unspecified. Treat it as guidance and ask the IRB about it.

PDPL 151/2020 does reach us, and is stricter than the GDPR in three respects. Its personal-data definition expressly covers data determining "psychological identity". Its sensitive category covers "psychological, mental, or physical health, or genetic, biometric or financial data", and Article 12 requires a **licence from the Personal Data Protection Centre** on top of written explicit consent, with imprisonment and fines to EGP 5,000,000 for breach. And it contains **no scientific-research exemption of any kind** — the words for "research", "studies" and "university" appear nowhere in the law or its executive regulations. Article 5(6) relaxes purpose limitation for not-for-profit statistical or educational processing but does not disapply consent or the Article 12 licence.

The executive regulations landed on 2 November 2025 with a compliance deadline of 31 October 2026. That is roughly three months away. If a study touching physiological, biometric or psychological data is contemplated, the DPO (`DPO@aucegypt.edu`) is the right first contact, not the IRB.

Design consequence. Our existing refusal of the biometric stream is not merely prudent under this regime; it is the difference between needing a state licence and not. And anonymisation, not pseudonymisation, is the only route out of scope — a linkable participant identifier keeps the data personal.

This is a reading of statutory text by a non-lawyer and has not been reviewed by counsel. It is now stated as such in P2 §7.1 and in the §9 limitations.

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4. What I recommend, and what I need from you

Phase IV as originally scoped — a primary study producing the data for P3 — remains impossible without collecting it. No amount of searching changes that, and I would rather say so than dress a secondary analysis in its clothes.

But the choice is no longer binary. Three options, in the order I would rank them:

A. Secondary-analysis paper (no participants, ~2 sessions). Run the architecture against BIRAFFE2, FOODEEG and Urban Appraisal. Validate calibration, abstention and order-invariance on real human data for the first time. Report plainly that the threat-confound claim is untouched. This is a real contribution — it converts "we have never run this on a human" into "we have run this on 282 people across three corpora" — and it is honest about what it does not show.

B. Hand-check SWELL-KW first (~1 session), then decide. If the topic-interest ratings are in the deposit, option A gains a genuine preference-contradiction result at N = 25 and gets substantially stronger. Low cost, real upside.

C. Primary study. Protocol, AUC IRB, CITI certification, CAPMAS permit, PDPL analysis, and — on the evidence above — the FOODEEG rating-then-choice structure crossed with a social-pressure manipulation in the Sowden et al. mould. This is the only route to the threat-confound claim. It is months of calendar time before a single participant is recruited, most of it waiting on approvals.

My recommendation is B then A, with C prepared in parallel as a protocol rather than executed — because a written, IRB-ready protocol is itself a publishable artefact and it is the thing P2's reviewers will ask

whether you have.

What I need from you: which of A, B or C, and whether you want me to draft the IRB packet in the meantime. I can do everything in A and B without you. C needs your CITI certification and a supervisor or department name, neither of which I can supply.

Shadow Mirror — Session 15: The Formation Model, and the Four Things It Will Not Say

Author of record: Omar Moawad · **The American University in Cairo** · **Session date:** 25 July 2026
Deliverables: `shadow_mirror/formation.py`, Invariant XI, `tests/test_v12_formation.py` (47 tests), `shadow_mirror` v1.2.0, this report.

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1. The request, and why it splits four ways

The request was to embed a feature that would *"determine and understand who people probably are according to their will, desire and need — where they come from, how mature they are, what they still need to be mature enough, what their eventual purpose probably is"*, with the explicit caveat *"I would not say judge people"*.

That caveat is the right instinct, and it turns out to be load-bearing. The request is not one feature but five, and they have different answers:

Ask	Answer	Why
will, desire, need	**Built.** Instruments the person completes; profiles with abstention.	Validated self-report exists for all three.
where they come from	**Refused unless declared.**	Inferring origin is inferring special-category data.
how mature they are	**Refused as a rank; offered as a profile.**	The rank *is* the construct in every stage instrument.
what they still need	**Refused; replaced by questions.**	It is a directive about identity — Invariant X.
eventual purpose	**Refused.**	No validated construct exists. Any output would be invented.

The refusals are not missing features. They are the feature, and they are what makes the rest of it safe to build.

---

2. The finding that decided the design

I put eight developmental instruments through verification. The question I asked was narrow: *is there an honest way to represent developmental maturity that does not collapse a person into a single rank?*

The answer came back sharper than expected:

> ***You cannot keep the construct and discard the rank.*** *Loevinger's Washington University Sentence Completion Test produces a Total Protocol Rating. Kegan's Subject-Object Interview produces one position on one ordered scale, delivered with a written defence of that single score. Cook-Greuter's MAP reports a "Center of Gravity (current stage)". These are not incidental summaries of richer data — the ordered position is the theory's claim. Reporting a profile from a stage instrument would misrepresent the instrument.*

So the stage construct is dropped, not softened. What replaces it is **dimensional profiles from instruments that never claimed a stage in the first place**:

- **3D-WS** (Ardelt 2003) — three dimensions, and its published criterion for high wisdom is *conjunctive*: ≥4.00 on **each** dimension independently, which is a profile test rather than an aggregate. It is not a stage theory and claims no developmental sequence. That is precisely why it is admissible and Loevinger is not.
- **MEPSI** (Darling-Fisher & Leidy 1988) — eight psychosocial themes. It measures attributes *associated with* Erikson's stages — how much trust or generativity a person reports — and never claims stage attainment.

One idea was worth stealing from an instrument we refuse. **DIT-2 distinguishes "consolidated" from "transitional"** — it can say a person is *between* framings rather than forcing a point estimate. That is the most intellectually honest feature in the whole set, and it is the same move this architecture already makes when it holds a contested hypothesis open instead of resolving it.

---

3. What was built

`shadow_mirror/formation.py`

Two routes in, and no third. A person **declares** something, or they **self-complete an instrument**. Everything else raises.

```
m = FormationModel("omar", deployment=DeploymentContext(context=SELF), ledger=led)
m.declare("background", "moved countries at 12")
m.administer("MLQ", [6,7,7,5,6,4,6,7,2,7], population="Egypt")

m.profile()              # 8 dimensions across 2 instruments, no total
m.open_questions()       # 4 invitations
m.eventual_purpose()     # FormationRefusal: PURPOSE_ATTRIBUTION
```

Eleven named refusal grounds, each carrying an explanation the person can read rather than a silent absence:

`INFERRED_ORIGIN` · `GLOBAL_RANKING` · `PRESCRIPTIVE_DEFICIT` · `PURPOSE_ATTRIBUTION` · `UNLICENSED_INSTRUMENT` · `EXTERNAL_SCORING` · `RATER_REQUIRED` · `UNCONSENTED_INSTRUMENT` · `NOT_SELF_COMPLETED` · `NORMS_ABSENT` · `POPULATION_UNVALIDATED` · `DEPLOYMENT_CONTEXT`

The refusing methods **exist rather than being absent**, deliberately. `eventual_purpose()` is a real method that raises with a real explanation, because a caller who finds no such attribute learns nothing, and a caller who reads the refusal learns why.

Person overrides, always

You chose this and it is implemented literally. A correction from the person wins for display, immediately, without argument. The contradicted statement is **marked contested and retained** — not deleted, because deleting it would remove the person's ability to see what the system had thought and on what basis.

```
declared:  motivation = "seeking approval"   [contested_by: 2026-07-25T…]
override:  motivation = "curiosity"          ← displayed
```

The instrument registry, and the licence refusal

Ten instruments registered, including the four that exist **only so they can be refused by name**. Refusing something by name is more useful than not having it: `administer("MAP", …)` returns the reason, the $1,000-per-assessment cost and the two-year certification requirement, rather than a `KeyError`.

Two design decisions here are worth defending because they cost functionality:

No item text is bundled with the software. The registry carries scoring rules, citations, licence status and caveats. The researcher supplies the licensed instrument. This keeps the package distributable and makes

the licence status honest instead of decorative.

An unconfirmed licence is a refusal, not a warning. Verification found that most instruments in this area are author-held with *no published permission statement at all* — the Claremont Purpose Scale, MEPSI, 3D-WS and ASTI all fall in this category. Administering one on an assumption exposes a real researcher to a real risk. Only **MLQ** and **BPNSFS** carry quoted, published permission statements, and only those two are administrable by default. Anything else requires `confirm_licence(instrument_id, evidence)` where `evidence` must be **written text describing who granted permission, when, and in what form** — the same reasoning as `DeploymentContext.safety_exception_basis`. A boolean can be set by accident; a sentence has to be written on purpose.

PVQ-RR is registered and still cannot be administered, because the 19-value item mapping was not successfully read during verification — `scholarworks.gvsu.edu` returned HTTP 403. Its `subscales` dict is deliberately empty and the refusal message says *"cannot be scored without inventing the mapping"*. Guessing it would have put an unverified number into a person's result, which is ground rule 1.

Invariant XI — declared-only personhood

```
XI. No attribute about a person exists in a formation model unless that person
    declared it or self-completed the instrument it came from; and the model
    exposes no route to a global rank, a prescribed deficit, or an attributed
    purpose.
```

The decision procedure **calls each refusing method and requires a `FormationRefusal` with the matching ground**. Two failure modes are caught: a method that returns a value, and a method that has been deleted. The second matters — *absence is not refusal*, because a caller cannot read an explanation off a missing attribute.

Two negative tests prove the check is not vacuous. Subclass the model so `eventual_purpose()` returns `"to become a great engineer"` ☐ **XI fails**. Delete `maturity_level` entirely ☐ **XI fails, and for a different stated reason.**

---

4. Something the tests found that I had not designed

`test_repeated_administration_warns_against_reading_development_into_it` failed on first run. The cause was not the assertion — it was that I had timestamped a session **before** the consent grant, and `ConsentLedger.streams_granted_at` correctly answered that nothing was under grant at that instant.

The system had already refused retroactive consent, and I had not written that rule.

It falls out of asking *what was granted at time T* rather than *is there a grant*, which is the GDPR Art. 7(1) demonstrability query the ledger was built for. Backdating a session to before the person agreed is precisely the move Art. 7(1) exists to prevent, and the hash chain makes it undoable rather than merely discouraged.

I kept the finding as an explicit test — `test_consent_cannot_be_applied_retroactively` — with the docstring recording that it was found by a failing test rather than by design. This is now the third time in this programme that a property emerged from measurement rather than intent, and it is a better argument for the architecture than anything written in prose.

---

5. Honest limitations

Not one of these instruments has been validated in Egypt. The Arabic evidence that exists is Lebanese (MLQ, CPS), Omani, Saudi, Palestinian, Qatari and Jordanian (BPNSFS). The model records `population_validated: False` for an Egyptian respondent and does not pretend otherwise. A registry test asserts that no instrument claims Egypt among its validated populations, so this cannot be quietly relaxed.

The Arabic MLQ is not the MLQ. Fekih-Romdhane et al. (2024) found the 10-item two-factor model fit inadequately in Lebanon and dropped items 9 and 10 — which removes the only reverse-scored item and changes the Presence range from 5–35 to 4–28. The caveat is attached to the spec and travels with every score. It must be registered as a separate instrument before any Arabic administration.

BPNSFS's need-frustration construct is under live challenge. Murphy et al. (2023) argue the satisfaction/frustration distinction is probably a method artifact of item keying direction. The caveat travels with the profile, and a test asserts that it does.

Self-report is self-report. 3D-WS measures *self-perceived* wisdom, and a person low in reflective capacity may be least able to report it. No instrument here publishes a reliable change index, so `instrument_trajectory()` explicitly warns that change between administrations may be measurement error, mood, or regression to the mean.

The refusals are checkable; the phrasing is not. Invariant XI verifies that the refusing methods refuse. It cannot verify that `open_questions()` never drifts into prescription — that is guarded by a keyword test, which is the same syntactic-proxy weakness already conceded for Invariant X.

---

6. Why this belongs in P2

P2 currently argues that refusal is a safety mechanism using a **simulated** threat confound. This session supplies something better: a **real request, from a real person, refused for stated reasons, with a decision procedure and negative tests proving the refusal is not decorative.**

The proposed §4.4, *"A request the architecture declines"*, would report: the request as made; the four-way split; the finding that stage constructs cannot be de-ranked without misrepresentation; the eleven refusal grounds; Invariant XI with its two negative tests; and the retroactive-consent property that emerged from a failing test.

It also strengthens the §7.1 argument written earlier today. Psychometric responses are expressly named in Egypt's PDPL Art. 1 as sensitive data, and the PDPL grants no research exemption — so the psychometric stream is gated at the deployment context *and* the consent ledger, and refused outright under `INSTITUTIONAL` and `UNDECLARED`. Reasoning about an employee's needs, values or self-reported maturity on behalf of their employer is the exact case Art. 5(1)(f) addresses, and it is the case this project has committed not to serve.

---

7. State

Package	**v1.2.0**
Tests	**371 passed, 2 xfailed** (was 324; +47)
Invariants	**11**, all executable, 18 negative tests
Instruments registered	10 — 2 administrable, 4 licence-blocked, 4 refused by construction
Refusal grounds	12, each with a readable explanation
New defects	none

Still open, unchanged from Session 14: your corresponding email; the Phase IV decision (A / B / C); and the fact that nobody outside this project has run the code.

New, and it is a real one: if the formation model is ever used with a person, the instruments must be obtained under written permission first. Four of the six useful ones have no published licence statement. That is four emails, and they are not emails I can send.

Shadow Mirror — Session 16: Reconciling the Papers to Eleven Invariants

Author of record: Omar Moawad · **The American University in Cairo** · **Session date:** 25 July 2026
Deliverables: patched P1 and P2, `INSTRUMENT_PERMISSION_EMAILS.md`, defect D-34.

Session 15 added an eleventh invariant. P1 said "ten" in seven places. Fixing that properly turned up a second problem I was not looking for.

---

1. D-34 — the negative-test count was overstated, in our favour

Both papers reported **"16 negative tests"** for Invariants I–X. Counting them exactly:

`test_i_catches_an_explanation_with_no_evidence`	negative
`test_ii_catches_a_dropped_provenance_parent`	negative
`test_iii_catches_a_transition_with_no_triggering_evidence`	negative
`test_iv_catches_an_explanation_that_hides_its_uncertainty`	negative
`test_v_catches_a_competitor_being_dropped`	negative
`test_vi_catches_a_stale_conclusion_left_unannotated`	negative
`test_vii_catches_an_explanation_with_no_assumptions_stated`	negative
`test_viii_catches_an_inquiry_that_justifies_nothing`	negative
`test_ix_catches_a_partial_write`	negative
`test_x_catches_verdict_phrasing`	negative
`test_x_catches_a_missing_question`	negative
`test_x_catches_a_missing_disclaimer`	negative
`test_ix_passes_with_rollback`	**positive control**
`test_vi_passes_when_revise_context_is_used`	**positive control**
`test_x_accepts_an_offer`	**positive control**
`test_shipped_reflective_prompts_satisfy_human_authority`	**positive control**

Twelve negative tests and four positive controls, not sixteen negative tests. With Invariant XI's two, the correct figures are **14 negative and 4 positive controls**.

The error is small and it is the kind that matters here. A paper whose contribution is that architectural claims should be checkable rather than asserted cannot miscount its own checks. The correction is stated explicitly in P1 §3.3 rather than made silently:

> *Earlier drafts of this paper reported "sixteen negative tests", which counted the four positive controls among them. The correction is small and we make it explicitly rather than silently, because a paper about checkable claims should not miscount its own checks.*

This is the second inflated figure the programme has found in its own manuscripts, after D-33's stale test count. Both were found by re-running rather than re-reading.

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2. Invariant XI got its own randomised harness

Invariants I–X are asserted across 200 randomised graph states. XI ranges over a *formation model* rather than a graph, so the existing generator could not reach it — which would have left XI as the one invariant asserted only by hand-built cases, exactly the position the other ten started from before this programme began.

`_random_model(seed)` builds a model from a random sequence of declarations, person overrides and **attempted** instrument administrations, under a randomly chosen deployment context, with a consent grant present 80% of the time. Most administrations therefore refuse.

That is the point. **XI must hold whatever the model has been through, including every path where each administration was declined.** It is asserted across 200 seeds, with 100 further randomised checks on two derived properties: that no model ever retains a non-self-completed administration, and that a person's override always wins for display.

Suite: **673 collected, 671 passing, 2 xfailed.**

---

3. What changed in the papers

P1 — abstract, contributions, the §3.1 invariant table, §3.2 (second harness), §3.3 (the D-34 correction), §3.6 (counts), Appendix B, and a new **§3.7, "An invariant that exists because of a request"**. That section reports the five-way split of the original request, why stage instruments cannot be de-ranked, and the design choice that deserves defending: the refusing methods **exist in order to refuse**, which inverts the usual advice to delete dead code. A caller who finds no `eventual_purpose` attribute learns nothing; a caller who calls it and reads why it declined learns what the architecture will not do and on what ground.

55 ☐ 58 references. 5,137 ☐ 6,037 words.

P2 — a new **§4.4, "A request the architecture declines"**. §6.1 measures a refusal we designed against a confound we simulated; §4.4 reports a real request, from a real person, declined four-fifths, on grounds a sceptical reviewer can verify by removing the refusal and watching the check fail.

50 ☐ 56 references. 5,987 ☐ 6,670 words.

Citation integrity machine-checked on both: zero dangling, zero orphaned, zero references without a resolvable identifier.

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4. The permission letters

Four drafted — Bronk (Claremont Purpose Scale), Ardelt (3D-WS), Darling-Fisher (MEPSI), Schwartz (PVQ-RR coding instructions). Two placeholders per letter: the corresponding email, and a department if one applies.

Two of them ask a substantive question alongside the request — whether an Arabic administration is advisable, and whether MEPSI's reported total-scale alpha of .97 undercuts reading its eight subscales independently. Instrument authors receive a great many permission requests; the ones worth answering are the ones that show the instrument has been read.

Session 17 subsequently made two of these four unnecessary by finding free substitutes.

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5. State at the end of the session

Package	v1.2.0 — 673 collected, 671 passed, 2 xfailed

Invariants	11, all executable
Defects	34 identified; D-34 new this session
P1	6,037 words, 58 refs
P2	6,670 words, 56 refs

Outstanding: the corresponding email; the Phase IV decision; four permission letters to send.

Shadow Mirror — Session 17: The Workaround Hunt

Author of record: Omar Moawad · **The American University in Cairo** · **Session date:** 25 July 2026
Instruction: *"create a to-dos pending draft at the very end of the paper to include all remaining pending work … THAT YOU CAN NOT FIND AN ONLINE ALTERNATIVE OR A WORK AROUND IT"*

That instruction inverts the usual order. A limitations section is normally written by listing what you did not do. This one had to be written by *searching for a way to do each thing first*, and recording only the residue. Eleven candidate items went in. **Six dissolved. Five survived.**

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1. What a search resolved

1.1 Two blocked instruments had free replacements all along

Sessions 15 and 16 left four instruments permission-blocked and listed four emails as the fix. Two of those emails turn out to be unnecessary.

Values. The PVQ-RR was blocked twice over — permission-gated, and its item-to-value mapping file returned HTTP 403. The **ESS Human Values Scale (PVQ-21)** measures the same construct and is free in a way that is unusually well documented:

- Items published in full, in GESIS ZIS Table 1 and in every ESS source questionnaire.
- **The item-to-value mapping is published** in ZIS Table 2 — the exact thing the 403 was hiding.
- Two independent grants. ESS ERIC: *"European Social Survey documentation is licensed under CC BY-SA 4.0"* — no NonCommercial clause, no NoDerivatives clause, so translation is permitted. And GESIS ZIS: *"The measurement instrument documented here can use free of charge for non-commercial research purposes."* **Schwartz is a named author of the ZIS record**, so the second grant is the instrument author's own.
- **An official Arabic version already exists**, fielded by ESS in Israel and hosted by the B. I. Cohen Institute at Tel Aviv University. It is Levantine Arabic rather than Egyptian, so dialect review and cognitive pretesting are still required — but adapting a professionally TRAPD-translated instrument is a smaller and far more defensible job than translating from English.

Cost of the substitution: **10 basic values instead of 19 refined ones**, from 2–3 items each. If a downstream distinction ever needs societal versus natural universalism, this instrument cannot supply it. Scoring requires ipsatising — subtract each respondent's mean across all 21 items before comparing profiles — because values are relative by construction.

One implementation detail worth stating because it is easy to get backwards: ESS codes **1 = "very much like me"**, so a low raw score means high importance. All 21 items are reverse-scored in our registry so that a high subscale score means the value matters more, and a test asserts it.

Wisdom. The 3D-WS was permission-gated. The **revised Adult Self-Transcendence Inventory** (Koller, Levenson & Glück 2017) is **CC BY 4.0**, and the reason that matters is structural: **Michael**

Levenson, author of the original ASTI, is a named co-author and therefore one of the copyright holders granting the licence. It is dimensional across five subscales, and the authors argue explicitly against a total — *"if some items measure non-attachment while others measure self-knowledge, and these two constructs can occur independently, then summing up across all items is not informative"* — which is exactly the commitment this architecture already makes.

Construct drift is real here and is declared in the caveats. The ASTI measures *self-transcendence*, not wisdom. Non-attachment and peace-of-mind overlap Ardelt's reflective and affective dimensions reasonably; there is **no analogue of her cognitive dimension**, knowledge of the limits of knowing. It substitutes for the dimensional *form*, not for the construct.

A trap avoided. The Brief Wisdom Screening Scale looked like an easier escape — it is CC BY. But its Table 6 labels each item by source scale, and the sources include Ardelt's 3D-WS. Glück et al. are not Ardelt and cannot license her items. **Adopting the BWSS to escape Ardelt's permission gate walks straight back into it.** The same shape defeats the Arabic Claremont paper: CC BY article, but Bronk is not among its authors, so its licence does not license his items.

1.2 "Coverage is unmeasured" became a number, and the number is middling

Both papers previously conceded that nothing measured whether the randomised harness exercises the state space in any principled sense. Mutation testing measures precisely that, and it took an hour.

5,114 mutants. 2,319 killed. 1,549 survived. Mutation score 59.9%. A further 1,244 sit in code with no covering test at all — 909 of those in a command-line interface that has never had tests and carries no claim in either paper.

Survivors concentrate in the engine (404), the pattern layer (278), **the invariant checks themselves (206)** and the formation model (195). A sampled inspection shows most are label and message mutations rather than behavioural ones, so the raw score understates the suite.

But one class was not cosmetic, and it is the finding of the session. Mutating an invariant's own identifier from `"I"` to a nonsense string **survived**. So did replacing its human-readable name with `None`. No test asserted that a check returns the invariant it claims to be — meaning this paper's central contribution, a catalogue of eleven *named* architectural properties, could have been silently renamed and nothing would have noticed. Two tests now close it, one asserting per-check identity and one asserting that `check_all` returns exactly `I` through `XI` without duplicates.

That is the fourth time in this programme a property has been found by measurement rather than by intent, and it is the most embarrassing of the four, because it is a gap in the checking apparatus itself.

We report 59.9% rather than tuning it first. A mutation score is a lower bound on test strength and an upper bound on nothing, and 59.9% is middling for a package that makes conformance claims.

1.3 The other three

Independent execution had a route we had not looked for. Venue artifact tracks are genuinely closed — **none of TiiS, FAccT or AIES operates one** an author can enter at submission — but **CODECHECK** is live, free, accepts submissions on the basis of a preprint alone with no venue tie, and ends in a Zenodo DOI certificate naming the person who ran the code. It certifies execution rather than correctness, which is exactly the claim we need. It is an action, not a limitation.

Distribution-free coverage guarantees (D-23) were blocked on labelled outcomes. Two of the three Route A corpora supply an item-matched choice that can contradict a stated preference, so this becomes available under the secondary analysis — for the preference model only. The safety claim is untouched.

Two dead ends worth recording so nobody re-walks them. ReScience C and the ML Reproducibility Challenge both publish work written *by* the replicator; neither has any mechanism to accept an artifact and find someone to run it. And JOSS changed its scope on 5 January 2026 to require *demonstrated* research use

— *"aspirational statements about future use are not sufficient"* — so it now presupposes the deadlock it used to break.

---

2. What survived: the five irreducible items

These are now **Appendix C** of P1 and **Appendix B** of P2, each with a companion table showing what a search resolved, so a reader can tell genuine blockage from mere inconvenience.

1. The threat-confound claim requires primary collection. ~40 datasets assessed. The four-way intersection — stated stance, contradictable choice, time-locked arousal, coercion manipulation — does not exist publicly, and the failure is structural rather than incidental.

2. No instrument in the battery is validated in an Egyptian sample. Confirmed Arabic validations are Lebanese, Saudi, Omani, Jordanian, Palestinian, Qatari, Moroccan. Four Egyptian-journal leads were identified by DOI but every `*.ekb.eg` host was unreachable, and Egyptian psychometric papers frequently build local instruments rather than translating, so relevance cannot be assumed. **A colleague with Egyptian institutional access resolves this in minutes; open search does not.**

3. Two constructs have no free replacement. There is no openly licensed eight-theme Eriksonian inventory — the two structurally correct candidates are both permission-gated, and going upstream from one to the other does not escape the gate. The free material covers two of eight themes and yields single scores. For purpose beyond the MLQ, nothing free was found.

4. No party outside this project has executed the software. The venue route is verified closed. CODECHECK is available and unused as of writing, so the paper claims the code is inspectable, not that it has been independently run.

5. Invariant X and the question generator are syntactic proxies. We know of no decision procedure that settles whether a sentence claims authority over a person, and we do not think one follows from more patterns. A research problem, not an unfinished task.

Explicitly excluded from the appendix, per your instruction: two unimplemented operators recorded as strict `xfail`s, the linear read path, a candidate drift state, and the 1,549 surviving mutants. None is blocked on anything external. They are simply not done, and including them would dilute a list whose whole value is that every entry has been searched for and not found.

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3. State

Package	v1.2.1 — **679 passed, 2 xfailed** (was 671)
Instruments	13 registered, **5 administrable** (was 2): MLQ, BPNSFS, **ESS-HVS**, **ASTI-R**, LGS
Permission emails still needed	**2, not 4** — Bronk (CPS) and Darling-Fisher (MEPSI)
Mutation score	**59.9%**, weakest modules named
P1	8,783 words, 58 refs, new §10.1 and Appendix C
P2	9,441 words, 56 refs, new §9.1 and Appendix B

Not done this session, and next: the Route A secondary-analysis pipeline against BIRAFFE2, FOODEEG and Urban Appraisal, and the AUC IRB packet. Both were scoped and neither was built — the workaround hunt took the session, and it was the right trade, because two of its findings changed what the other two sessions should contain.

One operational note. I can write the Route A loaders against each corpus's published data dictionary, but I cannot pull the archives themselves into this environment. Those three downloads are yours, and they are three commands; I will supply them with the pipeline.

Shadow Mirror — Session 18: The Route A Pipeline and the AUC IRB Packet

Author of record: Omar Moawad · **The American University in Cairo** · **Session date:** 25 July 2026
Deliverables: `shadow_mirror/route_a.py`, `tests/test_route_a.py` (28 tests), `AUC_IRB_PACKET.md`, package v1.3.0.

The final session of the programme. Route A was chosen: a secondary analysis of three existing corpora, no new participants, alongside a written IRB protocol for the primary study that would be needed to test the claim no public data can reach.

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1. The loaders, and why they are written the way they are

Schemas were read from the **actual archives**, not from the papers describing them. That distinction earned its keep immediately, because **all three corpora contradict their own documentation**:

- **FOODEEG's** `task-pairedchoice_beh.json` documents 87 trials per block. That is true for `sub-006` and false for the other 116 participants, who have 60.
- **Urban Appraisal's** `dataset_description.json` says CC0 and its README says CC BY 4.0. Its sidecar spells a scale key differently from the data file.
- **BIRAFFE2's** *Scientific Data* paper prints `GAME-EXP-PAST-YEAR`; the file spells it `GAME-EXO-PAST-YEAR`.

So the files are the authority, and a schema mismatch raises `SchemaError` rather than warning. The error message prints expected columns, found columns, and the difference, because the realistic time to discover this is six months after someone re-downloaded a newer version.

1.1 Three design rules, each of which cost functionality

A loader that cannot verify its schema refuses. Not a warning. The failure mode being prevented is a silent misread, which produces numbers rather than errors.

Undocumented coding is a refusal, not a default. This is the important one, and it concerns Urban Appraisal. The archive gives anchor wording — *"valence: happy – unhappy"* — but **nowhere states which end of the numeric scale is which**. No `Levels` in the sidecar, nothing in the README, nothing in `participants.json`. A wrong polarity **does not fail**. It inverts every result that depends on sign and looks entirely normal. So `load_urban_appraisal` raises `PolarityUnknown` until a direction is passed explicitly, and records what was assumed in the returned report:

```
load_urban_appraisal(root, polarity={"valence": +1, "arousal": +1})
```

No data means no numbers. Every analysis entry point returns a structured `no_data` record rather than a plausible one, and a test asserts it. Nothing in the module can produce a figure without the archives present.

1.2 Nine file traps, each with a test

Trap	Consequence if unhandled
BIRAFFE2 is semicolon-delimited	A default comma read yields one column and no error
BIRAFFE2 abstention tokens are `''`, `NaN` **and** `None`	Abstentions silently become missing data

Trap	Consequence if unhandled
FOODEEG Session 1 uses Windows path separators, Session 2 POSIX, for the same images	`os.path.basename` will not strip a backslash on POSIX; the cross-session join fails
Every FOODEEG `.tsv` has a byte-identical `.csv` twin	A glob on `sub-*/beh/*` doubles every row
FOODEEG encodes the choice twice — `respKey` and `stimChosen_path`	Disagreement between them means the file is not what we think; the loader raises
FOODEEG applied **no RT timeout** — real values reach 66.7 s	Untrimmed outliers dominate any latency analysis
Urban Appraisal ratings live inside a semicolon-delimited string in `events.tsv`	There is no ratings file; a loader expecting one finds nothing
Urban Appraisal's beauty scale contains a raw `0x1A` byte, and the sidecar drops the character entirely	An exact-literal match finds neither spelling
Urban Appraisal's `sub-51` uses English scale labels where 62 others use German	German-only matching silently drops 336 of that participant's ratings

1.3 What the analyses can and cannot establish

Can: preference contradiction (FOODEEG records a rating and a later choice on the same items, so a choice can disagree with the person's own stated preference); abstention behaviour (BIRAFFE2 records when a participant left the rating widget at its default rather than moving it — a person declining to answer, recorded as such, which is the closest thing to a real abstention signal in any public affective corpus); order-invariance against real observations rather than generated ones.

Cannot: the threat confound. The contradiction result carries its own disclaimer in the returned object:

> *A contradiction here is a choice that disagrees with the person's own earlier rating of the same item. It is NOT evidence of a hidden preference and NOT evidence of compliance under pressure — this corpus contains no manipulation that could separate those.*

A test asserts that disclaimer is present, and another asserts that no corpus in the registry claims to support the threat-confound claim.

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2. The IRB packet

A complete draft protocol for the primary study: a two-session, within-subjects design where Session 1 collects private ratings and Session 2 collects forced choices under three social contexts — private, observed, and a displayed majority — with EDA, ECG and response latency throughout. The discriminating comparison is between contradictions arising without social pressure (candidate hidden preference) and contradictions arising only under a manufactured majority with elevated arousal and longer latency (candidate compliance).

Everything requiring a fact only the author holds is marked `[YOU]` and left blank. I will not invent a supervisor, a sample size, a start date, or a committee that has not met.

Three findings that change the sequence

Expect full board review, not expedited. AUC's expedited category covers "behaviour/cognition studies without subject manipulation or stress". §3 of the protocol is a social-pressure manipulation with incomplete disclosure. Submitting as expedited would cost a review cycle.

The PDPL licence and the CAPMAS permit must start before the IRB, not after. The study processes psychological *and* biometric data, which under Law 151/2020 Art. 12 requires a licence from the Personal Data Protection Centre in addition to written explicit consent — with no research exemption. Separately, AUC's own IRB page warns that its approval does not imply CAPMAS approval and that the IRB "is not equipped to handle matters related to CAPMAS".

Only key destruction takes the dataset out of PDPL scope. Pseudonymised data remains personal data. Until the code-to-identity key is destroyed, every obligation continues to apply. That makes the key-destruction milestone the most consequential line in the data management plan.

Two things written deliberately

The deception is stated plainly rather than buried in a methods paragraph, with the argument for why no alternative design preserves the construct, what participants are told in advance, and why the risk is low. Consent obtained at the start is treated as insufficient: at debriefing, after the true design is explained, participants are asked **again** whether their data may be retained, and one who declines has it destroyed and keeps full compensation.

A pre-committed null-result statement. The study tests an assumption the analysing software already makes, which is a real confirmation-bias risk. The protocol commits in advance that if the physiological and latency channels do *not* separate the two mechanisms, that finding is reported as prominently as a positive one — and notes that such a result would be more valuable than the one we hope for, because the architecture's headline safety result depends on those channels carrying signal.

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3. State

Package	**v1.3.0** — 709 collected, **707 passed**, 2 xfailed
New module	`route_a.py` — 3 loaders, 3 analyses, 12 documented corpus quirks
New tests	28, against fixtures reproducing the real file structure
Sessions	**18 of 18 complete**

What sits with the author: three dataset downloads (two are small — BIRAFFE2 needs only its 35 MB procedure archive, Urban Appraisal only its 16 MB git metadata); the Urban Appraisal rating polarity, from the BeMoBIL analysis code; CITI certification and a supervisor if submitting as a graduate student; and two remaining instrument permissions.

The pipeline runs the moment the archives arrive. It will not run before then, and it will not pretend to.

Part VII — Submission material

AUC IRB Packet — Draft

Study title: Distinguishing latent preference from compliance under social pressure: a behavioural and psychophysiological study **Principal Investigator:** Omar Moawad, The American University in Cairo **Version:** Draft 1 · 25 July 2026 · **NOT YET SUBMITTED**

How to read this document

This is a complete draft, not a form filled in. Everything I could write without inventing anything is written. Everything that requires a fact only you hold is marked `[YOU]` and left blank — I will not guess at a supervisor's name, a start date, a sample size you have not agreed to fund, or a committee that has not met.

Six things must happen before this is submitted, and three of them are not IRB matters:

1. `[YOU]` **CITI certification.** *Social and Behavioral Research, Stage 1, Basic*. Mandatory at AUC since 1 January 2013, and the certificate number goes on the form.
2. `[YOU]` **A supervisor**, if you are submitting as a graduate student. AUC permits graduate students to be PI but requires a named supervisor. Undergraduates may not be PI at all.
3. `[YOU]` **Confirm the current IRB chair and submission route.** AUC's IRB policy PDF and its live web page name different chairs, and the PDF describes a paper process that an online portal has superseded. Email `irb@aucegypt.edu` before preparing anything final.
4. **CAPMAS.** AUC's own IRB page warns that IRB approval does *not* imply CAPMAS approval and that the IRB "is not equipped to handle matters related to CAPMAS". This is a separate national permit for survey and interview research conducted in Egypt, with its own timeline. Start it early.
5. **PDPL.** This study processes psychological *and* biometric data. Under Law 151/2020 Art. 12 that requires a licence from the Personal Data Protection Centre **in addition to** written explicit consent. There is no research exemption. Contact `DPO@aucegypt.edu` before the IRB, not after.
6. **Expect full board review, not expedited.** AUC's expedited category covers "behaviour/cognition studies without subject manipulation or stress". §3 of this protocol is a social-pressure manipulation with a partial-deception element. Submitting as expedited would be optimistic and would cost a review cycle.

1. Purpose and scientific justification

1.1 The question

When a person states one preference and then acts against it, systems that model people routinely resolve the discrepancy in favour of the action — the revealed-preference assumption. That inference has a failure mode. Approach under stated reluctance is produced by hidden preference. It is also produced by compliance under perceived pressure. The two are not distinguishable from the surface behaviour, and the cost of confusing them is asymmetric: telling someone their compliance was really their own wish is a specific harm, and it is one no existing system measures.

This study measures how often that confusion occurs in real behaviour, and whether physiological and latency channels separate the two mechanisms.

1.2 Why it cannot be answered with existing data

Approximately forty public datasets were assessed. The question requires four measurements on the same person and the same item: a stated stance, a choice that can contradict it, an arousal signal time-locked to that choice, and a manipulation that varies social pressure. **That intersection does not exist in any public corpus.** Datasets with good physiology are passive-viewing paradigms where nothing can contradict a rating; datasets with real choice lack autonomic recording, self-report, or both. Two published studies used nearly the right design and deposited no data.

A secondary analysis of three open corpora is proceeding in parallel and can establish calibration, abstention behaviour and order-invariance on 282 existing participants. **It cannot address the question above.** That is the entire justification for collecting new data, and if the IRB judges the justification insufficient, the correct outcome is that this study does not run.

1.3 Prior work establishing the constructs

Appeasement under traumatic entrapment (Cantor & Price, 2007, *ANZJP*), tonic immobility as an evolved defence (Marx et al., 2008), and the acute stress response as a spectrum rather than a dichotomy (Bracha, 2004) establish that compliance-under-threat is a recognised phenomenon. **This study does not claim to induce or measure any of those clinical constructs.** It induces ordinary social pressure of the kind used routinely in conformity research, and the clinical literature is cited only to locate why the distinction matters.

---

2. Design

Within-subjects, two sessions, separated by 7–14 days.

Session 1 — private rating. Participants rate each of 40 items privately, on a 0–100 scale, for how much they would want to choose it. No other person sees these ratings, and participants are told so truthfully. Session 1 establishes the stated stance.

Session 2 — choice under varied social context. Participants make forced choices between pairs of the same items. Each pair is presented in one of three conditions, randomised within participant:

Condition	What the participant sees	Purpose
Private	Choice recorded privately	Baseline: contradiction without social pressure
Observed	Choice is visible to the researcher	Mild evaluative presence
Majority	A displayed distribution of prior participants' choices favours one item	Social pressure toward a specific option

The **Majority** condition is the manipulation. The critical trials are those where the displayed majority favours the item the participant rated *lower* in Session 1. A choice of the majority-favoured item on such a trial is a contradiction of their own stated preference under pressure.

Physiological recording throughout Session 2: electrodermal activity and electrocardiography, plus response latency on every trial.

The discriminating comparison. Contradictions arising in the Private condition are candidate hidden preferences. Contradictions arising only in the Majority condition, accompanied by elevated arousal and longer latency, are candidate compliance. The study tests whether those two classes separate on the physiological and latency channels — which is precisely the separation the architecture assumes and has never tested against people.

2.1 Sample size

[YOU] **To be set after a power analysis you have agreed to.** I will not put a number here that has not been justified and funded. The analysis will be pre-registered before any data is collected, including the stopping rule, so that the sample size cannot become a researcher degree of freedom.

2.2 Stimuli

[YOU] Item domain to be chosen. Everyday consumer items are the appropriate default: they support genuine preferences without touching health, politics, religion, sexuality, or anything else that would make a contradiction sensitive. **Do not use food items if any participant screening reveals disordered eating**, and see §5.3.

---

3. The manipulation, stated plainly

Participants are shown a distribution of prior participants' choices. **On a subset of trials that distribution is constructed rather than observed.** This is incomplete disclosure — participants are not told during the session which distributions are constructed.

I want to be direct with the committee about this rather than bury it in a methods paragraph.

Why incomplete disclosure is necessary. The construct under study is behaviour under social pressure. A participant told in advance which trials carry a manufactured majority has no social pressure on those trials, and the manipulation measures nothing. There is no alternative design that preserves the construct.

Why the risk is low. The pressure induced is of the kind present in ordinary social life — seeing what other people chose. No participant is told they are wrong, evaluated, judged, or observed by a person in the room during the Majority condition. There is no confederate, no simulated rejection, and no feedback about the participant's own performance or character. The literature on conformity paradigms of this kind reports transient mild discomfort and no lasting effect.

What participants are told in advance. The consent form states that *"some information about other participants' choices shown during the study may not be accurate, and the full details of the design will be explained to you at the end."* This is disclosure of the existence of incomplete disclosure, which is the standard the committee should hold us to and which we meet.

Debriefing is full, written, in-session, and includes an explicit second consent to retain the data now that the participant knows the true design (§6.3).

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4. Participants

Inclusion. Adults 18 or over, able to give informed consent, sufficient English or Arabic for the materials.

Exclusion. Anyone reporting a cardiac condition, a skin condition affecting the recording sites, or current pregnancy — all three concern the physiological sensors rather than the psychological content. [YOU] Confirm these against whatever equipment you use.

No vulnerable populations are recruited. No minors, no prisoners, no persons with impaired capacity to consent, and no one in a dependent relationship with the PI. **If you teach, your own students may not participate**, and the recruitment materials must say so, because a student choosing whether to comply with an authority figure is exactly the confound this study is about.

Recruitment. [YOU] Route to be specified. Compensation must be set at a level that does not constitute undue inducement, and must be **paid in full regardless of withdrawal** — a participant who loses money by stopping is a participant under pressure to continue, which would be an ethical failure peculiarly relevant to this study's own subject matter.

---

5. Risks and mitigations

5.1 Psychological

Risk: transient discomfort from acting against a stated preference, or from learning at debriefing that a majority display was constructed.

Mitigation: no personal feedback of any kind is given during the session; no participant is told their choices were inconsistent, suggestible, or revealing; withdrawal is available at every point without explanation; debriefing explains the design and states explicitly that behaviour under social influence is ordinary and not a defect.

What participants are never told: any interpretation of their own data. [YOU] If a participant asks what their results mean, the answer is that individual results are not interpretable and the study is about aggregate rates. This is true, and it is also the commitment the software enforces — the architecture behind this study refuses to issue a verdict about a person.

5.2 Physical

Minimal. Surface electrodes only, no stimulation of any kind. Skin irritation from the gel is the realistic risk and is rare and transient.

5.3 Screening risk

[YOU] If food stimuli are used, screen for disordered eating and exclude accordingly, and be aware that the screening instrument itself needs justification. Non-food stimuli avoid this entirely and are recommended.

5.4 Data risk

Addressed in full in §7. The short version: this study generates special-category data under two regimes simultaneously, and the design response is to collect less rather than to protect more.

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6. Consent

6.1 Informed consent form — draft text

> ***What this study is about.*** *We are studying how people make choices, and how the situation they are in affects those choices. It takes two sessions of about 45 minutes each, one to two weeks apart.* ***What you will do.*** *In the first session, you will privately rate a set of items for how much you would want to choose them. Nobody else sees these ratings. In the second session, you will make choices between pairs of those items while we record your heart activity and skin conductance using small surface sensors. The sensors only record; they do not deliver anything.* ***Something you should know in advance.*** *During the second session you may see information about what other participants chose.* ***Some of this information may not be accurate.*** *We will explain exactly how the study worked at the end of the second session, and you can decide then whether we may keep your data.* ***What we record.*** *Your ratings, your choices, how long each choice took, and your heart and skin conductance signals. We do not record your name against any of this — see below.* ***Sensitive data.*** *Your heart and skin conductance recordings are biometric data, and information about your psychological state is sensitive personal data under Egyptian Law 151 of 2020. We process them only with your explicit written consent, and only for the purpose described here.* ***You can stop at any time.*** *You do not need to give a reason, nothing happens as a result, and* ***you keep the full compensation whether or not you finish.*** *You can also ask us to delete your data at any point up to* `[YOU: date or milestone]`*, after which it will have been anonymised and we will no longer be able to find it to delete it.* ***What we will never do.*** *We will not tell you what your choices mean about you. We will not give you an individual result or interpretation, because individual results from this study are not interpretable. We will not share your data with your employer, your university, or anyone else outside the research team.* ***Questions or concerns.*** `[YOU: PI email]` *· AUC IRB: irb@aucegypt.edu · AUC Data Protection Officer: DPO@aucegypt.edu*

6.2 Consent is recorded, not assumed

Consent is captured as a **written signature and additionally recorded in an append-only, hash-chained ledger** with a separate grant per data stream — ratings, choices, physiological. Withdrawal is a first-class event in the same ledger, and a withdrawal receipt is issued. This exists so that the GDPR Art. 7(1) demonstrability question, *what was this person consenting to at this instant*, has an answer that is checkable rather than asserted.

One property worth flagging to the committee because it was not designed and was found by testing: the ledger **refuses retroactive consent**. A session timestamped before the consent grant is rejected, because the query is "what was under grant at that moment" and the answer is nothing. Backdating is not discouraged; it is impossible.

6.3 Second consent at debriefing

Because the majority display involves incomplete disclosure, consent obtained at the start is not sufficient. At debriefing, after the true design is explained, the participant is asked again whether their data may be retained. **A participant who declines has their data destroyed and keeps full compensation.** The refusal is recorded in the ledger; the data is not.

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7. Data management

7.1 What is collected, and what is deliberately not

Collected	Not collected
Ratings, choices, response latencies	Name, contact details, or student/employee ID in the research dataset

Collected	Not collected
EDA and ECG signals	Video, audio, or facial imagery
Age band, gender if volunteered	Nationality, religion, ethnicity, socioeconomic background

Nothing about a participant's background is inferred. The architecture used to analyse these data refuses to infer origin, and refuses to produce any global rating or developmental ranking of a person. That refusal is enforced by a decision procedure with negative tests, not by policy.

7.2 Identifiers

A participant code links Session 1 to Session 2. The code-to-identity key is held **separately from the research data**, on `[YOU: specify — AUC-managed encrypted storage]`, accessible only to `[YOU: named persons]`.

The key is destroyed at `[YOU: milestone]`**.** This is the single most consequential line in this packet, because **pseudonymised data is still personal data under the PDPL**, and the PDPL grants no research exemption. Only genuine anonymisation takes the dataset out of scope. Until the key is destroyed, every PDPL obligation continues to apply.

7.3 Legal basis

- **PDPL Law 151/2020.** Explicit written consent under Art. 12, **plus** a licence from the Personal Data Protection Centre for sensitive-data processing. Executive Regulations in force 2 November 2025; compliance deadline **31 October 2026**. There is no scientific-research exemption in either the law or the regulations — the words for "research", "studies" and "university" do not appear in them.
- **Law 214/2020** on clinical medical research does **not** apply. Its Article 1 subject matter is therapeutic, pharmaceutical, surgical, nutritional, preventive and diagnostic interventions. This study is none of those.
- **This is a reading of statutory text by a non-lawyer and has not been reviewed by counsel.** It is offered so the committee can correct it, not as a compliance opinion.

7.4 Sharing

Anonymised trial-level data will be deposited publicly `[YOU: repository]` after the key is destroyed. **Raw physiological time series will not be deposited without a further specific consent**, since continuous ECG carries re-identification risk that a summary statistic does not.

Analysis code is already public and its results are reproducible from a single command.

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8. What this study will not claim

Recorded here because the committee should be able to hold us to it.

- **No clinical construct is measured.** Not trauma, not a stress disorder, not a trauma response typology. Social pressure in a laboratory is not entrapment, and the study will not describe it as though it were.
- **No individual participant is characterised.** The output is a rate across people, and no participant receives an interpretation of their own behaviour.
- **No claim of generalisation beyond the sample.** In particular, none of the instruments used in the wider project has been validated in an Egyptian sample, and any that appear here will be described as first-in-population.
- **No deployment claim.** This is a research instrument. It is not suitable for clinical, educational, employment, credit, insurance or policing use, and the software refuses to run in institutional deployment contexts.

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9. Timeline

[YOU] Dates. What the sequence must be:

1. CITI certification, DPO consultation, PDPL licence application — **before** IRB submission
2. CAPMAS permit application — in parallel, separate track, expect it to be the long pole
3. IRB submission ☐ reviews sit on Thursdays; allow one month, longer for full board
4. Pre-registration of hypotheses, sample size and analysis plan — **before** the first participant
5. Data collection
6. Key destruction at the pre-registered milestone
7. Public deposit of the anonymised dataset

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10. Attachments to prepare

Attachment	Status
Protocol (§1–§9 above)	drafted
Informed consent form, English	drafted (§6.1)
Informed consent form, Arabic	[YOU] — needs professional translation, not mine
Debriefing script	[YOU] — drafted on request; needs the final design
Recruitment materials	[YOU]
CITI certificate	[YOU]
Data management plan	§7 above; may need AUC's own template
PDPL licence application	[YOU] — via the DPO
CAPMAS application	[YOU] — separate track
Instrument permissions	2 outstanding: Claremont Purpose Scale, MEPSI

A closing note on the conflict of interest in this design

The study tests an assumption the analysing software already makes. That is a real risk of confirmation bias, and the mitigations are: pre-registration of the analysis before data exists; deposit of the anonymised data so a third party can reach a different conclusion; and a pre-committed statement that a null result — that the physiological and latency channels do **not** separate the two mechanisms — will be reported as prominently as a positive one.

The architecture's own headline safety result depends on those channels carrying signal. If this study shows they do not in real people, that finding is more valuable than the one we hope for, and it should be published either way.

Four permission emails

Send these before the formation model touches a real participant. Each is short on purpose — instrument authors receive a lot of these, and the ones that get answered say what you want, what you will do with it, and what you will not do with it, in that order.

Before sending, fill in: your corresponding email address, and your department or programme at AUC if you have one. Both are marked […]. Do not send with the placeholders in.

A note on the fourth: it is not a permission request. The PVQ-RR is already free under a Creative Commons licence; what is missing is a file that would not download.

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1 — Claremont Purpose Scale

To: kcbronk@cgu.edu **Subject:** Permission to use the Claremont Purpose Scale in a non-commercial research study

Dear Dr Bronk,

I am a researcher at The American University in Cairo working on a software architecture for reflective human–AI interaction. Part of the system lets a person voluntarily complete validated self-report instruments and then reason over their own answers — it does not infer psychological constructs from behaviour, and it is designed specifically to refuse doing so.

I would like to include the Claremont Purpose Scale, and I could not find a permission statement for it, so I am asking directly rather than assuming.

What I would do: administer the 12 items as published, score the three subscales separately (following Veazey et al.'s 2023 recommendation against a total score), and cite the 2018 *Research in Human Development* paper. What I would not do: redistribute the items with the software, modify or abridge the scale, or use it commercially. The software ships scoring rules and citations only; the items would be supplied at deployment by whoever holds permission to use them.

If you would prefer the Arabic version, I would be glad to hear whether the items from Noureddine et al. (2025) can be made available — our participants would be Egyptian, and I am aware that the Lebanese validation does not straightforwardly transfer.

Would you be willing to grant permission for non-commercial academic use? I am happy to share results if that would be useful.

With thanks, Omar Moawad The American University in Cairo [corresponding email]

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2 — Three-Dimensional Wisdom Scale

To: ardelt@ufl.edu **Subject:** Request for the 3D-WS for a non-commercial research study

Dear Dr Ardelt,

I am writing to request a copy of the Three-Dimensional Wisdom Scale, following the instruction on your webpage to contact you directly.

I am a researcher at The American University in Cairo building an architecture for reflective human–AI interaction. It lets a person voluntarily complete self-report instruments and reason over their own responses; it does not infer anything about them from behaviour.

The 3D-WS is in the design for a specific reason worth stating. I went looking for a way to represent developmental maturity without collapsing a person into a single rank, and found that the stage instruments will not permit it — the ordered position is the construct, not a summary of it. The 3D-WS is the instrument I found that reports dimensions rather than a stage, and whose own criterion for high wisdom is conjunctive across the three dimensions rather than an average. That distinction is doing real work in the design, and I wanted you to know it is the reason for the request.

I would score the three dimensions separately and would not compute or report an overall wisdom score. The items would not be redistributed with the software, which ships scoring rules and citations only. Use is non-commercial and academic.

Could you send the instrument, and let me know of any conditions attached? I would also welcome your view on whether an Arabic administration is advisable — I have not been able to locate an Arabic validation, and I would rather say so in the write-up than imply coverage that does not exist.

With thanks, Omar Moawad The American University in Cairo [corresponding email]

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3 — Modified Erikson Psychosocial Stage Inventory

To: Cynthia S. Darling-Fisher, University of Michigan School of Nursing *(find the current address via the School of Nursing directory — I did not want to guess at it)* **Subject:** Permission to use the MEPSI in a non-commercial research study

Dear Dr Darling-Fisher,

I am a researcher at The American University in Cairo working on a system that lets people complete validated self-report instruments and reflect on their own responses. I would like to include the Modified Erikson Psychosocial Stage Inventory, and I could not locate a distributor or a permission statement, so I am writing to you directly.

The MEPSI is in the design because it measures the psychosocial attributes associated with Erikson's stages rather than claiming stage attainment. That distinction matters to us: the system is built to refuse producing a single developmental level for a person, and an instrument that reports eight themes rather than one position is compatible with that commitment where a stage instrument would not be.

Two practical questions, if you have time for them:

1. Would you grant permission for non-commercial academic use? The items would not be redistributed with the software — it ships scoring rules and citations only.
2. Could you confirm the response format? I have seen the instrument described with different numbers of scale points and I do not want to hard-code the wrong one. I would rather ask than assume.

I would also value your view on the reported total-scale alpha of .97. It reads high enough to suggest a strong general factor, which would bear on whether the eight subscales can carry an interpretation independently — which is exactly how I would want to use them.

With thanks, Omar Moawad The American University in Cairo [corresponding email]

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4 — PVQ-RR coding instructions *(not a permission request)*

To: the corresponding author listed on the ORPC repository entry, or Prof. Shalom H. Schwartz **Subject:** PVQ-RR coding and analysis instructions — repository files returning HTTP 403

Dear Professor Schwartz,

I am a researcher at The American University in Cairo, using the revised Portrait Values Questionnaire under the ORPC repository's Creative Commons licence.

I am not writing to request permission — the licence covers what I need. The problem is access: the supplementary files on `scholarworks.gvsu.edu`, including the PVQ-RR coding and analysis instructions that carry the item-to-value mapping, return HTTP 403 to automated retrieval, and I have not been able to obtain the mapping table.

I would rather ask you for it than reconstruct it. Our software refuses to score an instrument whose item mapping has not been verified against a primary source, precisely so that no unverified number ends up in a participant's result, and the PVQ-RR is currently registered in that refusing state.

Could you point me to a working copy of the coding instructions, or send the mapping directly? I would also be grateful to know whether an Arabic version validated in an Egyptian sample exists — I can see that Oman

and Israel Arab groups are in the 49-group psychometrics, and I would like to describe the coverage accurately rather than generously.

With thanks, Omar Moawad The American University in Cairo [corresponding email]

After a reply arrives

Record it in the model rather than in your memory. The licence gate takes written evidence, not a flag:

```
m.confirm_licence(
    "CPS",
    "Email from K. C. Bronk, 14 Aug 2026, granting non-commercial academic use; "
    "no redistribution of items. Saved at docs/permissions/cps-bronk-2026-08-14.eml",
)
```

An empty string raises. That is deliberate — a boolean can be set by accident, and a sentence has to be written on purpose.

If an author declines, or does not reply, the instrument stays refused and the study proceeds without it. That is a worse study and an honest one, and it is the right trade.

Part VIII — Outstanding work

Shadow Mirror — Outstanding Work

What is not done, what is blocked, and what would change the conclusions

Omar Moawad · *The American University in Cairo* **Companion appendix to the unified research paper draft · Package** `shadow_mirror` **v1.3.0 · 25 July 2026**

How to read this document

This is the appendix a reader should check before believing anything in the main draft.

It exists because the programme that produced Shadow Mirror ran under a rule that turns out to be unusual: **nothing is recorded as pending until a search for a substitute has come back empty.** A pending list assembled that way is the residue of a search rather than a wishlist, and it is only worth that if the searches are recorded too. So §2 lists what a search *resolved*, immediately after §1 lists what it did not.

Four categories, and the distinction between them is the point:

§1	**Blocked**	Searched for; no substitute exists. Needs collection, permission, or another party
§2	**Resolved**	Was blocking; a search found a substitute. Recorded with the cost of the substitution
§3	**Newly found**	Identified while preparing the unified draft. Two are blocking for submission
§4	**Ordinary**	Simply not done. Nothing external blocks any of it

A defect that is merely inconvenient does not belong in §1, and putting it there would dilute a list whose whole value is that every entry has been hunted for and not found.

1. Blocked: no available substitute

OW-1 · The threat-confound claim requires primary data collection

The central safety claim — that a reflective system can decline to read threat as desire — is measured against a simulator. Validating it against people requires four things measured on the same person and the same item:

1. a stated stance,
2. a behavioural choice that can contradict it,
3. an arousal channel time-locked to that choice, and
4. a coercion manipulation.

Approximately forty public datasets were assessed across affective computing, approach-avoidance paradigms, ecological momentary assessment, experimental economics and social conformity. **That four-way intersection does not exist publicly.**

The failure is structural rather than incidental. Corpora with good physiology are passive-viewing paradigms in which nothing can contradict a rating. Corpora with real choice lack autonomic recording, self-report, or

both. Two published studies used almost exactly the right design and deposited no data.

No workaround exists. Only collection.

OW-2 · No instrument in the battery is validated in an Egyptian sample

Arabic validations were confirmed for Lebanese, Saudi, Omani, Jordanian, Palestinian, Qatari and Moroccan samples. **None is Egyptian.**

Four candidate Egyptian-journal papers were identified by DOI, but every `*.ekb.eg` host was unreachable from the working environment, so their contents are unknown. Egyptian psychometric papers also frequently construct local instruments rather than translating Western ones, so their relevance cannot be assumed even if retrieved.

Any use of this battery in Egypt is first-in-population work and must be described as such in the protocol and in any resulting paper.

Partially resolvable by a colleague with Egyptian institutional access; not resolvable by open search.

OW-3 · Two instrument constructs have no free replacement

A licence hunt found free, published-mapping substitutes for two of four blocked instruments (§2). It did not for the other two.

- **An eight-theme Eriksonian inventory.** No openly licensed instrument exists. The two structurally correct candidates are both permission-gated, and going upstream from one to the other does not escape the gate. The best free material covers two of eight themes and yields single scores rather than subscales.
- **Purpose beyond the Meaning in Life Questionnaire.** No free substitute was identified.

Resolvable only by author permission, which is a letter, not a search.

OW-4 · No party outside this project has executed the software

This was the largest risk identified by the programme's own adversarial review, and it was searched for specifically.

- The **venue route is closed.** Neither ACM TiiS, nor ACM FAccT, nor AAAI/ACM AIES operates an artifact-evaluation or badging track that an author can enter at submission time.
- **Archival deposit** satisfies an availability criterion but verifies no execution.
- **Two replication communities** publish work written *by* the replicator and therefore cannot be pointed at an artifact.
- **One community code-check service** matches the need exactly — free, accepts submissions on a preprint alone, issues a citable certificate naming the person who ran the code. **It has not been used.**

The claim the work can currently make is that the code is inspectable, not that it has been independently run. Note also that this certifies execution, not scientific correctness.

OW-5 · Invariants X and XI are syntactic proxies for semantic properties

Invariant X detects verdict phrasing by pattern. The accompanying question generator is guarded against prescriptive drift by a keyword test. **Both raise a floor and neither measures the property.**

Whether an output usurps a person's authority over their own self-understanding is not a matter of which pronouns it contains. The check has already produced one documented false positive on real output, and it will produce false negatives: a sentence can be a verdict without matching any pattern in the list.

We know of no decision procedure that settles whether a sentence claims authority over a person, and we do not think one follows from more patterns. **This is a research problem, not an unfinished task** — which is why it is stated rather than left for a passing check to imply otherwise.

2. Resolved by a search — recorded so §1 reads as a residue

Was blocked	Resolved by	Cost of the substitution
Values instrument permission-gated, item mapping unreachable	The ESS Human Values Scale, whose items and item-to-value mapping are both published, under CC BY-SA documentation terms plus the instrument author's own non-commercial grant, with an already-fielded Arabic version	10 basic values instead of 19 refined, from 2–3 items each
Wisdom instrument permission-gated	The revised Adult Self-Transcendence Inventory, CC BY 4.0, dimensional, whose authors argue explicitly against a total score	Construct drift: it measures self-transcendence and has no analogue of the cognitive dimension
Harness coverage unmeasured	Mutation testing	None; it replaced an admission with a number and a work list
Invariant identity untested	Found *by* the mutation run — mutating an invariant's identifier produced surviving mutants — and closed with two tests	None
Distribution-free coverage guarantees blocked on labelled outcomes	Available under the secondary-analysis route, where two of three corpora supply an item-matched choice that can contradict a stated preference	Applies to the preference model only; the safety claim is untouched
Independent execution	A community code-check service that accepts a preprint alone and issues a citable certificate	Certifies execution, not scientific correctness. **Not yet used — see OW-4**

Two of the four drafted instrument-permission letters became **unnecessary** as a result of this search.

3. Newly identified while preparing the unified draft

These were found by assembling the unified draft against the shipped package and checking each number rather than transcribing it. Both blocking items are of the same kind: a document asserting something the archive does not support.

OW-6 · The benchmark scripts and their outputs are not in the archive · blocking

Both source papers state that every number is produced by a named script that a reviewer can run. The archive as assembled contains the package, the papers, the session reports, the figures and the submission material — but **not** the scripts or the data they wrote:

Missing	Supports
`run_benchmarks.py`, `results.json`	Every safety rate, the ablation, refusal calibration, inquiry regret, median-split parity
`conformance_v10.py`	The prototype-versus-package conformance comparison (4/7 vs 7/7)
`experiment_d23.py`	Identifiability and offset pooling
`experiment_ambivalence.py`	Ambivalence precision, pre- and post-fix
`analyze_v030_vs_v040.py`, `v030_vs_v040.json`	Calibration ECE and AUC(uncertainty ☐ error)
`make_paper_figures.py`	All six figures — the PDFs and PNGs exist; their generator does not

Consequence: the headline safety figures cannot be re-derived from what is on hand. They are carried on the authority of the session reports and the two source papers, and the unified draft's Appendix A says so explicitly rather than implying reproducibility it cannot demonstrate.

Action. Locate these files in the working environment that produced them and commit them; or regenerate and re-verify every number they support before submission. Until one of those happens, the data-availability statement in both papers overclaims.

OW-7 · The two source papers carry pre-Session-12 figures · blocking

Four figures were superseded after P1 and P2 were drafted, and both papers still carry the old values:

Where	Paper still says	Current package
P1 §7, P2 §9	Write-then-consult path is quadratic — 3.8× per doubling, 1121 ms at 800 pairs	**Linear.** ≈2.0× per doubling; 5.24 ms at 800 pairs — a 214× improvement
P1 §8	Identifiability at δ = 0.45, eight topics: 61.3%	**97.0%** under soft-assignment EM, which also refuses below three topics
P2 §8	No append-only consent record exists; withdrawal produces no receipt	**Both exist.** Hash-chained ledger; `withdrawal_receipt()`
P2 §8	No deployment-context mechanism; enforcement ratio five of twelve	**Mechanism exists.** Ratio **six of twelve**

Action. Propagate the corrections into `P1_ACM_TiiS_draft.md` and `P2_AIES_FAccT_draft.md`, or withdraw those files in favour of the unified draft. Submitting either as it stands would publish four figures the author knows to be stale — which would breach ground rule 3 of the programme.

OW-8 · Test-count drift across documents

Three documents state three different totals: the archive README says 707 passing; P1 says 673 collected and 671 passing, with a breakdown of 563 invariant checks, 14 negative tests and 4 positive controls; Session 12 says 324. The first is correct for v1.3.0 and was reproduced while preparing the unified draft (`pytest -q` ☐ **707 passed, 2 xfailed**). The others are correct for the versions they describe.

The per-category breakdown has **not** been recomputed for the current suite, so the unified draft does not restate it.

Action. Recompute from `pytest --collect-only` and state the breakdown once, in one place, with the version it belongs to.

OW-9 · Nine references sit outside the verified set

Ground rule 6 of the programme: *no citation enters a manuscript unless confirmed against a live source with a resolvable identifier.* `bibliography.json` records 105 manuscript references verified in Session 14, with `entries_without_resolvable_identifier: 0` and the note *"Every manuscript reference confirmed against a live source in Session 14."*

The papers' reference lists are longer than that set:

	Listed in the paper	In the verified set	Outside it
P1	58	55	**3** — entries [56]–[58]
P2	56	50	**6** — entries [51]–[56]

Those nine reference-list entries are **six unique sources** — three are cited in both papers. All six are psychometric instrument sources: Loevinger's ego-development measure, the Kegan subject-object interview guide, Ardelt's three-dimensional wisdom scale, the Modified Erikson Psychosocial Stage Inventory, the Meaning in Life Questionnaire, and the Basic Psychological Need Satisfaction and Frustration Scale. All were introduced in **Session 15**, which built the formation model and Invariant XI — *after* the Session 14 verification pass, and they were never added to it.

This is not a claim that any of the nine is wrong; they are well-known instruments and each carries an identifier in the reference list. It is a claim that the bibliography's own summary line overstates its coverage, and that under ground rule 6 the nine are unconfirmed until someone confirms them.

Action. Verify the nine against live sources and fold them into `manuscript_references_verified_session_14`, updating `counts`; or mark them explicitly as Session-15 additions pending verification. `papers/make_bibliography.py` emits every entry with a `keywords = {verified}` flag so the distinction survives into the `.bib`.

OW-10 · Two release decisions the repository now forces

Neither is a research defect. Both are consequences of the archive having been committed, and both need a decision before anything is made public.

The conversation transcript is in the repository. `archive/CONVERSATION.md` and `archive/TARIK_MESSAGES.md` hold the full working transcript — roughly 150,000 characters of human-authored text, including the original brief. The repository is currently **private**, so this is storage rather than publication. It stops being true the moment the repository is made public, or a zip of it is attached to a preprint or an artifact submission. The data-availability statements in both papers promise a public repository. **Action.** Decide before publishing: keep the transcript in a private repository and publish a code-only mirror, move it behind a separate access-controlled deposit, or publish it deliberately as a provenance record. Publishing it by omission is the one option to avoid.

The v1.3.0 archive sits on a branch, not on `main`. It was pushed to `shadow-mirror-v1.3.0-archive` rather than merged, so `main` still carries v0.3.0 — 34 tests, none of the papers, and none of the modules the drafts describe. Any repository URL quoted in a submission would currently resolve to the old package. **Action.** Review and merge, then confirm that the URL in the availability statements points at the merged state.

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4. Ordinary unfinished work

Nothing here is blocked on anything external. It is simply not done.

Item	Status
Causal reflection operator — intervention, counterfactual query	Declared gap, strict `xfail`. No do-operator exists. Presented as research agenda, not as a component
Knowledge consolidation operator — promotion from belief to revisable knowledge	Declared gap, strict `xfail`. No promotion criterion exists
Read path still O(n) per consultation	Open. A read-heavy pattern (one consultation per twenty writes) still scales at 3.1× per doubling. Anchored-decay accumulators would fix it
1,549 surviving mutants of 5,114 — mutation score 59.9%	Open. Concentrated in the inference engine (404), the pattern layer (278), the invariant checks themselves (206) and the formation model (195)
909 mutants in an untested command-line interface	Open. Carries no claim in any paper, which is why it is last
Candidate belief state for monotonic drift	Proposal. "Your position moved steadily" was computed and discarded
Conformal calibration for credible sets	Blocked on labelled outcomes — i.e. on a human study. A set claiming 90% mass contains the truth 60.5% of the time at $\delta = 0.45$
Encryption at rest	Specified, unimplemented
Multimodal acquisition	Specified, unimplemented
Reflective execution engine with event queues	Specified, unimplemented

Item	Status
Any user interface beyond a command line	Specified, unimplemented

5. Before submission

Item	Who can close it
Corresponding email, and a department if one applies	**You.** The only gap in the author block
Repository URL in both availability statements	**You**
Venue selection for the ethics paper — AIES or FAccT	**You.** Check current calls; dates move annually
OW-6 — recover or regenerate the benchmark scripts	Whoever holds the working environment that produced `results.json`
OW-7 — propagate corrections into P1 and P2, or withdraw them	**You**, editorially
OW-10 — decide whether the conversation transcript ships publicly	**You.** Decide before the repository is made public, not after
OW-10 — merge `shadow-mirror-v1.3.0-archive` into `main`	**You.** Until then a quoted repository URL resolves to v0.3.0
Three dataset downloads for the secondary-analysis route	Mechanical. Two are small — 35 MB and 16 MB
Urban Appraisal rating polarity	From the BeMoBIL analysis code. The loader refuses until supplied
Two instrument permissions — Claremont Purpose Scale, MEPSI	Letters to the instrument authors. Two of the original four are no longer needed
CITI certification and a named supervisor	Only if the primary study proceeds
AUC IRB packet — `[YOU]` markers	**You.** The packet is drafted and **not submitted**

6. What would change the conclusions

Stated separately from everything above, because these are the items that would move the work rather than merely extend it. Each is phrased so that a null result is still a result.

1 · A human-subjects study. Research objective O4 — *does any of this change how a person reasons about themselves?* — is unanswered. Every claim about reflection, comprehension or trust is currently a design intention with no evidence behind it. The architecture could be entirely sound and the interaction could still fail, and nothing collected so far would detect that.

2 · Primary collection on the four-way intersection (OW-1). The safety result is a property of a simulator that encodes the authors' own theory of how threat and preference differ in observable signal. Real data could refute the *ordering* of the methods, not merely the magnitude of the rates — and the ordering is the claim.

3 · A third party running the conformance suite against a different substrate (OW-4). The decision procedure has been shown to separate two implementations, which rules out vacuity. It has not been shown to be portable to a graph database, an event-sourced store, or a logic programming substrate. Until someone unconnected to this work runs it, *decidable* is demonstrated and *decidable usefully* is not.

4 · A decision procedure for semantic authority (OW-5). Invariants X and XI raise a floor they cannot measure. A real answer here would change what the strongest invariant in the catalogue is actually

worth.

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7. What is deliberately not in this document

Two things, so their absence is not read as an oversight.

Speculative extensions. Physiological sensing beyond the single arousal proxy, hierarchical Bayesian models, causal inference over the graph, LLM-assisted reflection: these appear in earlier drafts as "future research". They are not listed here because listing an idea next to a blocked dependency implies the two are comparable, and they are not. An idea is not outstanding work until something is waiting on it.

Applications. Mental-health support, education, coaching, organisational psychology. Earlier drafts listed these as potential applications. The regulatory analysis withdrew the education-pilot recommendation outright, and the software refuses to run in institutional deployment contexts. Naming application domains for a system with no human-subjects evidence and an active prohibition against two of those domains would be the kind of claim this appendix exists to prevent.